Charge + Light Foundation Model for MicroBooNE Open Data: Comparison with Reference Algorithms

Technical Note — Round 2

Haiwang Yu*

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Abstract

We compare our multi-modal (TPC charge + PMT light) masked-autoencoder foundation model, qlfm, with two kinds of reference. The first is the reconstruction already shipped in the MicroBooNE open data: Pandora's own outputs, and a gradient-boosted classifier trained on Pandora features. The second is four published foundation-model recipes, adapted to the same data: FM4NPP, Panda, PoLAr-MAE, and a self-distillation variant inspired by Panda v2.

All neural models get equal pretraining compute (about 4 node-hours each), identical task heads with identical late-fused light, and identical label budgets and test subsets. The downstream tasks are hit-level and event-level neutrino/cosmic separation, flavor classification, and energy regression.

With at least 1% of labels (3–7k events), the 21M-parameter qlfm-M is the best method on all four tasks. With a frozen backbone it reaches event AUC 0.999, flavor macro-F1 0.71, energy resolution 0.28 and hit F1 0.89. The best other foundation model, Panda, reaches 0.978, 0.68, 0.34 and 0.72; the Pandora+BDT reference reaches 0.976, 0.63 and 0.34, and Pandora's ν-slice hits reach F1 0.65.

With only 0.1% of labels, the Pandora+BDT reference beats every frozen foundation model on flavor and energy.

A per-event analysis of 80 easy and hard events (Appendix A) shows that the residual errors cluster in a few places:

- at the label boundaries (out-of-active-volume neutrinos; NC versus ν_μCC);
- at the tails of the energy spectrum;
- in events whose neutrino deposits only a few percent of the charge.

Round 2 used 37.6 Polaris node-hours.

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*Prepared with the assistance of Claude (Anthropic). Code: `https://github.com/HaiwangYu/fm4npp-lartpc`, branch `round2-references`, package `multi-modal-qlmatch/qlfm`. Tracking: GitHub issue #3. Round 1 (the qlfm model itself, ablations and scaling) is described in `docs/technote-round1.pdf` and is not repeated here.

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1 Summary of findings

1. **qlfm-M (multi-modal MAE, 21M) is the best method on all four tasks once ≥1% of labels are available.** Frozen backbone, 1% / 10% labels:

Task (metric)	qlfm-M CL	best other FM (Panda)	Pandora+BDT (+light)
T2 event ν/cosmic (AUC)	**0.999 / 0.999**	0.978 / 0.984	0.976 / 0.983
T3 flavor (macro-F1)	**0.713 / 0.755**	0.679 / 0.734	0.632 / 0.655
T4 energy (resolution, lower better)	**0.284 / 0.276**	0.343 / 0.302	0.342 / 0.308
T1 hit ν/cosmic (charge-weighted F1)	**0.892 / 0.894**	0.715 / 0.725	0.651 (ν slice)

2. **With very few labels (0.1%, about 260–720 events), Pandora+BDT is competitive.**
   - It beats every *frozen* foundation model on flavor (0.559 vs 0.531 for qlfm-M) and on energy (0.400 vs 0.418).
   - Fine-tuning qlfm-M overtakes it (0.640 and 0.385).
   - Low-dimensional reconstruction features are therefore a strong low-label prior.
   - Pandora+BDT saturates early, at flavor 0.66 and energy 0.31 at 10% labels.
3. **Panda (PTv3 + Sonata self-distillation, size-matched to 21M) is the strongest external recipe and the best charge-only representation.**
   - Without any light it still reaches T2 AUC 0.914, against 0.61–0.71 for the other charge-only backbones.
   - It is the only reference whose per-hit features beat Pandora's ν slice (0.715 vs 0.651).
4. **PoLAr-MAE and FM4NPP land between our small charge-only and charge+light models.**
   - With late light both reach T2 AUC 0.96.
   - Frozen, they fall below Pandora+BDT on flavor and energy, and below Pandora's ν slice on hits (0.54 and 0.46).
5. **Light matters for every backbone.**
   - The identical late-fused light head raises T2 from 0.61–0.91 (charge only) to 0.96–0.98.
   - qlfm's early fusion, with light inside the backbone, goes further: 0.999.
6. **Our self-distillation variant (qlfm-SD) underperforms MAE on the same backbone.** Frozen hit F1 is ≤0.15 and flavor about 0.45. The likely cause is a learning rate about 3× too low. Panda shows the objective itself works.
7. **Where the algorithms fail** (Sec. 5):
   - T2 errors concentrate on out-of-active-volume (out-of-AV) ν: 832 of the 1,208 events that at most one algorithm gets right are out-of-AV.
   - T3 errors are dominated by NC and ν_eCC events called ν_μCC.
   - T4 failures sit in the tails of the energy spectrum, because every regressor pulls predictions toward the mean.
   - T1 failures are events whose ν deposits only a few percent of the charge. Per event, qlfm-M reaches $F1_q \geq 0.8$ in 79% of events, against 51% for Pandora's ν slice and ≤25% for every other foundation model.

2 Data, tasks and metrics

All methods use the same corpus, splits, tasks and metrics as round 1.

Data. The corpus combines the MicroBooNE open-data inclusive overlay (753k events) and the ν_e overlay (77k), 830,807 events in total, in the HDF5 "NoWire" format: reconstructed wire hits, optical hits, Pandora tables and truth. It is split by (run, subrun) into train, val and test. Label pools `lab_f0.001/0.01/0.1` are nested random subsets of the train split.

Tasks and metrics.

T1 hit ν/cosmic. A hit is ν if its true ν energy deposit exceeds 0.05 MeV. Metric: charge-weighted F1 ($F1_q$) at threshold 0.5, over 3,000 test events.

T2 event ν presence. Three classes:

- 0, no-ν: $\leq$5 ν hits;
- 1, visible in-AV ν: vertex inside the active volume and $\geq$30 collection-plane ν hits;
- 2, out-of-AV ν: more than 5 ν hits otherwise.

Headline metric: AUC of class 1 versus class 0.

T3 flavor. ν_μCC, ν_eCC or NC, for events with a visible in-AV ν. Metric: macro-F1.

T4 energy. Regression of $\log E_\nu$. Resolution is half the 16–84% width of $E_{\text{pred}}/E_{\text{true}} - 1$; lower is better.

Test subsets for T2–T4 are 10,000 events drawn with a fixed seed. The Pandora references (Sec. 3.1) were scored on a 20,000-event superset drawn from the same permutation.

3 Algorithms and parameters

3.1 Pandora references (no neural network)

Pandora direct. Uses the reconstruction stored in the open-data HDF5 files as is. The "best ν slice" is the `pandoraPrimary` slice with the highest `nu_score`.

Task	Pandora-direct prediction
T1	Hits in the best ν slice are ν.
T2	Score = best `nu_score` clipped to $[-1, 1]$, or -1 if there is no slice. AUC from the score. For the event displays (3-class), score > 0.5 means class 1, else class 0. Pandora has no "out-of-AV" output.
T3	Best-slice `slice_pdg`: 14 $\rightarrow$ ν_μCC, 12 $\rightarrow$ ν_eCC, anything else (including no slice) $\rightarrow$ NC.
T4	Calorimetric: $\log E = a + b \cdot \log(1 + \sum q_Y)$, where $\sum q_Y$ is the summed collection-plane charge of the best slice. a and b come from a linear fit on all eligible *training* events, so this calibration uses 100% of labels.

Pandora + BDT. A classic "reconstruction + ML" analysis baseline. Its features are 32 charge features plus, in the "+light" variant, 8 summary features of the largest beam flash (Table 1). It is trained on exactly the same label pools as the neural models.

Parameter	Value
Model	scikit-learn `HistGradientBoostingClassifier(class_weight="balanced")` (T2, T3); `HistGradientBoostingRegressor` with squared error on $\log E$ (T4)
Boosting	`max_iter` 500, learning rate 0.1, early stopping on 15% of the label pool, patience 20
Trees	31 leaves max, `min_samples_leaf` 20, no L2, 255 bins (library defaults)
Training sets	`lab_f` 0.001 / 0.01 / 0.1 / 1.0 $\cap$ task eligibility; the same n_{train} as the neural runs (e.g. T3: 325 / 3,211 / 31,794 / 316,838)
Seeds	0, 1, 2 for label fractions $\leq$1%; 0 otherwise

Table 1: Pandora+BDT input features (per event; "best slice" as above).

Group	Features
Slices (6)	number of slices; number of ν-like slices; best `nu_score`; best slice is ν_e-like or ν_μ-like (2); $\log(1 + \text{flash-match score})$
Vertex (4)	best-slice vertex x, y, z; vertex inside the AV
PFPs (6)	PFP, track and shower counts; max / min / mean track–shower score
Hits (12)	best-slice hit count and $\log \sum q$ on U, V, Y; PFP-associated hit count and $\log \sum q$ on U, V, Y
Extents (4)	collection-plane time span and wire span; charge-weighted z; mean time
Light (8)	number of flashes; $\log(1 + \text{PE})$, time, PE-weighted z and y centroid, z RMS and number of lit PMTs of the largest beam flash; slice–flash Δz

3.2 qlfm-S / qlfm-M: multi-modal masked autoencoder (ours)

- **Tokens.** Charge hits are grouped into occupied (plane, 16-wire × 64-tick) cells; each cell becomes one token, positioned at its charge centroid. Each event also gets 32 PMT tokens, one per PMT, carrying its time profile.
- **Encoder.** A pre-LN Transformer attends jointly over charge and light tokens (early fusion).
- **Pretraining.** Masked autoencoding: a light decoder reconstructs the sub-cell charge pattern of masked charge tokens and the time profile of masked PMTs.

Parameter	qlfm-S CL	qlfm-M CL
Width / depth / heads	256 / 8 / 8	384 / 12 / 6
Backbone parameters	6.58M	21.79M
Steps (events seen)	17k (4.35M)	34k (8.70M)
Pretraining node-hours	0.72	2.05

Shared setting	Value
Block	pre-LN; FlashAttention-2 varlen (bf16); MLP ratio 4; GELU; no dropout or drop-path
Charge token	cell of 16 wires × 64 ticks; at most 4,096 per event in pretraining, 8,192 downstream
Per-hit features	in-cell offset (2), $\log(1 + q/200)$, rms/5 → MLP 4–64–128 (128-d hit embedding)
Cell embedding	PointNet max ‖ mean ‖ $(\log n_{\text{hit}}, \log \sum q)$ → MLP; plane embedding
Position	Fourier features (10 frequencies) of the charge centroid (drift cm, wire cm) + MLP(in-cell offset)
Light token	32 per event; $\log(1 + \text{PE})$ in 16 time bins (0–24 μs) + total; Fourier(PMT y, z) + PMT-ID
Charge masking	60%: half as cross-plane drift strips (256 ticks, $p = 0.3$), the rest random
Light masking	per event: 40% all PMTs, 40% half of the PMTs, 20% none
Decoder	width 192, 3 layers, 6 heads, learnable mask tokens
Targets / loss	charge: 4×8 sub-cell $\log \sum q$ + 2 statistics, standardized; light: 17-d profile; MSE_q + 1.0 MSE_L
Data	full train split (747k events), 4 GPUs × 64 = 256 events/step
Optimizer	AdamW $\beta = (0.9, 0.95)$, wd 0.05, lr 1.5×10^{-4}, 5% warmup + cosine, clip 1.0, bf16

qlfm-S C is identical to qlfm-S CL but has no light tokens.

3.3 qlfm-SD: self-distillation on qlfm tokens (Panda v2-inspired)

Panda v2 [5] has no public code, and its key hyperparameters are unpublished. We therefore test Panda-style self-distillation on the qlfm-M backbone instead, applying v2's hint to drop masked tokens and distil only the observable ones. The EMA teacher is the backbone used downstream.

Parameter	Value
Backbone	as qlfm-M (384 / 12 / 6, 21.79M), light tokens inside
Views	2 global (drift window covering 40–100% of the event's time span, masked, with light); 6 local (10–40%, unmasked, no light)
Masking	ratio 0.5 → 0.9 over the first 5%; half 256-tick strips, half random; masked tokens dropped
Head	MLP d–2048–2048–256, ℓ_2-normalized → 4,096 weight-normalized prototypes
Teacher	EMA 0.994 → 1 (cosine); sees the full event; Sinkhorn–Knopp (3 iterations); temperature 0.04 → 0.07; student temperature 0.1
Loss	cross-entropy on observed tokens; weight 0.5 split over the global views and 0.5 over the local views
Optimization	4 GPUs × 8 events; 61,922 steps (fixed by a 3.47 h budget; 1.98M events); AdamW (0.9, 0.95), wd 0.05; lr 1.875×10^{-5} (MAE batch scaling, about 3× below usual DINO scaling); clip 1.0; bf16
Pretraining	≈3.7 node-hours

3.4 FM4NPP port [3]

FM4NPP is a causal Mamba2 sequence model over individual detector hits. It is pretrained by k-nearest-next-neighbour (k-NNN) regression. We ported the released `MambaGPT` to `mamba_ssm` 2.2.5.

Three deviations were forced by 2D wire data:

- hits are serialized in a time-major raster;
- neighbours are restricted to the same plane;
- targets are *relative* in-plane offsets. Absolute coordinates were solved almost trivially by copying the input: loss 0.004 after 75 steps.

Parameter	Value
Architecture	$d = 256$, 12 blocks of $x + \text{Dropout}_{0.1}(\text{Mamba2}(\text{RMSNorm}(x)))$; d_state 16, d_conv 4, expand 2, head dim 64; 4.90M parameters (the published "m3" size)
Input embedding	fixed random projection of log-charge + fixed random projection of NeRF features (10 frequencies) of (wire/N_{wire}, tick/6400, plane/2)
Serialization	1 token per hit, at most 16,384 per event; sort by 8 drift slabs → plane → 8 wire boxes → wire → tick
Objective	$k = 30$ nearest same-plane hits later in the sequence; regress (wire cm, drift cm) offsets / 10 cm; MSE; fp32 linear head 256 → 60
Optimization	4 GPUs × 8 events (about 115k tokens/GPU); 64k steps (2.05M events, about 30G tokens); AdamW (0.9, 0.95), lr 2×10^{-4}, wd 0.01, 10% warmup + cosine, clip 0.1; bf16
Downstream features	softmax-weighted mix of all 12 layer outputs (trainable `adapter_mix`) + LayerNorm, per hit
Pretraining	3.62 node-hours

3.5 Panda [4]

Panda is a Sonata-style [6] self-distillation of a Point Transformer V3 on sparse point clouds. We use the `pimm` implementation. Each plane of each event is one 2D point cloud, with points at (wire×0.3 cm, drift cm, plane×10 cm) and energy equal to the hit integral.

Parameter	Value
Model	Sonata-v1m1 + PT-v3m2 encoder; depths (2,2,2,4,2), channels (32,64,128,256,320), heads (2,4,8,16,20), patch 256, strides 2; serializations Hilbert, Hilbert-transposed, Z, Z-transposed (shuffled); student drop-path 0.3; 20.57M backbone (size-matched; published 91M)
Input	4 channels (normalized x, y, z; log energy); grid 0.3 cm; NormalizeCoord centre (520, 130, 10) cm, scale 600; LogTransform 10–5000
Views / augmentation	2 global (40–100%) + 6 local (10–40%) crops, at most 30k points; in-plane rotation, flips, jitter, scale [0.9, 1.1], energy jitter 5%
Masking	patch 2 → 15 cm; ratio 0.5 → 0.9 (5% warmup)
Head	up-cast 2 levels (704-d) → MLP 4096 → 256 → 4,096 prototypes; Sinkhorn–Knopp; teacher temperature 0.04 → 0.07; student 0.1; EMA 0.994 → 1; loss weights mask 2/8, roll-mask 2/8, unmask 4/8
Optimization	batch 64 clouds (16/GPU); 1.84M clouds once (28,750 iterations, 0.44 s each); AdamW base lr 3.0×10^{-3} with block-wise decay 0.9; wd 0.04 → 0.2; OneCycle (5%, cosine); clip 3.0; bf16
Downstream features	teacher backbone, up-cast to the 0.3 cm voxels (704-d); each hit takes its voxel's feature
Pretraining	≈3.7 node-hours

3.6 PoLAr-MAE [2]

PoLAr-MAE is a masked point-cloud autoencoder. A ViT encodes point groups "patches", and the model reconstructs the positions and energies of masked groups. It uses the same plane clouds as Panda.

Parameter	Value
Encoder	ViT-S: 384-d, 12 layers, 6 heads, post-norm, position added at every layer, attention dropout 0.05, drop-path 0.25; 21.3M (decoder 384-d × 4 layers, 7.1M; total 29.0M)
Tokenizer	farthest-point sampling of 2048 candidates → C-NMS (overlap 0.72) → at most 1,024 groups per cloud; ball radius **4 cm** (published 1.5 cm); 32 points per group; mini-PointNet embedding
Objective	60% random group masking; Chamfer + per-point energy loss (weights 1, 1)
Optimization	batch 64 clouds; 1.67M clouds × 3 passes (78,281 iterations, 0.16 s each); AdamW lr 4×10^{-4}, wd 0.05; OneCycle (5%, cosine); clip 3.0; fp32
Downstream features	encoder without masking; each hit takes the token of the nearest group centre
Pretraining	≈3.7 node-hours

The 4 cm radius is needed because a 10 m × 3.5 m plane with about 26 cosmic tracks would need about 2k groups at 1.5 cm.

3.7 Common downstream interface and protocol (R2P)

Every backbone exposes token features z, an event index, and a hit-to-token map (plus a 128-d hit embedding for qlfm). Identical heads sit on top:

- **AttnPool** (T2–T4): LayerNorm, then 4 learned queries × 8 heads over all tokens of the event, then an MLP $4d$–256–n_{out}.
- **HitHead** (T1): [hit embedding (qlfm only) ‖ LN(token feature of the hit) ‖ pooled light] → MLP 128–128–1.
- **LateLight**: the qlfm PMT embedding (32 tokens per event), added to the pooled token set (T2–T4) or the hit feature (T1) of *every* backbone. qlfm-CL also has light inside the backbone (early fusion).

Parameter	Value
Modes	frozen at 0.1% (seeds 0, 1), 1%, 10%; fine-tuned at 0.1%, 1% (layer-wise lr decay 0.9); frozen 1% without late light (charge-only diagnostic)
Training sets (n_{train} at 0.1 / 1 / 10%)	T1 343 / 3,381 / 33,426; T2 722 / 7,139 / 71,567; T3 325 / 3,211 / 31,794; T4 262 / 2,669 / 26,801
Optimizer	AdamW; lr 10^{-3} (frozen), 2×10^{-4} (fine-tune); wd 0.05; 5% warmup $\times$ cosine; clip 1.0; bf16
Steps	clip(30 epochs, 400, 3000); batch 32 events (gradient accumulation 4×8 for memory-heavy runs)
Losses	T1: BCE weighted by $\log(1+q)$; T2/T3: class-weighted cross-entropy; T4: Huber ($\delta = 0.5$) on $\log E$
Selection	validation every 300 steps (2k events; 1k for T1); the best-on-val checkpoint is tested
Matrix	8 backbones $\times$ 4 tasks $\times$ 7 runs = 224 runs, all complete

4 Results

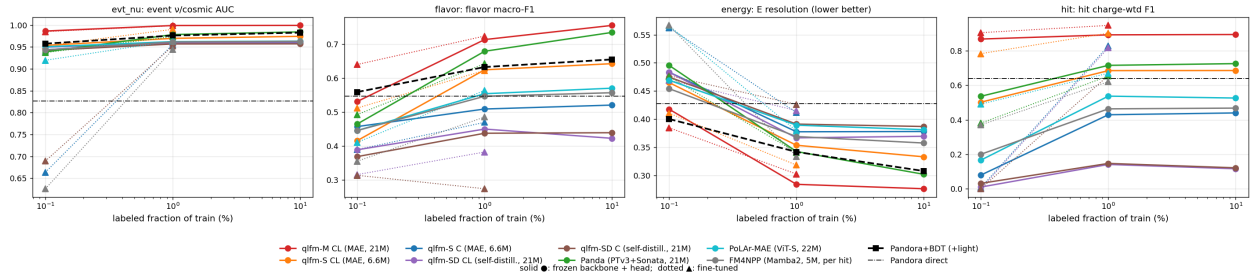


Figure 1: All four tasks versus label fraction. Solid lines with circles: frozen backbone + head. Dotted lines with triangles: fine-tuned. Black dashed: Pandora+BDT (+light). Dash-dot: Pandora direct.

In the tables below, $\pm$ is the spread over 2 seeds; every other entry is a single seed. † marks Panda "fine-tune" entries: these runs trained only the head, because the `pimm` teacher backbone has `requires_grad=False` (Sec. 7). "C-only diag" is the frozen run at 1% without the late-fused light; qlfm-CL backbones still contain light internally.

Table 2: T2 event-level ν/cosmic: AUC (in-AV ν vs no-ν).

Method	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	C-only diag
qlfm-M CL (MAE, 21M)	**0.986**±.002	**0.999**	**0.999**	**0.985**	**0.998**	0.999
qlfm-S CL (MAE, 6.6M)	0.953±.001	0.970	0.974	0.953	0.990	0.972
qlfm-S C (MAE, 6.6M)	0.943±.001	0.960	0.962	0.664	0.956	0.666
qlfm-SD CL (21M)	0.949±.000	0.959	0.959	0.940	0.959	0.605
qlfm-SD C (21M)	0.942±.000	0.957	0.957	0.690	0.952	0.609
Panda (21M)	0.938±.007	0.978	0.984	0.937†	0.974†	0.914
PoLAr-MAE (22M)	0.950±.001	0.962	0.963	0.920	0.964	0.660
FM4NPP (4.9M)	0.943±.000	0.961	0.963	0.626	0.944	0.712
Pandora+BDT C	0.926±.001	0.949	0.955			
Pandora+BDT CL	0.957±.003	0.976	0.983			
Pandora direct	0.826 (no training)					

Table 3: T3 flavor (ν_μCC / ν_eCC / NC): macro-F1.

Method	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	C-only diag
qlfm-M CL	0.531±.003	**0.713**	**0.755**	**0.640**	**0.724**	0.709
qlfm-S CL	0.416±.002	0.624	0.642	0.512	0.622	0.625
qlfm-S C	0.457±.000	0.509	0.520	0.389	0.469	0.429
qlfm-SD CL	0.389±.003	0.450	0.423	0.316	0.382	0.297
qlfm-SD C	0.369±.001	0.437	0.439	0.314	0.274	0.303
Panda	0.465±.001	0.679	0.734	0.492†	0.642†	0.666
PoLAr-MAE	0.446±.004	0.554	0.570	0.411	0.563	0.497
FM4NPP	0.445±.003	0.546	0.557	0.355	0.485	0.507
Pandora+BDT C	0.542±.007	0.619	0.641			
Pandora+BDT CL	**0.559**±.007	0.632	0.655			
Pandora direct	0.547 (no training)					

Table 4: T4 neutrino energy: resolution (lower is better).

Method	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	C-only diag
qlfm-M CL	0.418±.003	**0.284**	**0.276**	**0.385**	**0.303**	0.284
qlfm-S CL	0.465±.007	0.354	0.333	0.411	0.318	0.345
qlfm-S C	0.484±.001	0.378	0.378	0.563	0.412	0.441
qlfm-SD CL	0.483±.005	0.367	0.370	0.469	0.415	0.439
qlfm-SD C	0.475±.002	0.391	0.387	0.477	0.426	0.454
Panda	0.496±.001	0.343	0.302	0.476†	0.341†	0.423
PoLAr-MAE	0.469±.002	0.389	0.381	0.566	0.379	0.421
FM4NPP	0.454±.013	0.369	0.358	0.567	0.334	0.420
Pandora+BDT C	0.407±.005	0.355	0.324			
Pandora+BDT CL	**0.400**±.005	0.342	0.308			
Pandora direct	0.428 (calibrated on 100% of labels)					

Table 5: T1 hit-level ν/cosmic: charge-weighted F1. Pandora's ν-slice hits on the same 3k test events: $F1_q$ 0.651 (efficiency 0.784, purity 0.556).

Method	frozen 0.1%	frozen 1%	frozen 10%	FT 0.1%	FT 1%	C-only diag
qlfm-M CL	**0.868**±.000	**0.892**	**0.894**	**0.905**	**0.947**	0.891
qlfm-S CL	0.502±.005	0.685	0.685	0.783	0.902	0.675
qlfm-S C	0.080±.001	0.429	0.440	0.008	0.828	0.170
qlfm-SD CL	0.010±.001	0.141	0.116	0.001	0.818	0.000
qlfm-SD C	0.032±.003	0.147	0.122	0.001	0.657	0.004
Panda	0.536±.003	0.715	0.725	0.381†	0.653†	0.629
PoLAr-MAE	0.167±.025	0.537	0.526	0.491	0.670	0.351
FM4NPP	0.200±.013	0.463	0.468	0.371	0.620	0.238
Pandora ν slice	0.651 (no training)					

5 Event-level analysis and event displays

5.1 Per-event correctness and selection

To see *which* events each metric rewards or punishes, we score every test event for 8 algorithms: Pandora direct, Pandora+BDT (+light, 1% labels, seed 0), and the six foundation models qlfm-M CL, qlfm-S CL, Panda, PoLAr-MAE, FM4NPP and qlfm-SD CL. Each foundation model is frozen with a late-light head at 1% labels, seed 0.

An algorithm is *correct* on an event if:

- T2, T3: its predicted class (argmax) equals the truth;
- T4: $|E_{\text{pred}}/E_{\text{true}} - 1| < 0.2$;
- T1: its per-event $\text{F1}_q \geq 0.8$. For T1, "Pandora" means its ν-slice hits, and there are 7 algorithms because Pandora+BDT has no hit output.

For each task we then select:

- **10 easy events**, those the most algorithms get right;
- **10 hard events**, those the fewest get right.

Picks are spread round-robin over truth classes (T4: $E < 0.7$, 0.7–1.5, > 1.5 GeV). For T1, the hardest events overall all have a ν charge fraction below 1%, so 5 hard picks are taken from them and 5 from events with a ν charge fraction $\geq$5%.

Fig. 2 and Table 6 summarize agreement over the full test subsets.

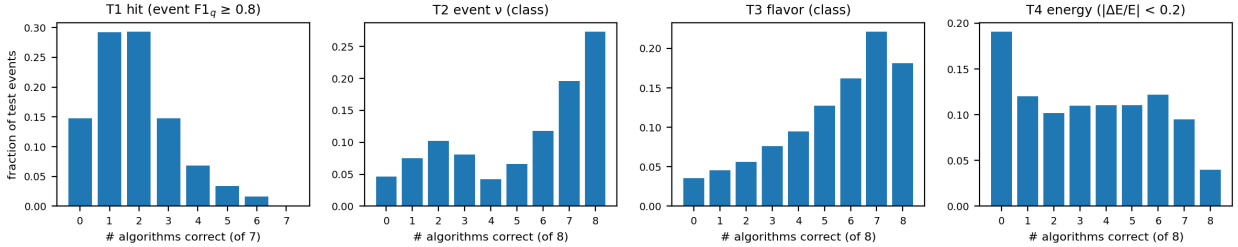


Figure 2: Distribution of the number of algorithms that are correct on each test event (T1: 3k events, 7 algorithms; T2–T4: 10k events, 8 algorithms).

Table 6: Per-event accuracy under the correctness rules above (T2: 3-class accuracy, including out-of-AV). "Hard": accuracy on the consensus-hard events, those where $\leq$1 algorithm is correct (T1: none correct). Their counts are 1,208 (T2), 810 (T3), 3,106 (T4) and 444 (T1).

	T1 hit		T2 event ν		T3 flavor		T4 energy	
Algorithm	all	hard	all	hard	all	hard	all	hard
qlfm-M CL	**0.786**	0	**0.755**	**0.242**	**0.800**	0.081	**0.485**	**0.094**
qlfm-S CL	0.212	0	0.677	0.030	0.710	0.033	0.430	0.054
Panda	0.249	0	0.727	0.136	0.742	**0.179**	0.434	0.082
PoLAr-MAE	0.066	0	0.668	0.006	0.638	0.022	0.403	0.008
FM4NPP	0.043	0	0.666	0.006	0.640	0.037	0.390	0.011
qlfm-SD CL	0.000	0	0.666	0.010	0.537	0.054	0.389	0.008
Pandora direct / ν slice	0.509	0	0.437	0.096	0.593	0.093	0.383	0.052
Pandora+BDT (+light)	–	–	0.692	0.094	0.712	0.060	0.451	0.076

5.2 What the hard events have in common

T2. Of the 1,208 consensus-hard events, 832 are out-of-AV ν (class 2), although that class is 34% of the test set. The foundation-model majority calls them no-ν (426) or in-AV ν (406). The reverse confusions, in-AV called out-of-AV (215) and no-ν called out-of-AV (108), are the next largest.

The class-0/2 boundary ($\leq$5 vs >5 ν hits) and the class-1/2 boundary (vertex inside the AV with $\geq$30 collection hits) are sharp in truth but nearly invisible in the data. Display T2-hard #1 is an example: a ν_μCC with its vertex 10 cm below the AV leaves at most 5 ν hits next to a beam-coincident flash. It is labelled no-ν, while every foundation model and the BDT call it out-of-AV ν.

The headline AUC (class 1 vs 0) avoids class 2 entirely, which is why it is near 1 while the 3-class accuracy is 0.67–0.76.

T3. The consensus-hard events are evenly split by truth: 211 ν_μCC, 283 ν_eCC and 316 NC. The dominant confusions are NC $\to \nu_\mu$CC (257), ν_eCC $\to \nu_\mu$CC (186) and ν_μCC $\to$ NC (151). These are the expected physics ambiguities: a short or absent muon track, π^0 showers, and electrons hidden under cosmics. Panda is the best on this hard subset (0.18), suggesting its point-level features keep finer topology than cell tokens.

T4. Consensus-hard events sit in the tails: the median E_ν is 1.31 GeV versus 1.14 overall, and the 10–90% range is 0.47–2.99 GeV versus 0.63–2.10. All regressors shrink toward the mean, so low-energy events are over-predicted and high-energy events under-predicted. The foundation-model mean is too high for 45% of these events.

T1. No event is correct for all 7 algorithms, because qlfm-SD never reaches $\mathrm{F1}_q \geq 0.8$ on any event. The 444 events that every algorithm misses have a median ν charge fraction of 2.6%, against 5.6% overall. A small ν (a few short tracks next to tens of cosmics) is the main failure mode for every method, Pandora included.

On typical events qlfm-M is clearly ahead. Its per-event median $\mathrm{F1}_q$ is 0.91, against 0.81 for Pandora's ν slice and 0.67, 0.64, 0.42, 0.32 and 0.01 for Panda, qlfm-S, PoLAr-MAE, FM4NPP and qlfm-SD.

5.3 Selected examples

These are referred to by their appendix labels.

T1 easy #1 A 1.6 GeV ν_eCC shower downstream (vertex $z = 910$ cm), carrying 16% of the event charge. The flash is centred on the vertex. Six of the seven algorithms reach $\mathrm{F1}_q \geq 0.94$. qlfm-SD (0.77) also tags parts of nearby cosmics, the diffuse $p(\nu)$ typical of its weak per-hit features.

T1 hard #7 A 1.6 GeV ν_μCC with its vertex near the top of the TPC ($y = 111$ cm), crossed by several cosmics. Pandora's ν slice is a neighbouring cosmic track ($\mathrm{F1}_q = 0$). qlfm-M recovers about half of the ν charge (0.45) and Panda about a quarter (0.27). qlfm-S reaches 0.16; PoLAr-MAE, FM4NPP and qlfm-SD find nothing.

T2 hard #1 Discussed above. A ν_μCC with its vertex just below the AV is labelled no-ν. The beam-coincident flash and a few ν hits lead every foundation model to "out-of-AV ν". This is a label-boundary case, not a detector-level failure.

T3 hard #1 A 0.5 GeV ν_μCC with its vertex 15 cm below the top of the AV. Only a few short ν tracks are visible, and all eight algorithms call it NC. Without a clearly identifiable muon, low-energy ν_μCC is topologically NC-like.

T4 hard #4 A 0.35 GeV ν_μCC at $z = 142$ cm lies next to a large, bright cosmic-induced deposit at the same drift time, and the beam flash covers both.

- Pandora's ν slice absorbs that deposit, giving $E = 1.87$ GeV (+429%).
- Every foundation model overestimates too, by +81% to +154%.
- qlfm-M's hit head isolates the ν hits well ($\mathrm{F1}_q = 0.86$), but its energy head is a separate attentive pool over all tokens and does not use that separation.

This suggests a task-coupled head that conditions energy on the predicted ν hits.

5.4 How to read the displays

Appendix A shows all 80 selected events, one per page. Each page contains:

- a header with the task, easy or hard, the number of correct algorithms, run/subrun/event, the truth (ν flavor, CC/NC, energy, vertex, task label) and each algorithm's task prediction (✓ / ×);

- **Charge**: every hit at (global channel, tick), coloured by $\log_{10}$ integral. U, V and Y are concatenated (channels 0–2399, 2400–4799, 4800–8255). Tick increases downward, i.e. away from the anode;
- **PMT view** (inspired by BEE): each PMT drawn at its (z, y) position. Circle area is proportional to that PMT's PE in the largest beam flash (or the summed optical hits if there is no flash). Grey rings mark all 32 PMTs; the star marks the true ν vertex;
- **eight hit panels**: Truth (ν = red), Pandora (ν-slice hits), and the six foundation-model hit heads (frozen, 1%, late light). Colour is $p(\nu)$ from blue (cosmic) to red (ν). Each title gives that algorithm's per-event $F1_q$. Dashed boxes mark the 1–99% extent of the true ν hits in each plane.

The hit panels are shown for every event, not only the T1 selection, so the event-level predictions can be compared with how each backbone separates ν from cosmic charge in the same event. The hit probabilities come from re-running the frozen 1% T1 heads on these 80 events (batch 4 instead of 32, bf16). The per-event $F1_q$ agrees with the original test predictions to within 0.01 for the qlfm models and FM4NPP, 0.03 for PoLAr-MAE and 0.06 for Panda.

6 Discussion

- **Versus the open-data reconstruction.**
  - Every foundation model with light beats Pandora's direct `nu_score` on T2 by a wide margin (0.94–0.999 vs 0.83).
  - On per-hit T1, only qlfm-M (all modes), qlfm-S CL (fine-tuned) and Panda (frozen, $\geq$1% labels) beat Pandora's ν slice.
  - Pandora+BDT is the tougher reference. Its 40 reconstruction features are very label-efficient but saturate early.
  - The BDT's own gain from light (+0.02–0.03 AUC on T2) shows light still carries information after reconstruction.
- **Versus other foundation-model recipes at equal compute.**
  - At matched size (about 21M) and node-hours, qlfm-M's multi-modal MAE with early light fusion gives the best frozen features on every task.
  - Panda is the clear runner-up and the best charge-only representation.
  - There is no charge-only qlfm model at 21M, so how much of qlfm-M's edge comes from early fusion versus the MAE objective versus the cell tokenizer is still open.
- **FM4NPP and PoLAr-MAE** learn useful but weaker features under these adaptations.
  - FM4NPP: plausible limits are causal per-hit serialization of cosmic-dominated 2D projections, the relative-offset objective, and its small (4.9M) size.
  - PoLAr-MAE: plausible limits are coarse 4 cm groups along long cosmic tracks.
- **qlfm-SD** gives poor per-hit and mediocre event features at this budget. Likely causes are the low learning rate, the drift-window crops, and dropping masked tokens without a mask token. A tuned qlfm-SD is a candidate for round 3.
- **Fine-tuning at 0.1% labels** (a few hundred events) is unstable for several backbones, e.g. FM4NPP T2 0.63 and qlfm-S C 0.66. Frozen probes are the more reliable low-label mode.

7 Limitations and caveats

- **Panda "fine-tune" is head-only.** The downstream Panda backbone is the `pimm` Sonata teacher, whose parameters `pimm` sets to `requires_grad=False`. The "FT" runs therefore trained the same 2.30M head parameters as the frozen runs, with the fine-tune learning rate. These entries (†) do not measure Panda fine-tuning. The frozen Panda results are unaffected.
- **PoLAr-MAE fine-tuning** also counts the MAE decoder as trainable. The decoder gets no gradient, so this inflates only the parameter count. Encoder drop-path (0.25) is active during fine-tuning.
- **Five runs hit the time limit before 3,000 steps**: FM4NPP T1 FT 1% (1,800 steps), FM4NPP T1 C-only (1,500), and PoLAr-MAE FT 1% on T1, T2 and T3 (2,400–2,700). Fine-tuned heavy backbones may be under-trained in general.
- **Gradient accumulation** (4×8) was used where memory required it. The optimizer batch is always 32.
- **Pandora test subsets.** The Pandora references are scored on a 20k-event test subset, a superset of the neural 10k subset drawn from the same permutation. The Pandora-direct energy calibration uses all training labels.
- **Scope of the comparison.** Every reference is an *adaptation* (2D planes, sizes, targets) trained for about 4 node-hours, far less than its published schedule; Panda, for example, saw about 18% of its published samples. The results compare these adaptations at equal compute, not the published models.
- **Statistics and data.** Most entries are single-seed; differences below about 0.02 are not significant. Pandora+BDT uses only the HDF5-level reconstruction. Round 1's overlay-only caveats apply.

8 Reproducibility and compute

- Pandora references: `python -m qlfm.data.pandora_features`, `scripts/pandora_refs.py`. Per-event predictions: `scripts/pandora_event_preds.py`.
- Pretraining:
  - `qlfm.train_pretrain` with `--method {mae,sd,fm4npp}`; FM4NPP runs in the `mamba_ssm` venv.
  - Panda and PoLAr-MAE: `qlfm/ref/pimm_train.py` with the configs in `qlfm/ref/pimm_configs/`, in the pimm venv.
- Downstream: `scripts/make_tasks_r2.py`, then `scripts/run_queue.sh`. Tables: `scripts/summarize_r2.py`.
- Event displays:
  - `scripts/r2_select_events.py` (selection) and `scripts/r2_agreement.py` (statistics);
  - `scripts/infer_hits.py` via `scripts/pbs/infer_hits.pbs` (per-hit inference);
  - `scripts/r2_event_displays.py` (rendering).
- Compute: 37.6 node-hours for round 2 (`docs/figures/round2_node_hours.md`), plus about 0.1 for the display inference.

References

[1] G. Cerati (for MicroBooNE), *MicroBooNE public data sets: a collaborative tool for LArTPC software development*, arXiv:2309.15362.

[2] *Particle trajectory representation learning with masked point modeling* (PoLAr-MAE), arXiv:2502.02558.

[3] *FM4NPP: a scaling foundation model for nuclear and particle physics*, arXiv:2508.14087.

[4] *Panda: self-distillation of reusable sensor-level representations for high energy physics*, arXiv:2512.01324.

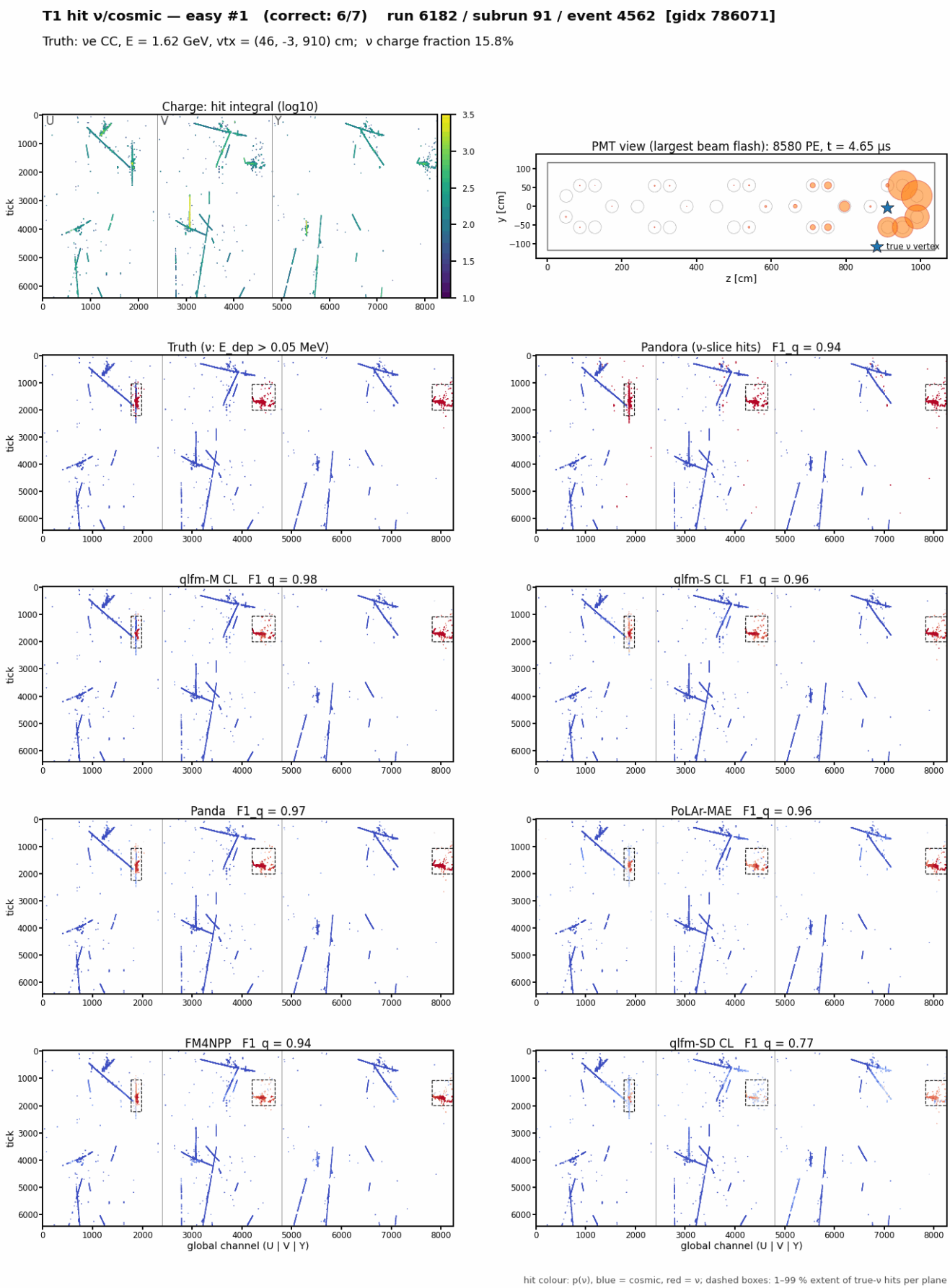
[5] *Panda Diplomacy: foundation model pre-training across particle imaging detectors*, arXiv:2609.00611.

[6] X. Wu et al., *Sonata: self-supervised learning of reliable point representations*, arXiv:2503.16429.

A Event displays

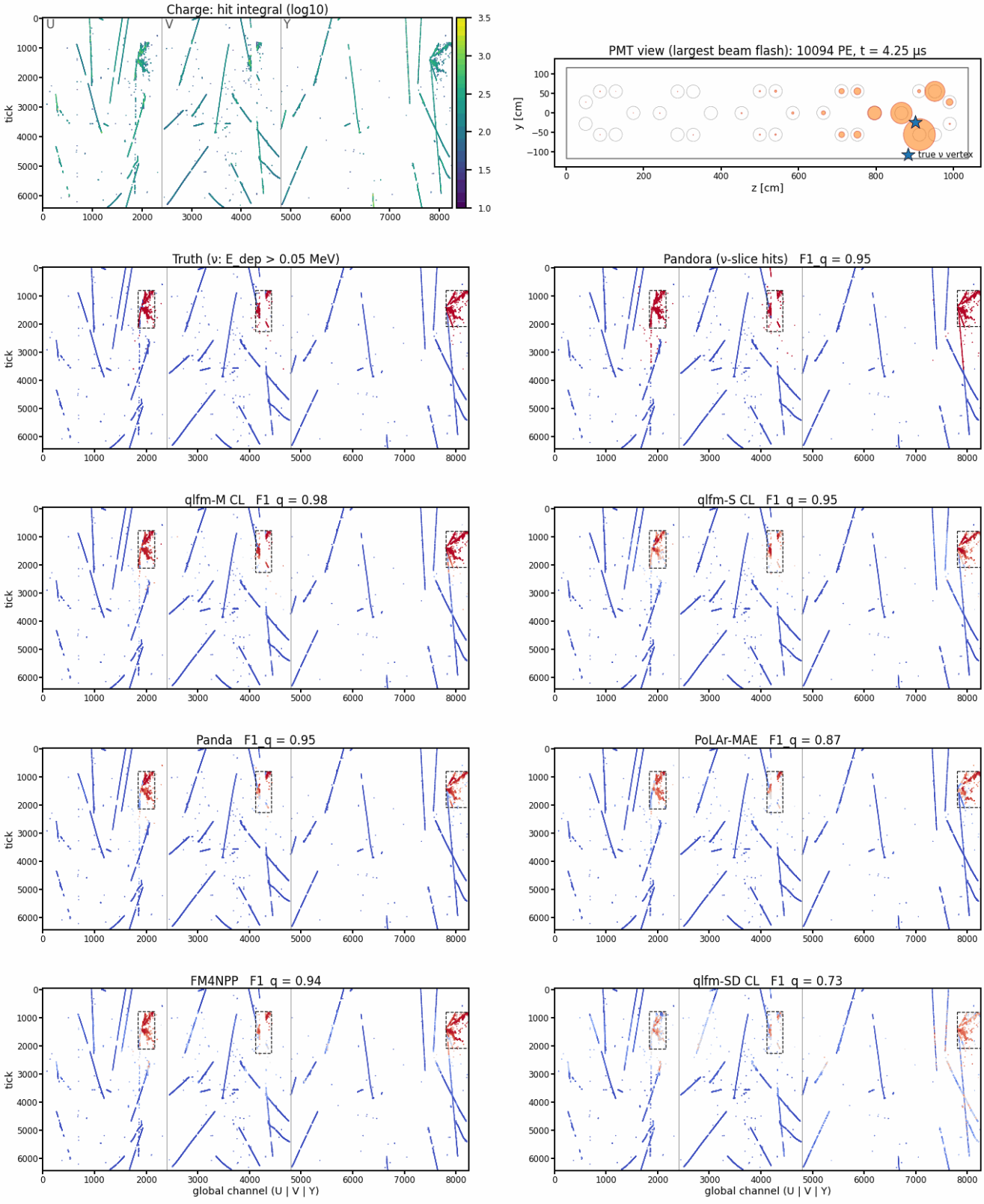
One page per selected event: for each task, 10 easy events (most algorithms correct) followed by 10 hard events (fewest correct). The selection rules are in Sec. 5. Hit panels use the frozen 1% T1 heads of every foundation model, regardless of the task the event was selected for.

A.1 T1: hit-level ν/cosmic



T1 hit ν/cosmic — easy #2 (correct: 6/7) run 7004 / subrun 535 / event 26791 [gidx 357251]

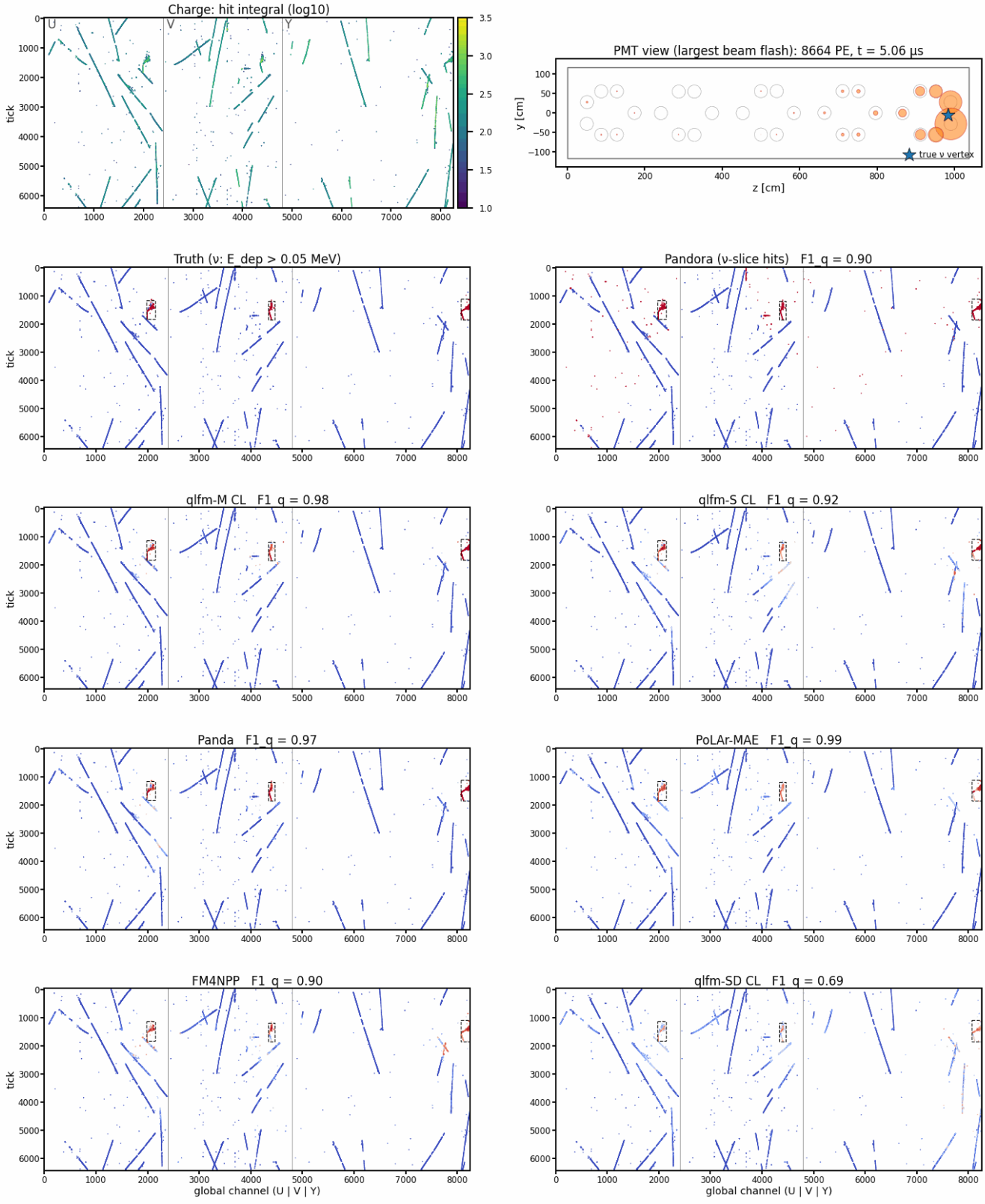
Truth: νμ CC, E = 3.36 GeV, vtx = (31, -24, 902) cm; ν charge fraction 28.3%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #3 (correct: 6/7) run 7014 / subrun 912 / event 45601 [gidx 767806]

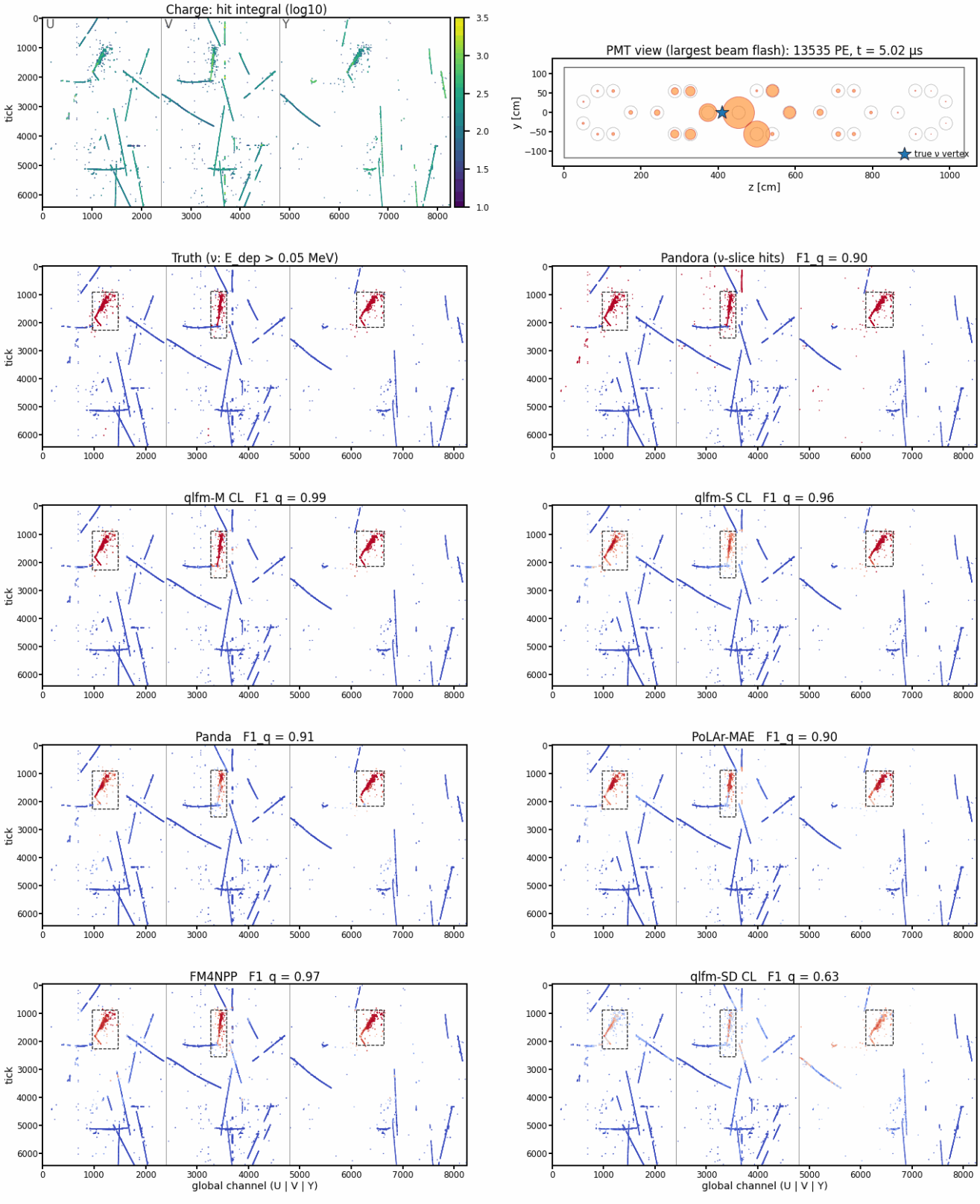
Truth: νe CC, E = 2.10 GeV, vtx = (39, -6, 984) cm; ν charge fraction 14.6%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #4 (correct: 6/7) run 7023 / subrun 422 / event 21139 [gidx 819059]

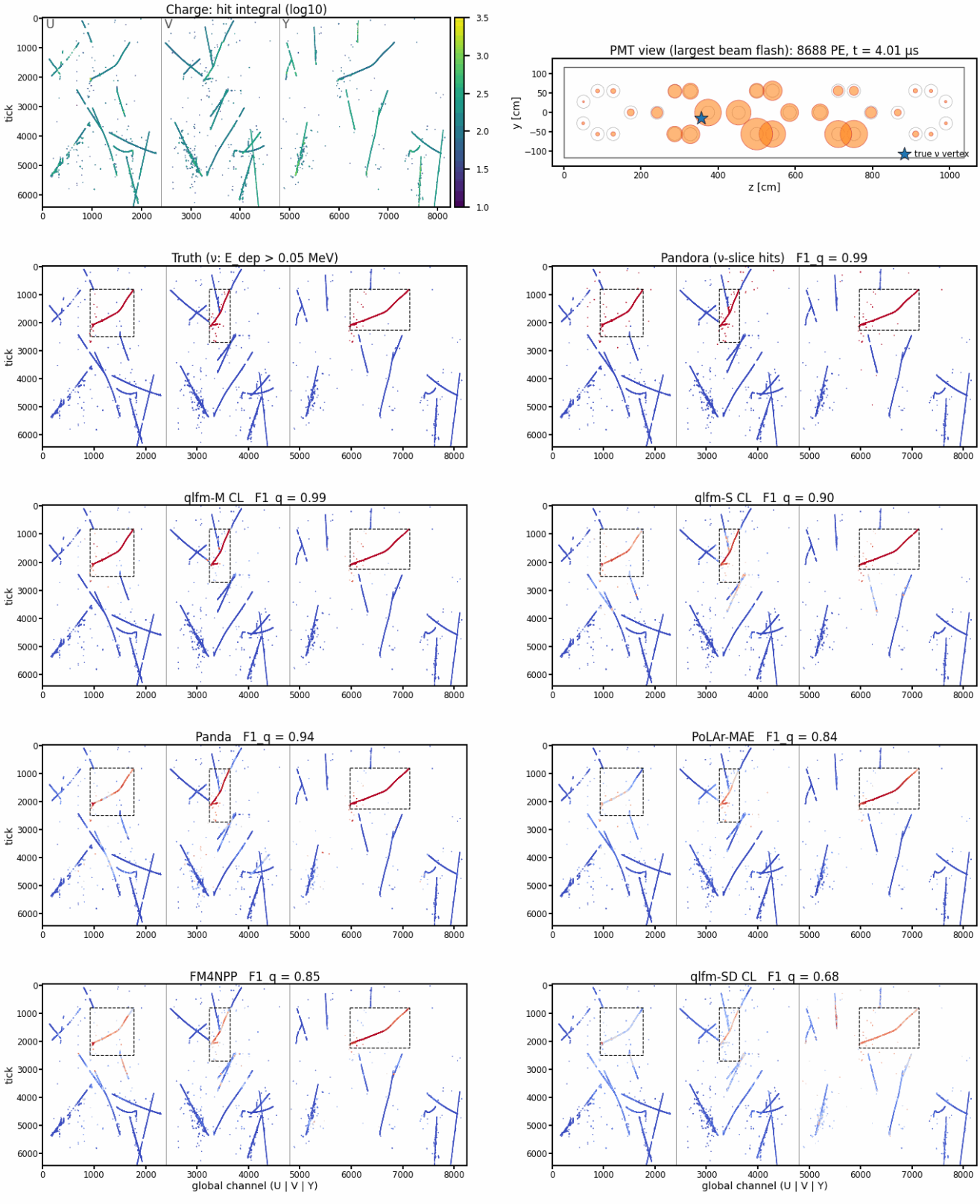
Truth: νe CC, E = 2.28 GeV, vtx = (53, 1, 409) cm; ν charge fraction 22.7%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #5 (correct: 6/7) run 6642 / subrun 105 / event 5252 [gidx 697965]

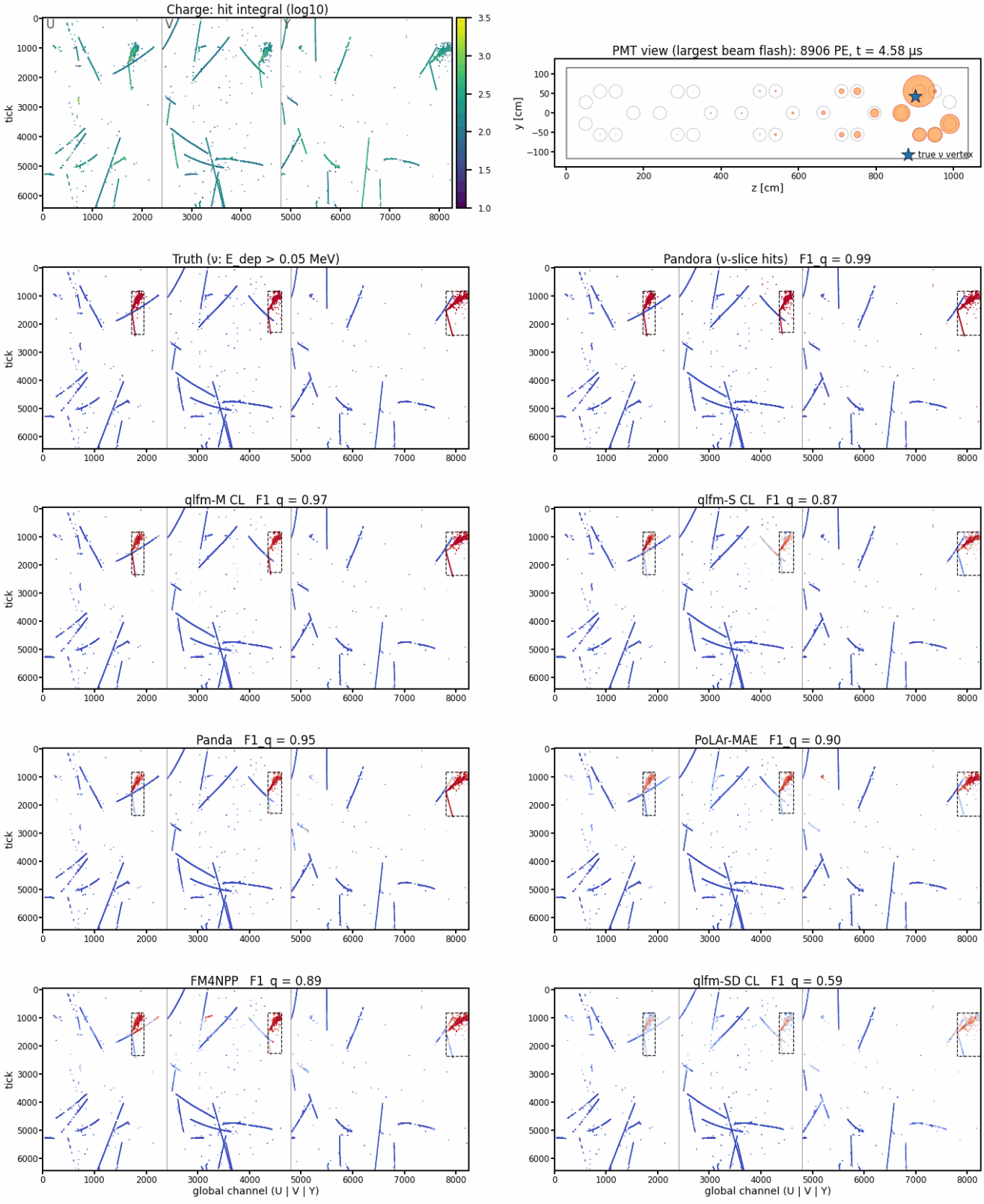
Truth: νμ CC, E = 1.62 GeV, vtx = (67, -15, 355) cm; ν charge fraction 13.6%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #6 (correct: 6/7) run 6768 / subrun 93 / event 4698 [gidx 775663]

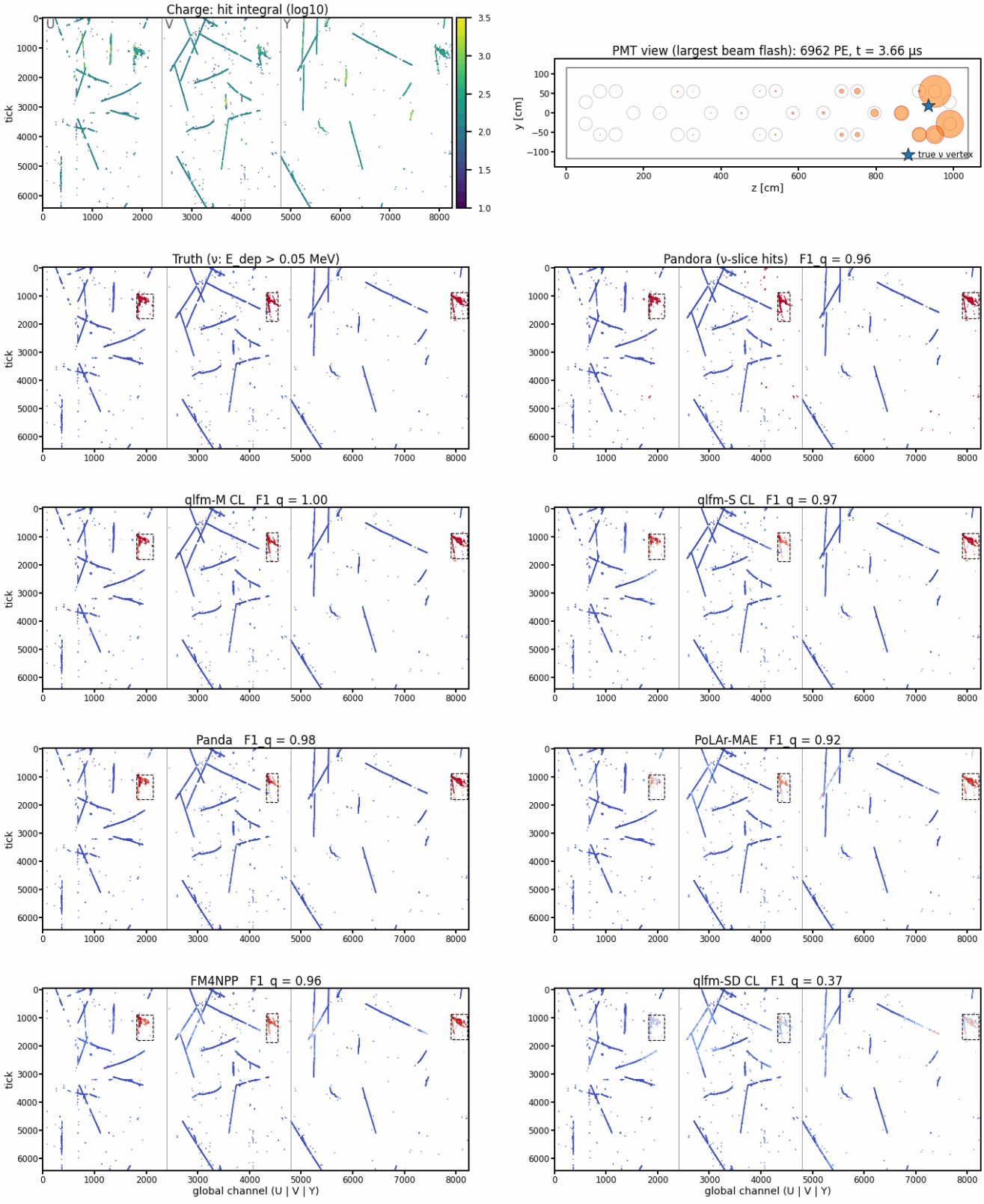
Truth: νe CC, E = 2.80 GeV, vtx = (38, 43, 902) cm; ν charge fraction 28.9%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #7 (correct: 6/7) run 7006 / subrun 447 / event 22386 [gidx 15276]

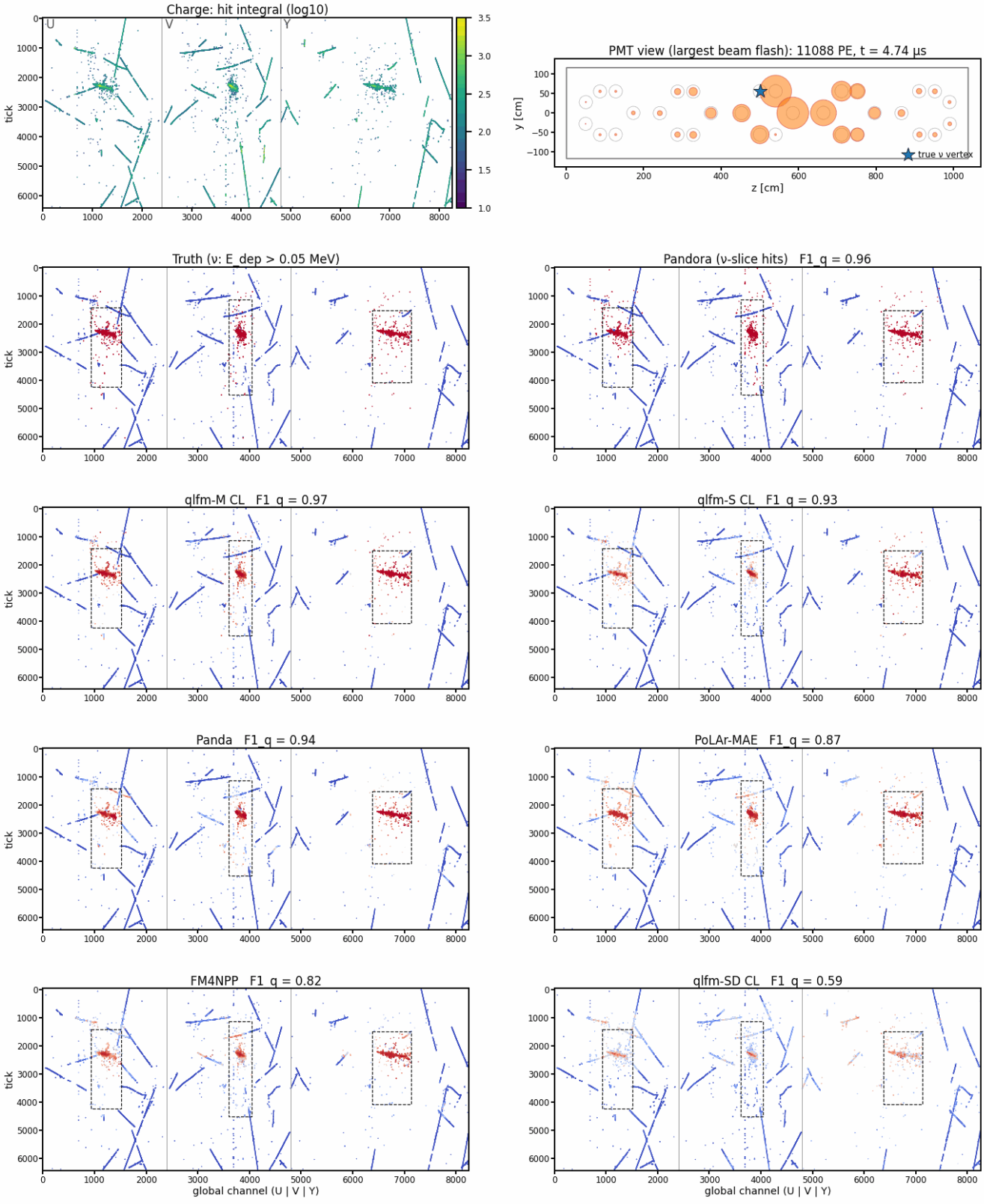
Truth: νμ NC, E = 1.44 GeV, vtx = (8, 18, 934) cm; ν charge fraction 13.6%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #8 (correct: 6/7) run 7020 / subrun 69 / event 3494 [gidx 773196]

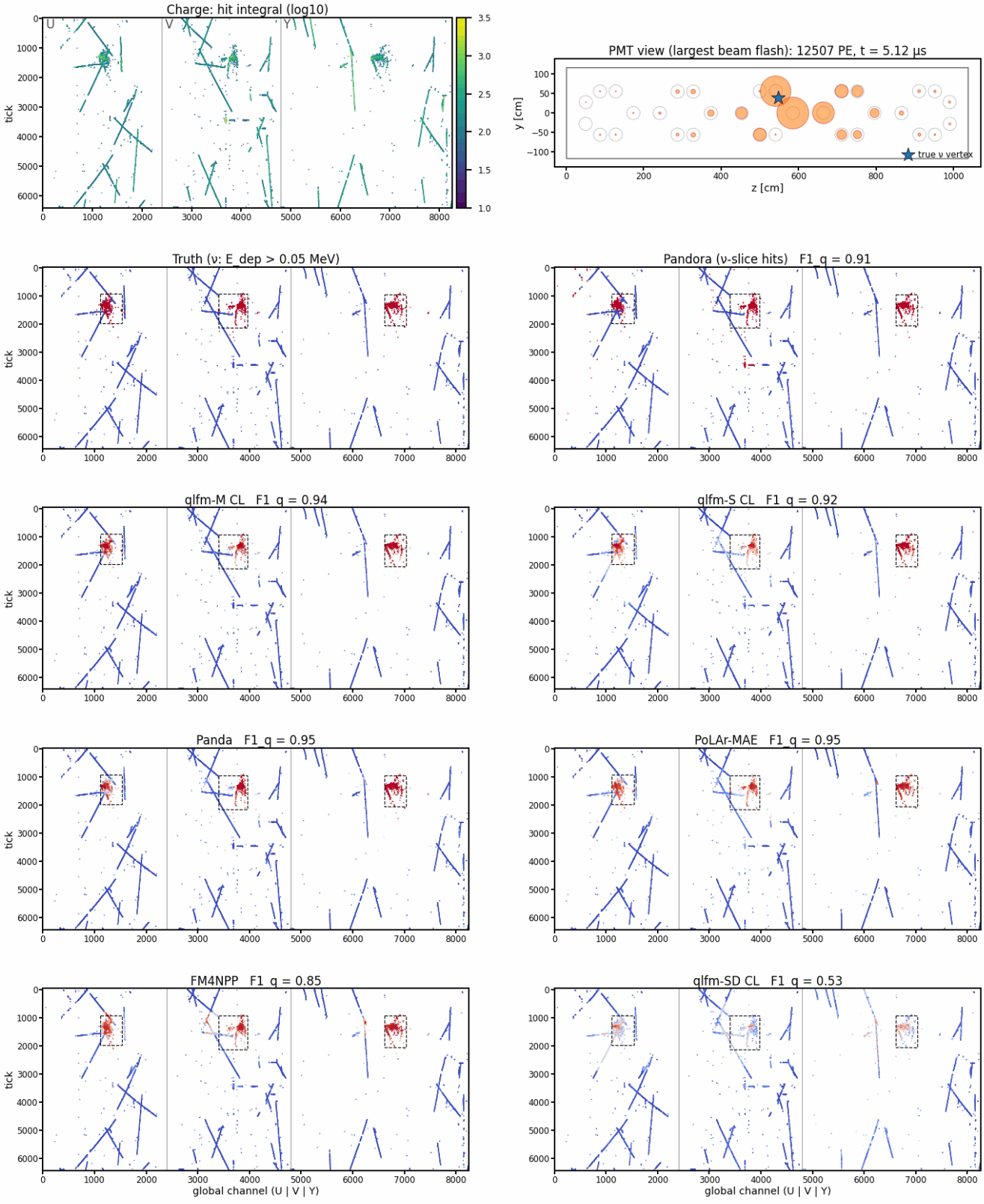
Truth: νe CC, E = 3.62 GeV, vtx = (75, 55, 502) cm; ν charge fraction 30.0%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #9 (correct: 6/7) run 7001 / subrun 541 / event 27075 [gidx 822654]

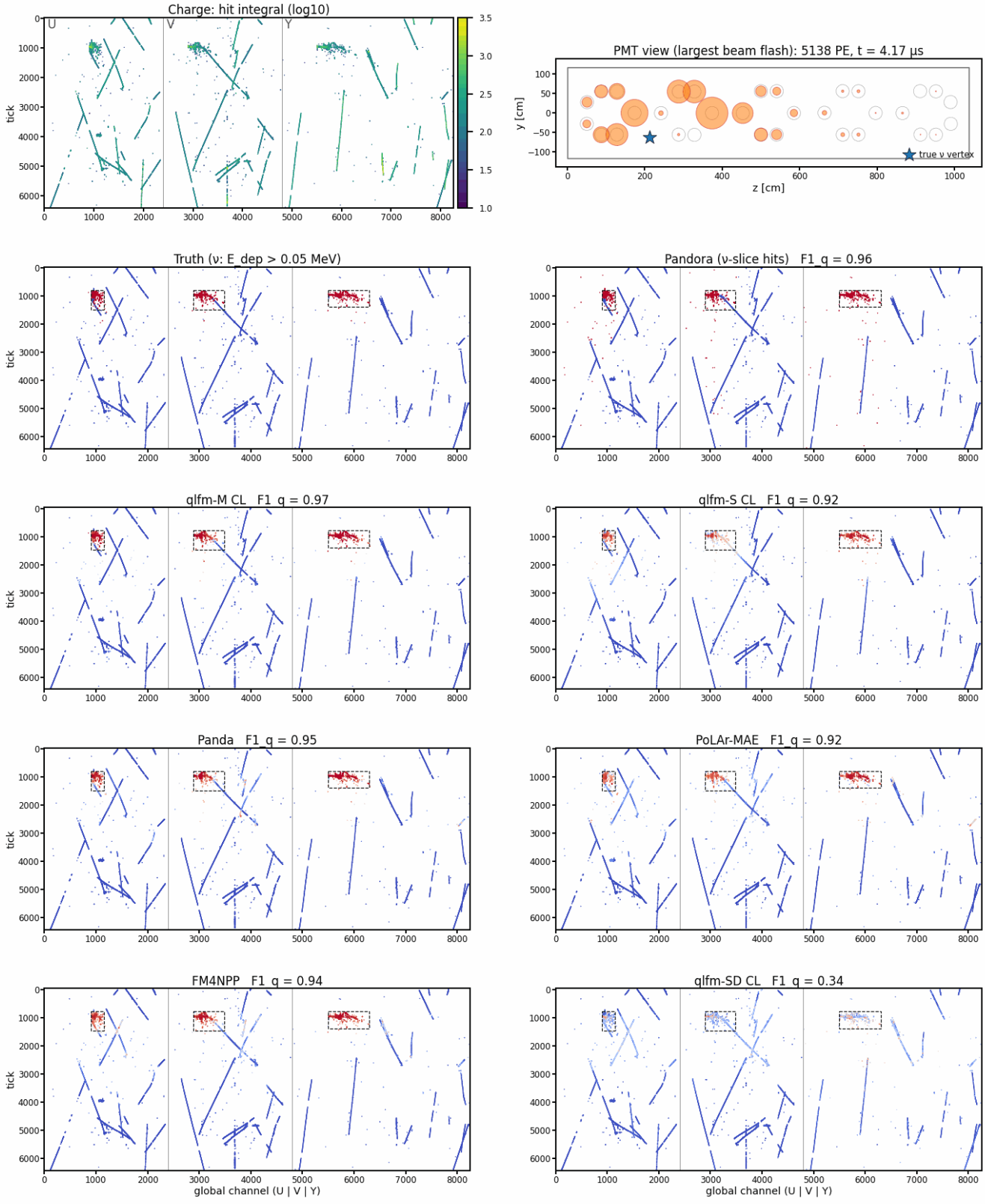
Truth: νe CC, E = 2.13 GeV, vtx = (27, 40, 548) cm; ν charge fraction 25.6%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — easy #10 (correct: 6/7) run 7026 / subrun 834 / event 41746 [gidx 755403]

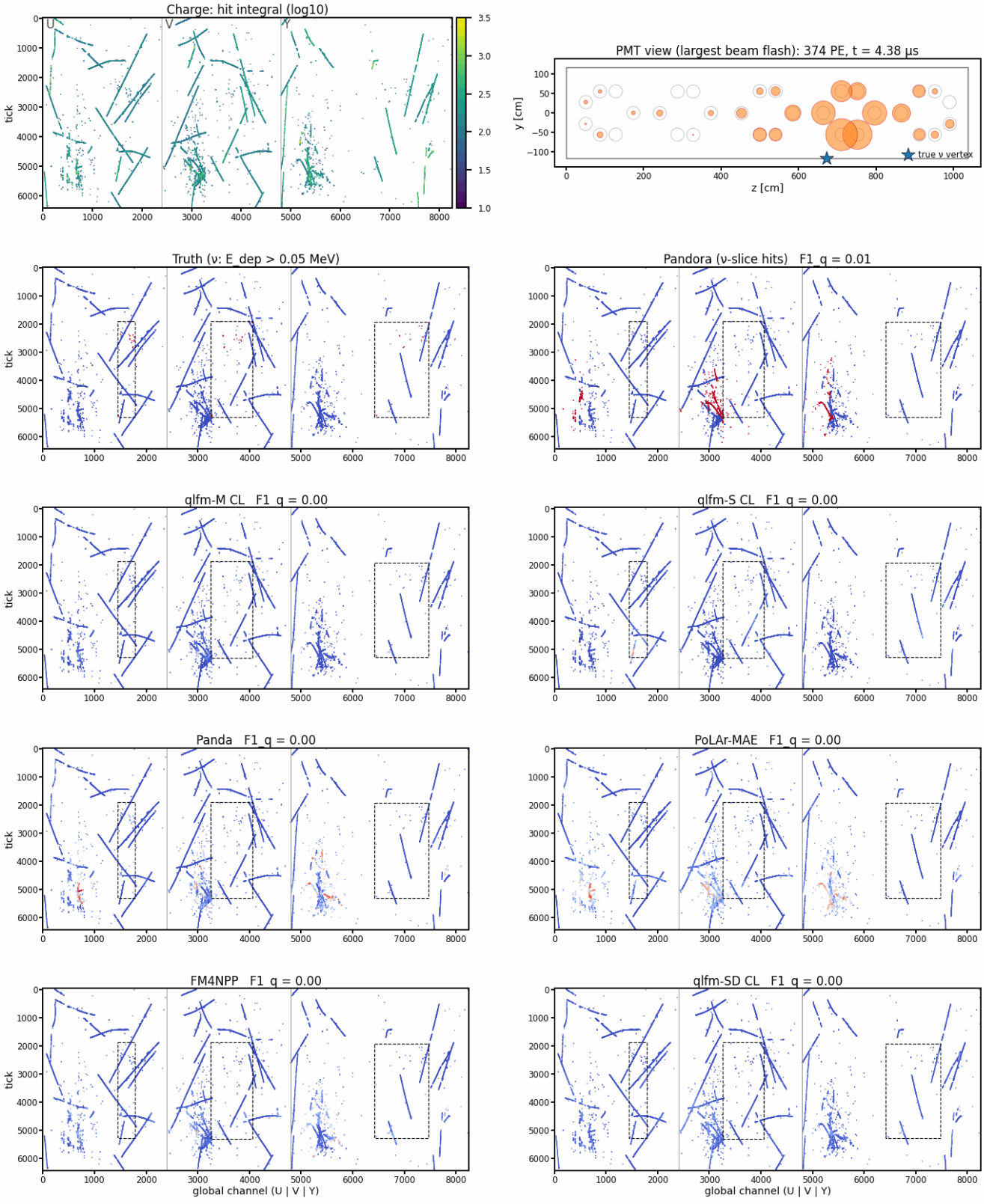
Truth: νe CC, E = 2.42 GeV, vtx = (8, -63, 213) cm; ν charge fraction 20.3%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #1 (correct: 0/7) run 5001 / subrun 74 / event 3734 [gidx 133774]

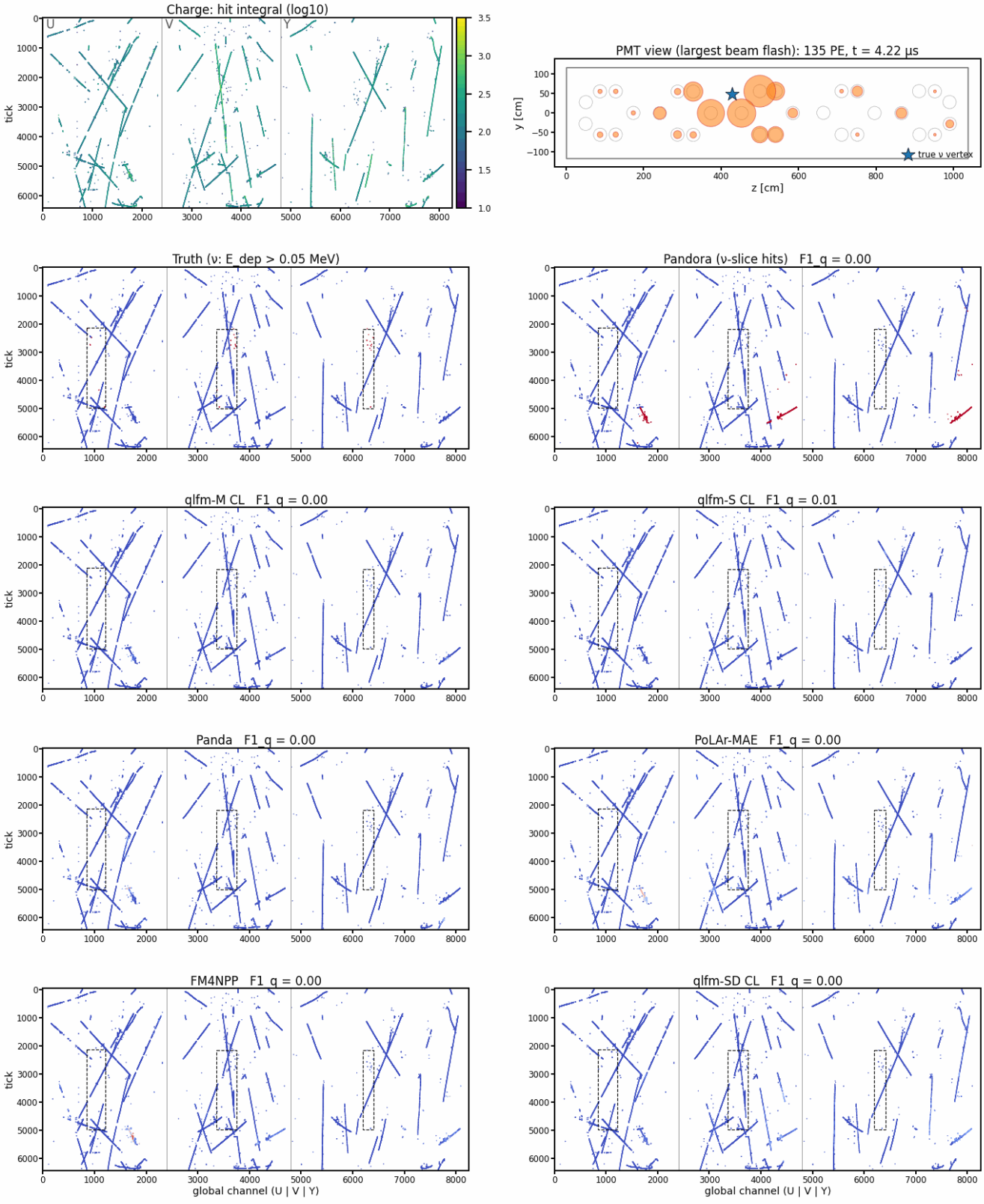
Truth: νμ NC, E = 1.04 GeV, vtx = (98, -116, 672) cm; ν charge fraction 0.3%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #2 (correct: 0/7) run 7010 / subrun 426 / event 21323 [gidx 17238]

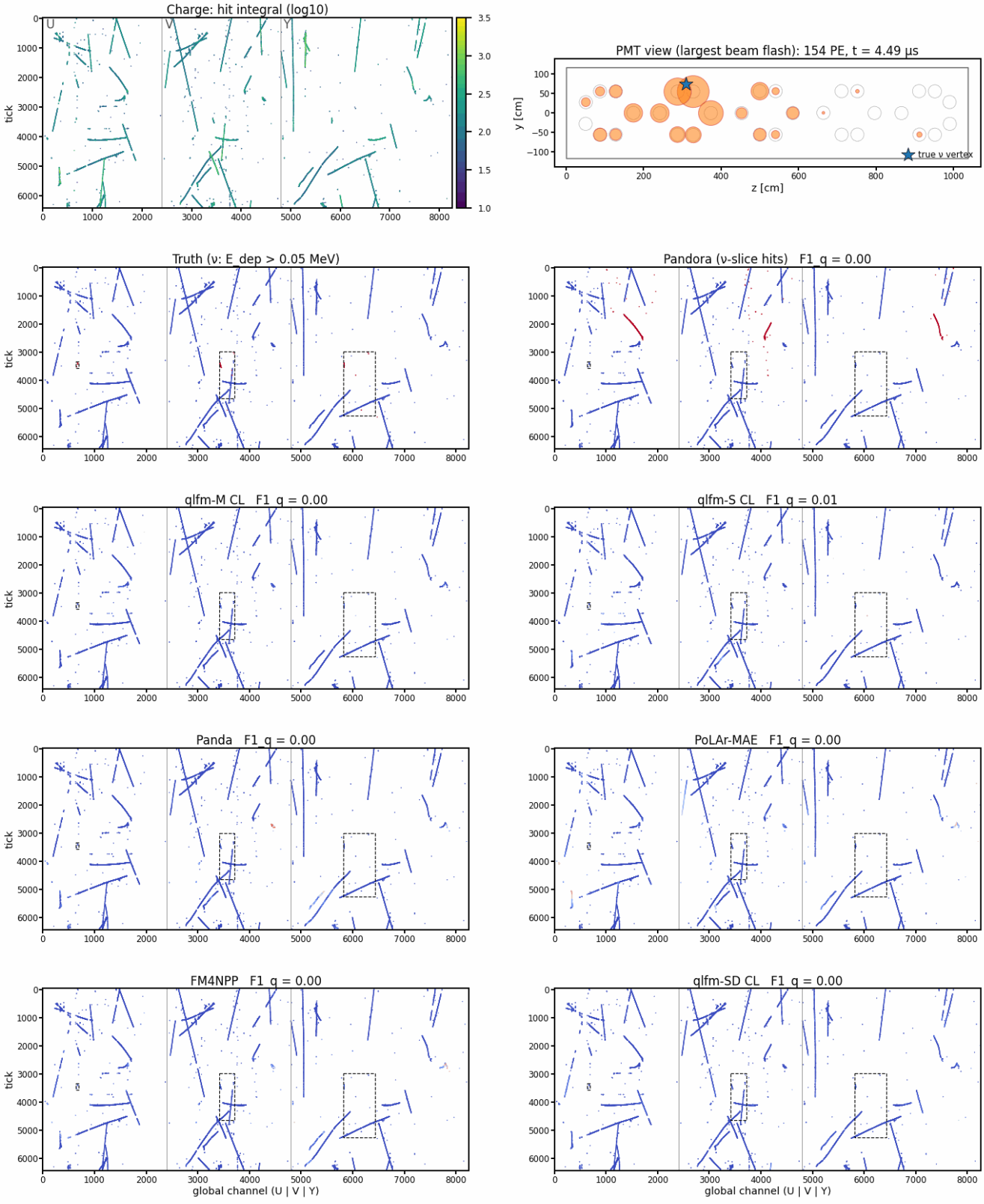
Truth: νμ NC, E = 0.72 GeV, vtx = (103, 48, 428) cm; ν charge fraction 0.2%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #3 (correct: 0/7) run 7012 / subrun 767 / event 38351 [gidx 623165]

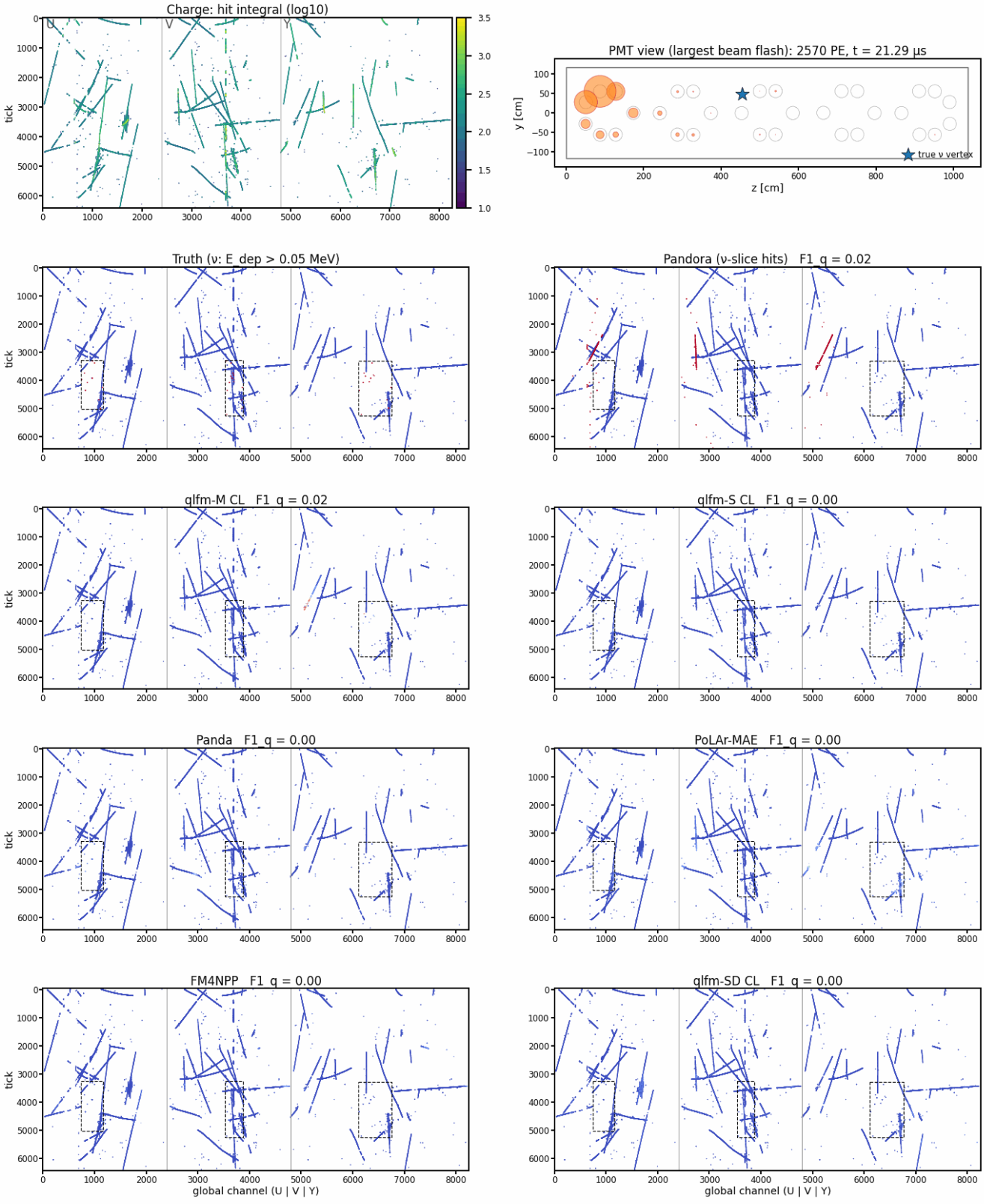
Truth: νμ NC, E = 1.48 GeV, vtx = (148, 74, 310) cm; ν charge fraction 0.6%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #4 (correct: 0/7) run 7020 / subrun 1179 / event 58960 [gidx 369666]

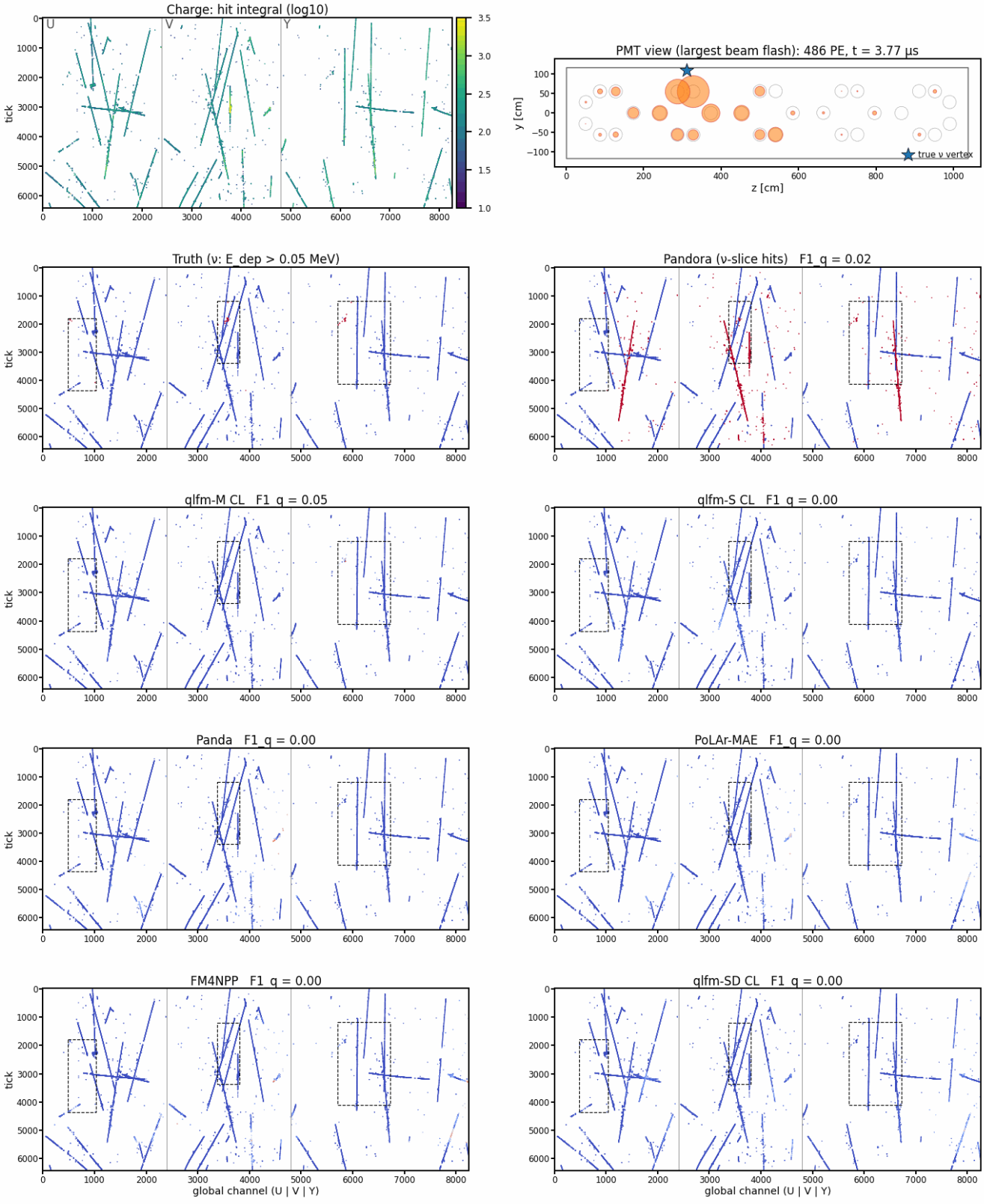
Truth: νμ NC, E = 0.92 GeV, vtx = (190, 48, 455) cm; ν charge fraction 0.4%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #5 (correct: 0/7) run 6529 / subrun 99 / event 4988 [gidx 289448]

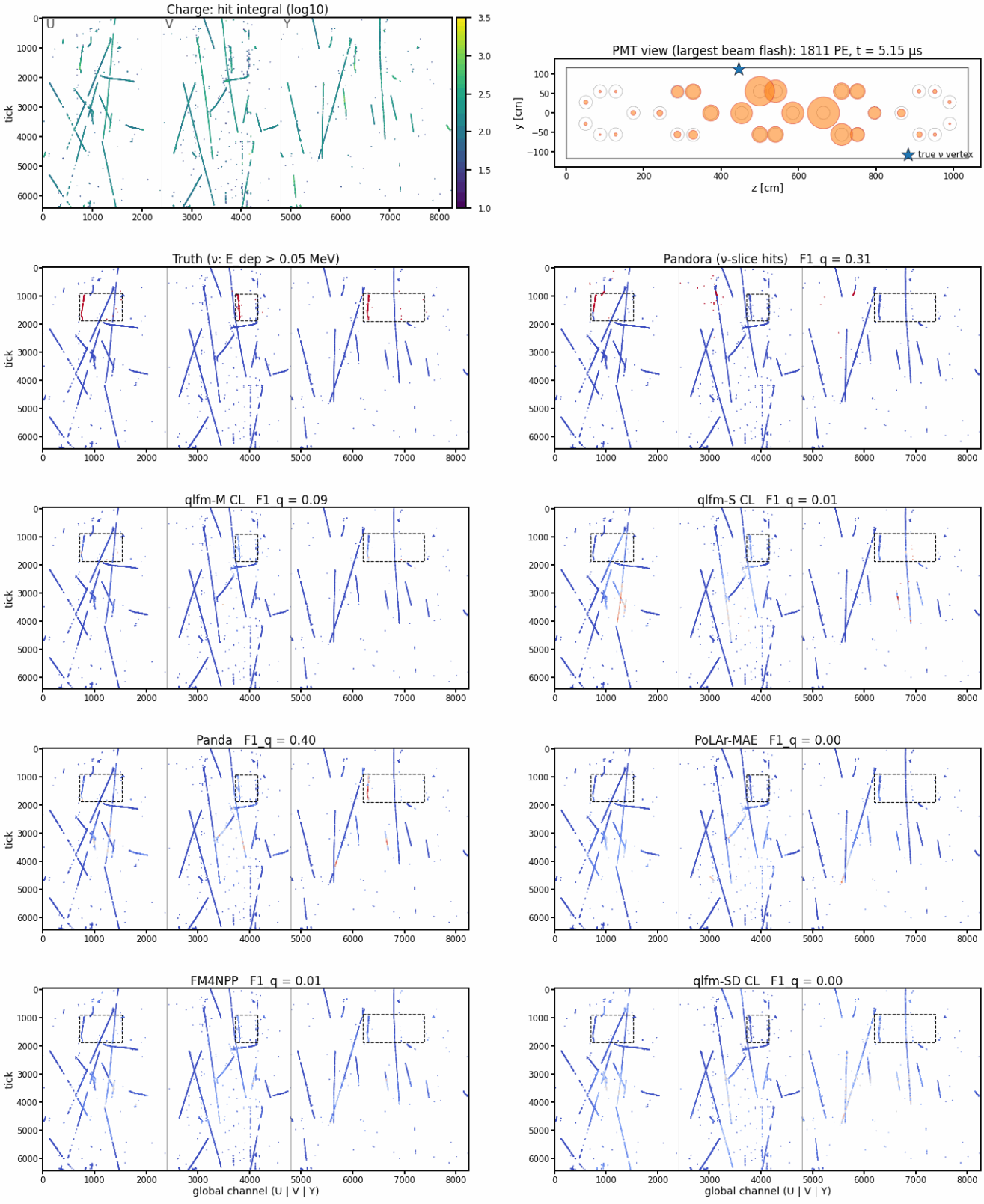
Truth: νμ NC, E = 0.56 GeV, vtx = (55, 111, 312) cm; ν charge fraction 0.4%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #6 (correct: 0/7) run 5575 / subrun 17 / event 887 [gidx 679506]

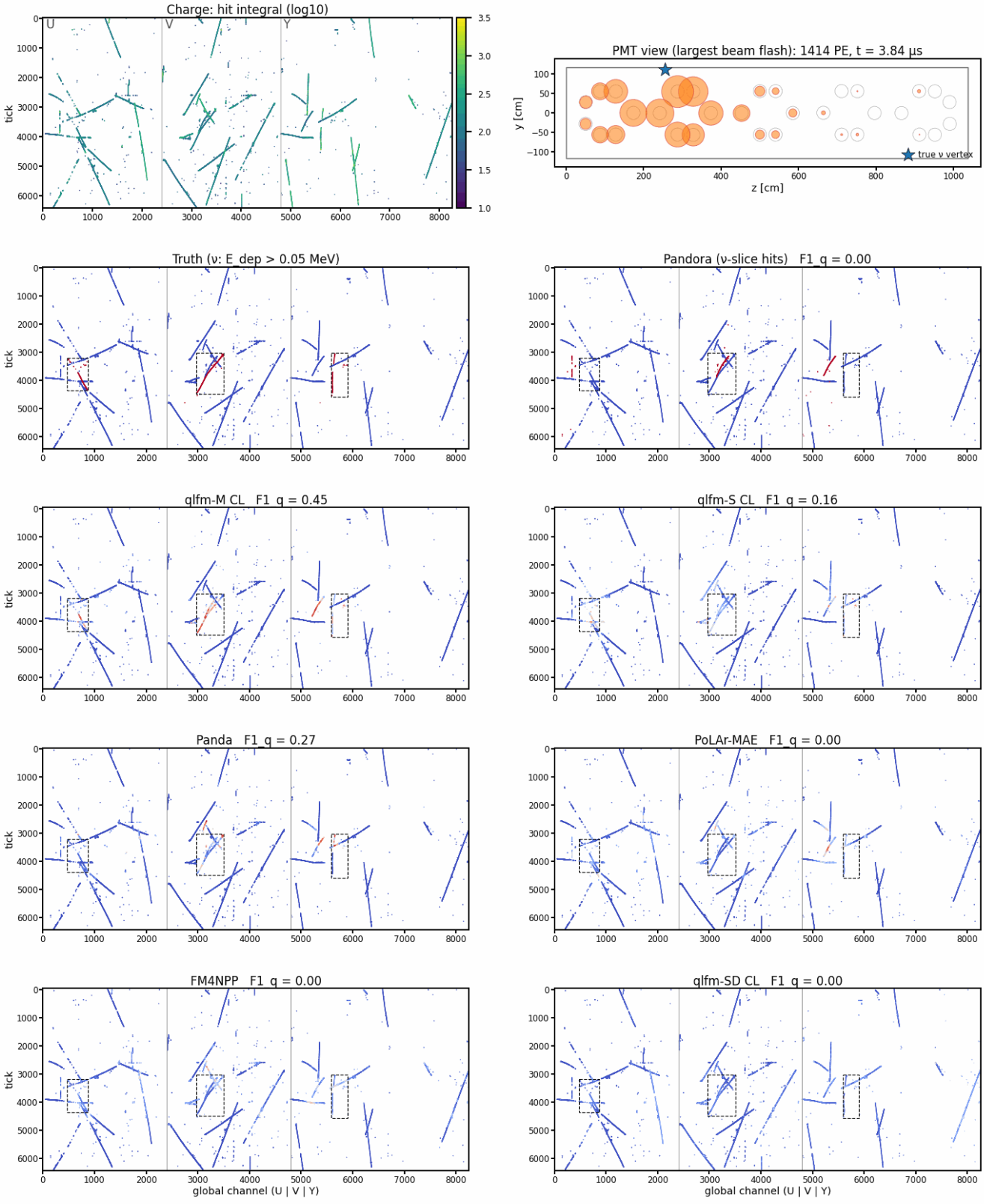
Truth: νμ CC, E = 1.36 GeV, vtx = (42, 114, 446) cm; ν charge fraction 5.2%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #7 (correct: 0/7) run 4998 / subrun 94 / event 4725 [gidx 18414]

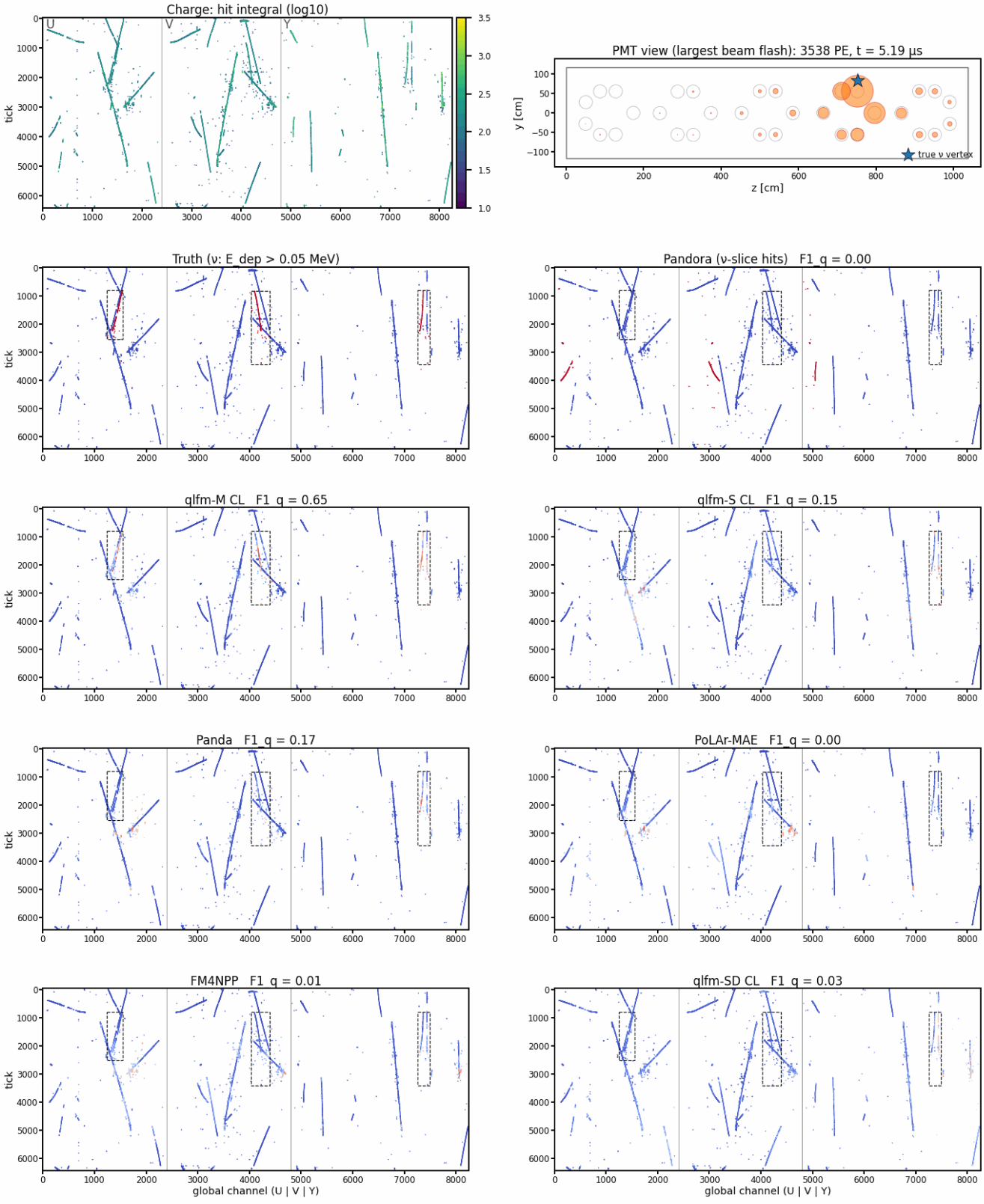
Truth: νμ CC, E = 1.63 GeV, vtx = (127, 111, 255) cm; ν charge fraction 6.2%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #8 (correct: 0/7) run 6046 / subrun 37 / event 1879 [gidx 687178]

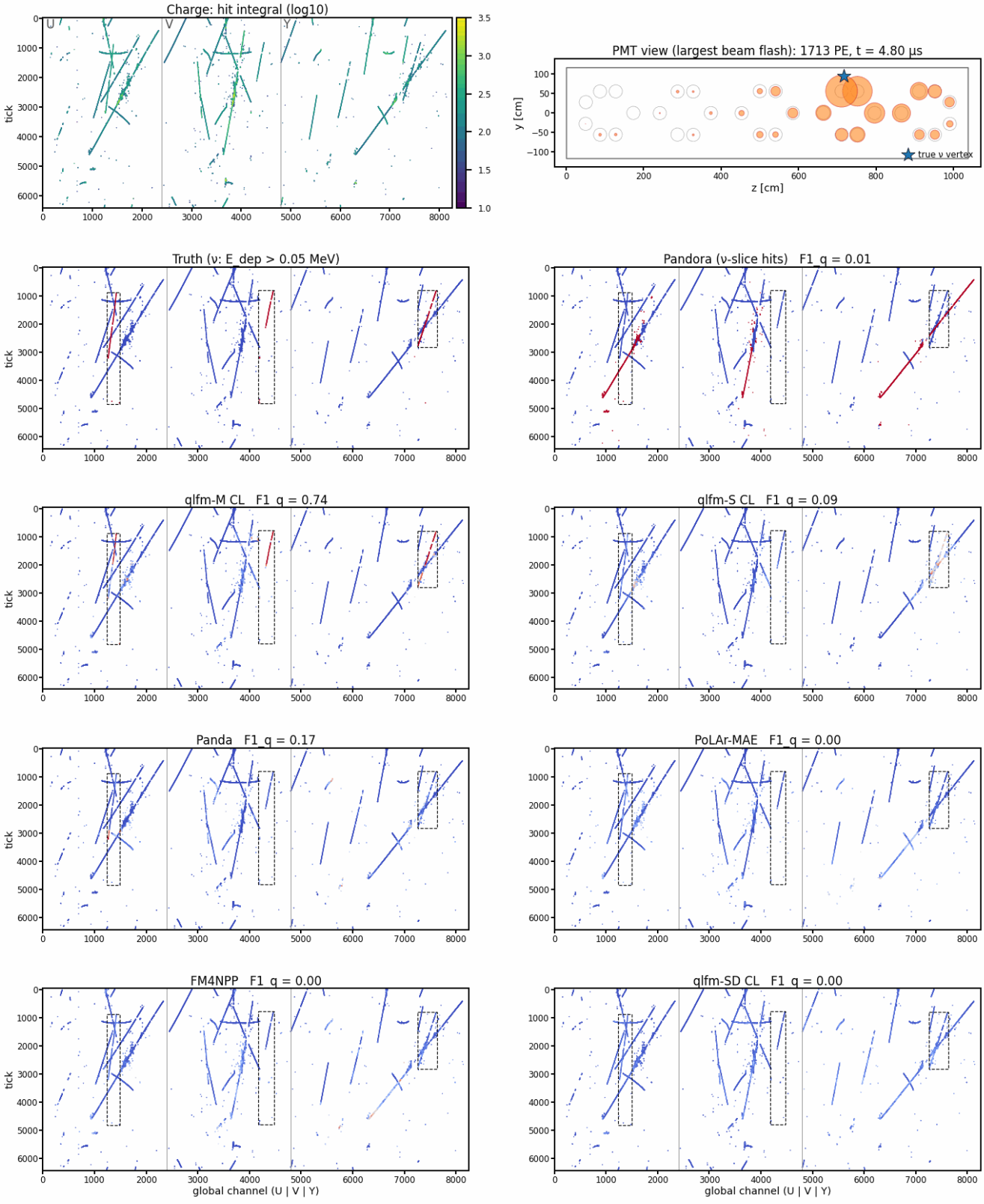
Truth: νμ CC, E = 0.60 GeV, vtx = (76, 83, 752) cm; ν charge fraction 6.0%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T1 hit ν/cosmic — hard #9 (correct: 0/7) run 7026 / subrun 788 / event 39406 [gidx 606534]

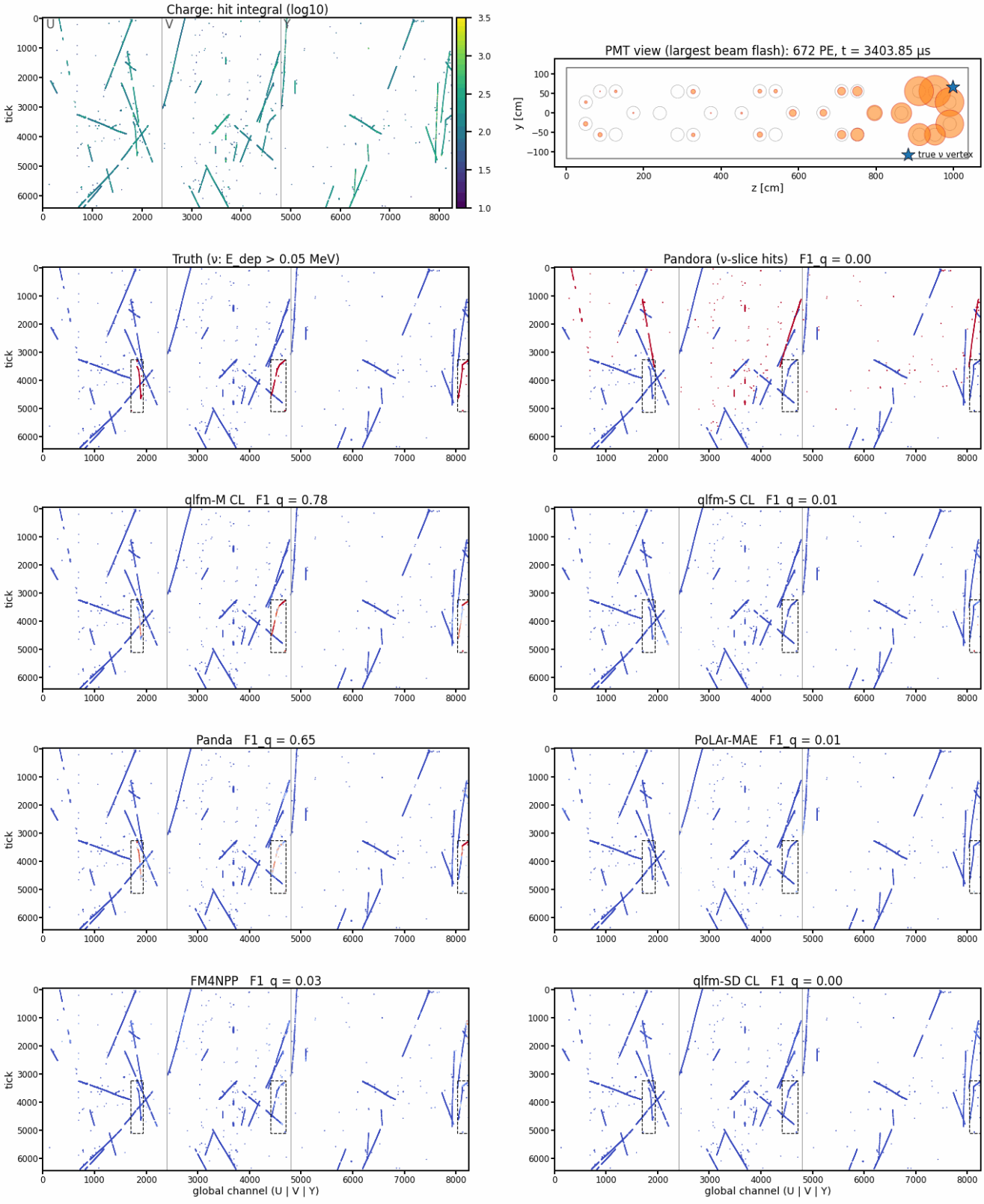
Truth: νμ CC, E = 1.32 GeV, vtx = (130, 96, 716) cm; ν charge fraction 5.1%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

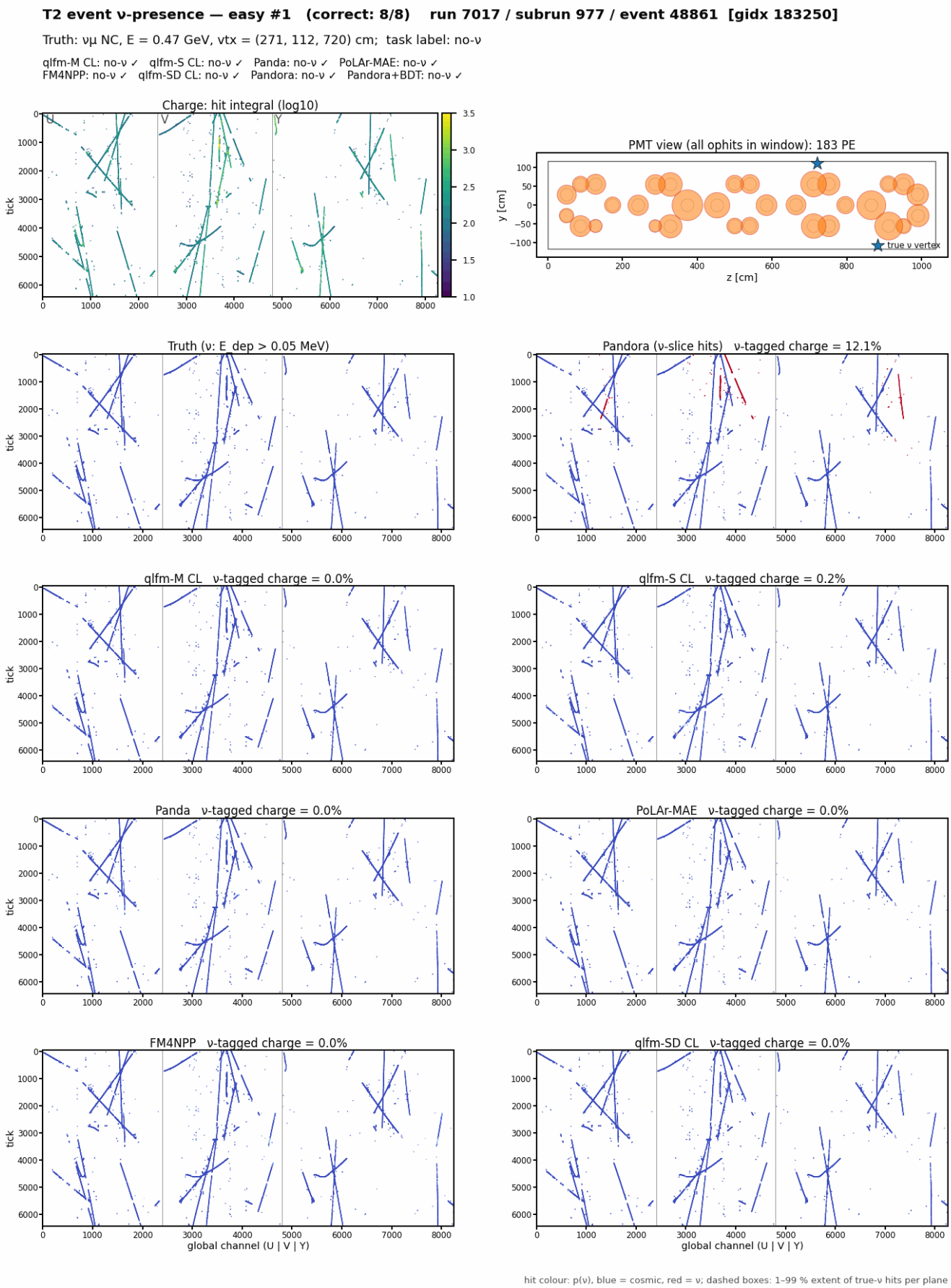
T1 hit ν/cosmic — hard #10 (correct: 0/7) run 7026 / subrun 400 / event 20044 [gidx 71076]

Truth: νμ CC, E = 3.80 GeV, vtx = (144, 68, 999) cm; ν charge fraction 6.2%



hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

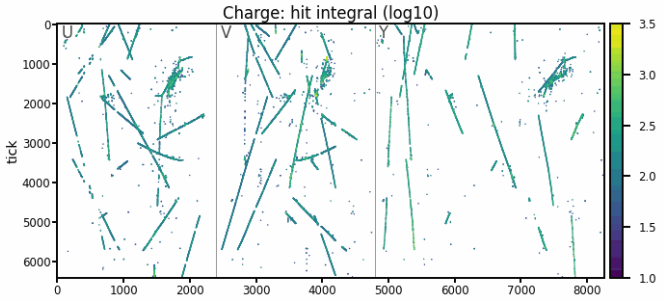
A.2 T2: event-level ν presence

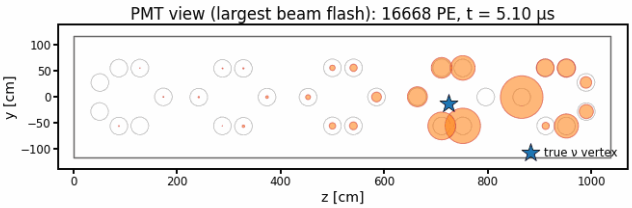


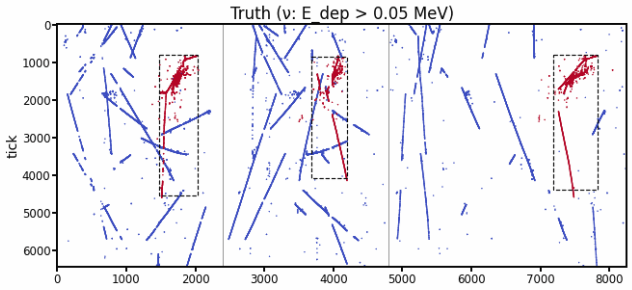
T2 event ν-presence — easy #2 (correct: 8/8) run 7049 / subrun 324 / event 16231 [gidx 425110]

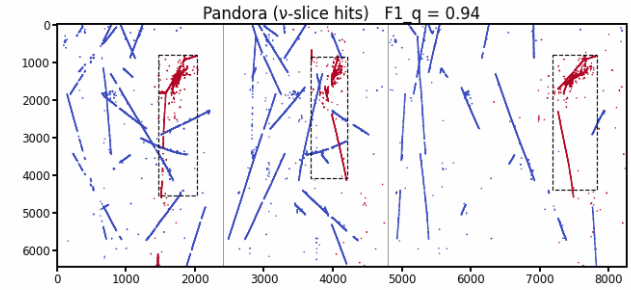
Truth: νμ NC, E = 5.71 GeV, vtx = (52, -13, 723) cm; task label: in-AV ν

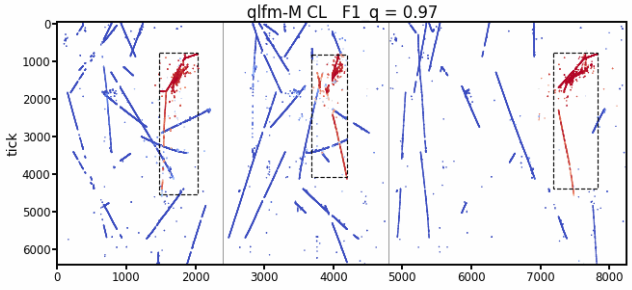
qlfm-M CL: in-AV ν ✓ qlfm-S CL: in-AV ν ✓ Panda: in-AV ν ✓ PoLAr-MAE: in-AV ν ✓
FM4NPP: in-AV ν ✓ qlfm-SD CL: in-AV ν ✓ Pandora: in-AV ν ✓ Pandora+BDT: in-AV ν ✓

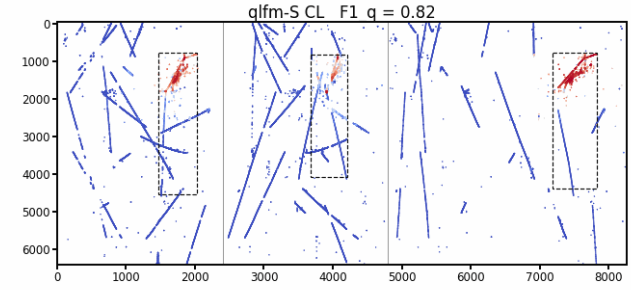


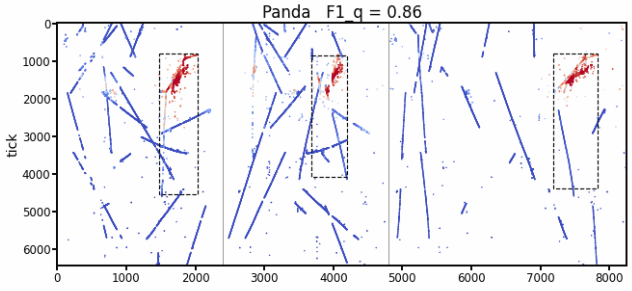


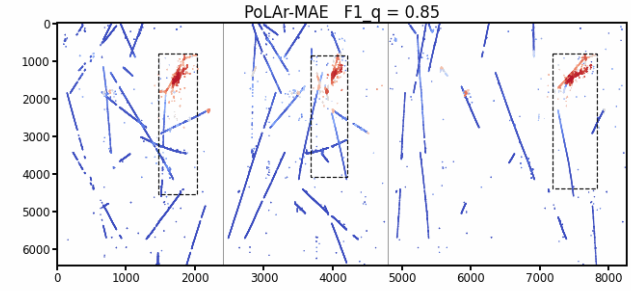


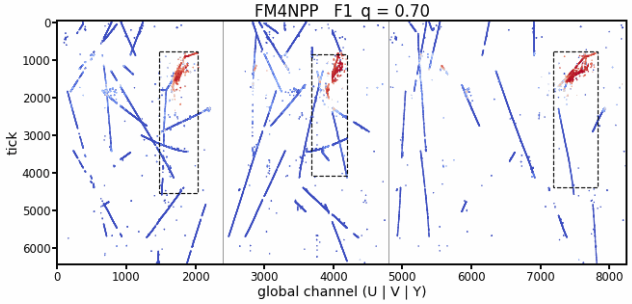


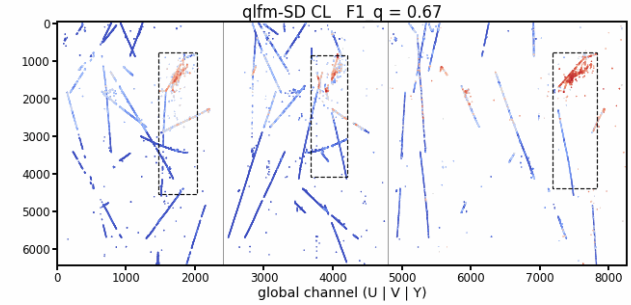










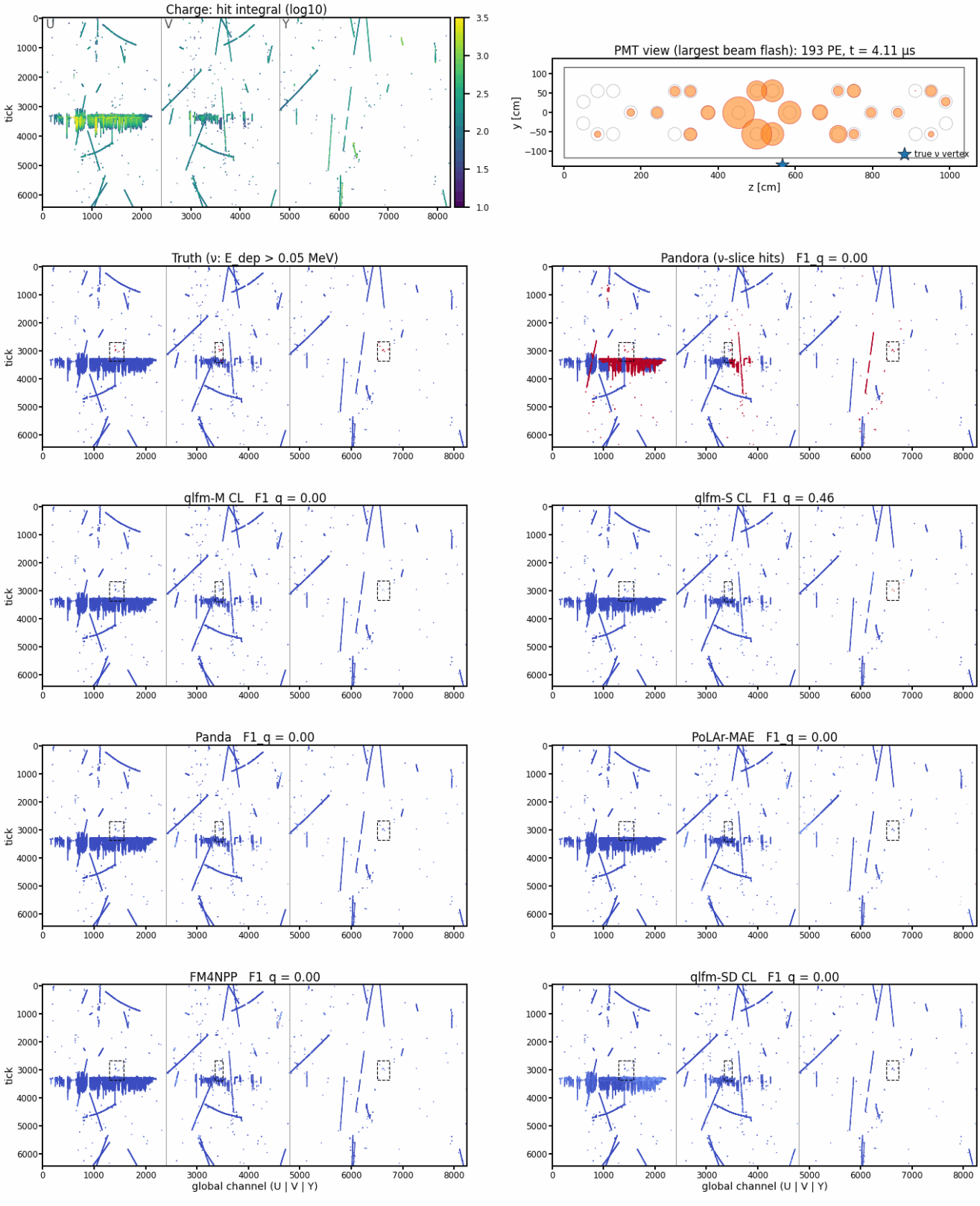


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T2 event ν-presence — easy #3 (correct: 7/8) run 6319 / subrun 143 / event 7176 [gidx 385370]

Truth: νμ CC, E = 0.78 GeV, vtx = (188, -135, 565) cm; task label: out-AV ν

qlfm-M CL: out-AV ν ✓ qlfm-S CL: out-AV ν ✓ Panda: out-AV ν ✓ PoLAr-MAE: out-AV ν ✓
FM4NPP: out-AV ν ✓ qlfm-SD CL: out-AV ν ✓ Pandora: no-ν ✗ Pandora+BDT: out-AV ν ✓

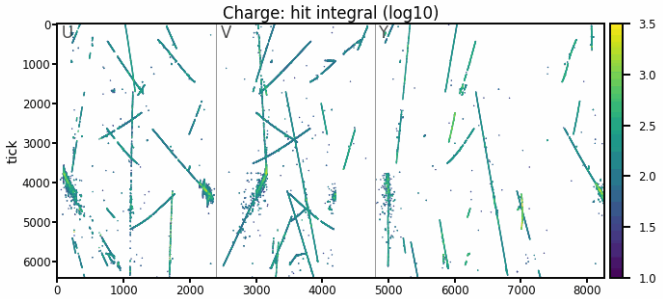


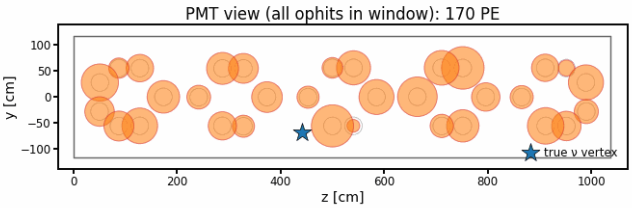
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

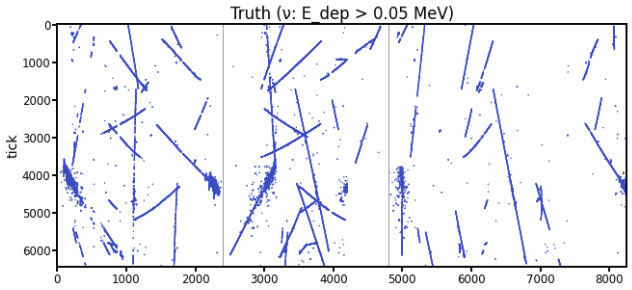
T2 event ν-presence — easy #4 (correct: 8/8) run 7054 / subrun 205 / event 10278 [gidx 30831]

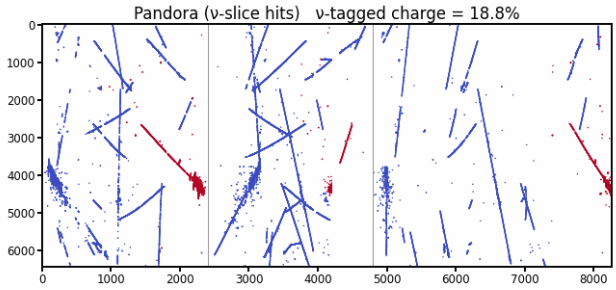
Truth: νμ CC, E = 0.69 GeV, vtx = (289, -69, 440) cm; task label: no-ν

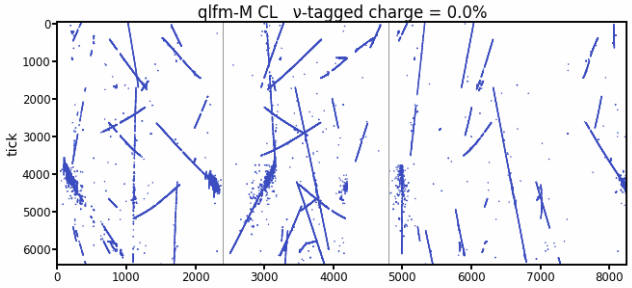
qlfm-M CL: no-ν ✓ qlfm-S CL: no-ν ✓ Panda: no-ν ✓ PoLAr-MAE: no-ν ✓
FM4NPP: no-ν ✓ qlfm-SD CL: no-ν ✓ Pandora: no-ν ✓ Pandora+BDT: no-ν ✓

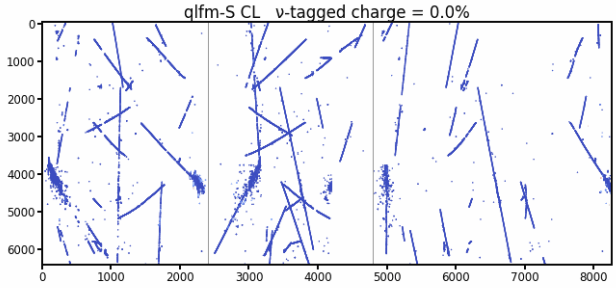


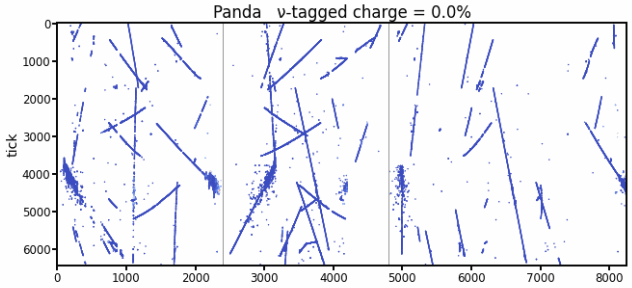


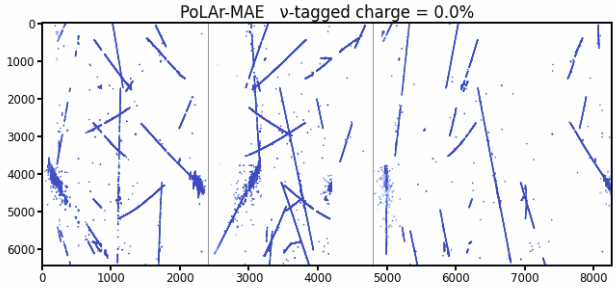


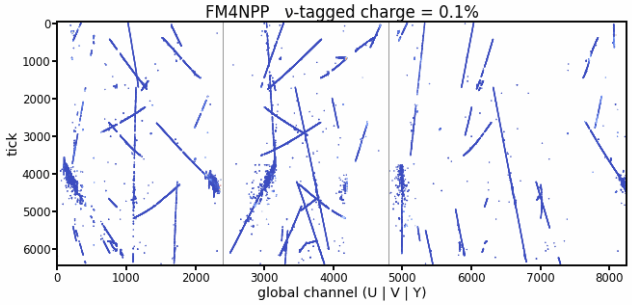


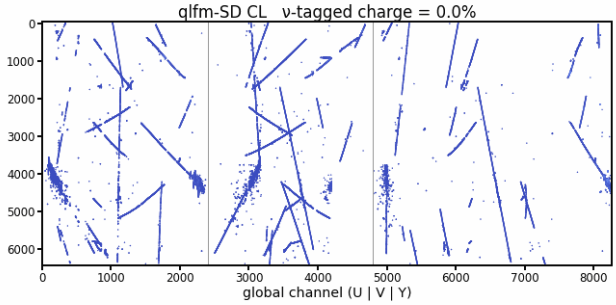










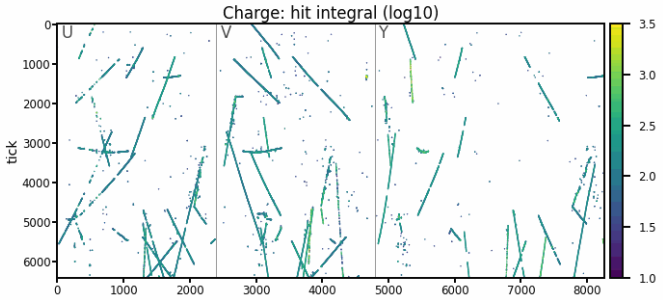


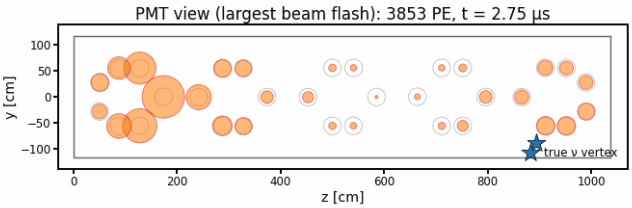
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

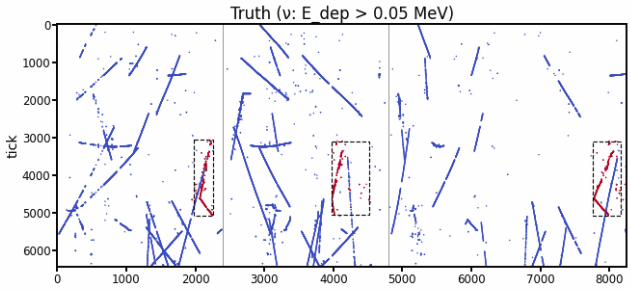
T2 event ν-presence — easy #5 (correct: 8/8) run 6417 / subrun 176 / event 8827 [gidx 827855]

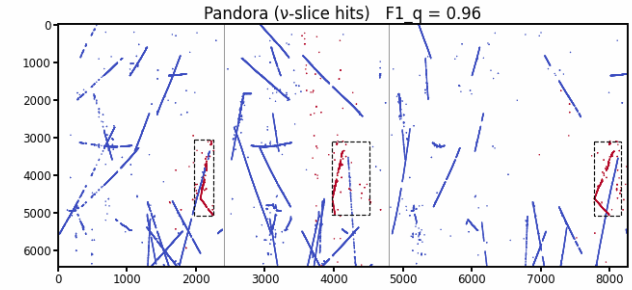
Truth: νe CC, E = 1.05 GeV, vtx = (207, -90, 894) cm; task label: in-AV ν

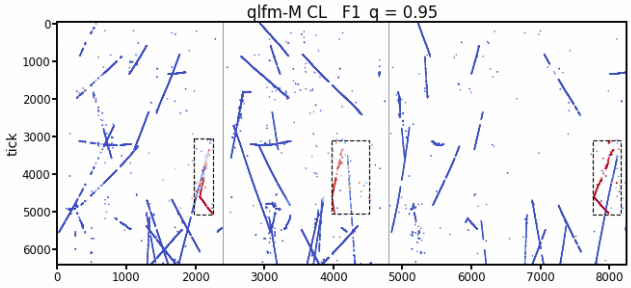
qlfm-M CL: in-AV ν ✓ qlfm-S CL: in-AV ν ✓ Panda: in-AV ν ✓ PoLAr-MAE: in-AV ν ✓
FM4NPP: in-AV ν ✓ qlfm-SD CL: in-AV ν ✓ Pandora: in-AV ν ✓ Pandora+BDT: in-AV ν ✓

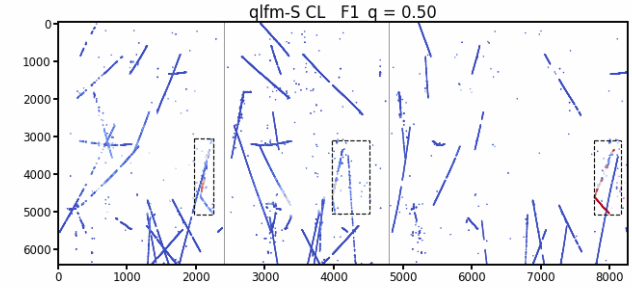


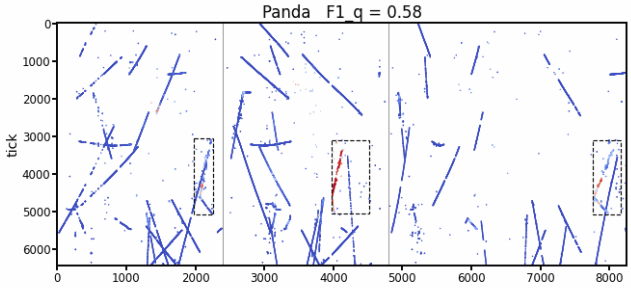


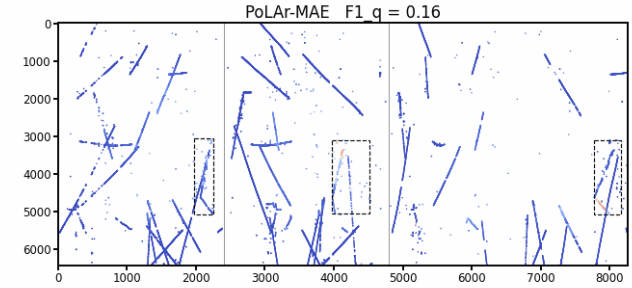


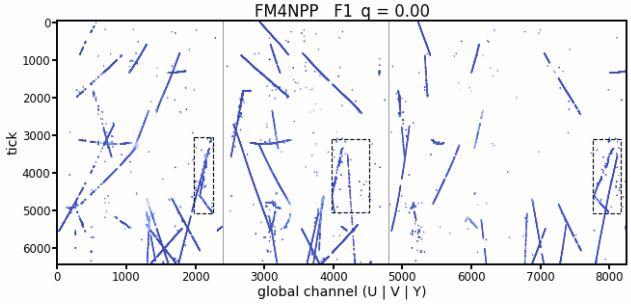


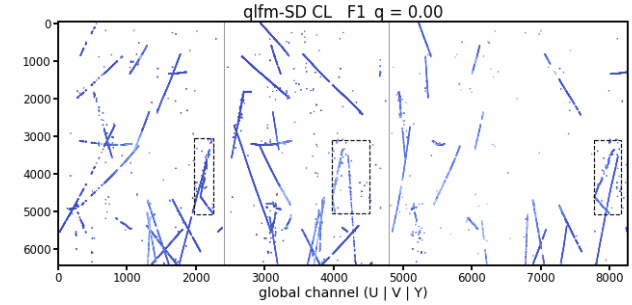










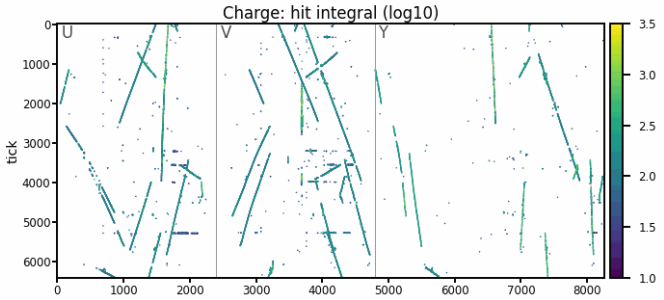


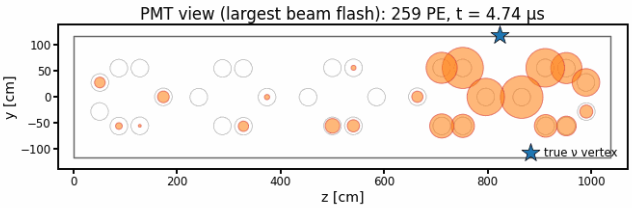
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

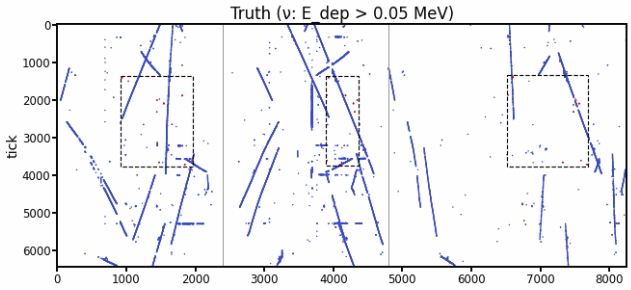
T2 event ν-presence — easy #6 (correct: 7/8) run 6855 / subrun 9 / event 472 [gidx 339486]

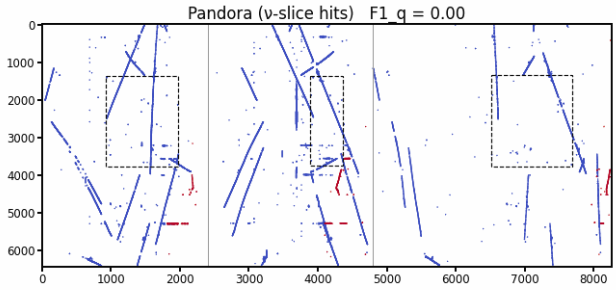
Truth: νμ CC, E = 0.38 GeV, vtx = (69, 120, 822) cm; task label: out-AV ν

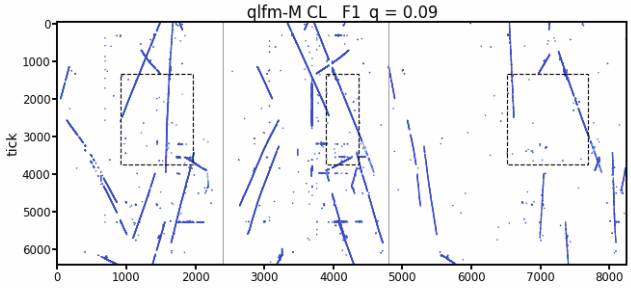
qlfm-M CL: out-AV ν ✓ qlfm-S CL: out-AV ν ✓ Panda: out-AV ν ✓ PoLAr-MAE: out-AV ν ✓
FM4NPP: out-AV ν ✓ qlfm-SD CL: out-AV ν ✓ Pandora: no-ν ✗ Pandora+BDT: out-AV ν ✓

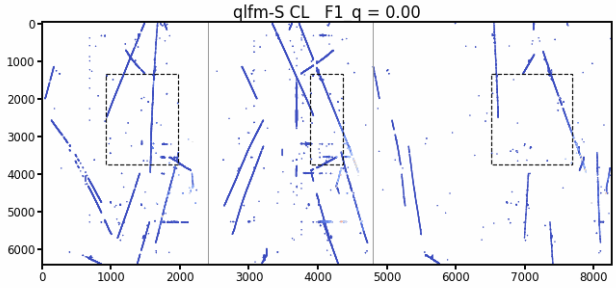


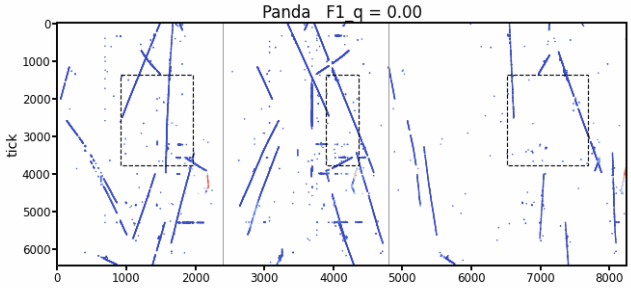


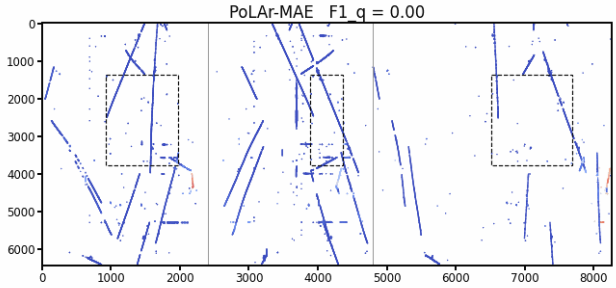


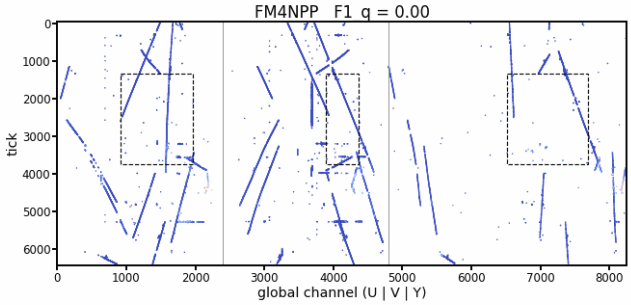


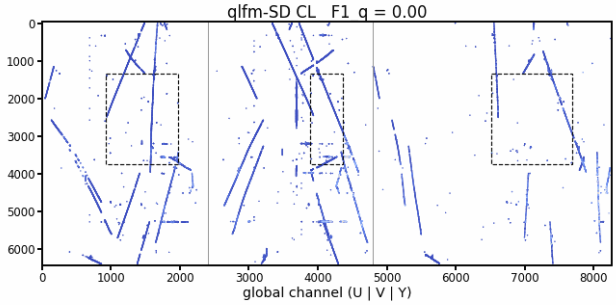










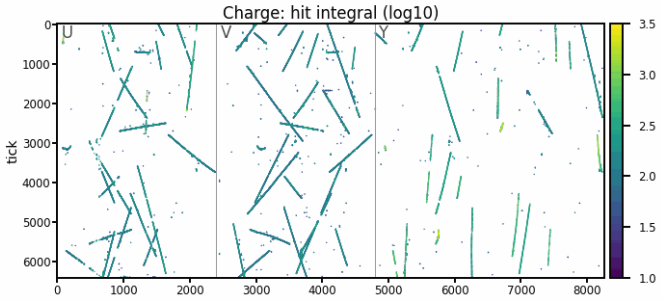


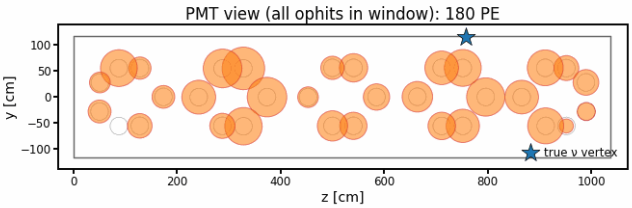
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

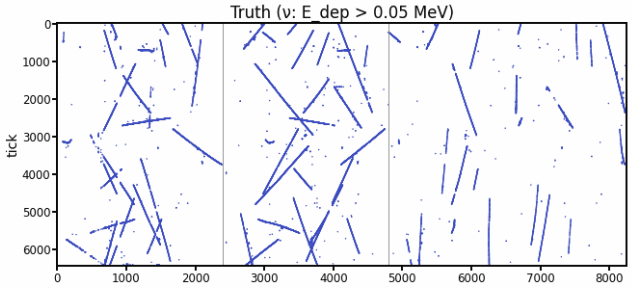
T2 event ν-presence — easy #7 (correct: 8/8) run 6953 / subrun 49 / event 2481 [gidx 308464]

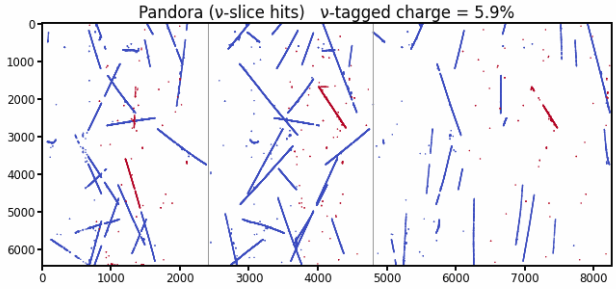
Truth: νμ CC, E = 0.85 GeV, vtx = (280, 116, 758) cm; task label: no-ν

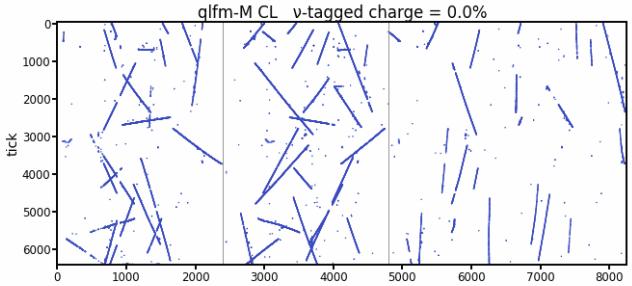
qlfm-M CL: no-ν ✓ qlfm-S CL: no-ν ✓ Panda: no-ν ✓ PoLAr-MAE: no-ν ✓
FM4NPP: no-ν ✓ qlfm-SD CL: no-ν ✓ Pandora: no-ν ✓ Pandora+BDT: no-ν ✓

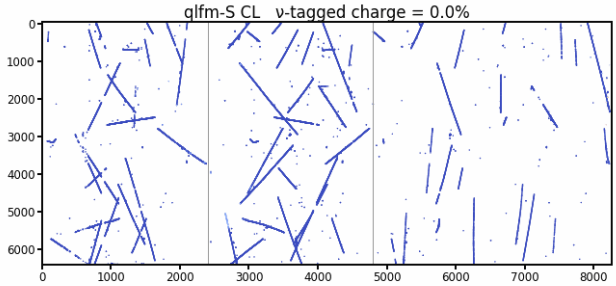


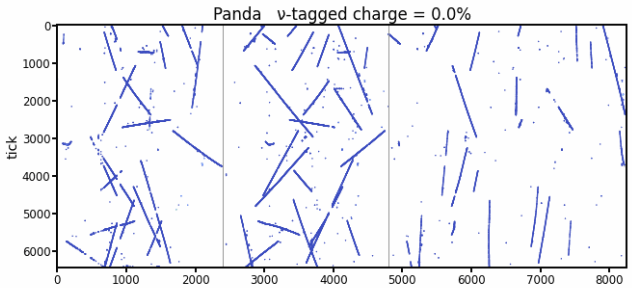


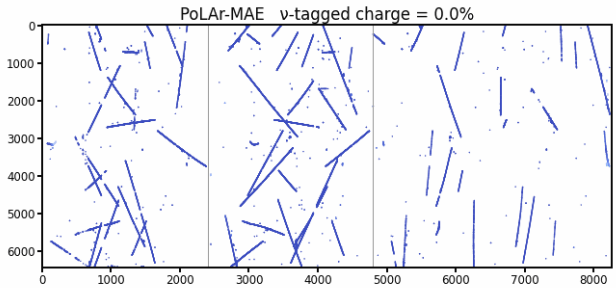


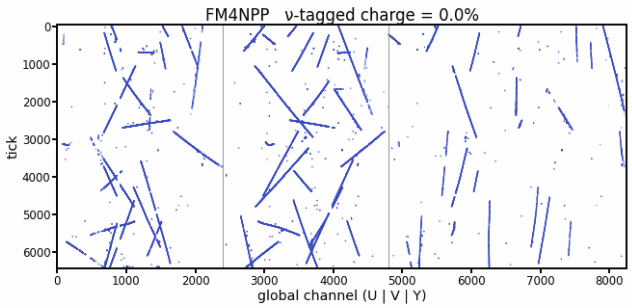


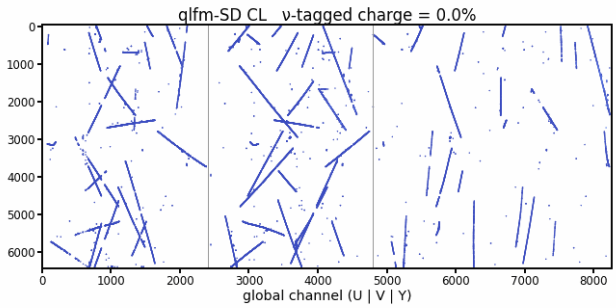










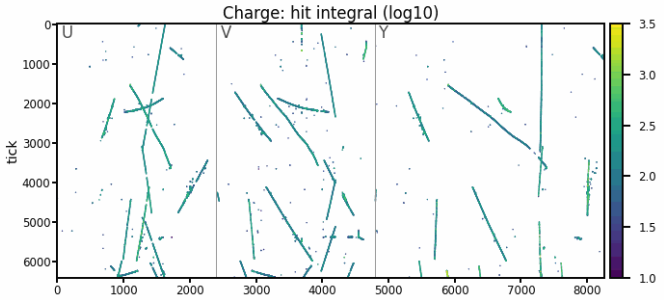


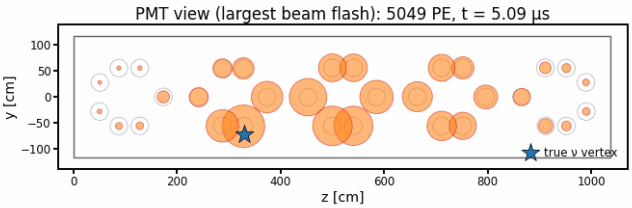
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

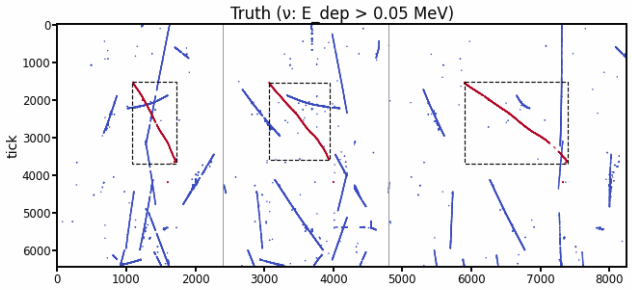
T2 event ν-presence — easy #8 (correct: 8/8) run 7645 / subrun 37 / event 1861 [gidx 500100]

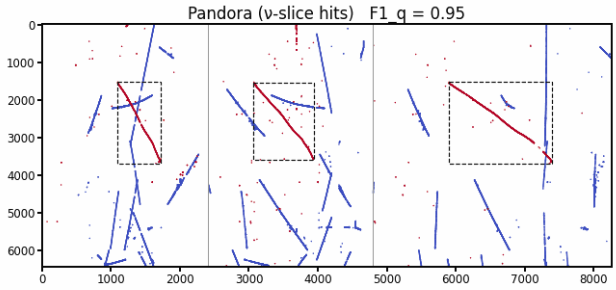
Truth: νμ CC, E = 1.23 GeV, vtx = (39, -73, 331) cm; task label: in-AV ν

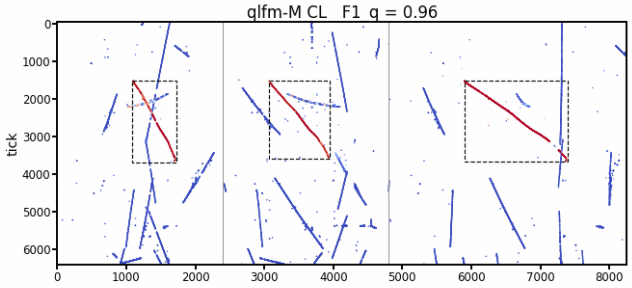
qlfm-M CL: in-AV ν ✓ qlfm-S CL: in-AV ν ✓ Panda: in-AV ν ✓ PoLAr-MAE: in-AV ν ✓
FM4NPP: in-AV ν ✓ qlfm-SD CL: in-AV ν ✓ Pandora: in-AV ν ✓ Pandora+BDT: in-AV ν ✓

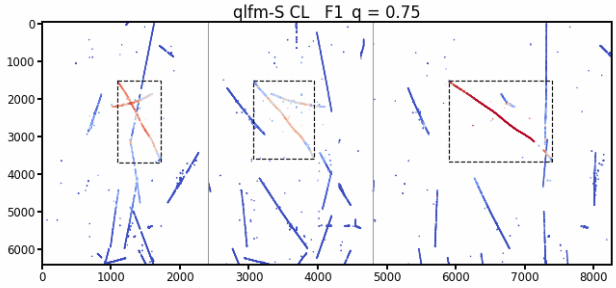


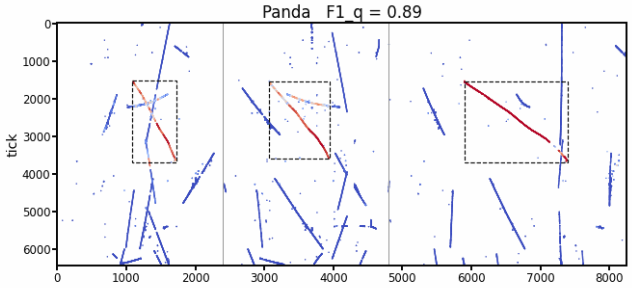


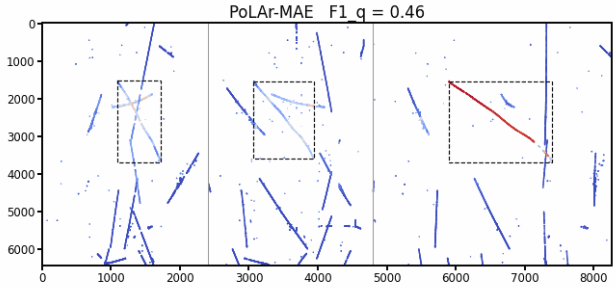


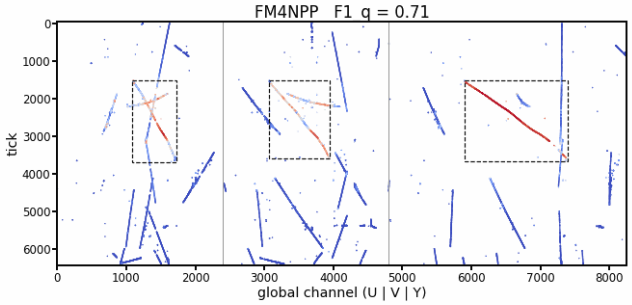


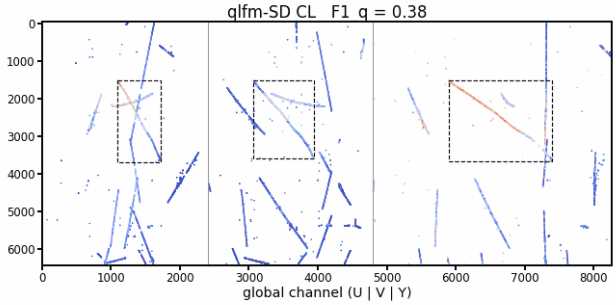










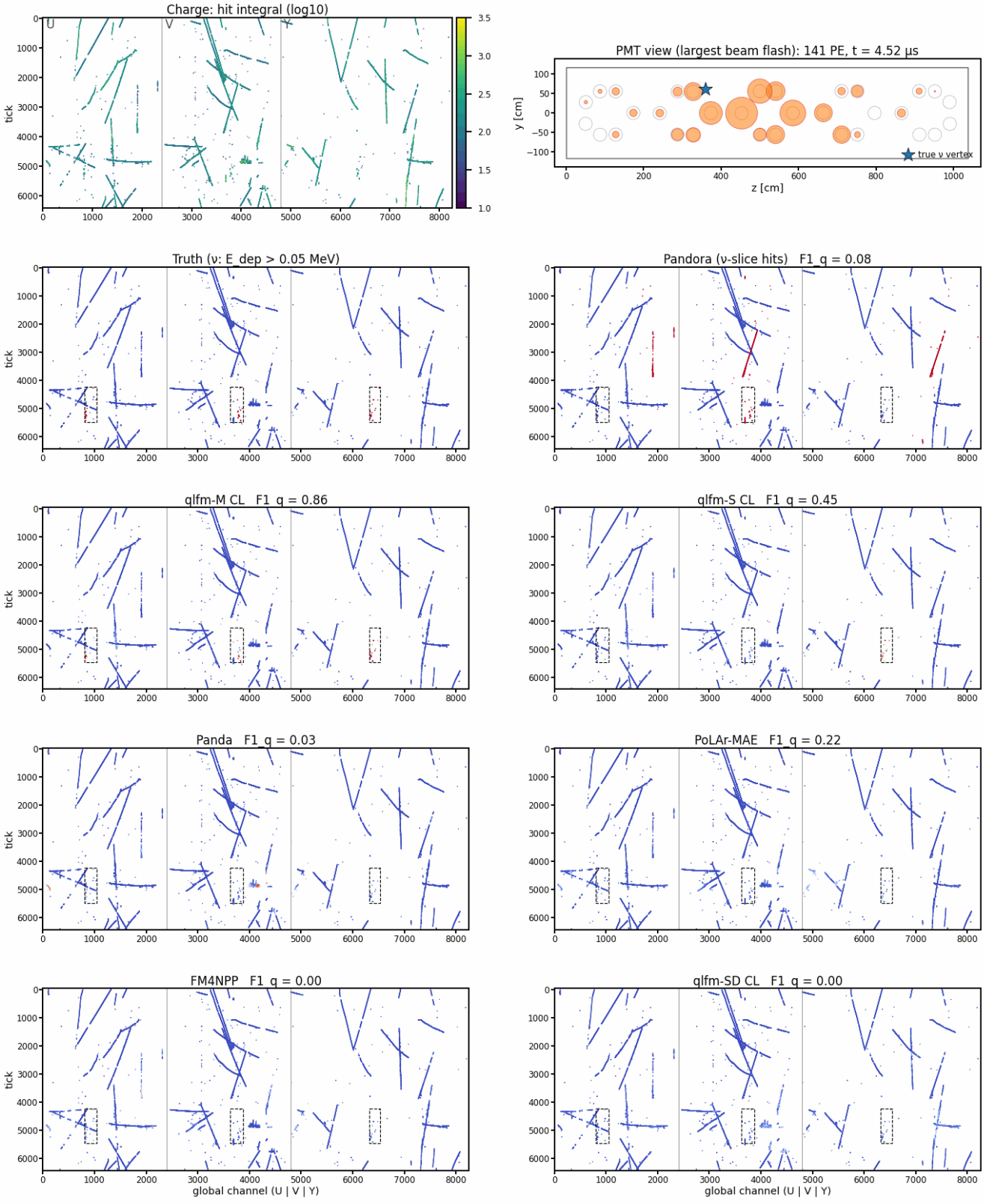


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T2 event ν-presence — easy #9 (correct: 7/8) run 7003 / subrun 1466 / event 73333 [gidx 344233]

Truth: νμ CC, E = 1.77 GeV, vtx = (263, 62, 360) cm; task label: out-AV ν

qlfm-M CL: out-AV ν ✓ qlfm-S CL: out-AV ν ✓ Panda: out-AV ν ✓ PoLAr-MAE: out-AV ν ✓
FM4NPP: out-AV ν ✓ qlfm-SD CL: out-AV ν ✓ Pandora: no-ν ✗ Pandora+BDT: out-AV ν ✓

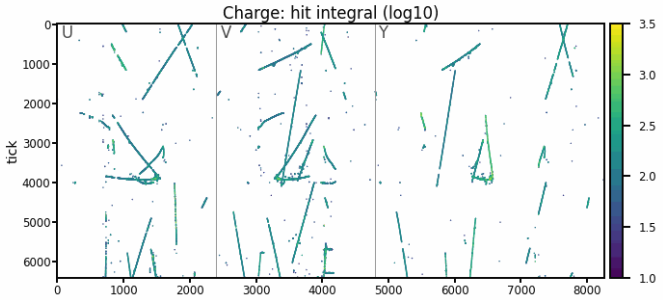


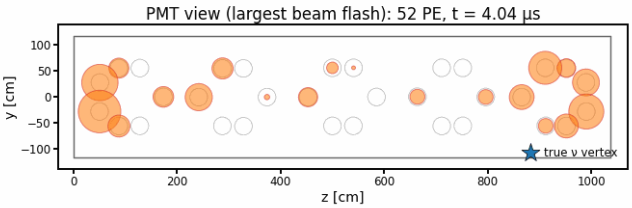
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

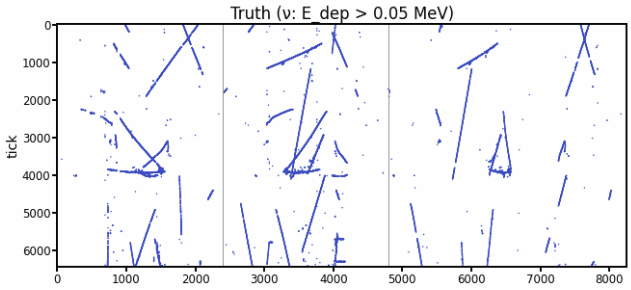
T2 event ν-presence — easy #10 (correct: 8/8) run 7023 / subrun 766 / event 38307 [gidx 639395]

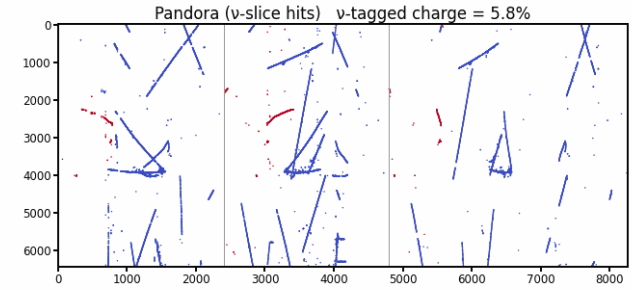
Truth: νμ CC, E = 0.56 GeV, vtx = (161, -86, 1088) cm; task label: no-ν

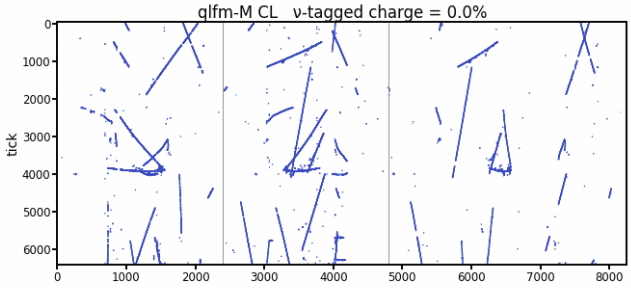
qlfm-M CL: no-ν ✓ qlfm-S CL: no-ν ✓ Panda: no-ν ✓ PoLAr-MAE: no-ν ✓
FM4NPP: no-ν ✓ qlfm-SD CL: no-ν ✓ Pandora: no-ν ✓ Pandora+BDT: no-ν ✓

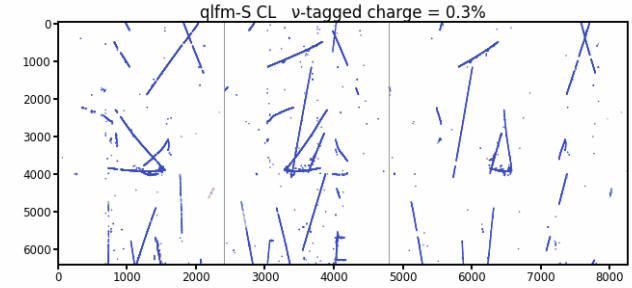


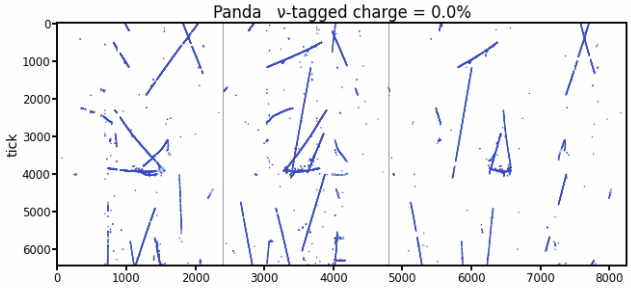


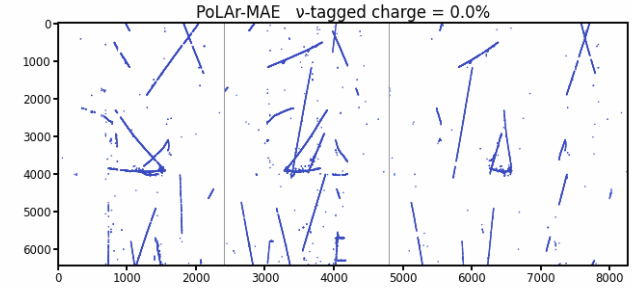


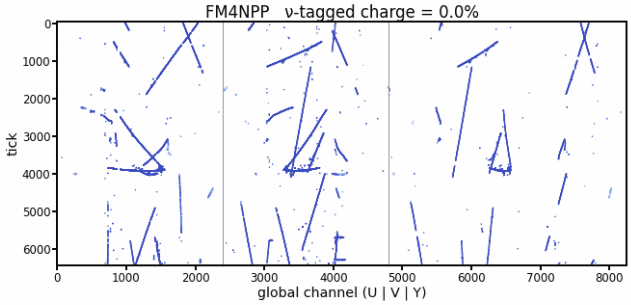


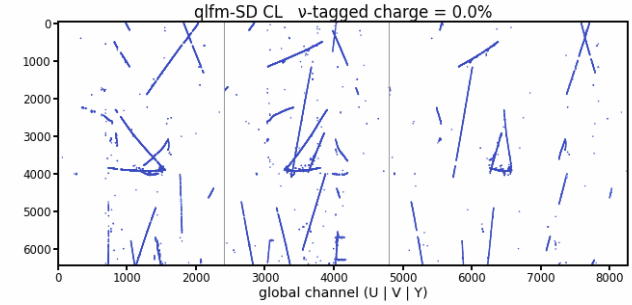










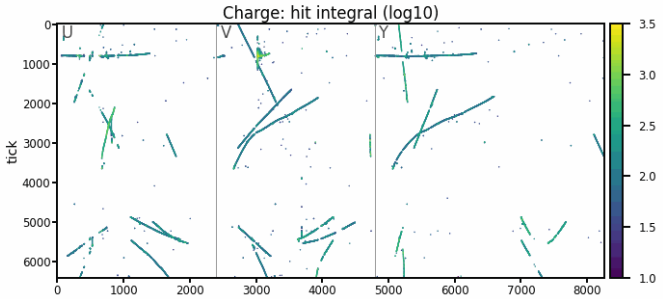


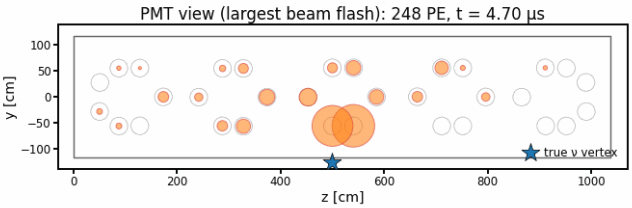
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

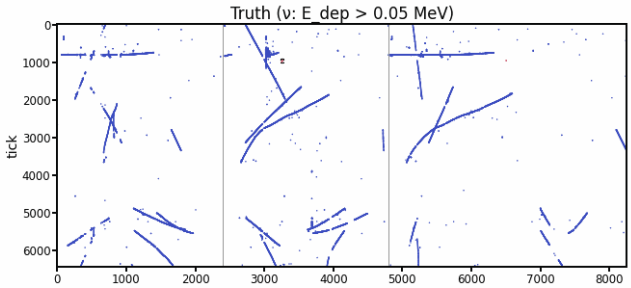
T2 event ν-presence — hard #1 (correct: 0/8) run 7049 / subrun 1647 / event 82371 [gidx 483463]

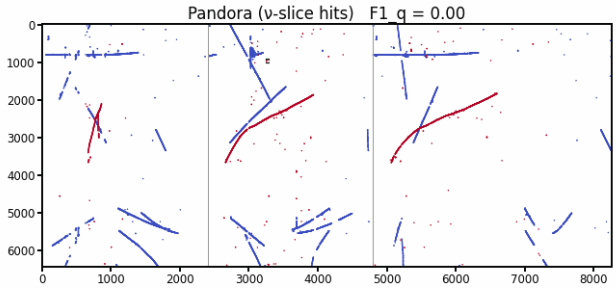
Truth: νμ CC, E = 0.86 GeV, vtx = (6, -127, 499) cm; task label: no-ν

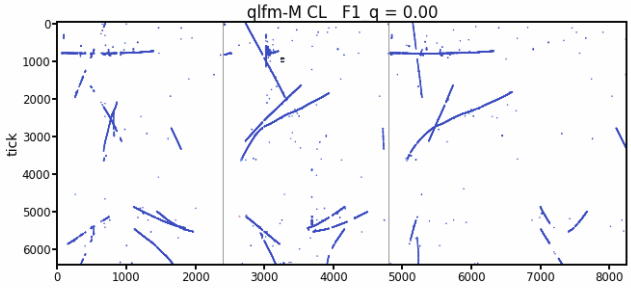
qlfm-M CL: out-AV ν ✗ qlfm-S CL: out-AV ν ✗ Panda: out-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: in-AV ν ✗ Pandora+BDT: out-AV ν ✗

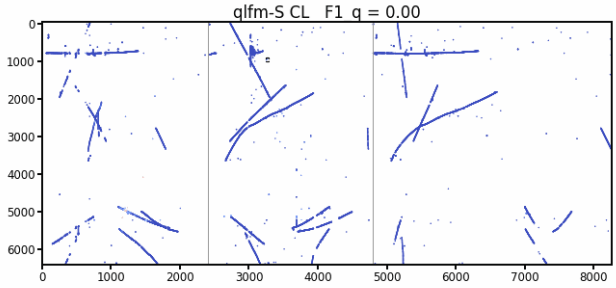


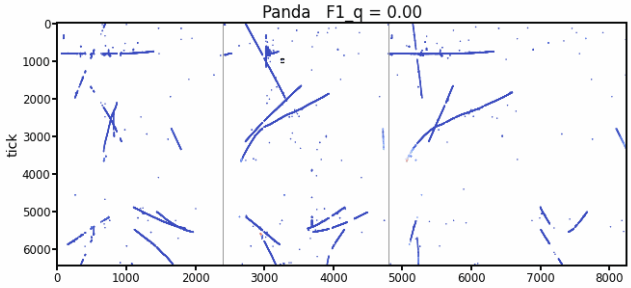


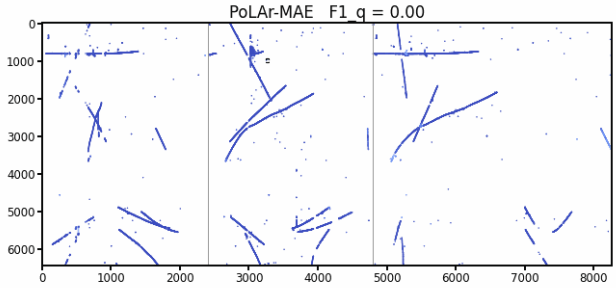


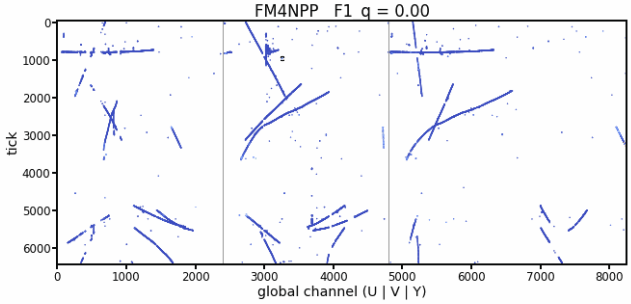


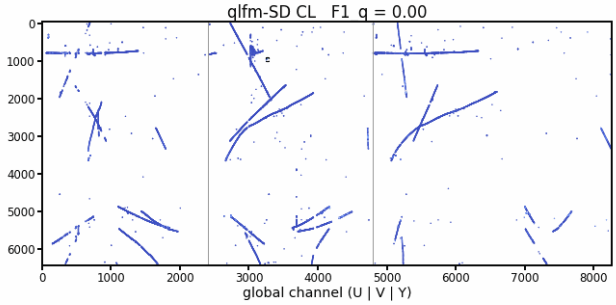










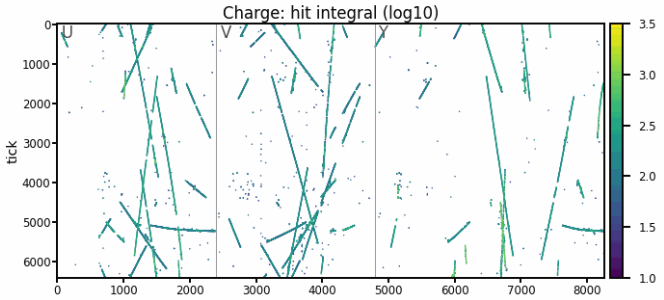


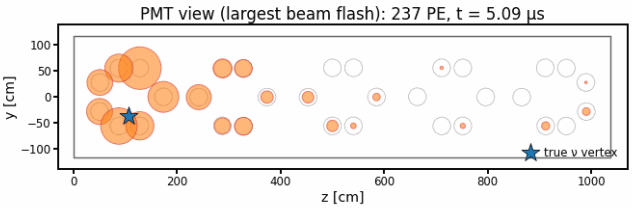
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

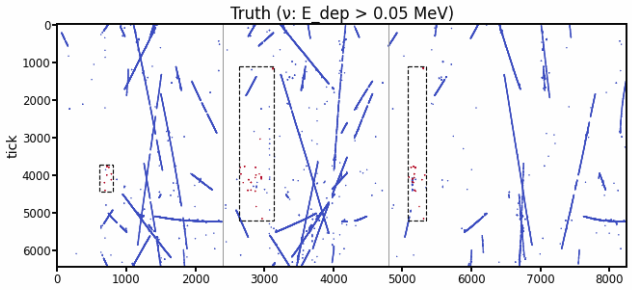
T2 event ν-presence — hard #2 (correct: 0/8) run 7047 / subrun 108 / event 5435 [gidx 291948]

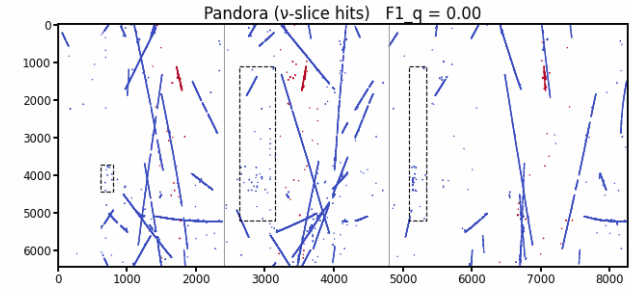
Truth: νμ NC, E = 0.50 GeV, vtx = (183, -37, 107) cm; task label: in-AV ν

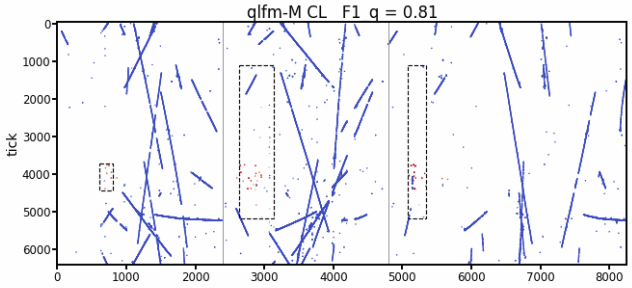
qlfm-M CL: out-AV ν ✗ qlfm-S CL: out-AV ν ✗ Panda: out-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: no-ν ✗ Pandora+BDT: out-AV ν ✗

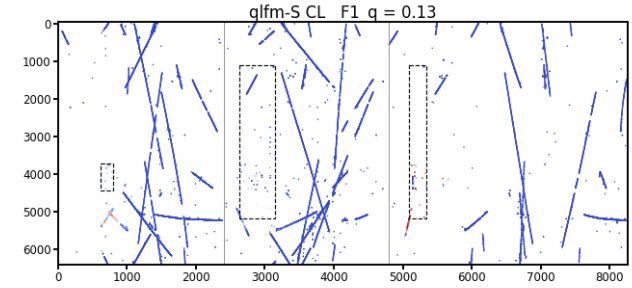


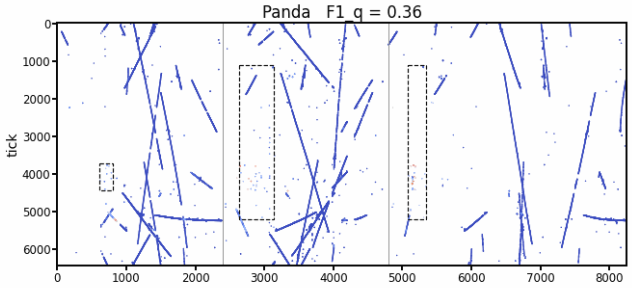


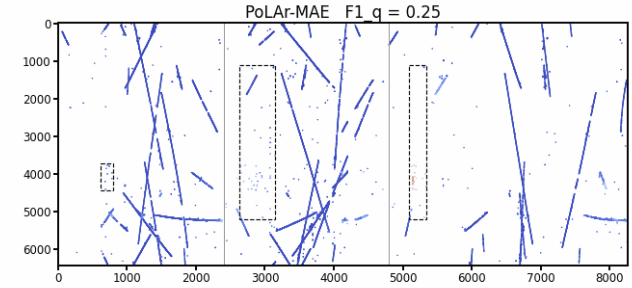


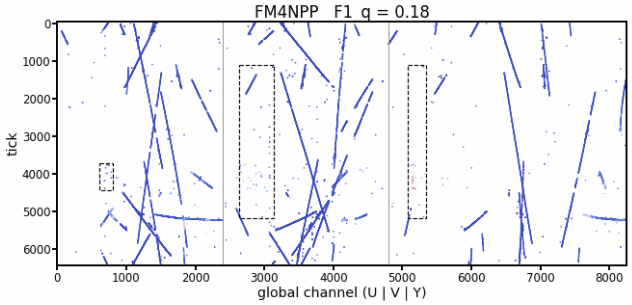


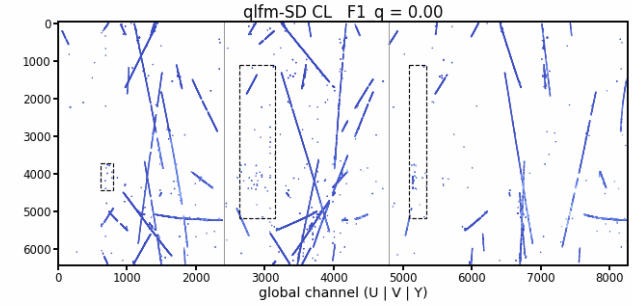










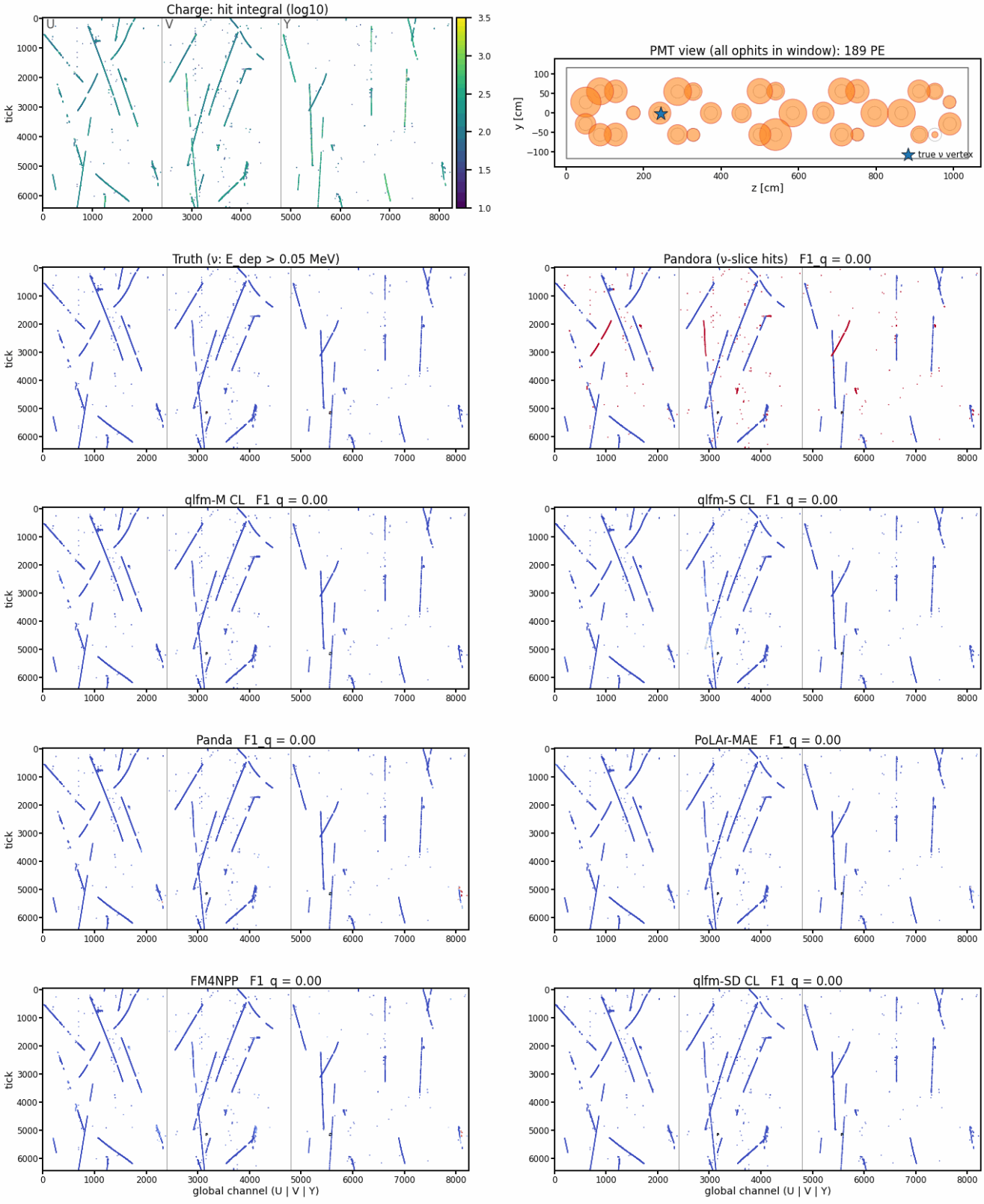


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T2 event ν-presence — hard #3 (correct: 0/8) run 7014 / subrun 1482 / event 74102 [gidx 430251]

Truth: νμ NC, E = 1.33 GeV, vtx = (319, -2, 245) cm; task label: out-AV ν

qlfm-M CL: no-ν ✗ qlfm-S CL: no-ν ✗ Panda: no-ν ✗ PoLAr-MAE: no-ν ✗
FM4NPP: no-ν ✗ qlfm-SD CL: no-ν ✗ Pandora: no-ν ✗ Pandora+BDT: no-ν ✗

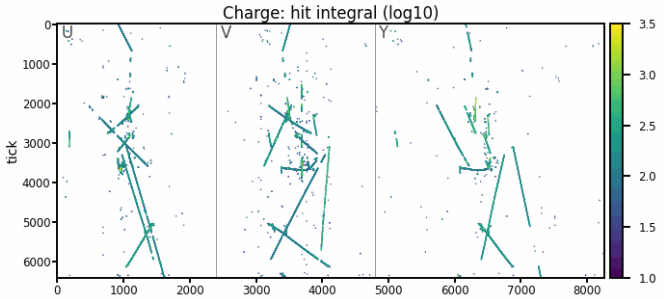


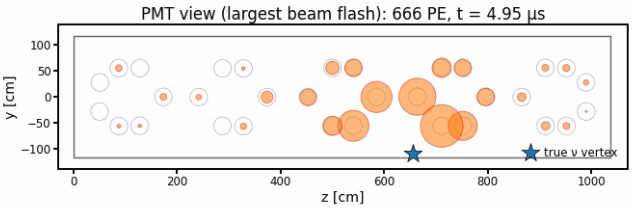
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

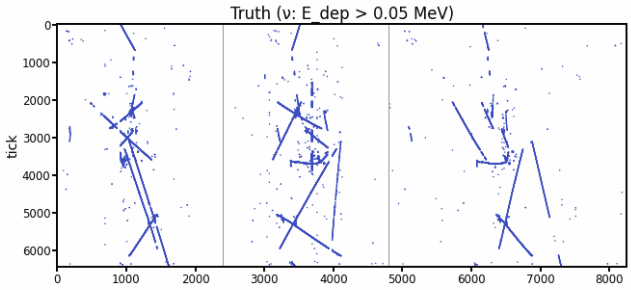
T2 event ν-presence — hard #4 (correct: 0/8) run 7021 / subrun 176 / event 8834 [gidx 466633]

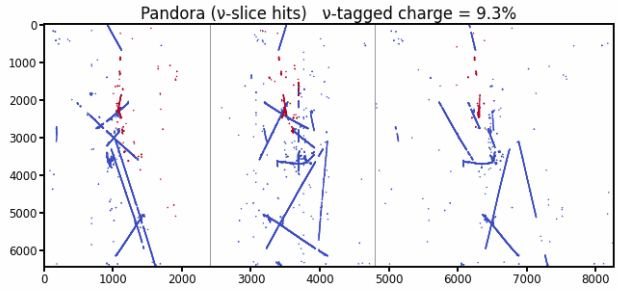
Truth: νμ CC, E = 0.63 GeV, vtx = (-9, -110, 655) cm; task label: no-ν

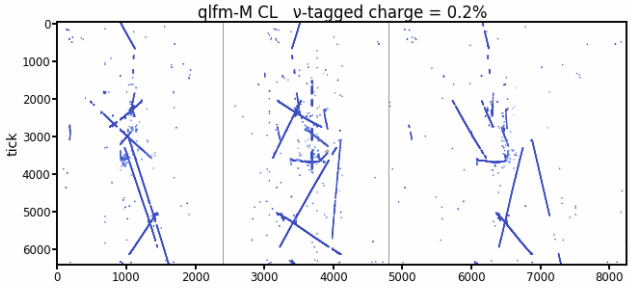
qlfm-M CL: out-AV ν ✗ qlfm-S CL: in-AV ν ✗ Panda: in-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: in-AV ν ✗ Pandora+BDT: out-AV ν ✗

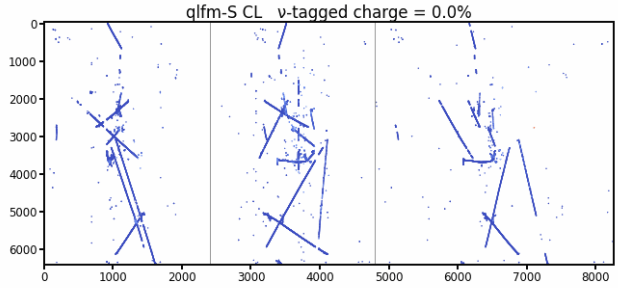


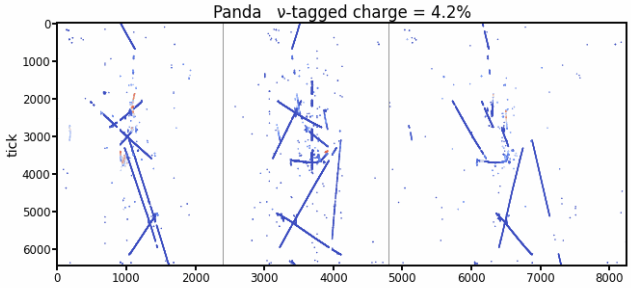


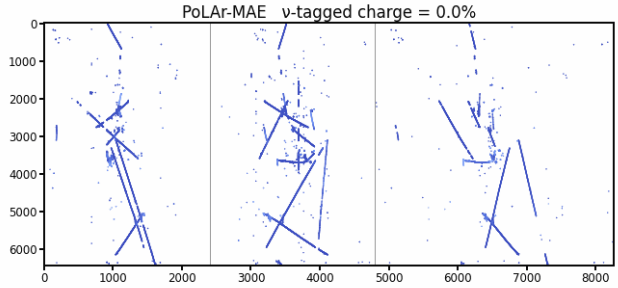


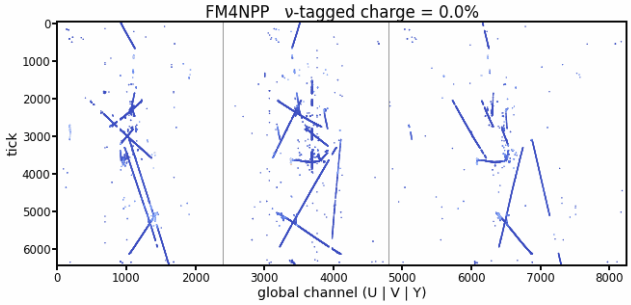


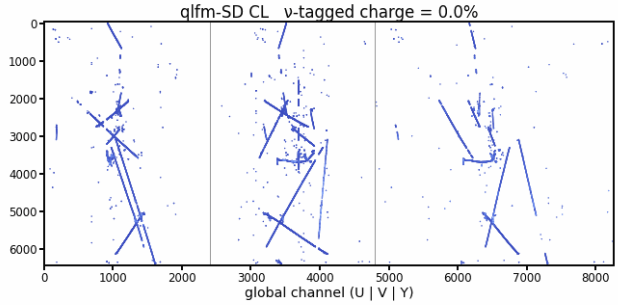










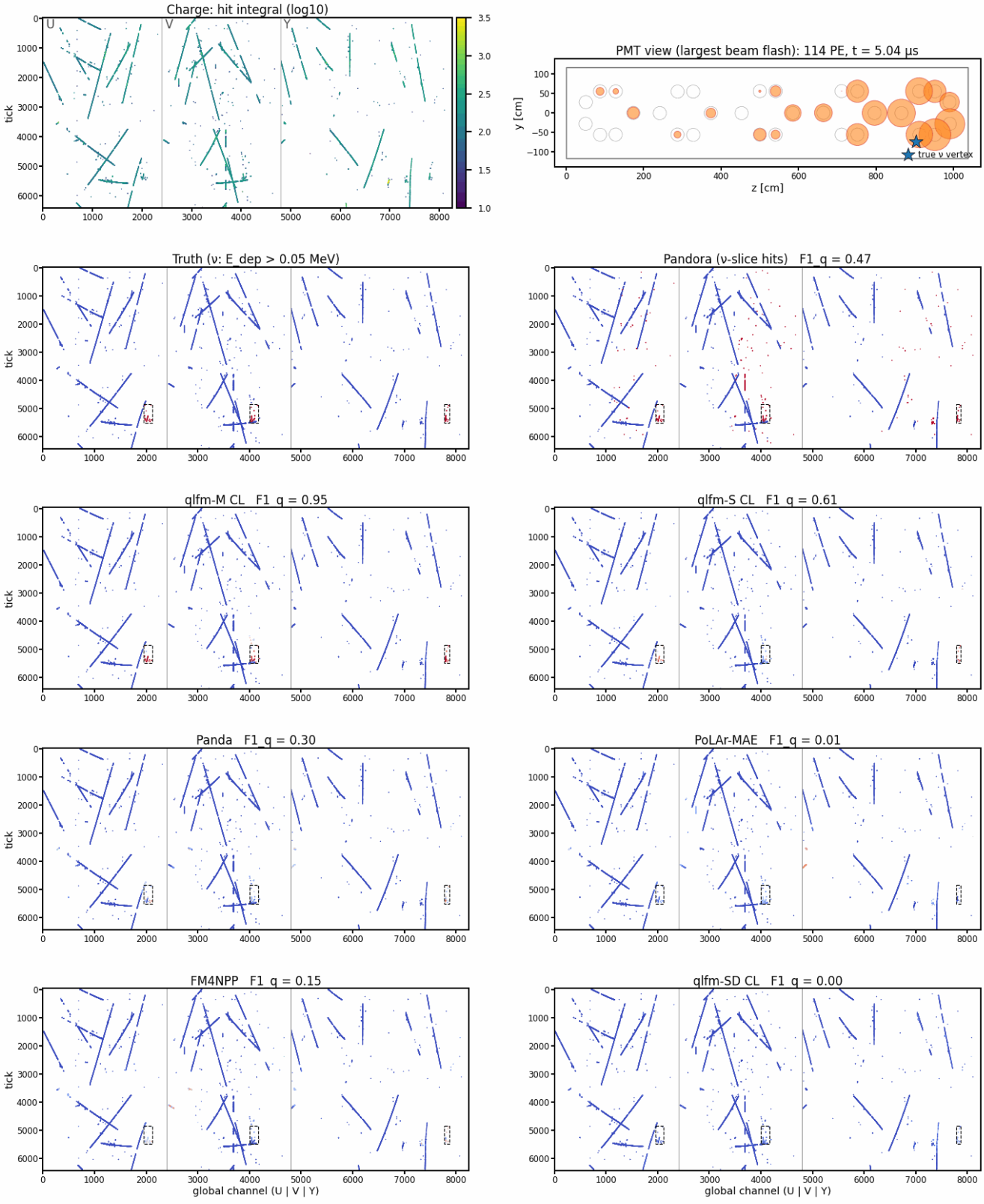


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T2 event ν-presence — hard #5 (correct: 0/8) run 7012 / subrun 553 / event 27662 [gidx 702728]

Truth: ν̄μ CC, E = 0.35 GeV, vtx = (244, -74, 903) cm; task label: in-AV ν

qlfm-M CL: out-AV ν ✗ qlfm-S CL: out-AV ν ✗ Panda: out-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: no-ν ✗ Pandora+BDT: out-AV ν ✗

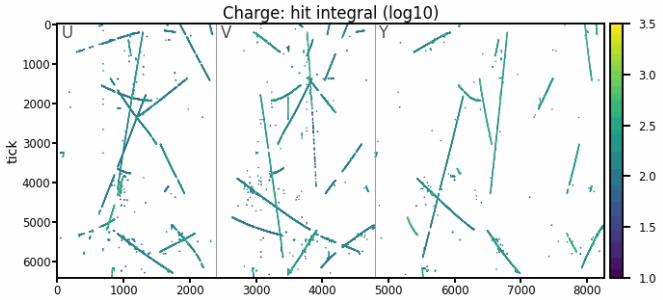


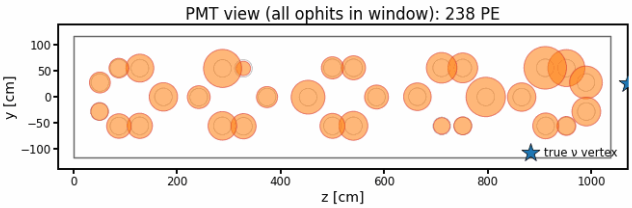
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

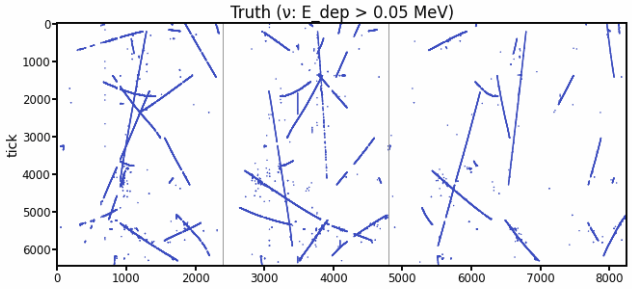
T2 event ν-presence — hard #6 (correct: 0/8) run 7017 / subrun 1239 / event 61985 [gidx 290390]

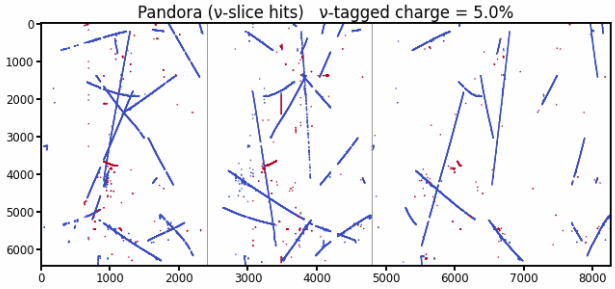
Truth: νμ CC, E = 1.29 GeV, vtx = (190, 27, 1071) cm; task label: out-AV ν

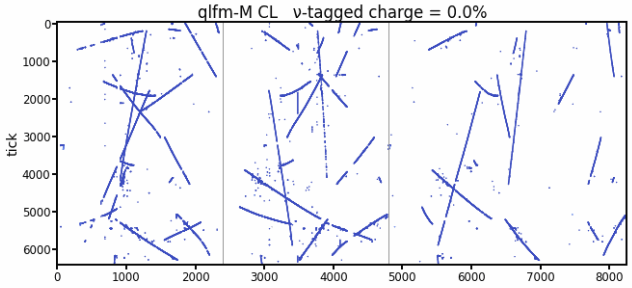
qlfm-M CL: no-ν ✗ qlfm-S CL: no-ν ✗ Panda: no-ν ✗ PoLAr-MAE: no-ν ✗
FM4NPP: no-ν ✗ qlfm-SD CL: no-ν ✗ Pandora: no-ν ✗ Pandora+BDT: no-ν ✗

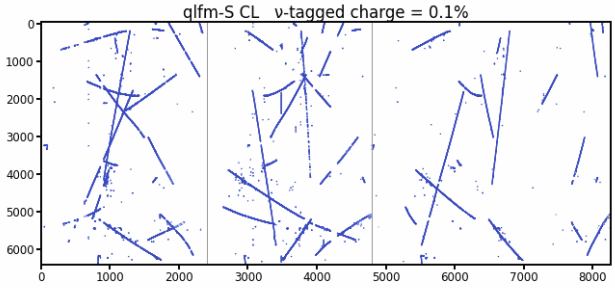


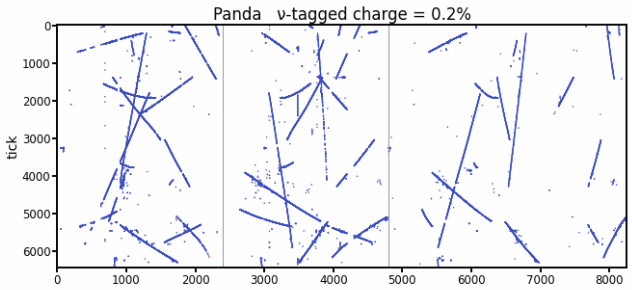


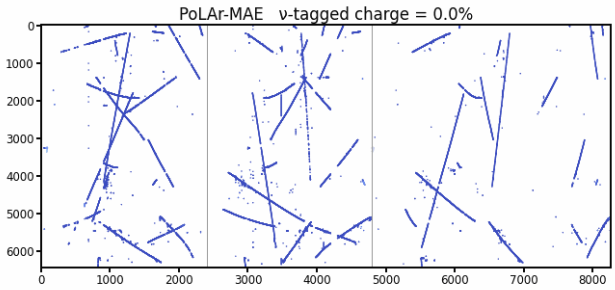


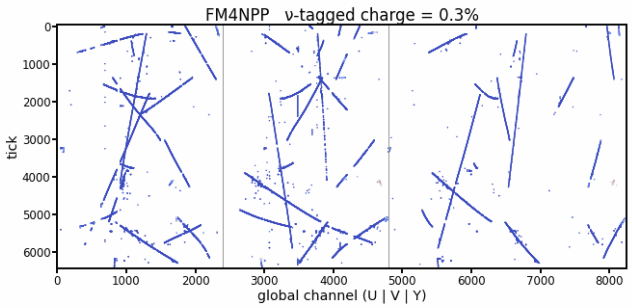


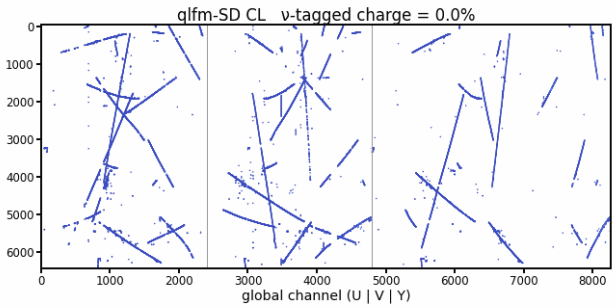










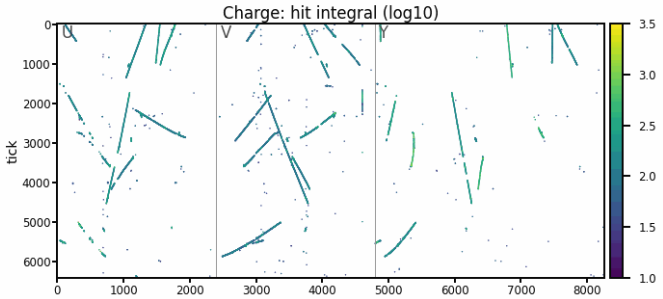


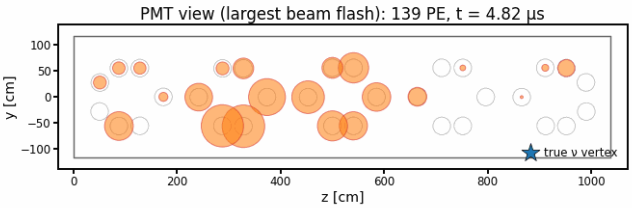
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

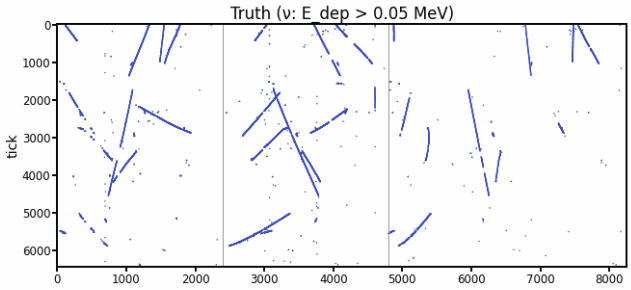
T2 event ν-presence — hard #7 (correct: 0/8) run 7004 / subrun 1211 / event 60593 [gidx 653993]

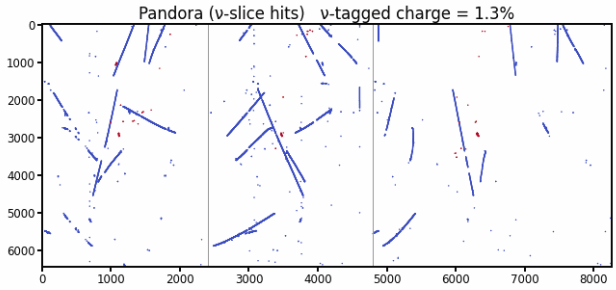
Truth: νμ CC, E = 0.86 GeV, vtx = (210, -164, 268) cm; task label: no-ν

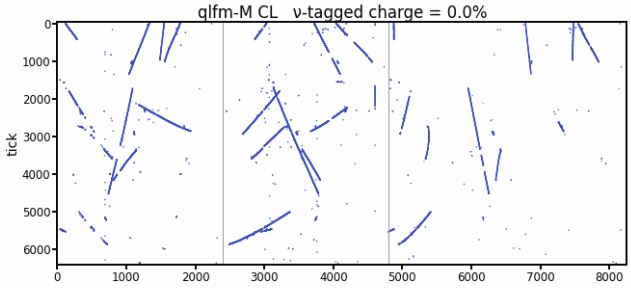
qlfm-M CL: out-AV ν ✗ qlfm-S CL: out-AV ν ✗ Panda: out-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: in-AV ν ✗ Pandora+BDT: out-AV ν ✗

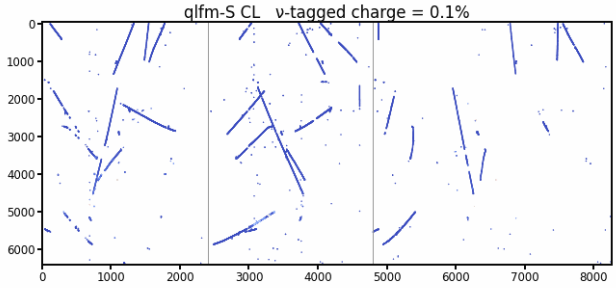


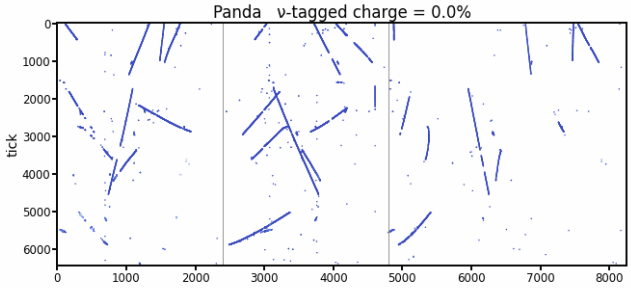


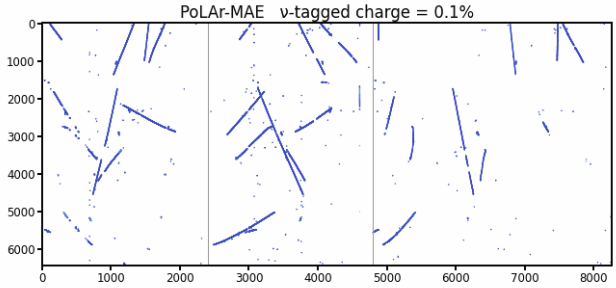


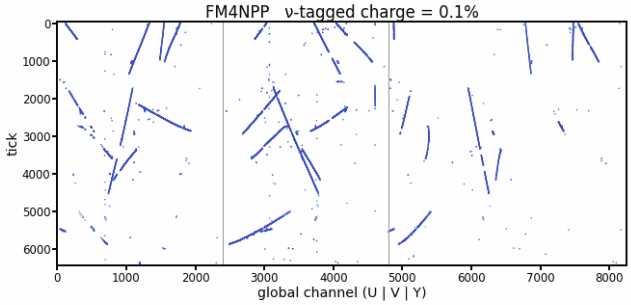


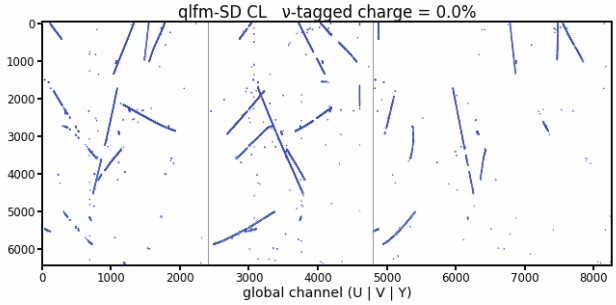










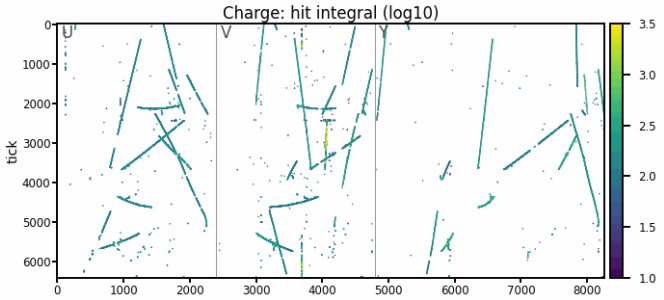


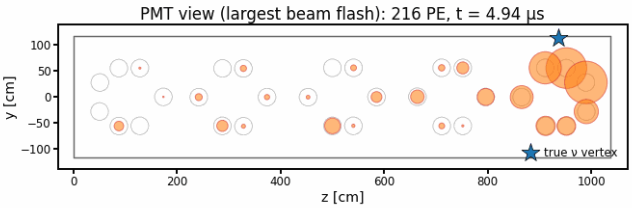
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

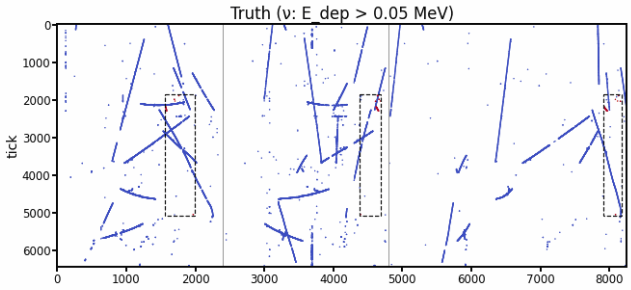
T2 event ν-presence — hard #8 (correct: 0/8) run 7049 / subrun 808 / event 40422 [gidx 575779]

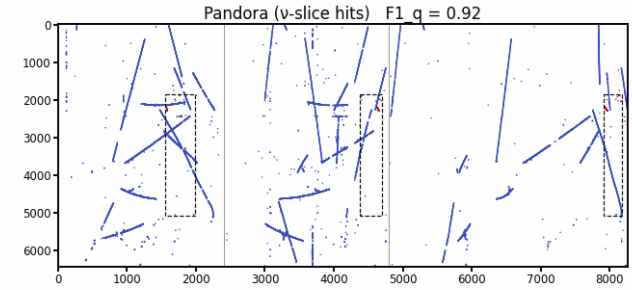
Truth: νμ NC, E = 0.97 GeV, vtx = (75, 114, 937) cm; task label: in-AV ν

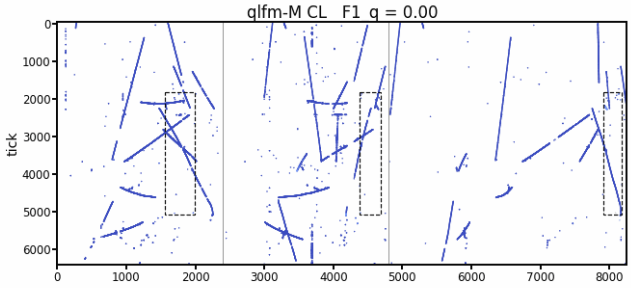
qlfm-M CL: out-AV ν ✗ qlfm-S CL: out-AV ν ✗ Panda: out-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: no-ν ✗ Pandora+BDT: out-AV ν ✗

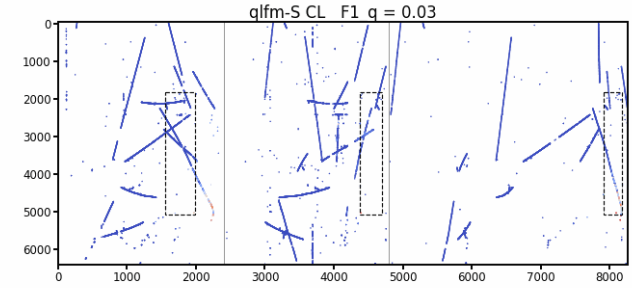


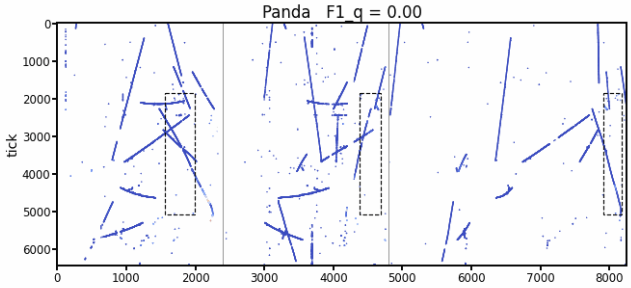


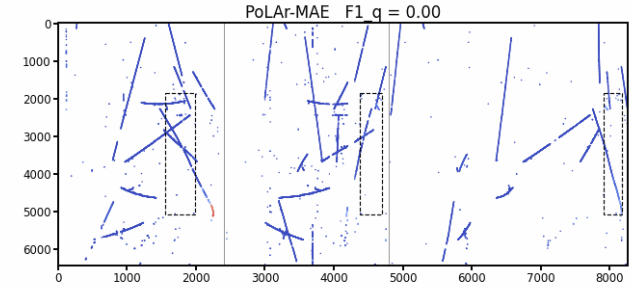


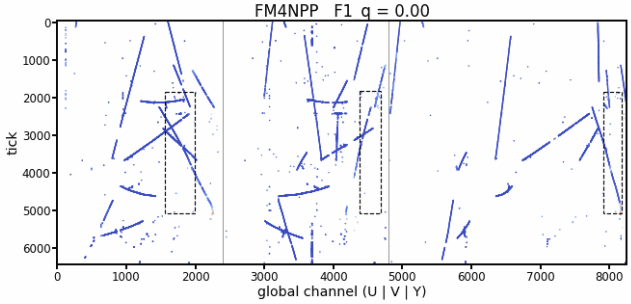


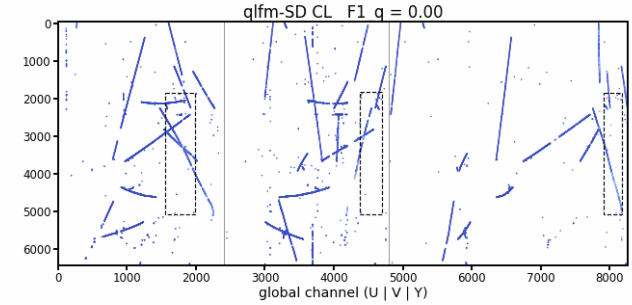










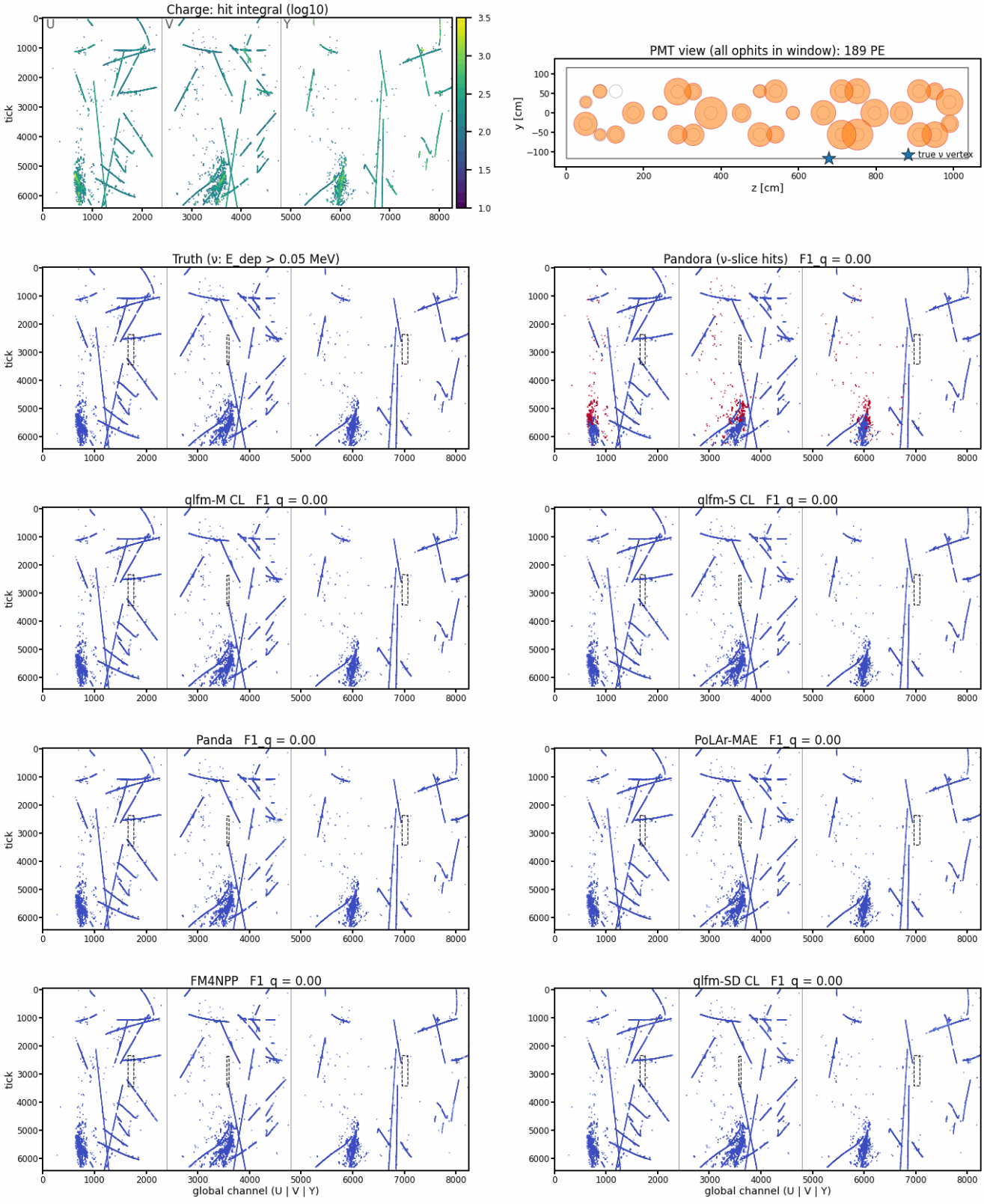


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T2 event ν-presence — hard #9 (correct: 0/8) run 7049 / subrun 1360 / event 68007 [gidx 737137]

Truth: νμ NC, E = 0.31 GeV, vtx = (118, -118, 678) cm; task label: out-AV ν

qlfm-M CL: no-ν ✗ qlfm-S CL: no-ν ✗ Panda: no-ν ✗ PoLAr-MAE: no-ν ✗
FM4NPP: no-ν ✗ qlfm-SD CL: no-ν ✗ Pandora: no-ν ✗ Pandora+BDT: no-ν ✗

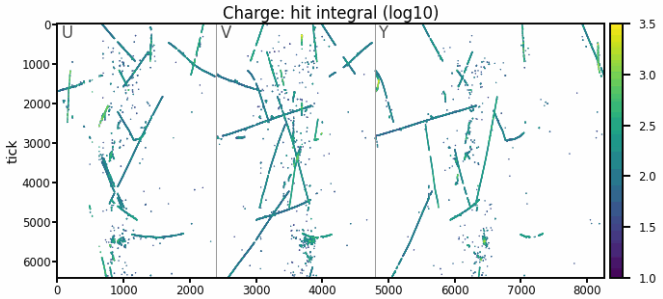


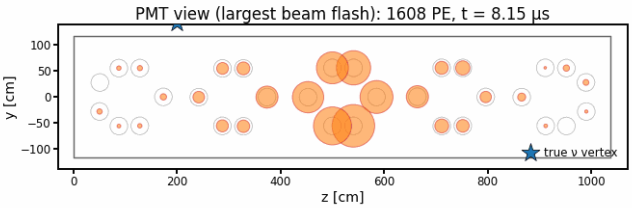
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

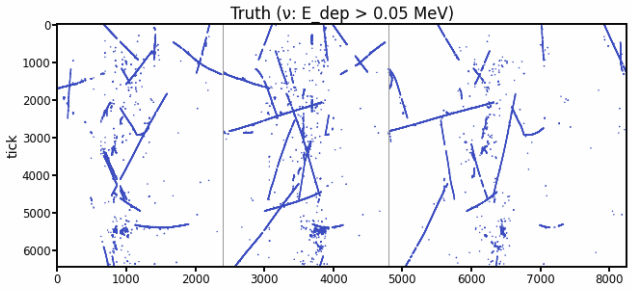
T2 event ν-presence — hard #10 (correct: 0/8) run 7135 / subrun 19 / event 967 [gidx 336854]

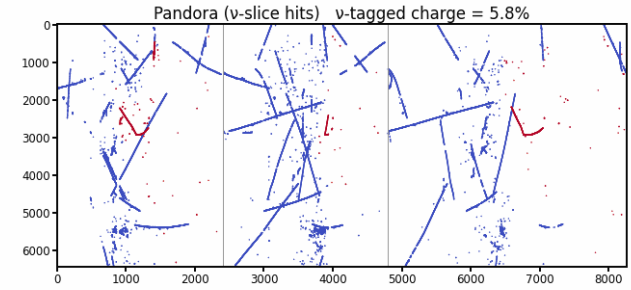
Truth: νμ CC, E = 0.75 GeV, vtx = (201, 144, 199) cm; task label: no-ν

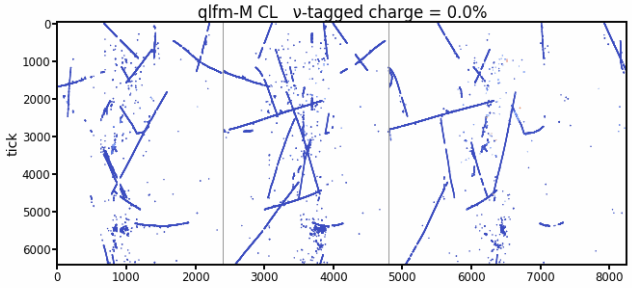
qlfm-M CL: out-AV ν ✗ qlfm-S CL: out-AV ν ✗ Panda: out-AV ν ✗ PoLAr-MAE: out-AV ν ✗
FM4NPP: out-AV ν ✗ qlfm-SD CL: out-AV ν ✗ Pandora: in-AV ν ✗ Pandora+BDT: in-AV ν ✗

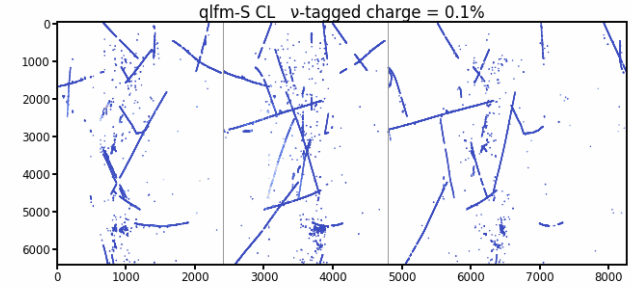


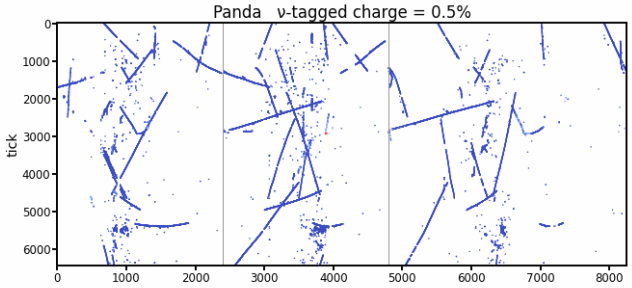


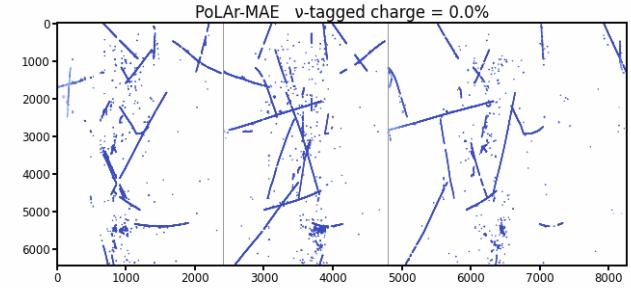


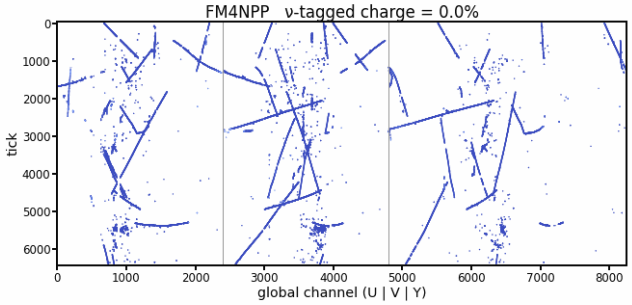


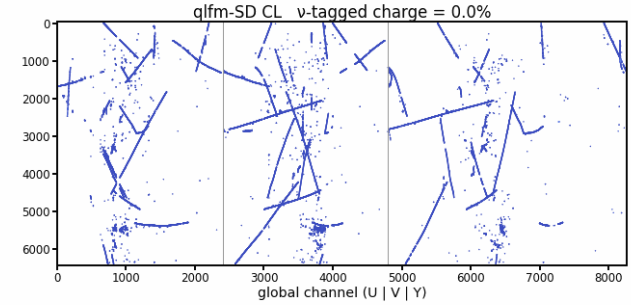






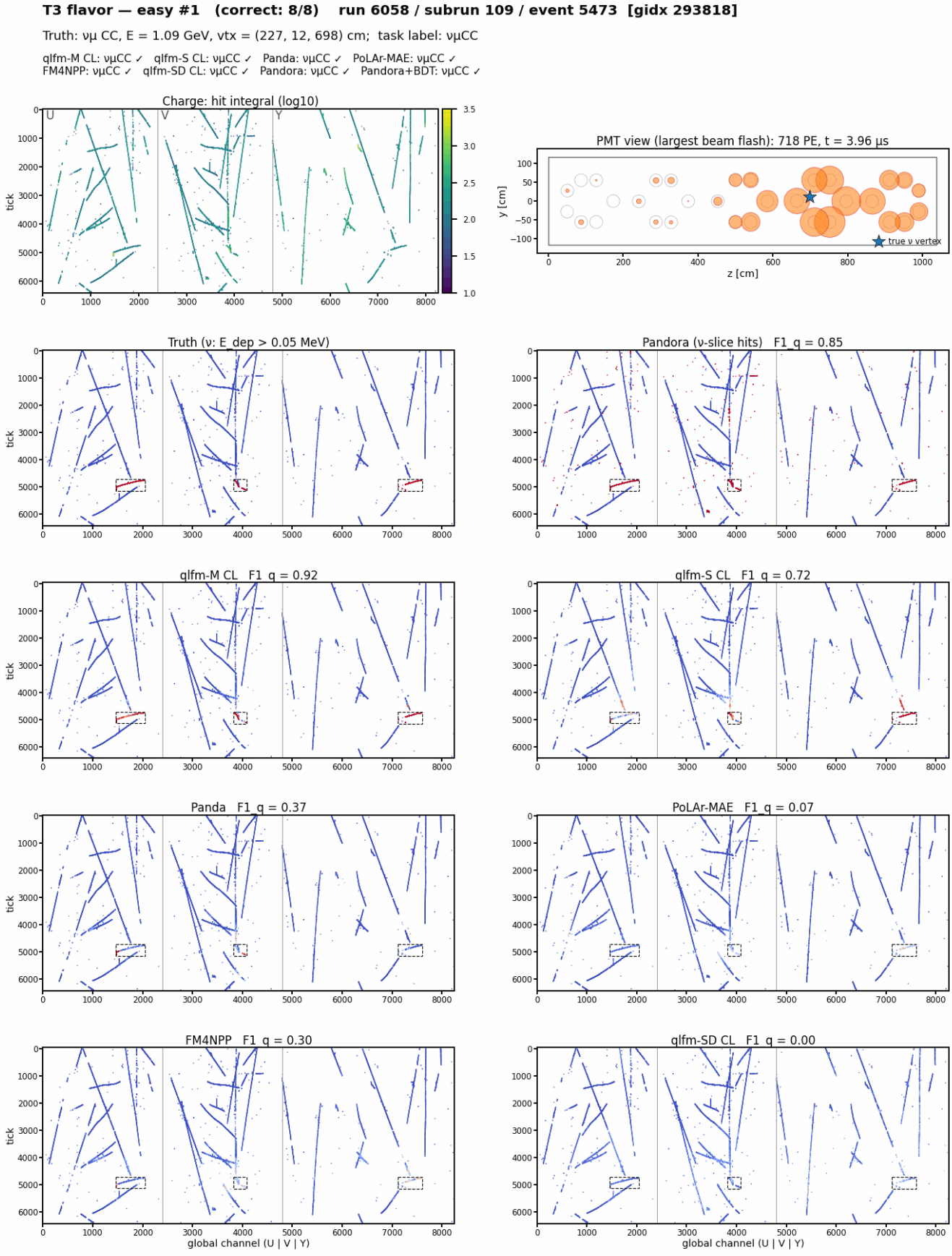






hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

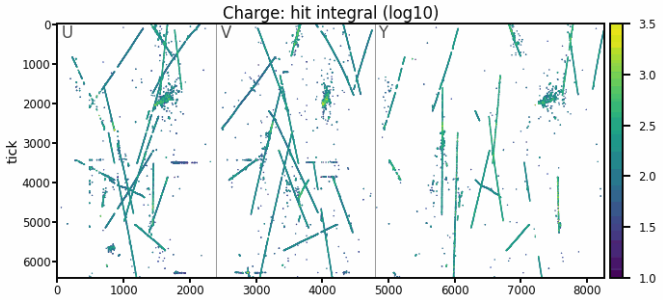
A.3 T3: flavor

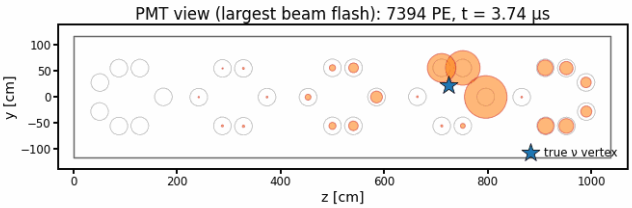

T3 flavor — easy #1 (correct: 8/8) run 6058 / subrun 109 / event 5473 [gidx 293818]
Truth: νμ CC, E = 1.09 GeV, vtx = (227, 12, 698) cm; task label: νμCC
qlfm-M CL: νμCC ✓ qlfm-S CL: νμCC ✓ Panda: νμCC ✓ PoLAr-MAE: νμCC ✓
FM4NPP: νμCC ✓ qlfm-SD CL: νμCC ✓ Pandora: νμCC ✓ Pandora+BDT: νμCC ✓
Charge: hit integral (log10)
U
V
Y
tick
PMT view (largest beam flash): 718 PE, t = 3.96 µs
y [cm]
z [cm]
true ν vertex
Truth (ν: E_dep > 0.05 MeV)
Pandora (ν-slice hits) F1_q = 0.85
qlfm-M CL F1_q = 0.92
qlfm-S CL F1_q = 0.72
Panda F1_q = 0.37
PoLAr-MAE F1_q = 0.07
FM4NPP F1_q = 0.30
qlfm-SD CL F1_q = 0.00
global channel (U | V | Y)
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

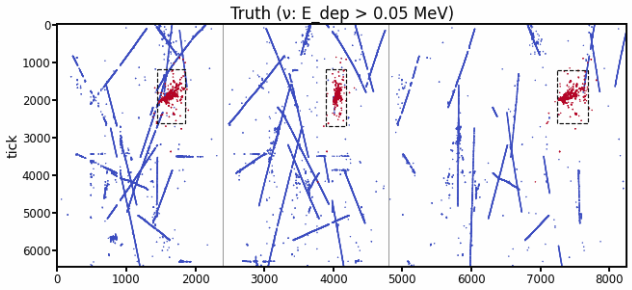
T3 flavor — easy #2 (correct: 8/8) run 5127 / subrun 44 / event 2249 [gidx 786390]

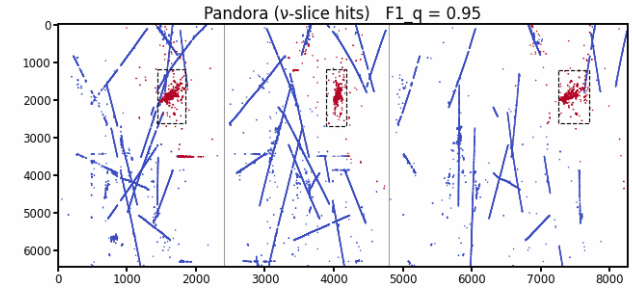
Truth: νe CC, E = 1.75 GeV, vtx = (65, 23, 725) cm; task label: νeCC

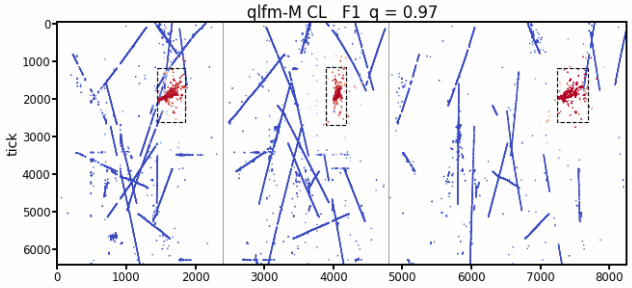
qlfm-M CL: νeCC ✓ qlfm-S CL: νeCC ✓ Panda: νeCC ✓ PoLAr-MAE: νeCC ✓
FM4NPP: νeCC ✓ qlfm-SD CL: νeCC ✓ Pandora: νeCC ✓ Pandora+BDT: νeCC ✓

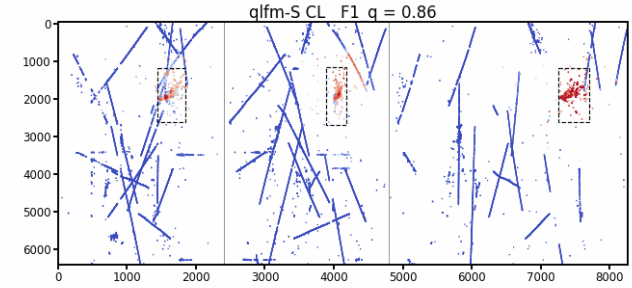


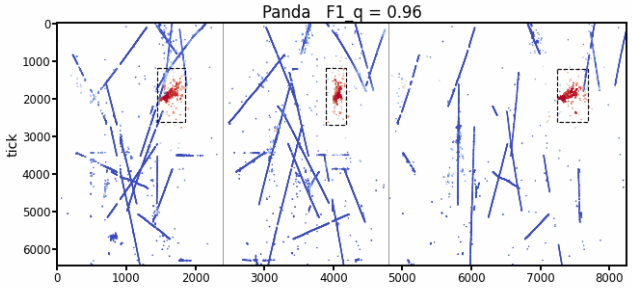


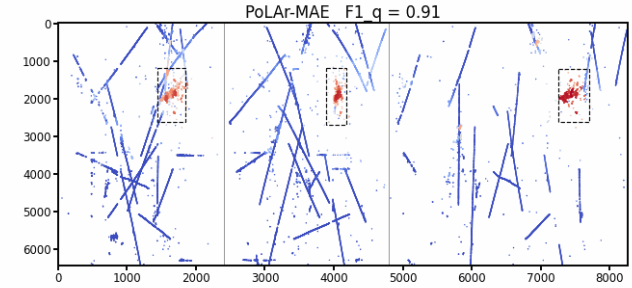


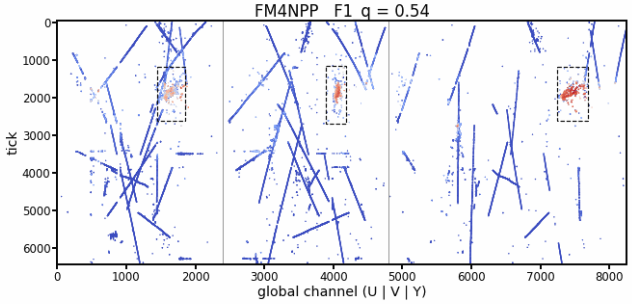


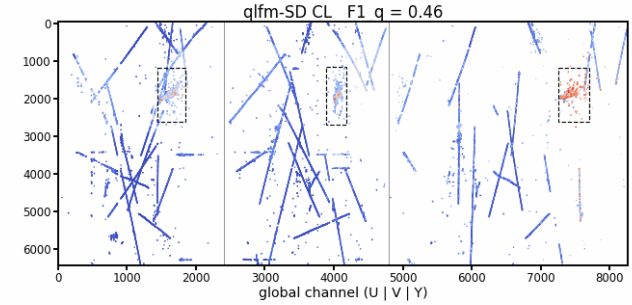










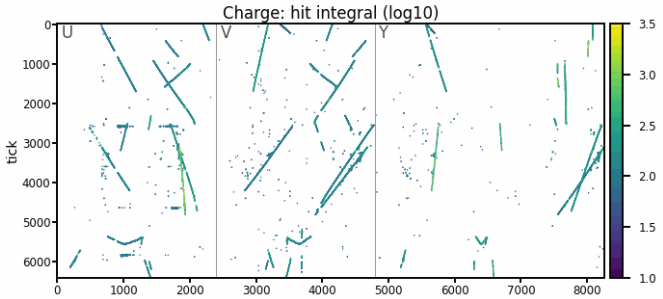


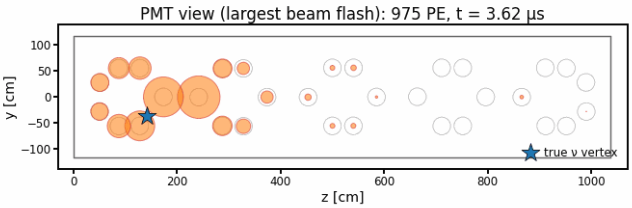
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

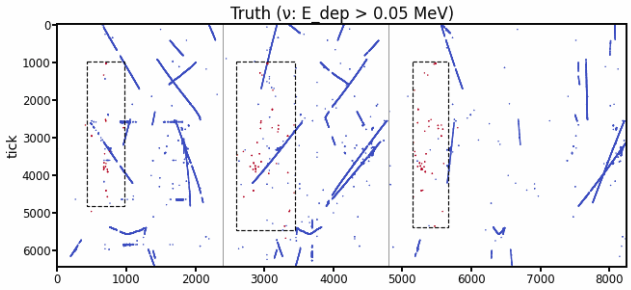
T3 flavor — easy #3 (correct: 8/8) run 5171 / subrun 8 / event 411 [gidx 199457]

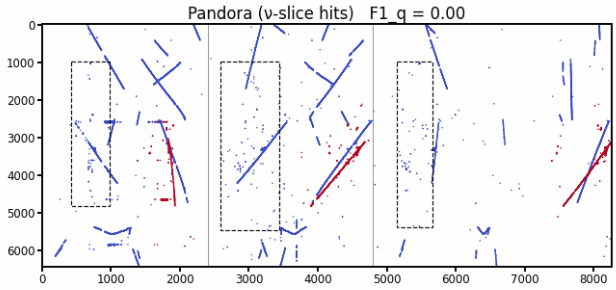
Truth: νμ NC, E = 1.04 GeV, vtx = (155, -38, 141) cm; task label: NC

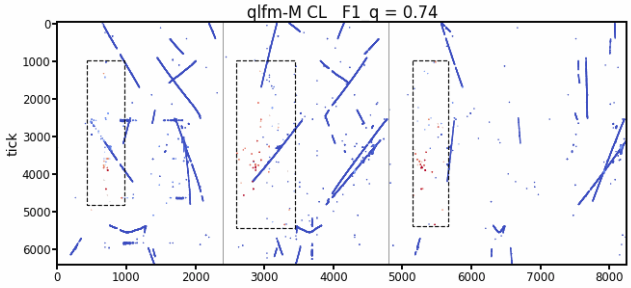
qlfm-M CL: NC ✓ qlfm-S CL: NC ✓ Panda: NC ✓ PoLAr-MAE: NC ✓
FM4NPP: NC ✓ qlfm-SD CL: NC ✓ Pandora: NC ✓ Pandora+BDT: NC ✓

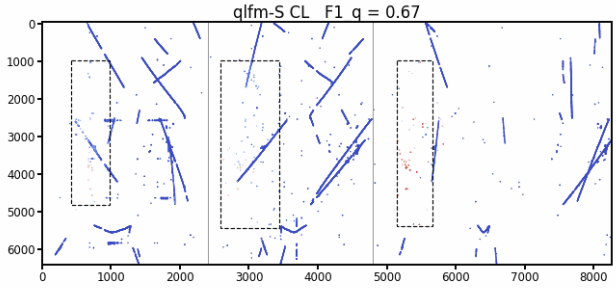


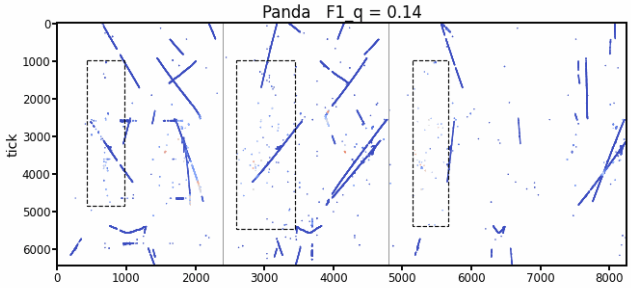


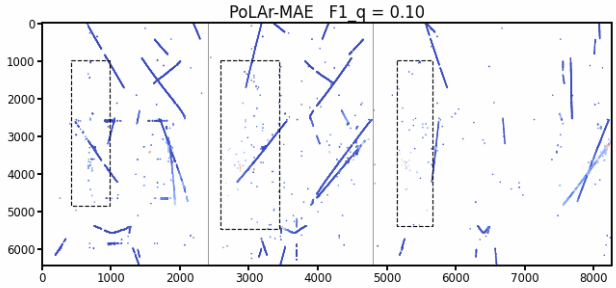


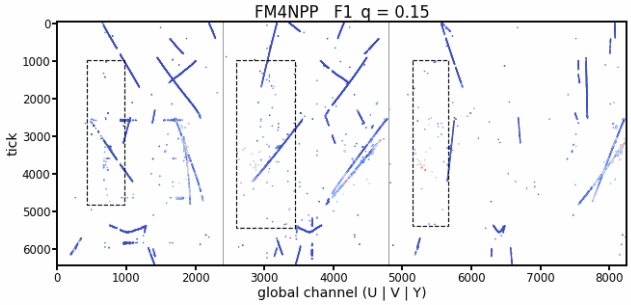


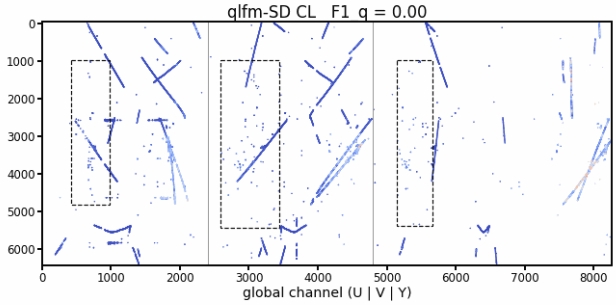










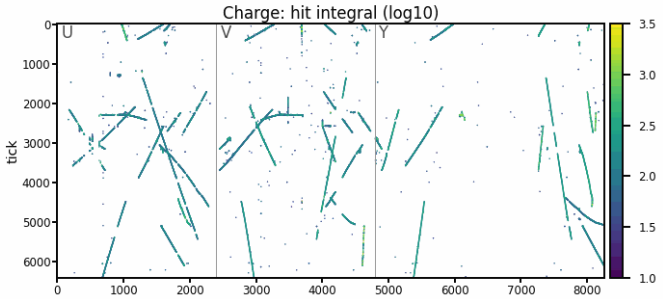


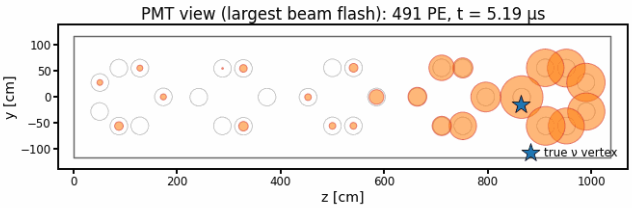
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

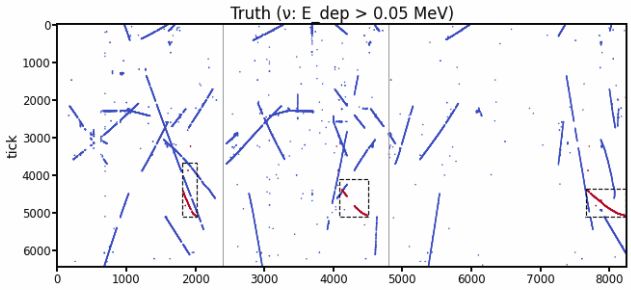
T3 flavor — easy #4 (correct: 8/8) run 7049 / subrun 640 / event 32046 [gidx 629920]

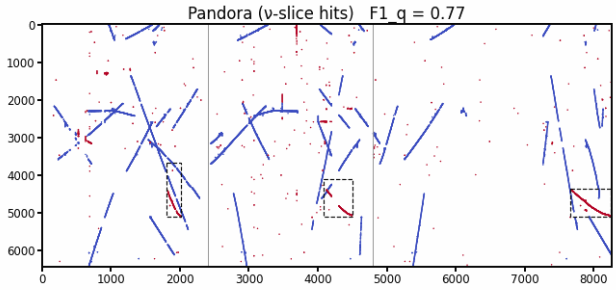
Truth: νμ CC, E = 0.71 GeV, vtx = (193, -15, 864) cm; task label: νμCC

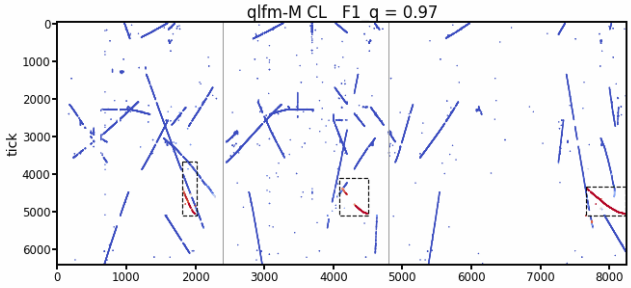
qlfm-M CL: νμCC ✓ qlfm-S CL: νμCC ✓ Panda: νμCC ✓ PoLAr-MAE: νμCC ✓
FM4NPP: νμCC ✓ qlfm-SD CL: νμCC ✓ Pandora: νμCC ✓ Pandora+BDT: νμCC ✓

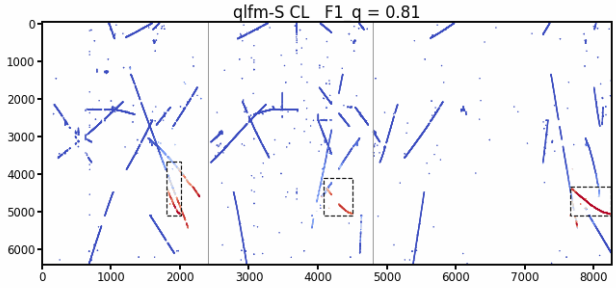


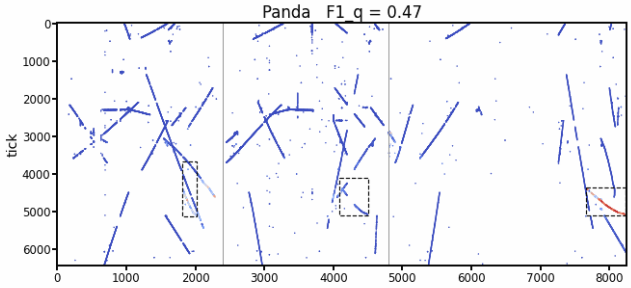


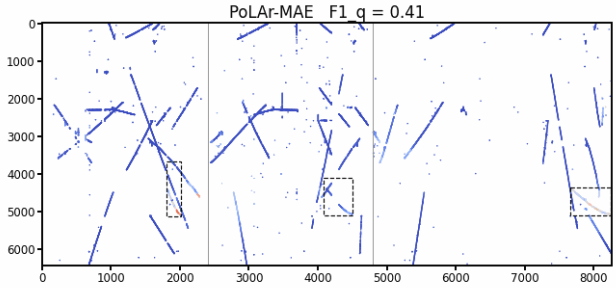


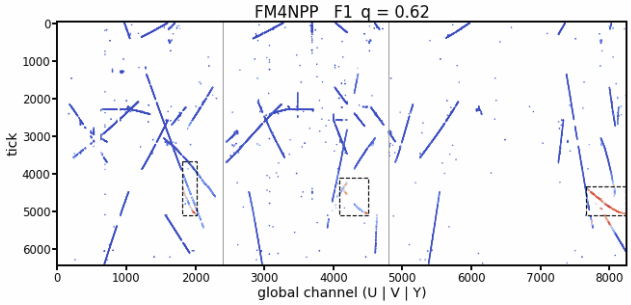


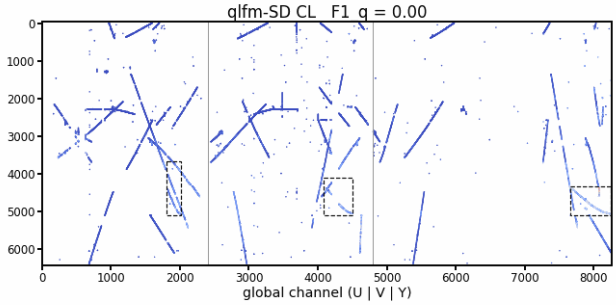










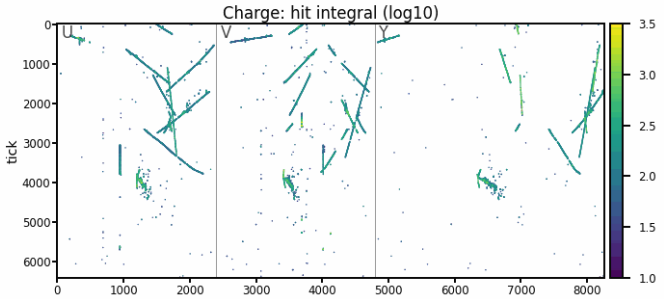


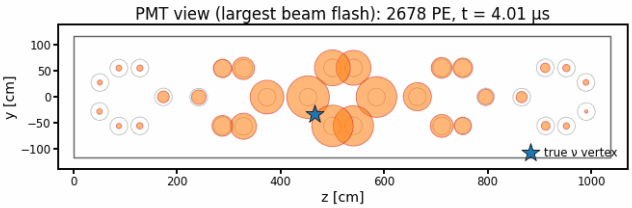
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

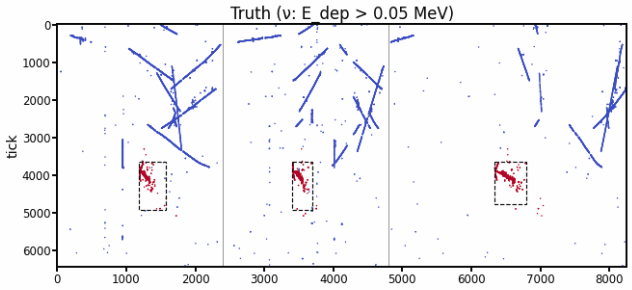
T3 flavor — easy #5 (correct: 8/8) run 7011 / subrun 126 / event 6338 [gidx 766125]

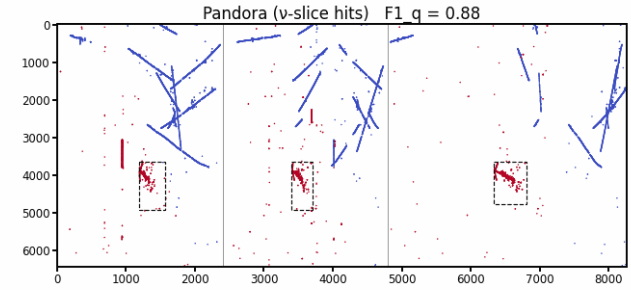
Truth: νe CC, E = 1.39 GeV, vtx = (164, -33, 465) cm; task label: νeCC

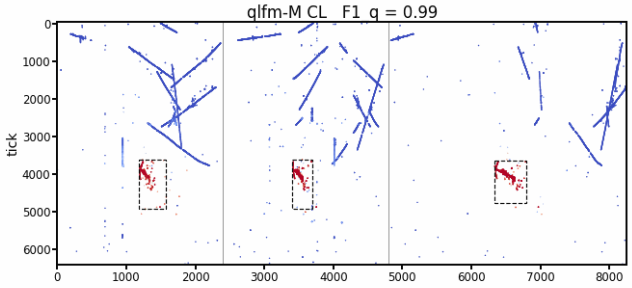
qlfm-M CL: νeCC ✓ qlfm-S CL: νeCC ✓ Panda: νeCC ✓ PoLAr-MAE: νeCC ✓
FM4NPP: νeCC ✓ qlfm-SD CL: νeCC ✓ Pandora: νeCC ✓ Pandora+BDT: νeCC ✓

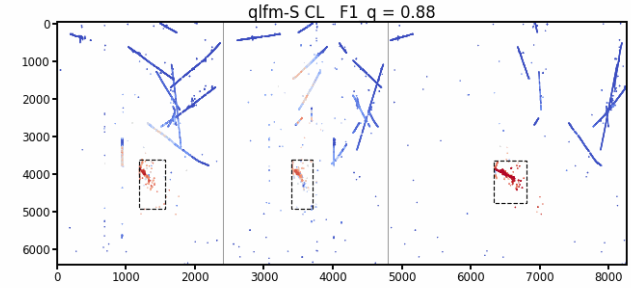


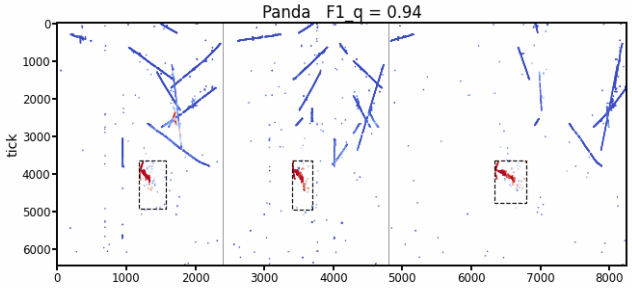


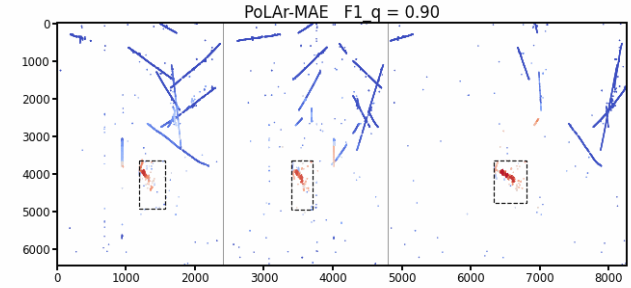


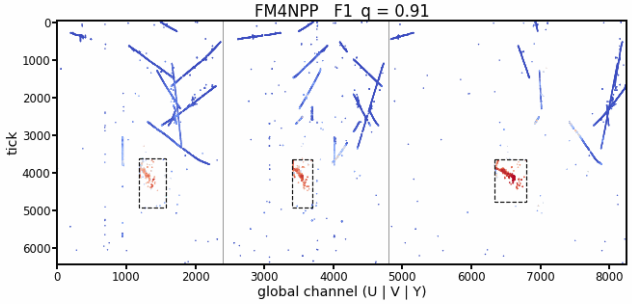


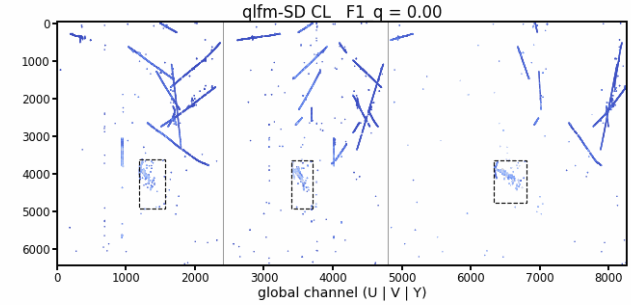










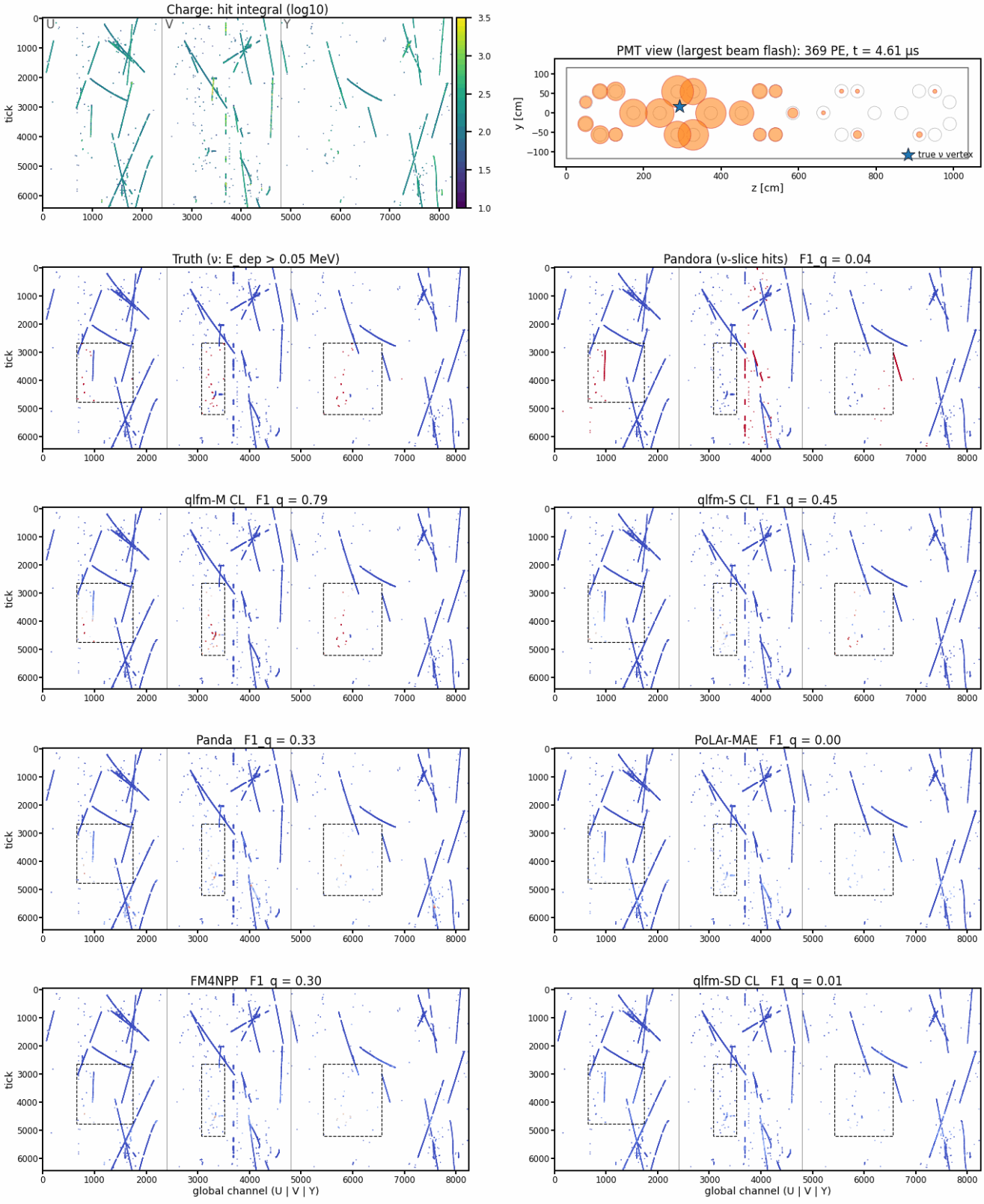


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T3 flavor — easy #6 (correct: 8/8) run 7001 / subrun 198 / event 9912 [gidx 549766]

Truth: νμ NC, E = 0.61 GeV, vtx = (179, 17, 293) cm; task label: NC

qlfm-M CL: NC ✓ qlfm-S CL: NC ✓ Panda: NC ✓ PoLAr-MAE: NC ✓
FM4NPP: NC ✓ qlfm-SD CL: NC ✓ Pandora: NC ✓ Pandora+BDT: NC ✓

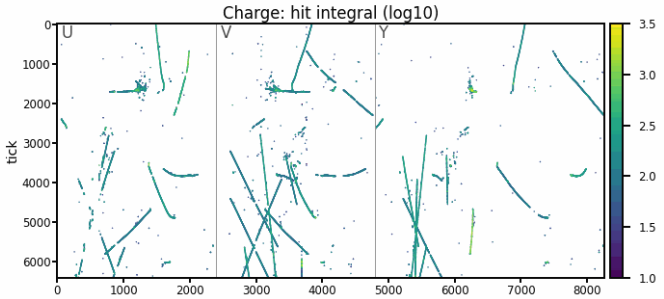


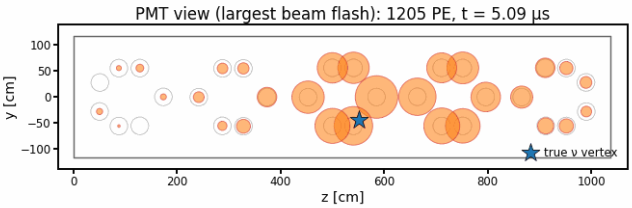
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

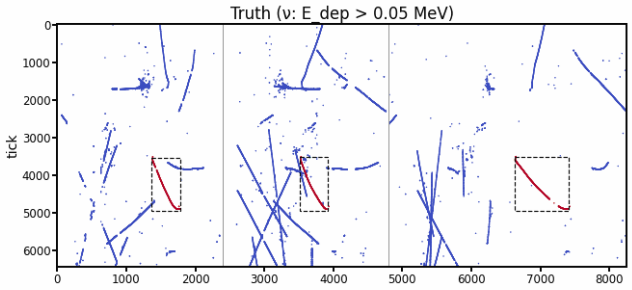
T3 flavor — easy #7 (correct: 8/8) run 7019 / subrun 147 / event 7373 [gidx 16064]

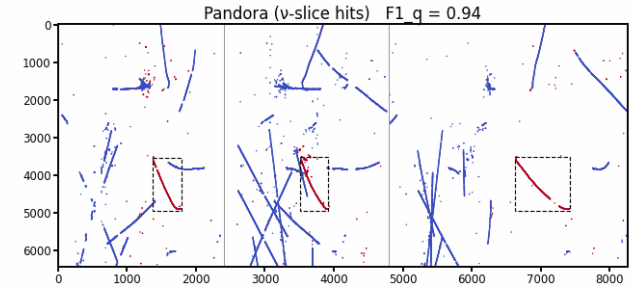
Truth: νμ CC, E = 0.80 GeV, vtx = (149, -45, 551) cm; task label: νμCC

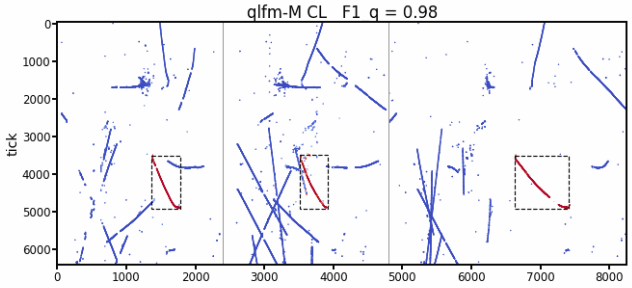
qlfm-M CL: νμCC ✓ qlfm-S CL: νμCC ✓ Panda: νμCC ✓ PoLAr-MAE: νμCC ✓
FM4NPP: νμCC ✓ qlfm-SD CL: νμCC ✓ Pandora: νμCC ✓ Pandora+BDT: νμCC ✓

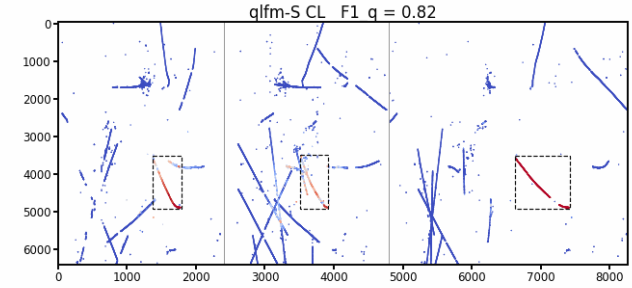


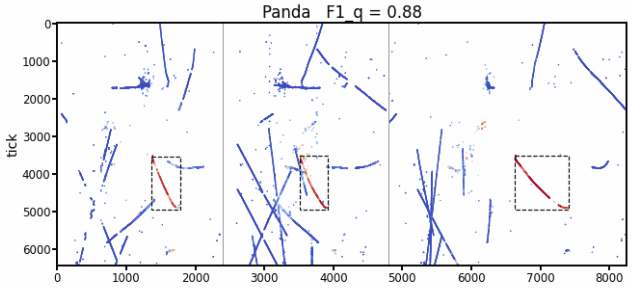


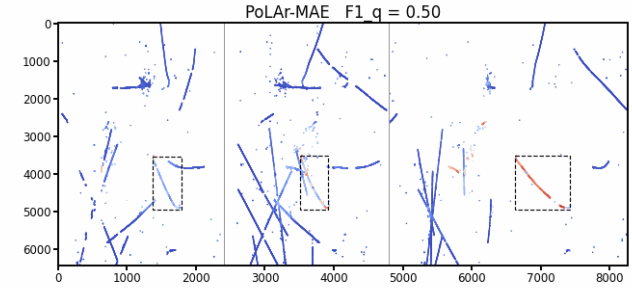


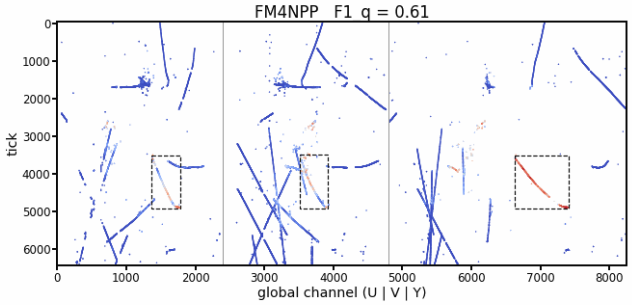


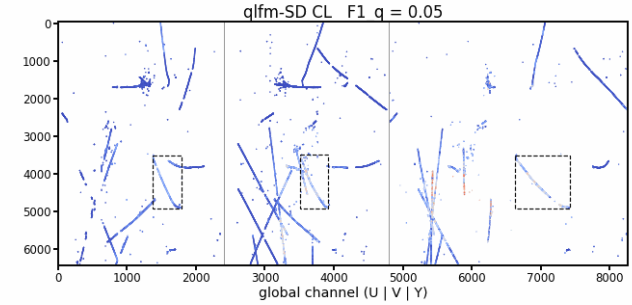










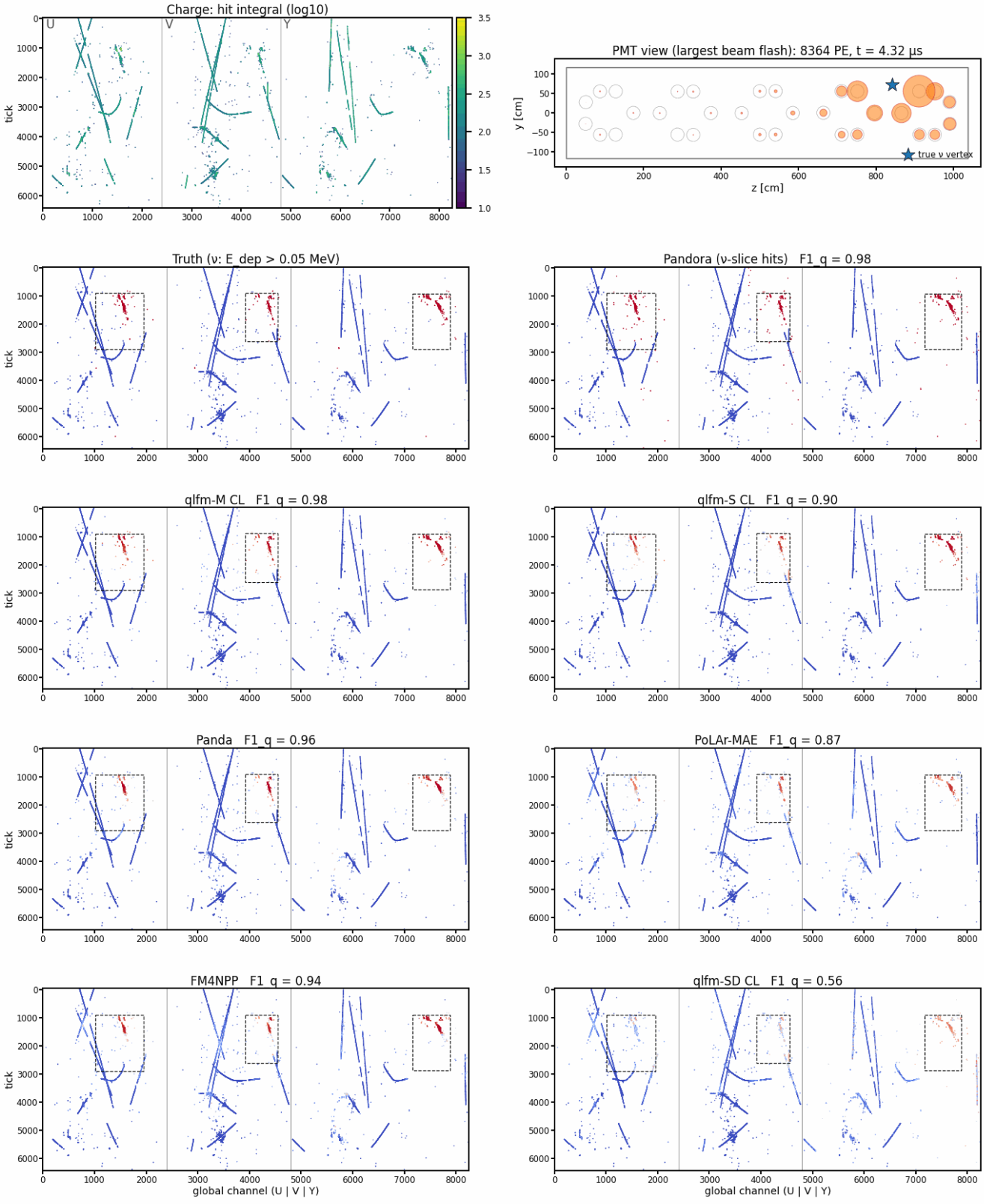


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T3 flavor — easy #8 (correct: 8/8) run 7026 / subrun 562 / event 28120 [gidx 805514]

Truth: νe CC, E = 2.02 GeV, vtx = (19, 73, 841) cm; task label: νeCC

qlfm-M CL: νeCC ✓ qlfm-S CL: νeCC ✓ Panda: νeCC ✓ PoLAr-MAE: νeCC ✓
FM4NPP: νeCC ✓ qlfm-SD CL: νeCC ✓ Pandora: νeCC ✓ Pandora+BDT: νeCC ✓

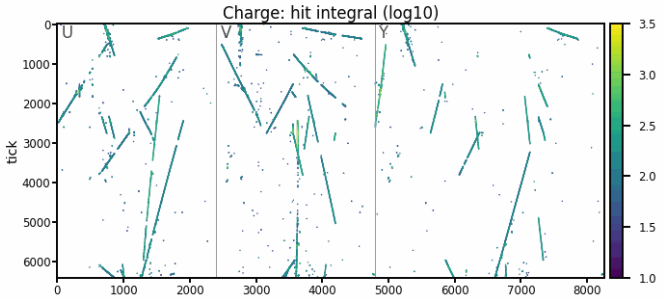


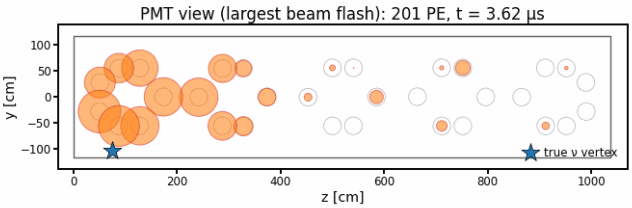
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

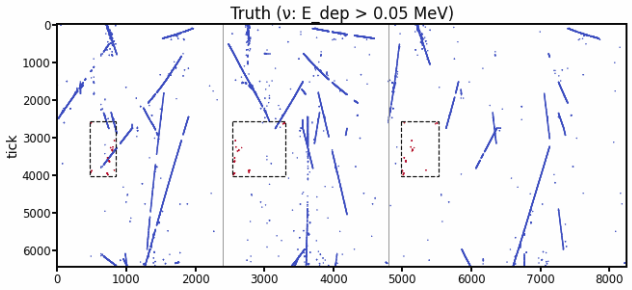
T3 flavor — easy #9 (correct: 8/8) run 7004 / subrun 699 / event 34986 [gidx 261003]

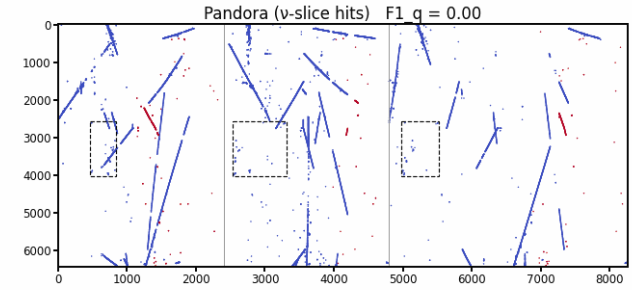
Truth: νμ NC, E = 1.24 GeV, vtx = (153, -103, 75) cm; task label: NC

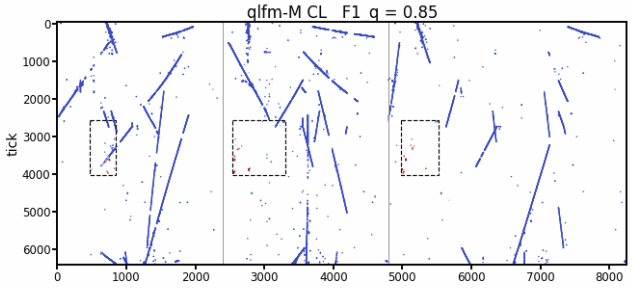
qlfm-M CL: NC ✓ qlfm-S CL: NC ✓ Panda: NC ✓ PoLAr-MAE: NC ✓
FM4NPP: NC ✓ qlfm-SD CL: NC ✓ Pandora: NC ✓ Pandora+BDT: NC ✓

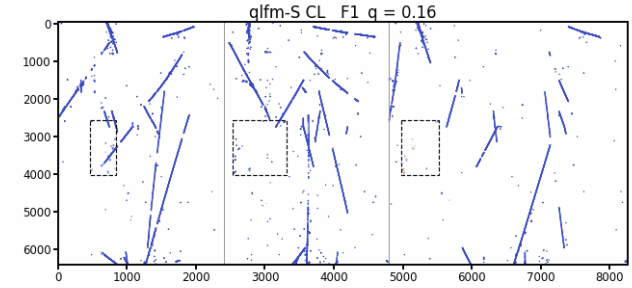


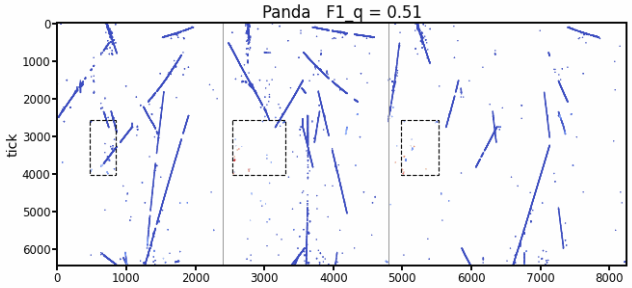


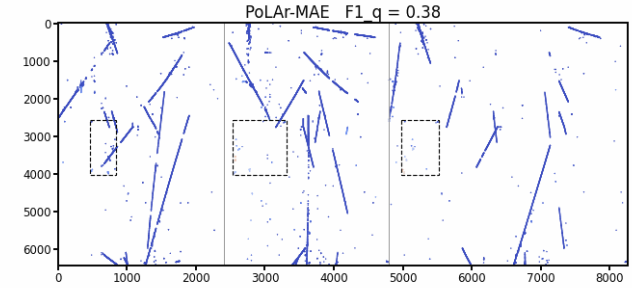


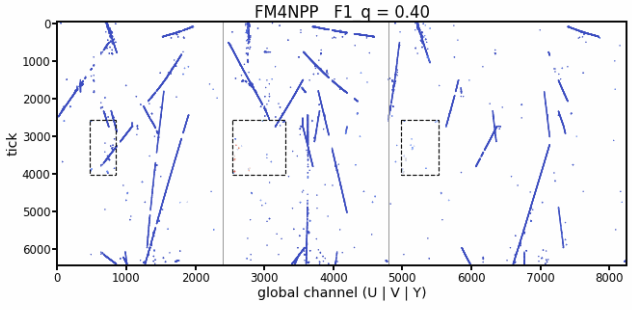


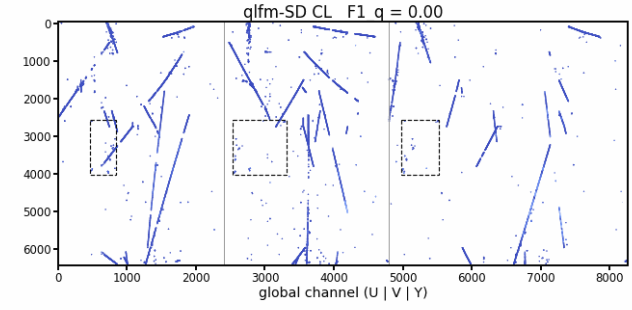










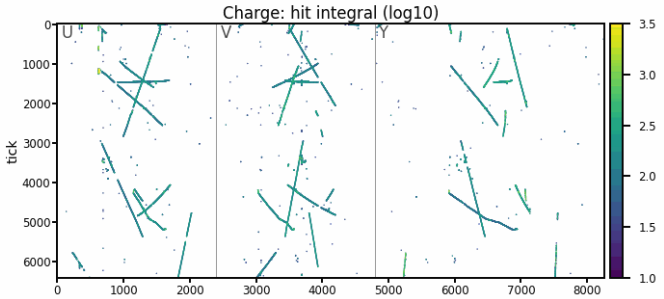


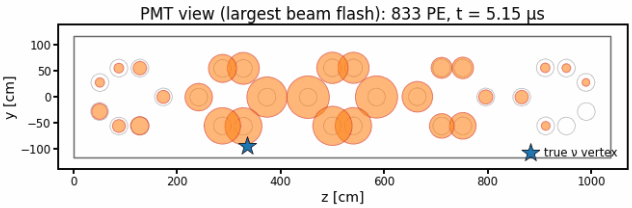
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

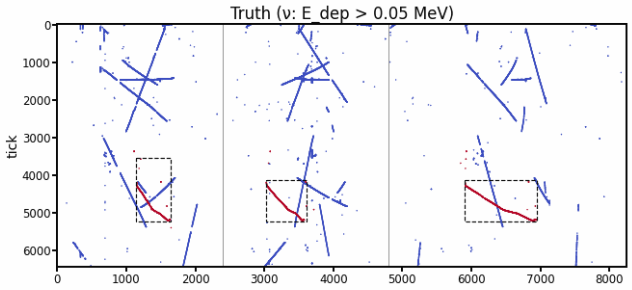
T3 flavor — easy #10 (correct: 8/8) run 7055 / subrun 79 / event 3968 [gidx 657729]

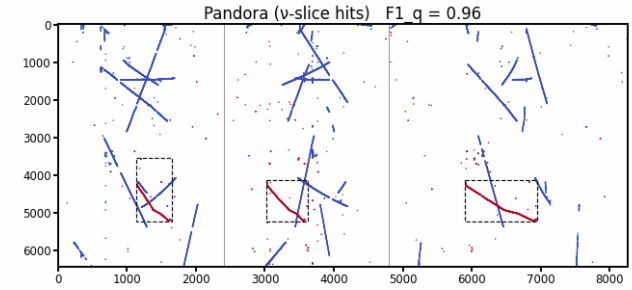
Truth: νμ CC, E = 0.92 GeV, vtx = (188, -95, 335) cm; task label: νμCC

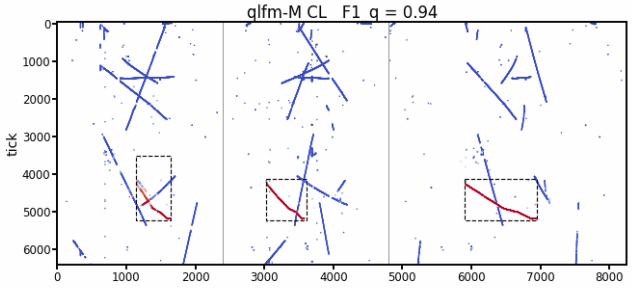
qlfm-M CL: νμCC ✓ qlfm-S CL: νμCC ✓ Panda: νμCC ✓ PoLAr-MAE: νμCC ✓
FM4NPP: νμCC ✓ qlfm-SD CL: νμCC ✓ Pandora: νμCC ✓ Pandora+BDT: νμCC ✓

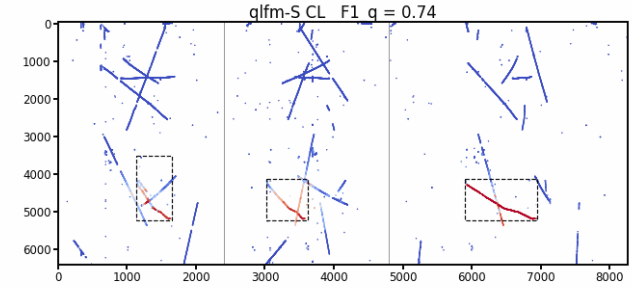


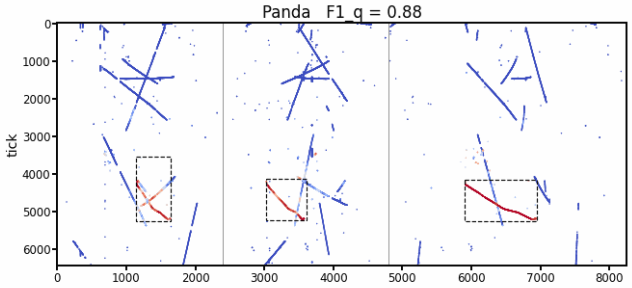


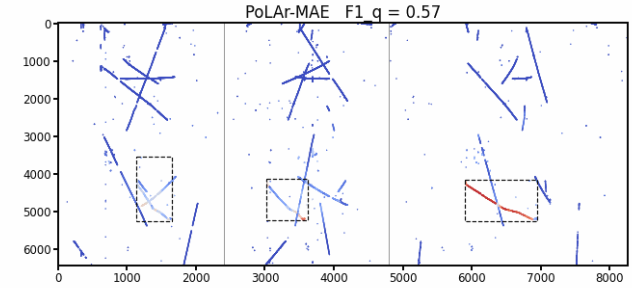


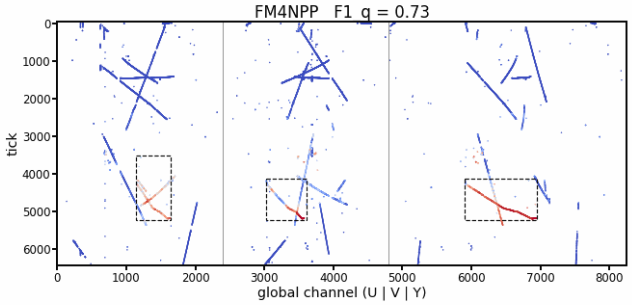


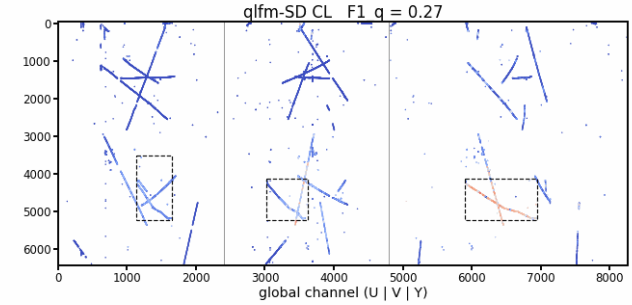










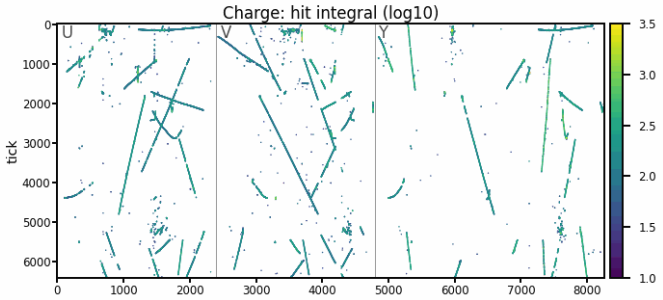


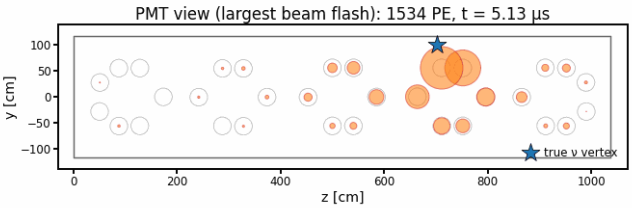
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

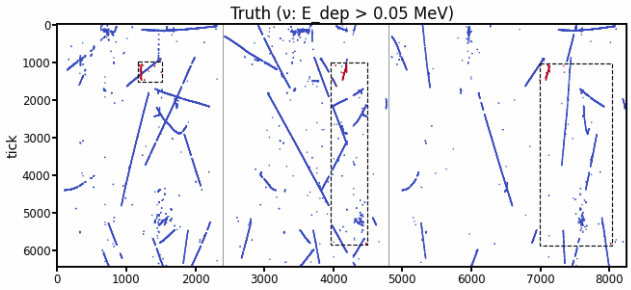
T3 flavor — hard #1 (correct: 0/8) run 7012 / subrun 767 / event 38385 [gidx 623198]

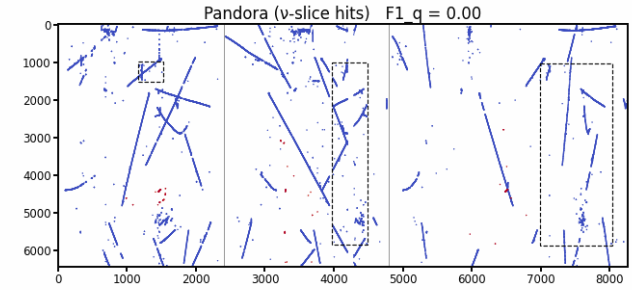
Truth: νμ CC, E = 0.50 GeV, vtx = (15, 101, 701) cm; task label: νμCC

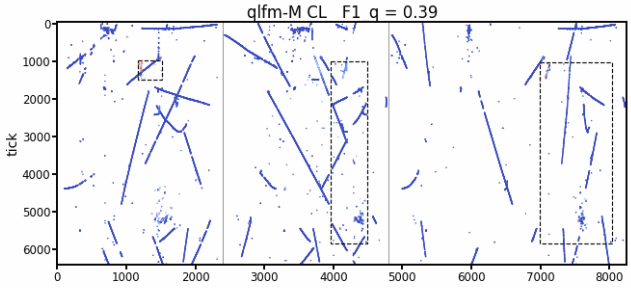
qlfm-M CL: NC ✗ qlfm-S CL: NC ✗ Panda: NC ✗ PoLAr-MAE: NC ✗
FM4NPP: NC ✗ qlfm-SD CL: NC ✗ Pandora: NC ✗ Pandora+BDT: NC ✗

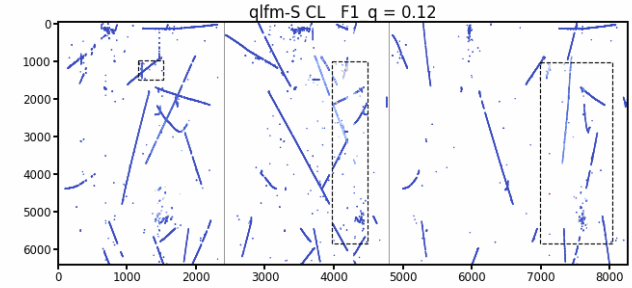


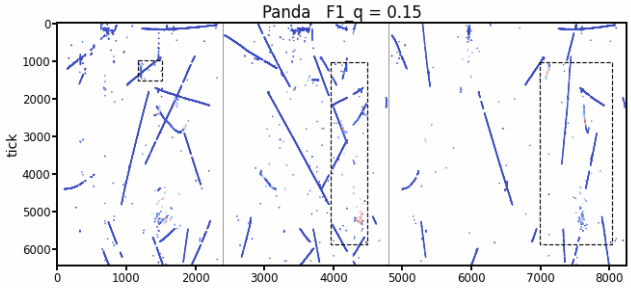


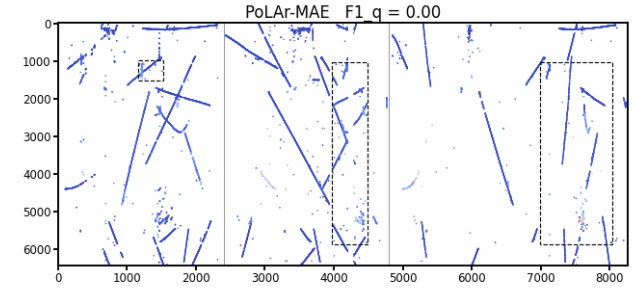


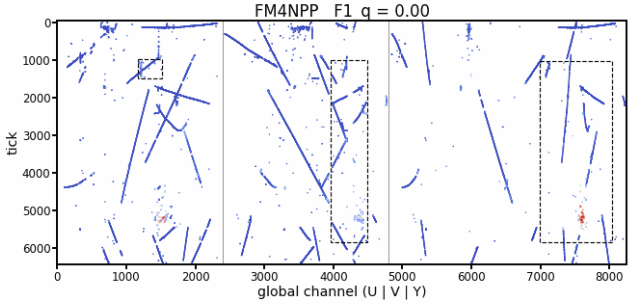


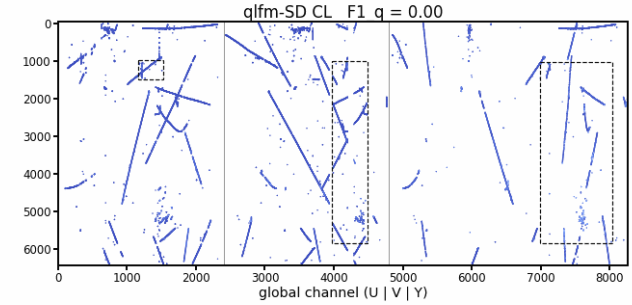










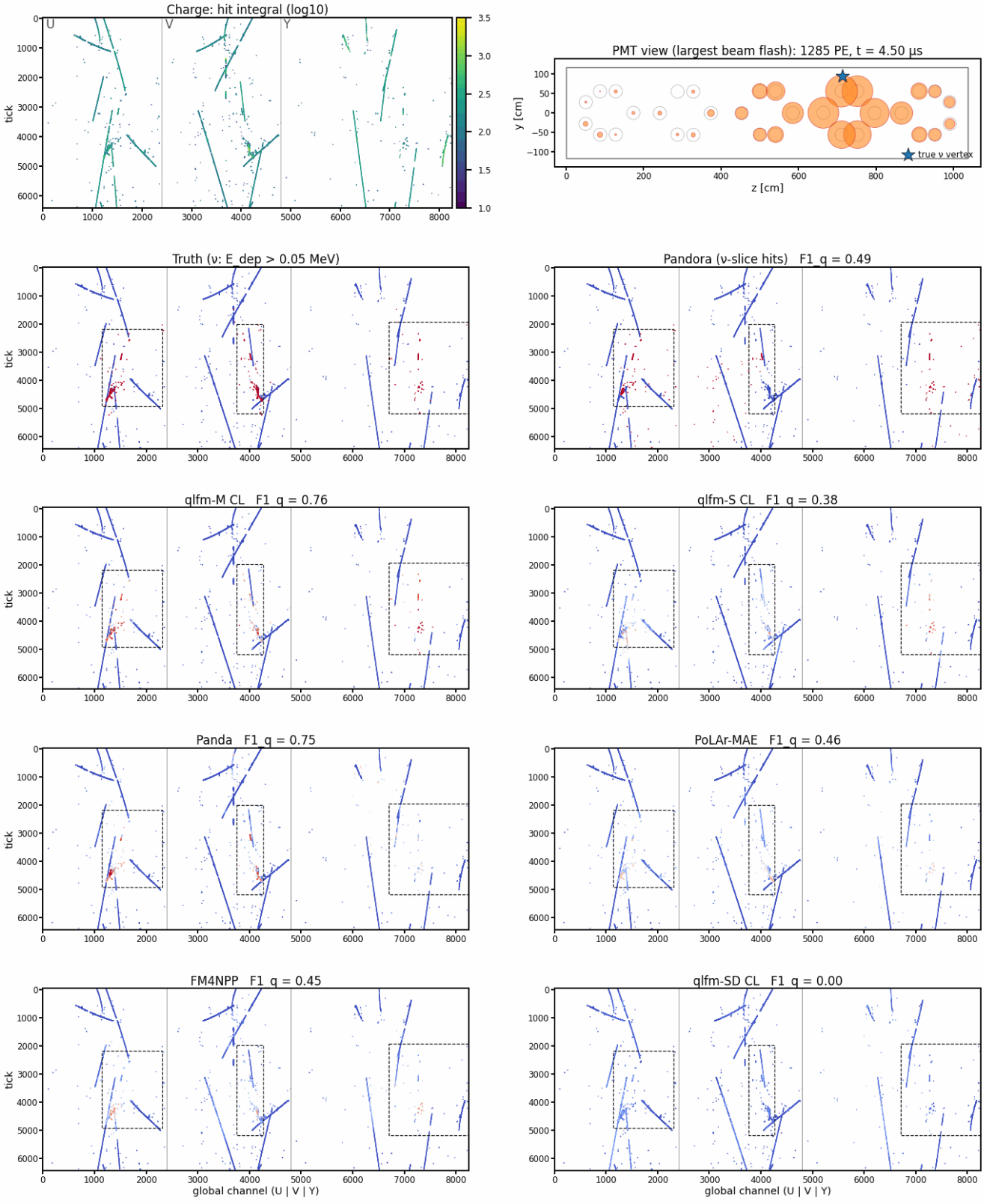


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T3 flavor — hard #2 (correct: 0/8) run 6589 / subrun 135 / event 6768 [gidx 775526]

Truth: νe CC, E = 1.63 GeV, vtx = (203, 94, 714) cm; task label: νeCC

qlfm-M CL: NC ✗ qlfm-S CL: νμCC ✗ Panda: NC ✗ PoLAr-MAE: νμCC ✗
FM4NPP: NC ✗ qlfm-SD CL: νμCC ✗ Pandora: NC ✗ Pandora+BDT: νμCC ✗

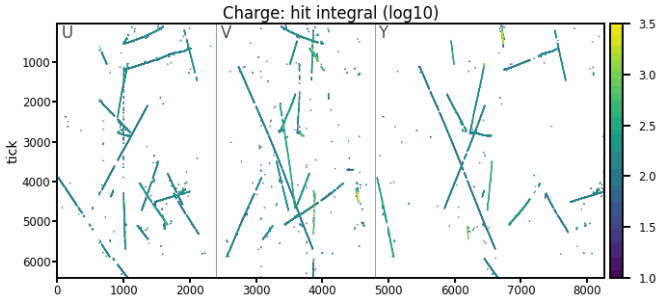


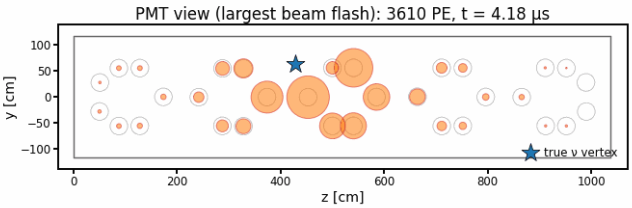
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

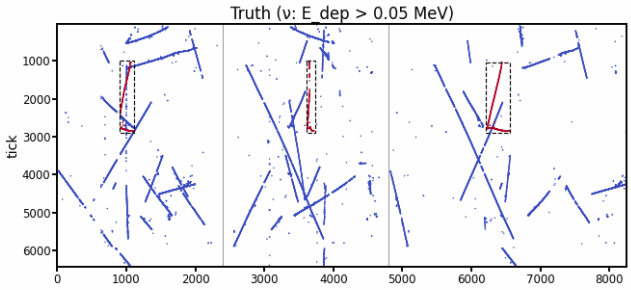
T3 flavor — hard #3 (correct: 0/8) run 7017 / subrun 437 / event 21897 [gidx 72662]

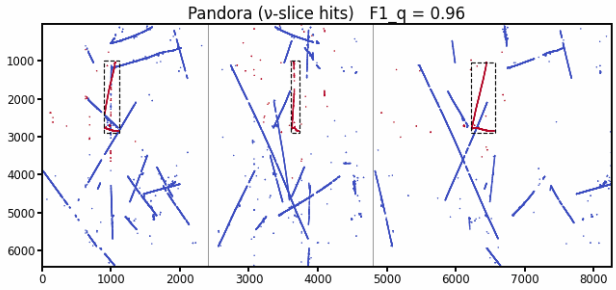
Truth: νμ NC, E = 1.24 GeV, vtx = (106, 63, 429) cm; task label: NC

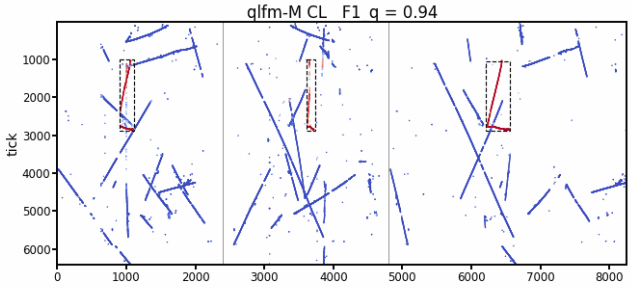
qlfm-M CL: νμCC ✗ qlfm-S CL: νμCC ✗ Panda: νμCC ✗ PoLAr-MAE: νeCC ✗
FM4NPP: νμCC ✗ qlfm-SD CL: νeCC ✗ Pandora: νμCC ✗ Pandora+BDT: νμCC ✗

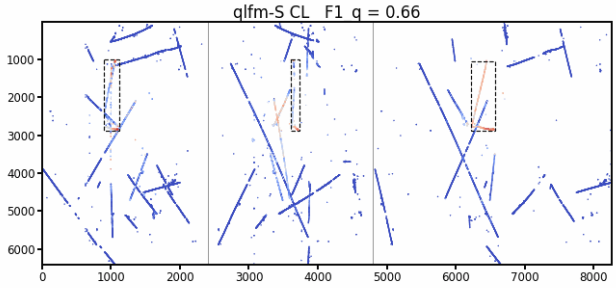


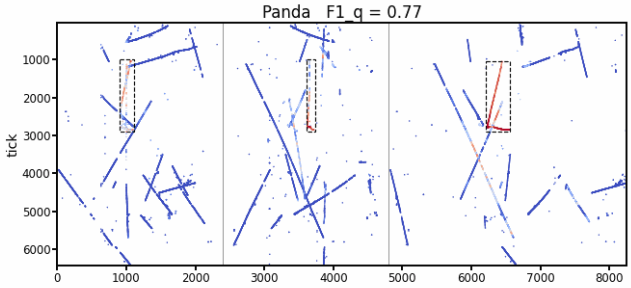


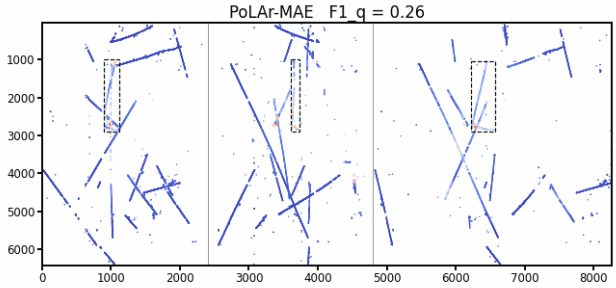


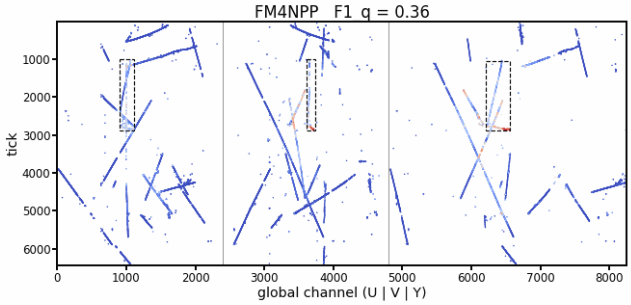


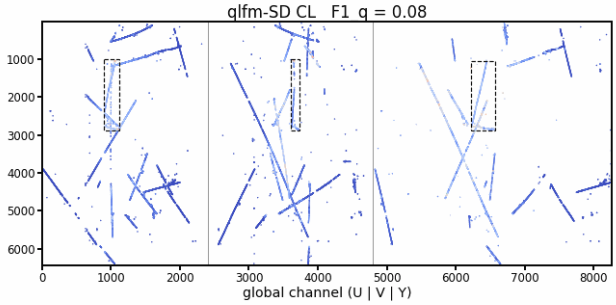










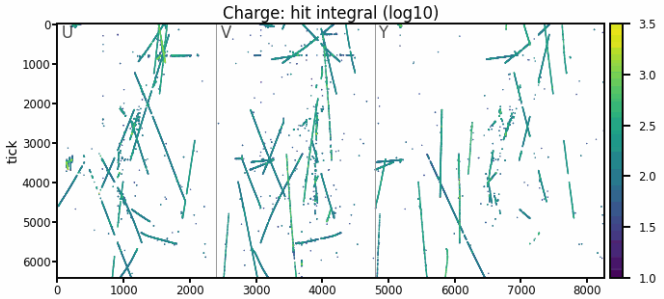


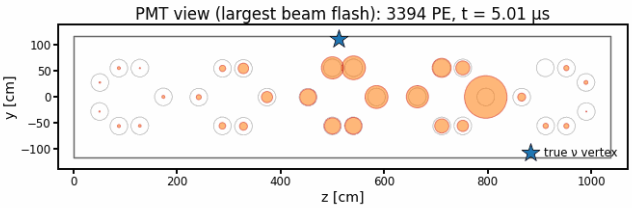
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

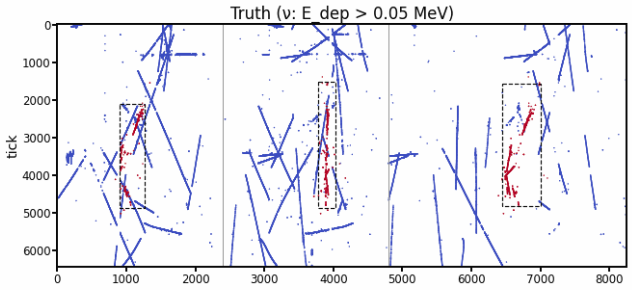
T3 flavor — hard #4 (correct: 0/8) run 6502 / subrun 64 / event 3242 [gidx 701292]

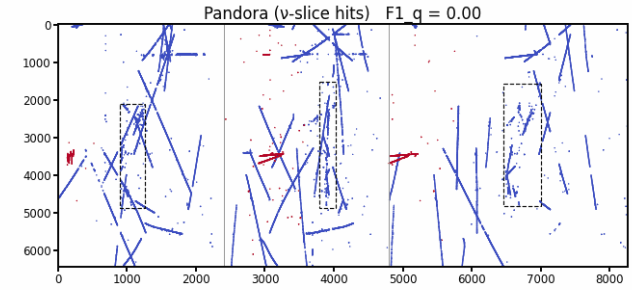
Truth: νμ CC, E = 2.65 GeV, vtx = (173, 112, 512) cm; task label: νμCC

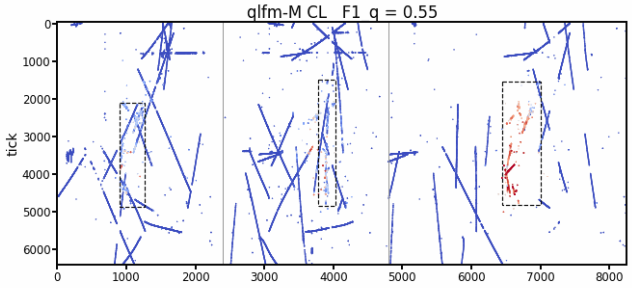
qlfm-M CL: νeCC ✗ qlfm-S CL: νeCC ✗ Panda: νeCC ✗ PoLAr-MAE: νeCC ✗
FM4NPP: νeCC ✗ qlfm-SD CL: νeCC ✗ Pandora: NC ✗ Pandora+BDT: νeCC ✗

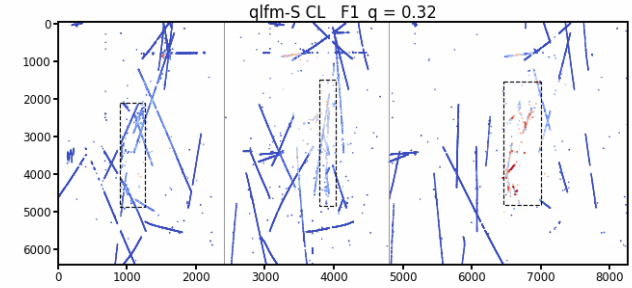


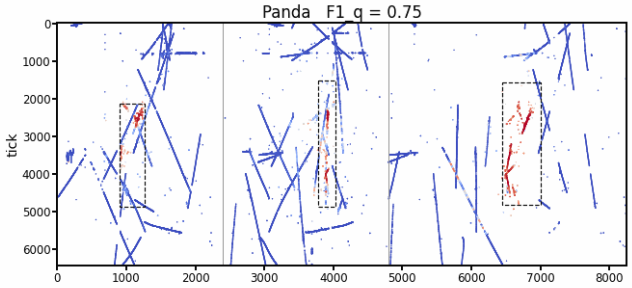


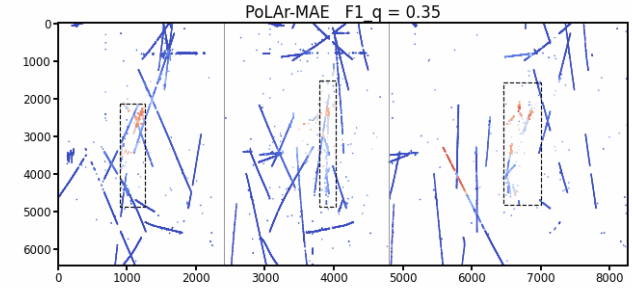


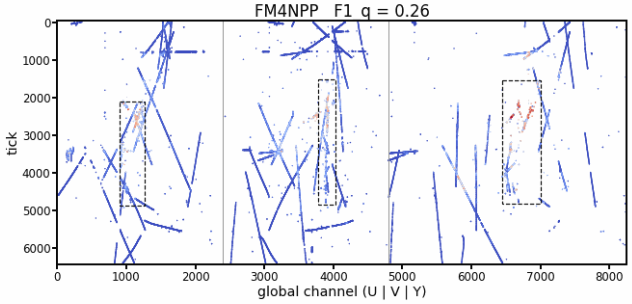


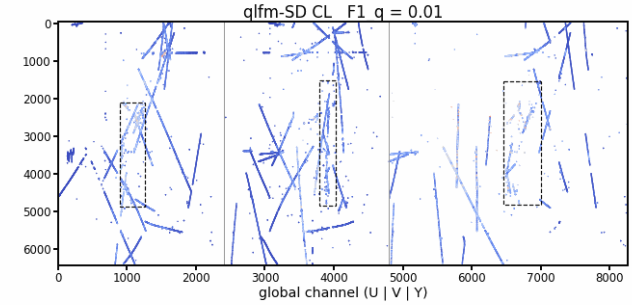










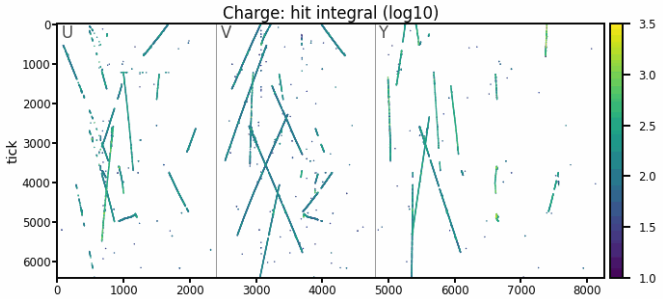


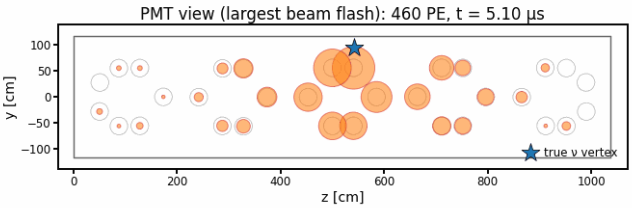
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

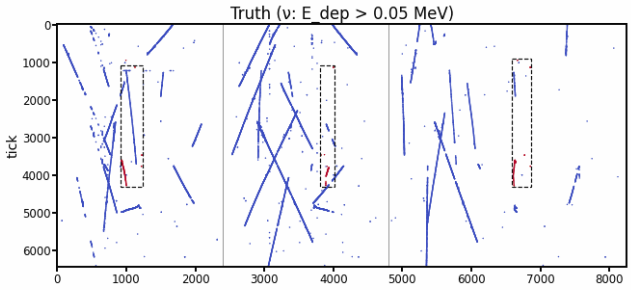
T3 flavor — hard #5 (correct: 0/8) run 6327 / subrun 62 / event 3118 [gidx 818845]

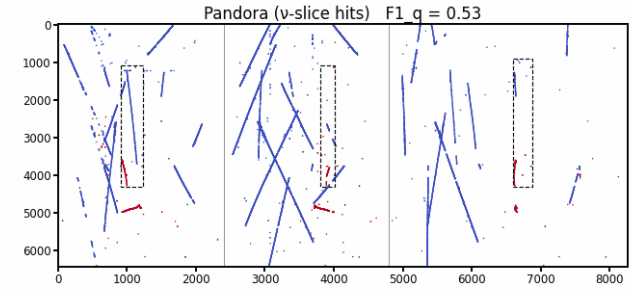
Truth: νe CC, E = 0.79 GeV, vtx = (174, 95, 542) cm; task label: νeCC

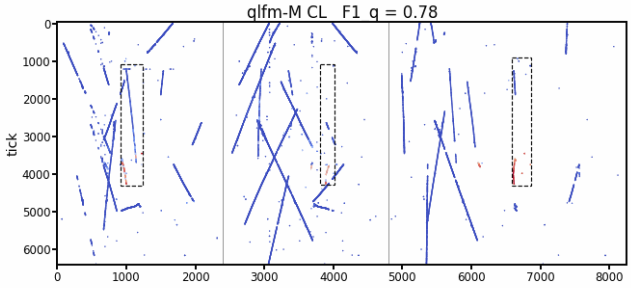
qlfm-M CL: νμCC ✗ qlfm-S CL: NC ✗ Panda: NC ✗ PoLAr-MAE: νμCC ✗
FM4NPP: νμCC ✗ qlfm-SD CL: NC ✗ Pandora: νμCC ✗ Pandora+BDT: νμCC ✗

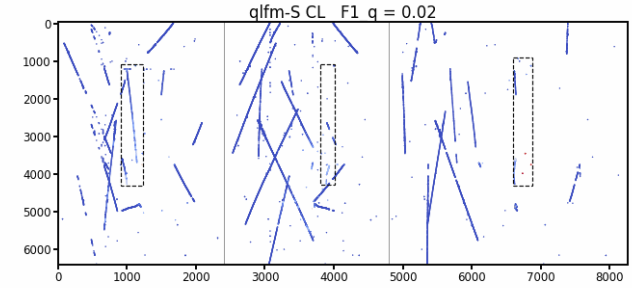


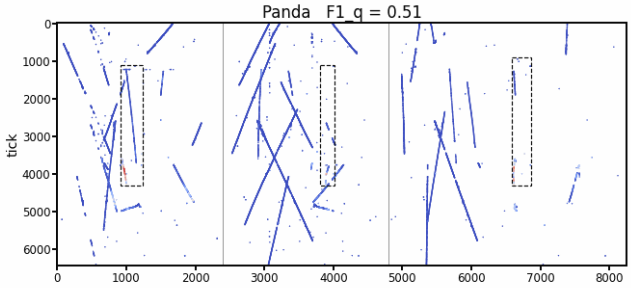


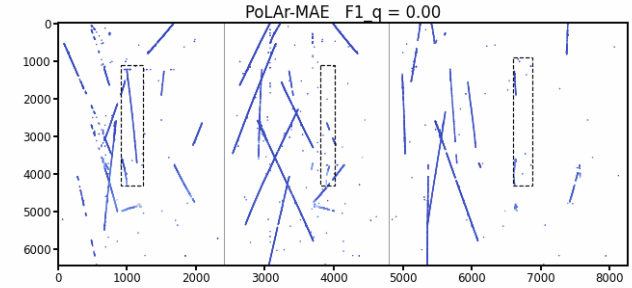


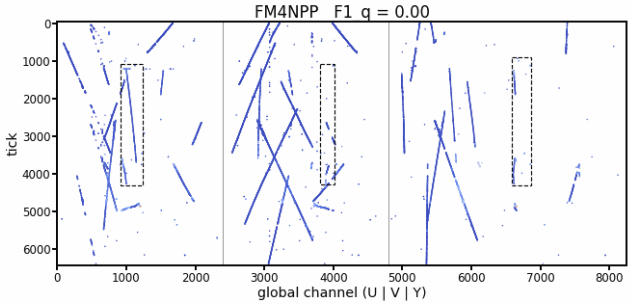


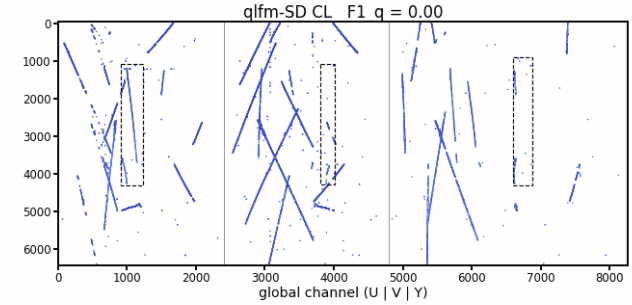










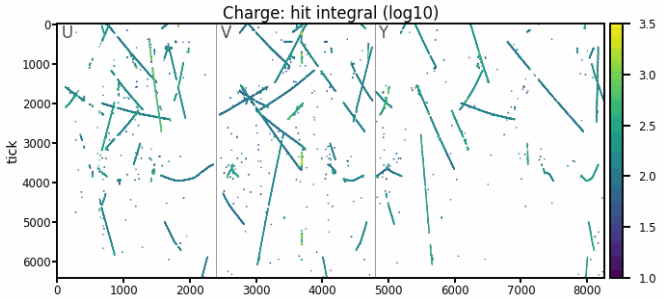


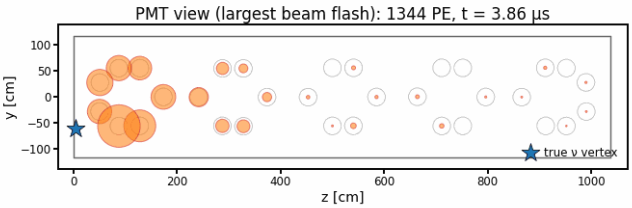
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

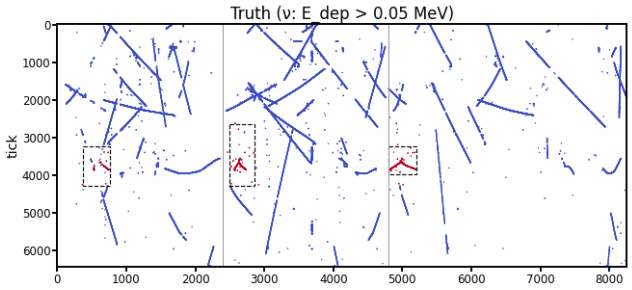
T3 flavor — hard #6 (correct: 0/8) run 7023 / subrun 256 / event 12845 [gidx 676829]

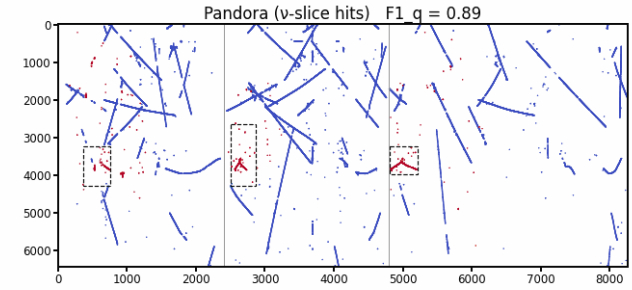
Truth: νμ NC, E = 1.42 GeV, vtx = (165, -62, 3) cm; task label: NC

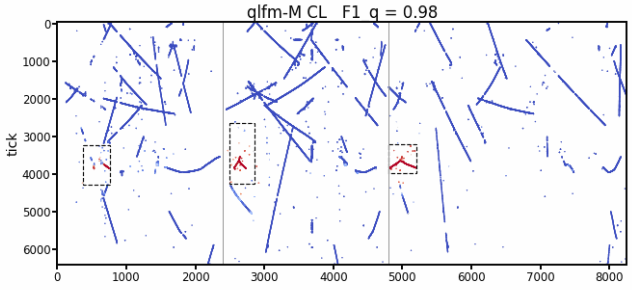
qlfm-M CL: νμCC ✗ qlfm-S CL: νμCC ✗ Panda: νμCC ✗ PoLAr-MAE: νμCC ✗
FM4NPP: νμCC ✗ qlfm-SD CL: νμCC ✗ Pandora: νμCC ✗ Pandora+BDT: νμCC ✗

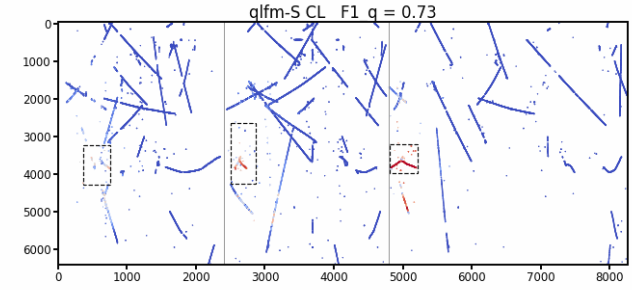


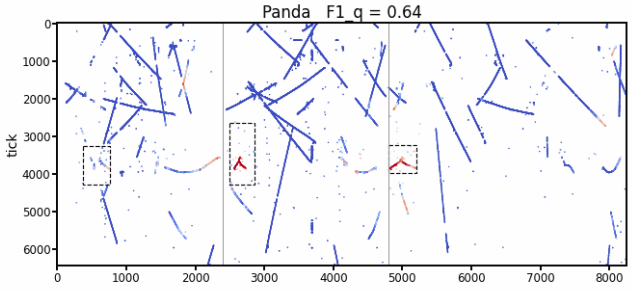


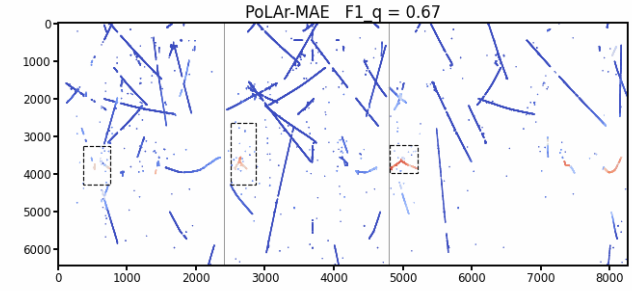


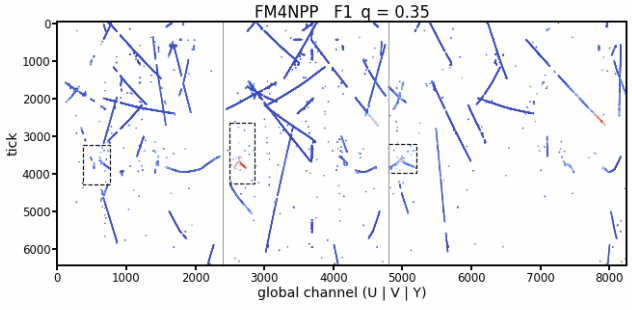


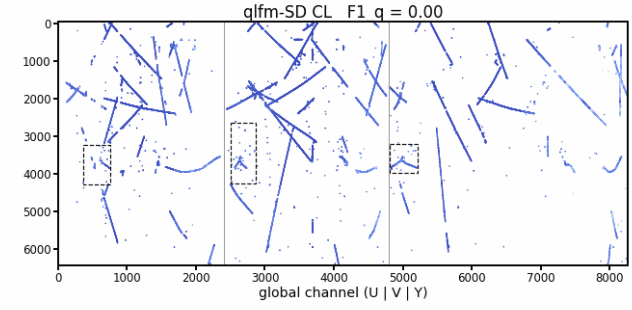










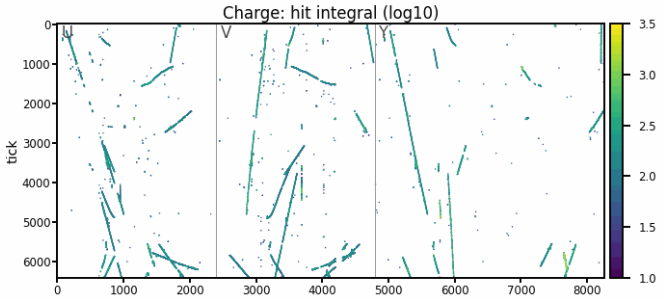


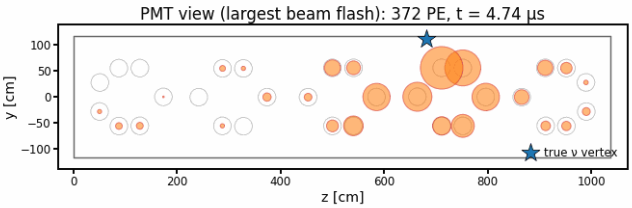
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

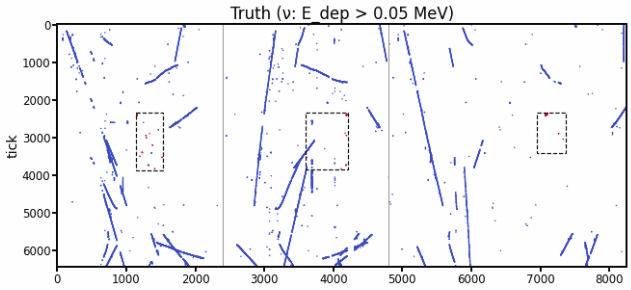
T3 flavor — hard #7 (correct: 0/8) run 7020 / subrun 1668 / event 83407 [gidx 357600]

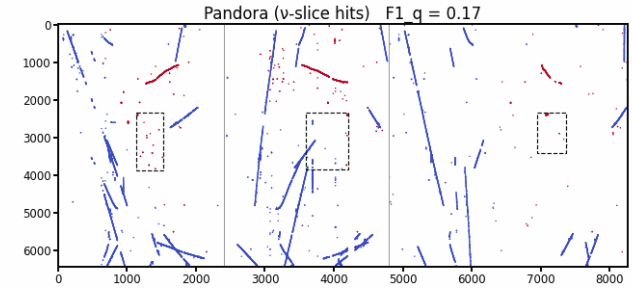
Truth: νμ CC, E = 1.20 GeV, vtx = (87, 112, 682) cm; task label: νμCC

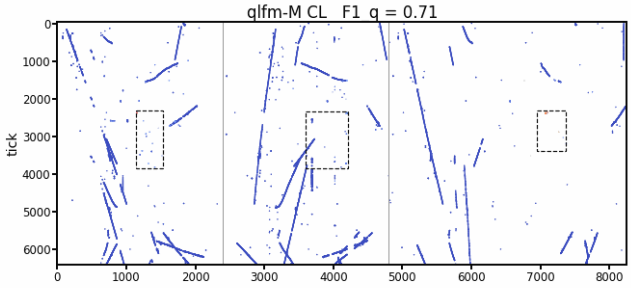
qlfm-M CL: NC ✗ qlfm-S CL: NC ✗ Panda: NC ✗ PoLAr-MAE: NC ✗
FM4NPP: NC ✗ qlfm-SD CL: NC ✗ Pandora: NC ✗ Pandora+BDT: NC ✗

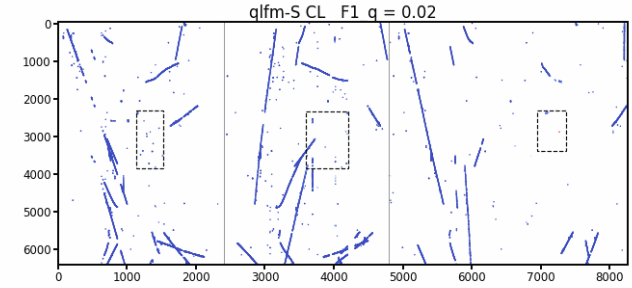


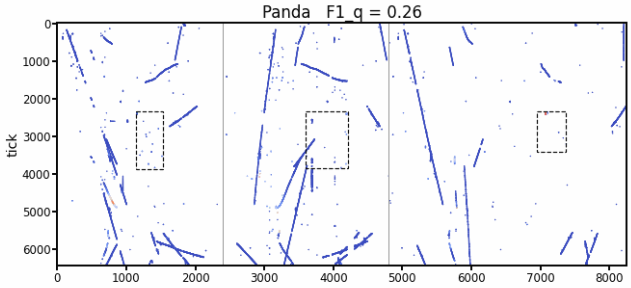


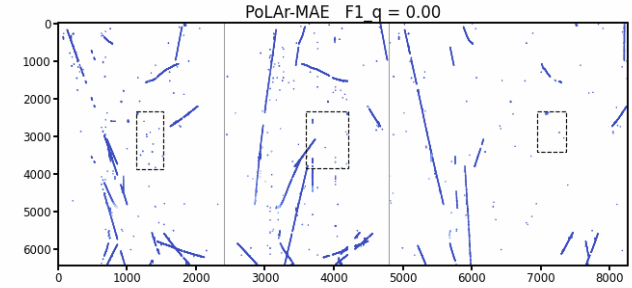


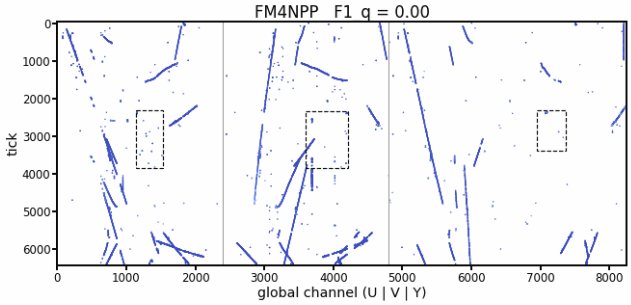


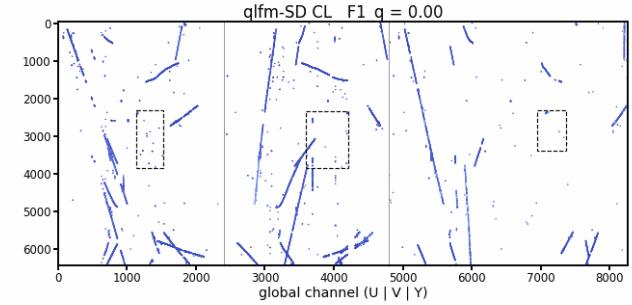










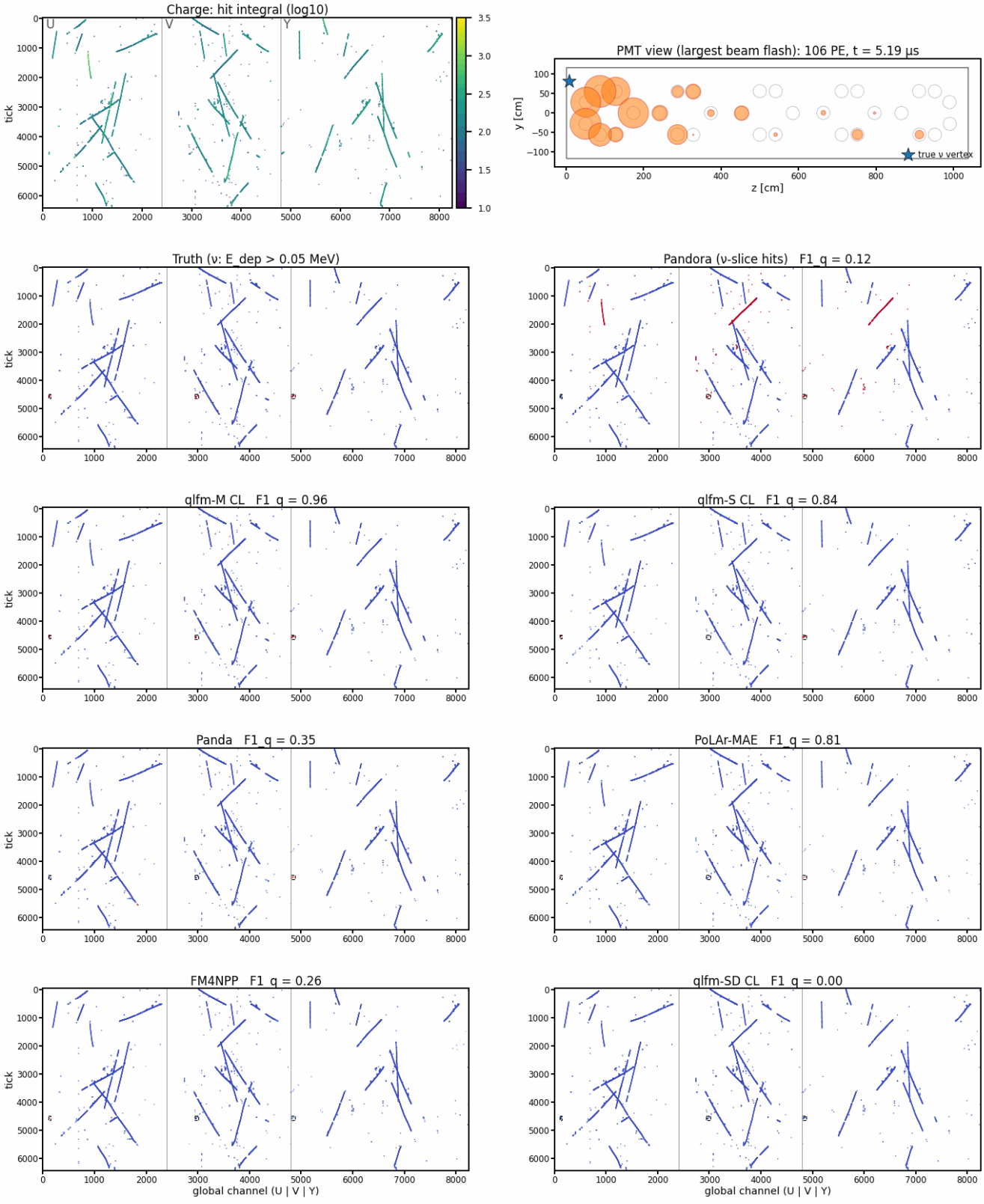


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T3 flavor — hard #8 (correct: 0/8) run 7018 / subrun 297 / event 14859 [gidx 829007]

Truth: νe CC, E = 0.41 GeV, vtx = (206, 82, 8) cm; task label: νeCC

qlfm-M CL: νμCC ✗ qlfm-S CL: νμCC ✗ Panda: NC ✗ PoLAr-MAE: NC ✗
FM4NPP: NC ✗ qlfm-SD CL: NC ✗ Pandora: NC ✗ Pandora+BDT: νμCC ✗

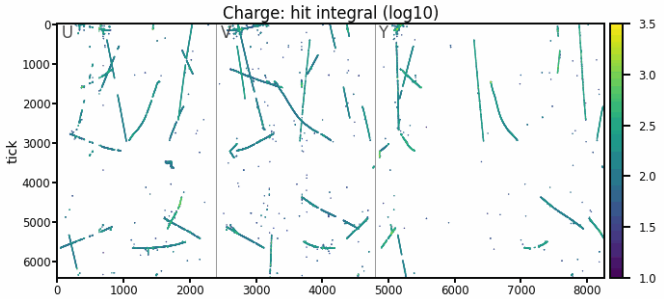


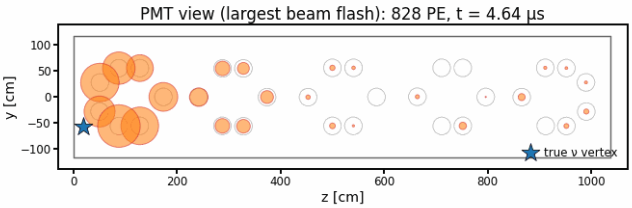
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

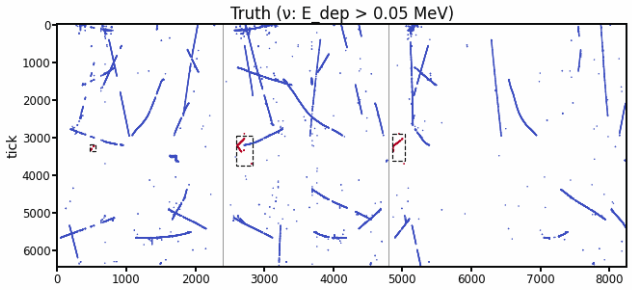
T3 flavor — hard #9 (correct: 0/8) run 7012 / subrun 767 / event 38381 [gidx 623194]

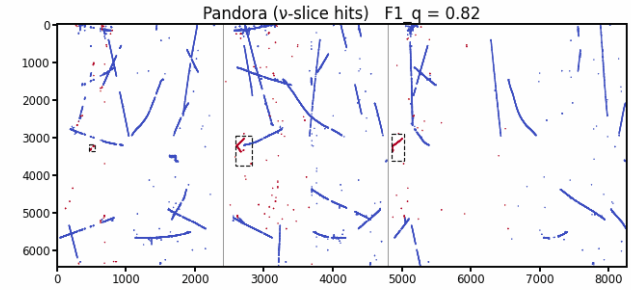
Truth: νμ NC, E = 1.14 GeV, vtx = (131, -57, 20) cm; task label: NC

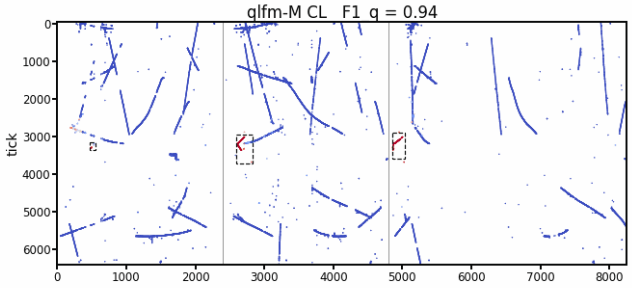
qlfm-M CL: νμCC ✗ qlfm-S CL: νμCC ✗ Panda: νμCC ✗ PoLAr-MAE: νμCC ✗
FM4NPP: νμCC ✗ qlfm-SD CL: νμCC ✗ Pandora: νμCC ✗ Pandora+BDT: νμCC ✗

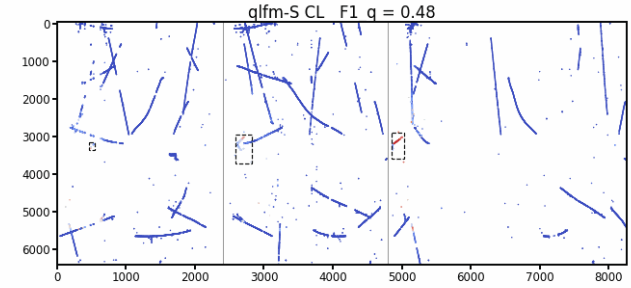


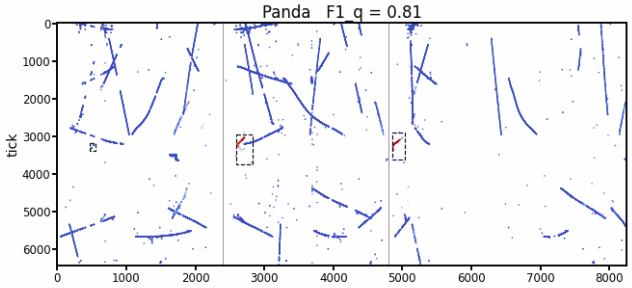


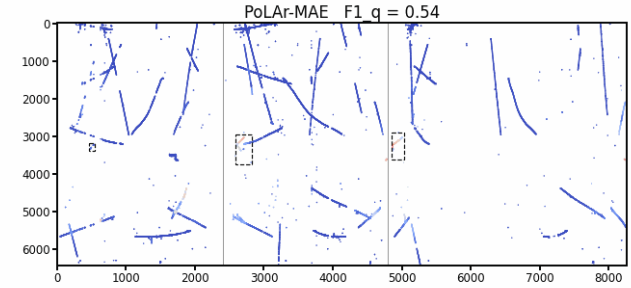


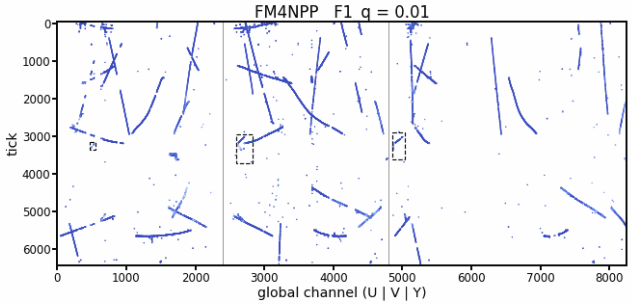


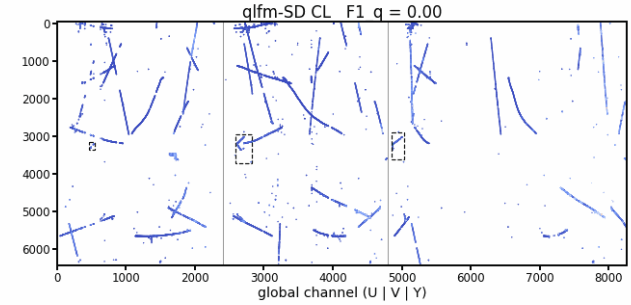










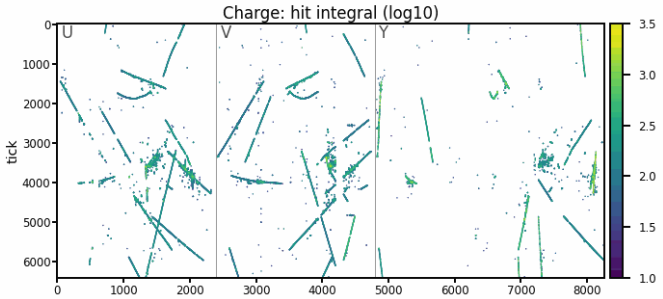


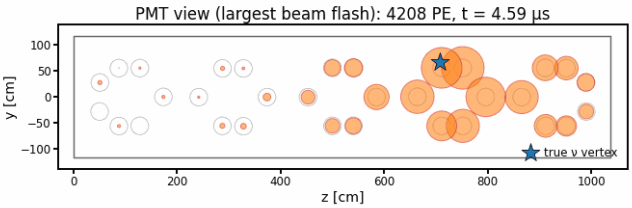
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

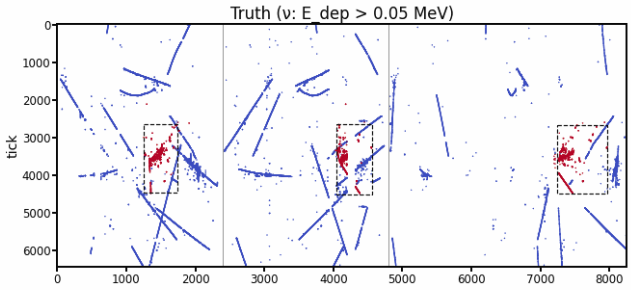
T3 flavor — hard #10 (correct: 0/8) run 7018 / subrun 190 / event 9508 [gidx 662504]

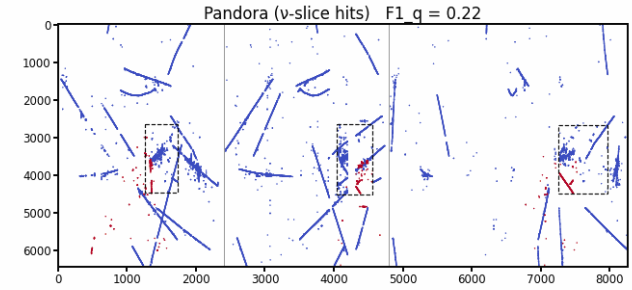
Truth: νμ CC, E = 3.35 GeV, vtx = (152, 66, 707) cm; task label: νμCC

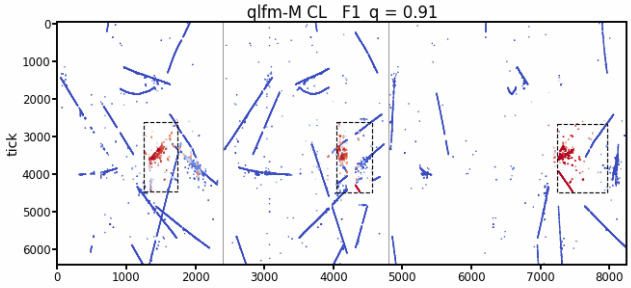
qlfm-M CL: νeCC ✗ qlfm-S CL: νeCC ✗ Panda: NC ✗ PoLAr-MAE: νeCC ✗
FM4NPP: νeCC ✗ qlfm-SD CL: νeCC ✗ Pandora: NC ✗ Pandora+BDT: νeCC ✗

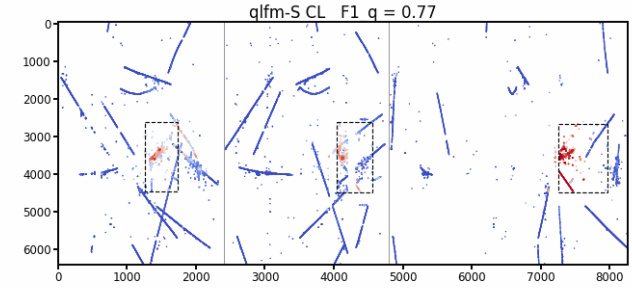


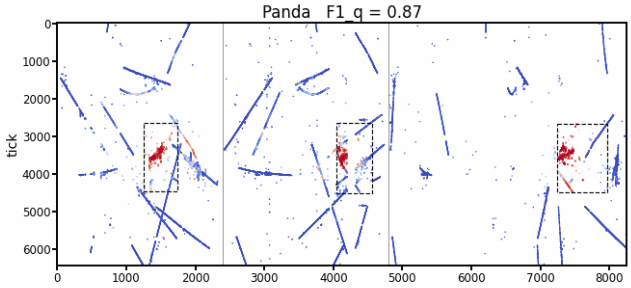


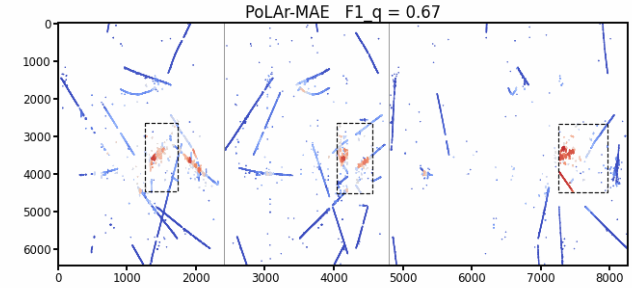


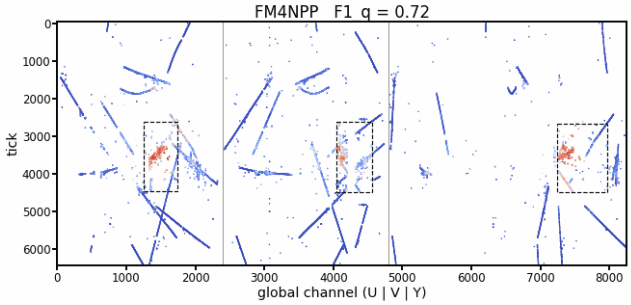


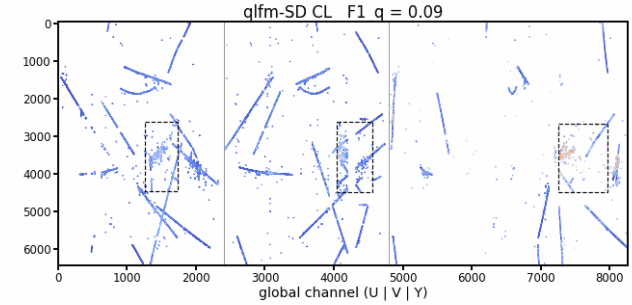






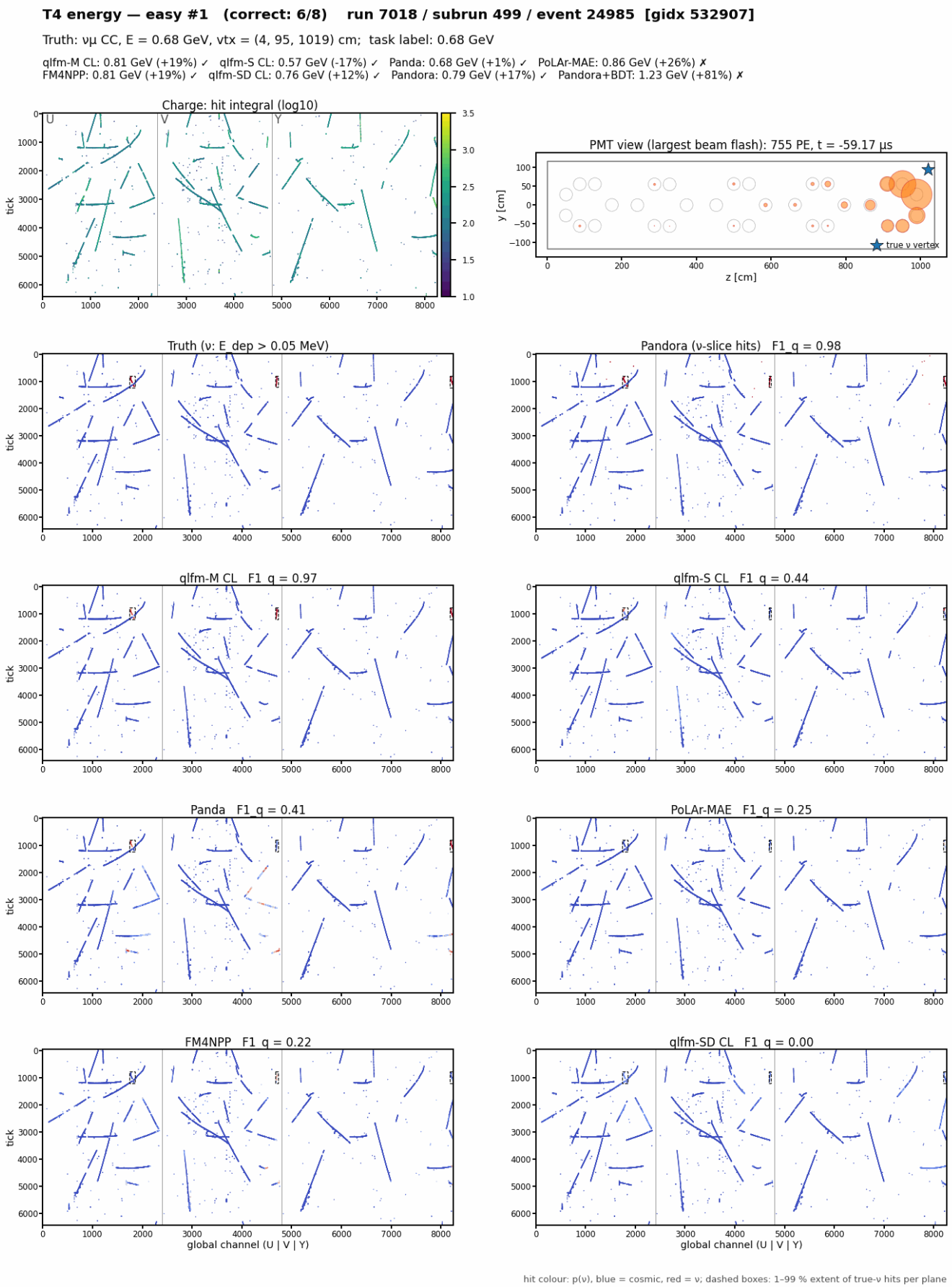






hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

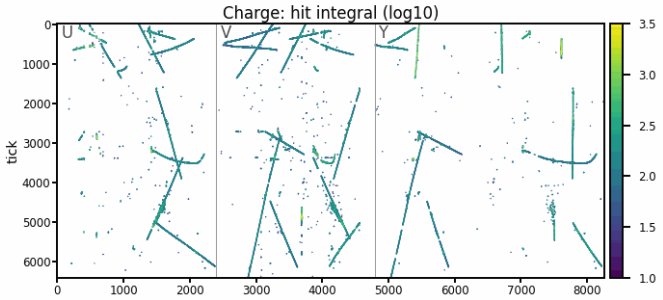
A.4 T4: neutrino energy

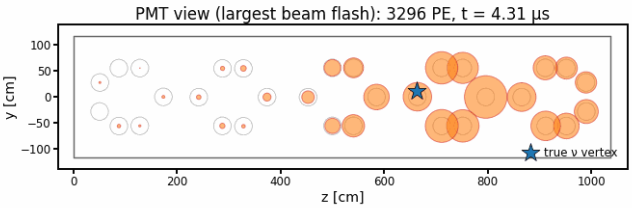


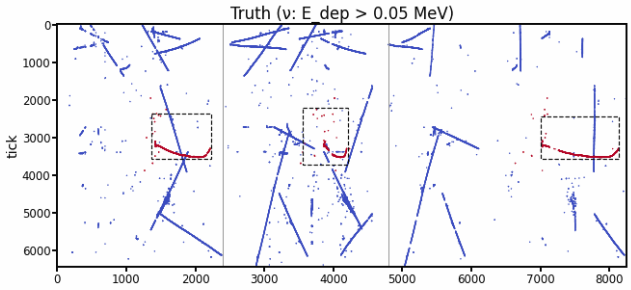
T4 energy — easy #2 (correct: 8/8) run 6798 / subrun 16 / event 838 [gidx 452460]

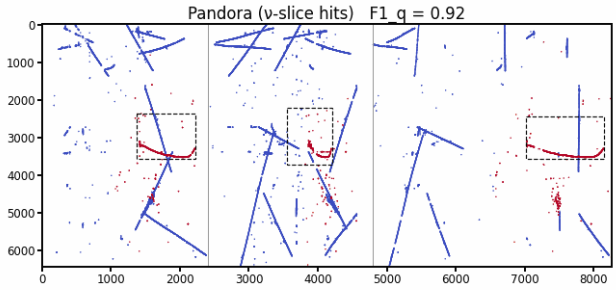
Truth: νμ CC, E = 1.36 GeV, vtx = (127, 10, 663) cm; task label: 1.36 GeV

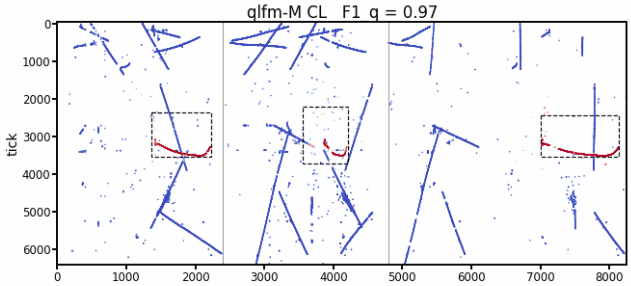
qlfm-M CL: 1.55 GeV (+14%) ✓ qlfm-S CL: 1.63 GeV (+19%) ✓ Panda: 1.25 GeV (-8%) ✓ PoLAr-MAE: 1.63 GeV (+20%) ✓
FM4NPP: 1.60 GeV (+18%) ✓ qlfm-SD CL: 1.47 GeV (+8%) ✓ Pandora: 1.40 GeV (+3%) ✓ Pandora+BDT: 1.52 GeV (+12%) ✓

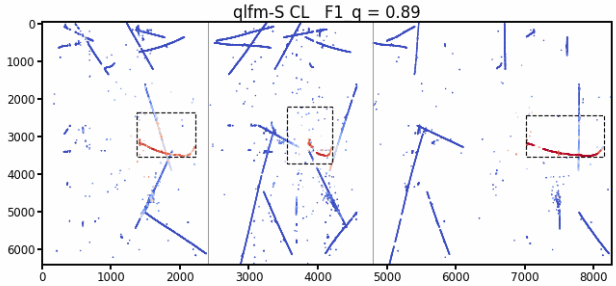


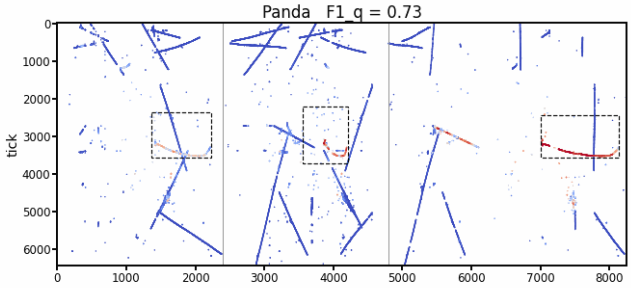


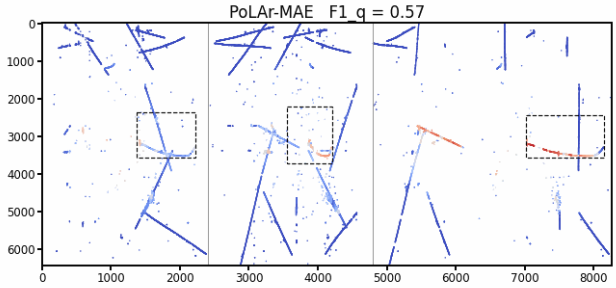


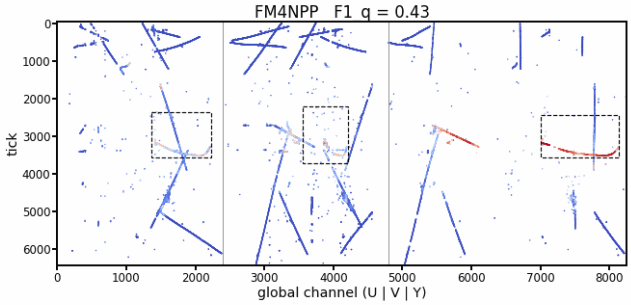


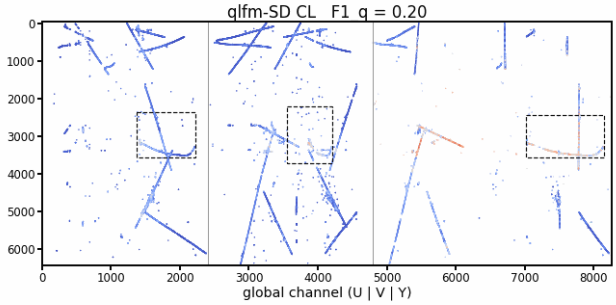










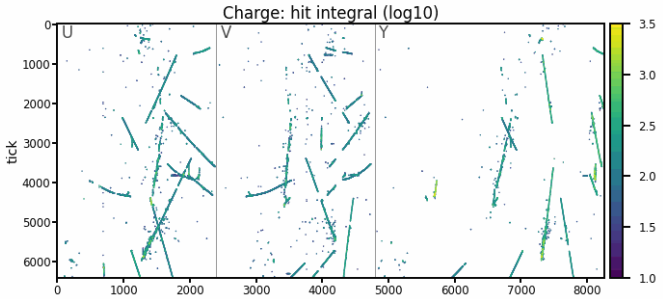


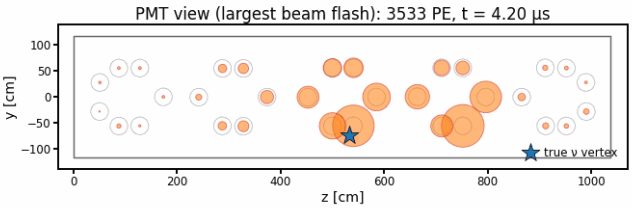
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

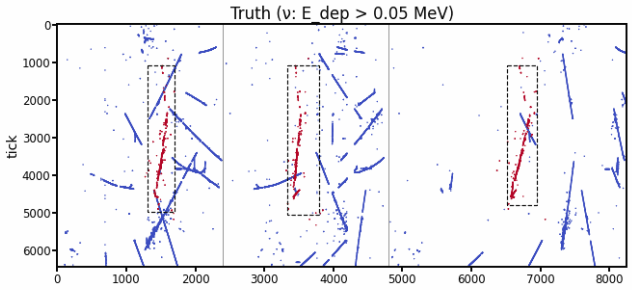
T4 energy — easy #3 (correct: 8/8) run 7049 / subrun 213 / event 10696 [gidx 807328]

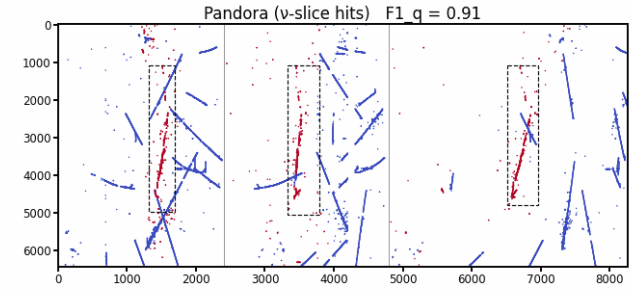
Truth: νe CC, E = 1.57 GeV, vtx = (204, -75, 532) cm; task label: 1.57 GeV

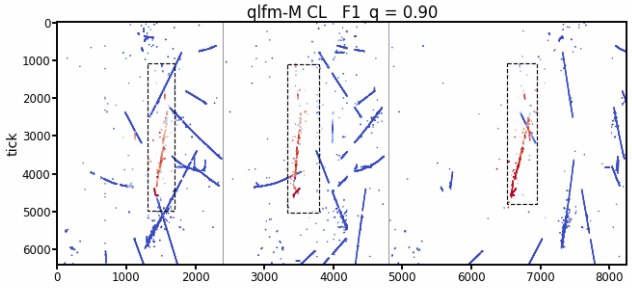
qlfm-M CL: 1.66 GeV (+5%) ✓ qlfm-S CL: 1.29 GeV (-18%) ✓ Panda: 1.61 GeV (+2%) ✓ PoLAr-MAE: 1.50 GeV (-5%) ✓
FM4NPP: 1.66 GeV (+6%) ✓ qlfm-SD CL: 1.26 GeV (-20%) ✓ Pandora: 1.47 GeV (-6%) ✓ Pandora+BDT: 1.75 GeV (+12%) ✓

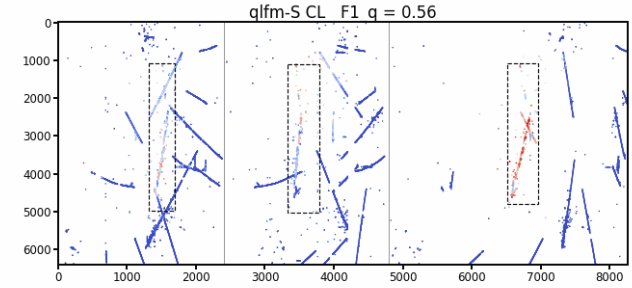


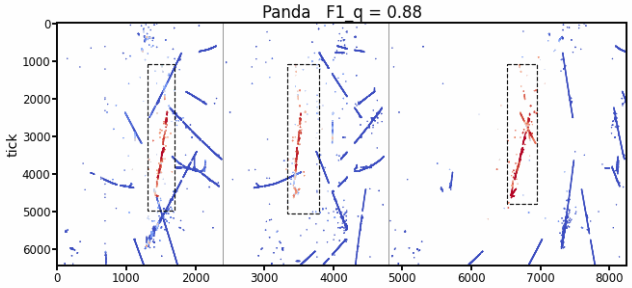


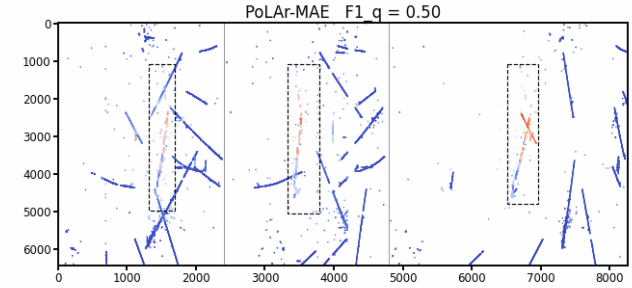


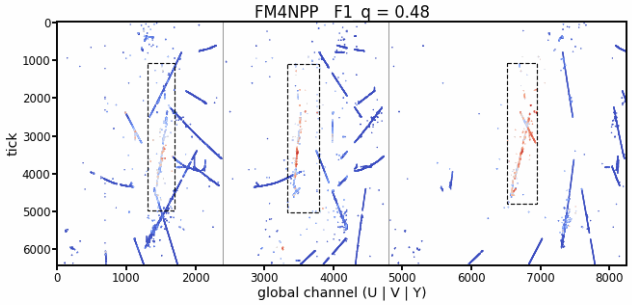


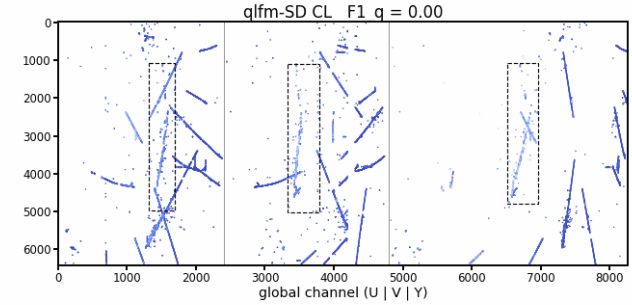










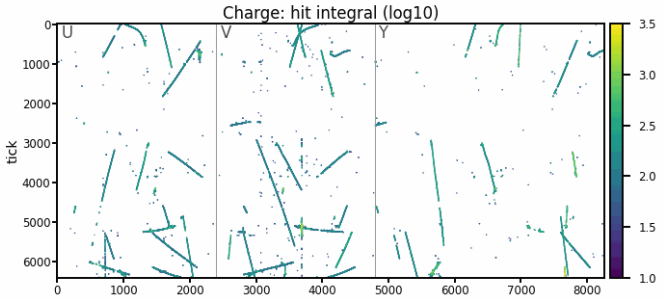


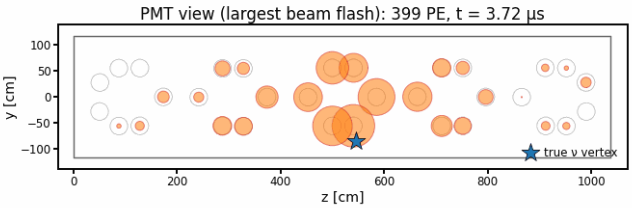
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

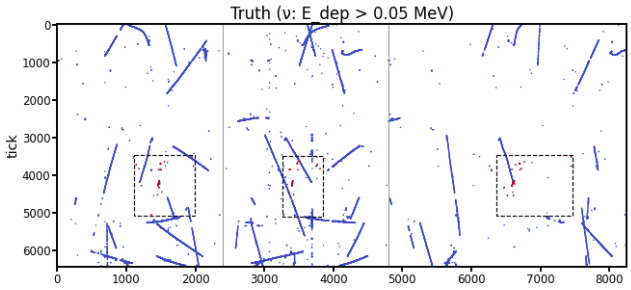
T4 energy — easy #4 (correct: 6/8) run 6829 / subrun 140 / event 7006 [gidx 311469]

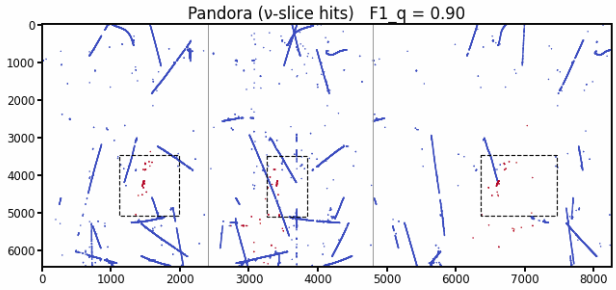
Truth: νμ CC, E = 0.69 GeV, vtx = (181, -86, 545) cm; task label: 0.69 GeV

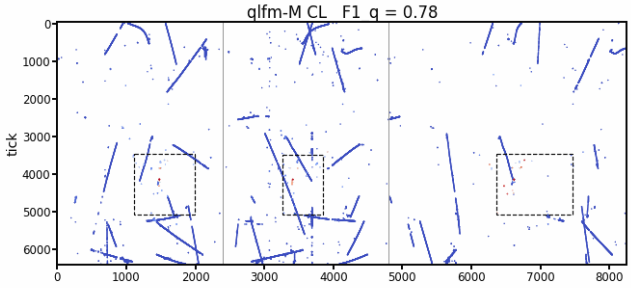
qlfm-M CL: 0.62 GeV (-9%) ✓ qlfm-S CL: 0.70 GeV (+2%) ✓ Panda: 0.55 GeV (-20%) ✗ PoLAr-MAE: 0.89 GeV (+31%) ✗
FM4NPP: 0.82 GeV (+20%) ✓ qlfm-SD CL: 0.82 GeV (+20%) ✓ Pandora: 0.81 GeV (+18%) ✓ Pandora+BDT: 0.74 GeV (+8%) ✓

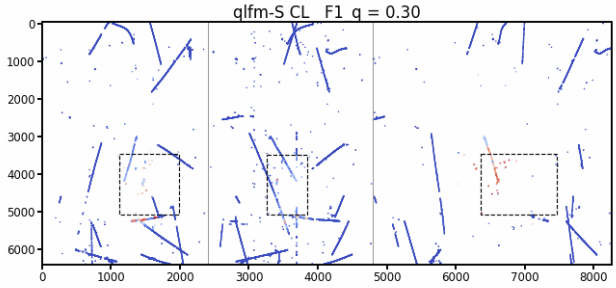


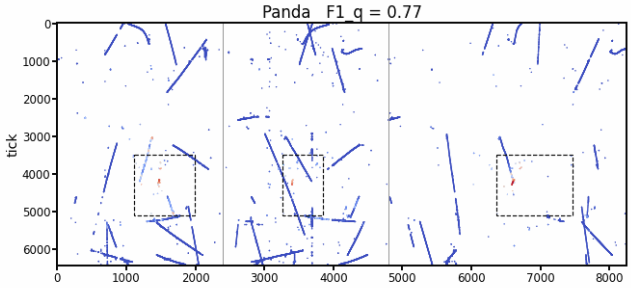


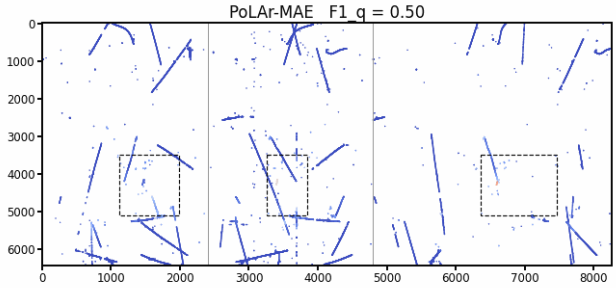


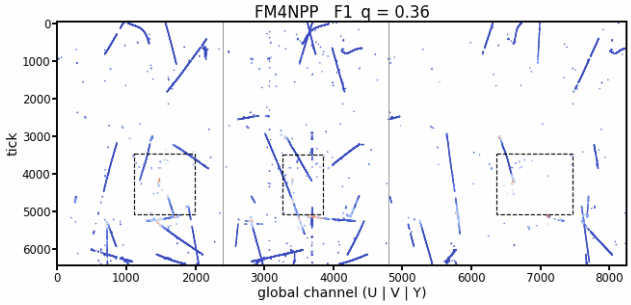


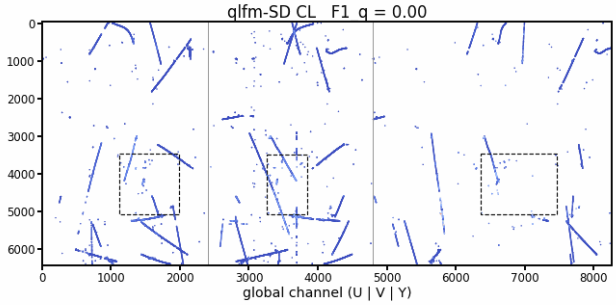










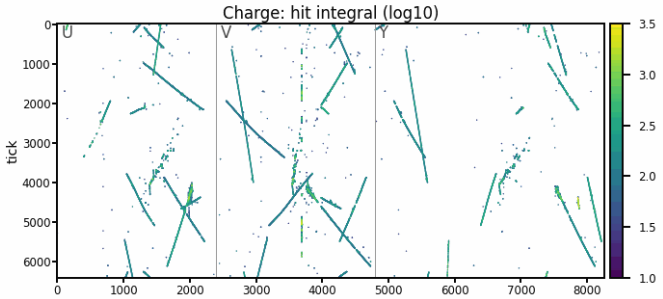


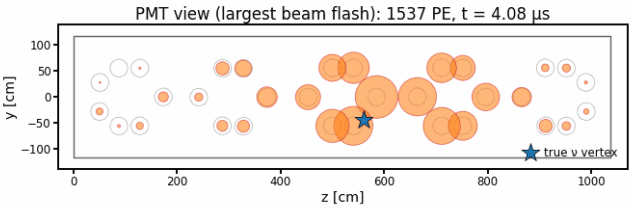
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

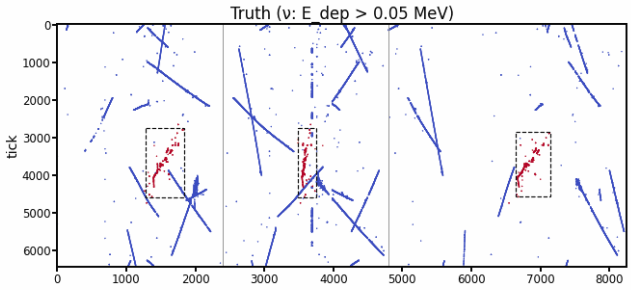
T4 energy — easy #5 (correct: 8/8) run 6577 / subrun 91 / event 4586 [gidx 759002]

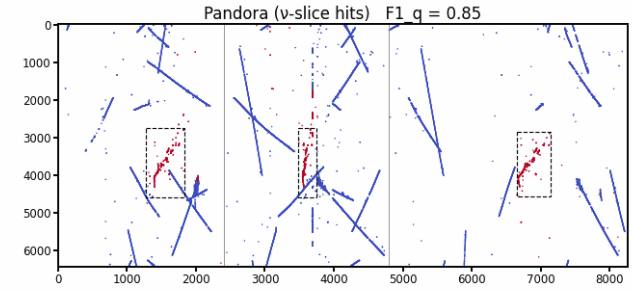
Truth: νe CC, E = 1.12 GeV, vtx = (174, -44, 561) cm; task label: 1.12 GeV

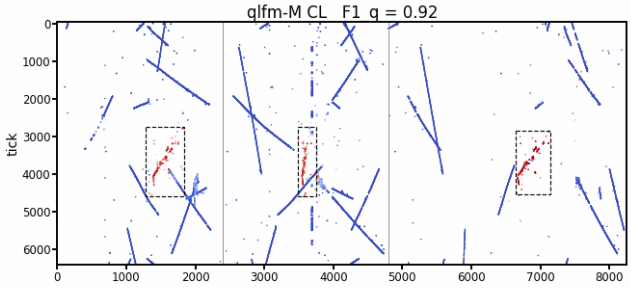
qlfm-M CL: 1.20 GeV (+7%) ✓ qlfm-S CL: 1.06 GeV (-5%) ✓ Panda: 1.21 GeV (+8%) ✓ PoLAr-MAE: 1.10 GeV (-2%) ✓
FM4NPP: 1.01 GeV (-9%) ✓ qlfm-SD CL: 1.12 GeV (+0%) ✓ Pandora: 1.27 GeV (+13%) ✓ Pandora+BDT: 0.98 GeV (-13%) ✓

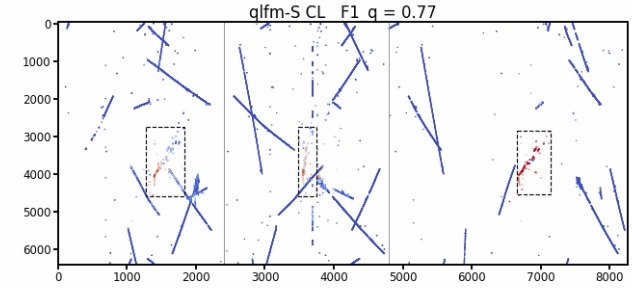


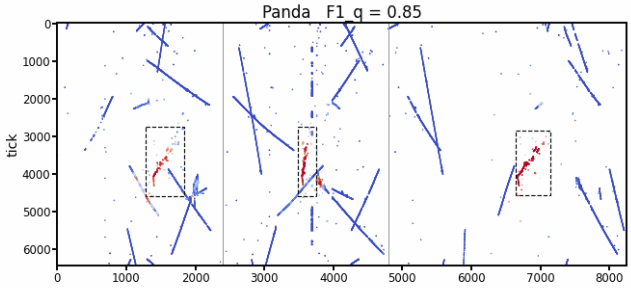


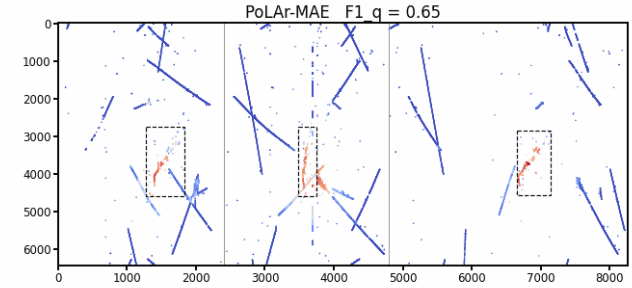


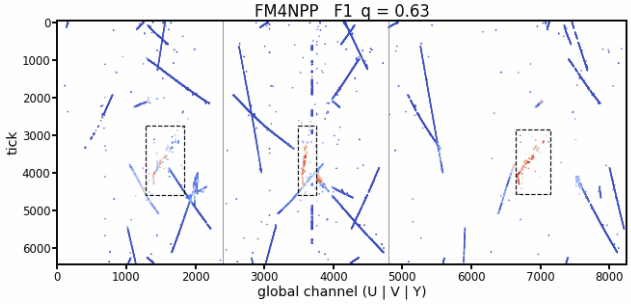


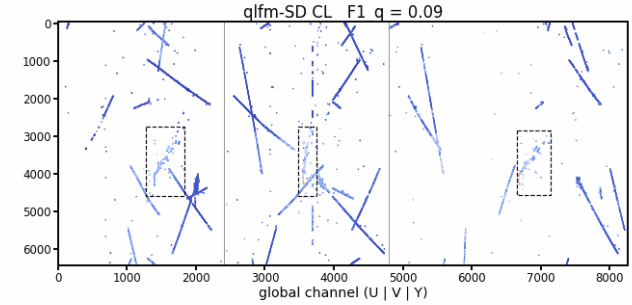










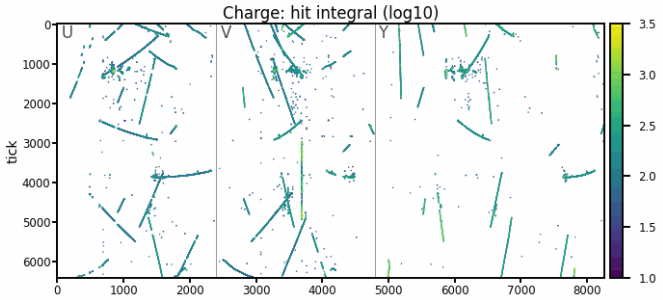


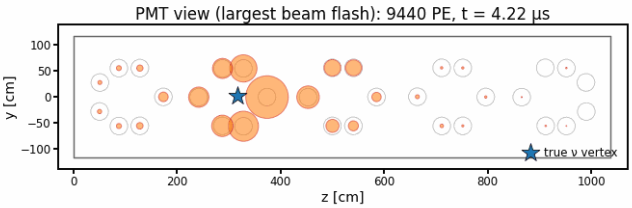
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

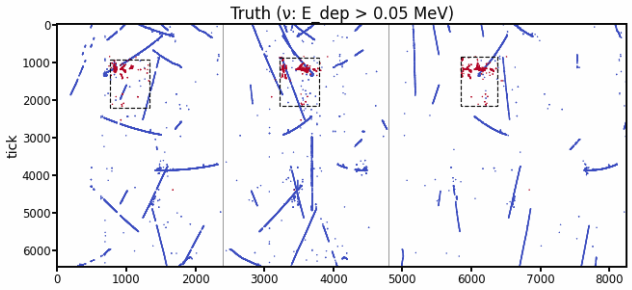
T4 energy — easy #6 (correct: 8/8) run 7017 / subrun 1166 / event 58320 [gidx 427339]

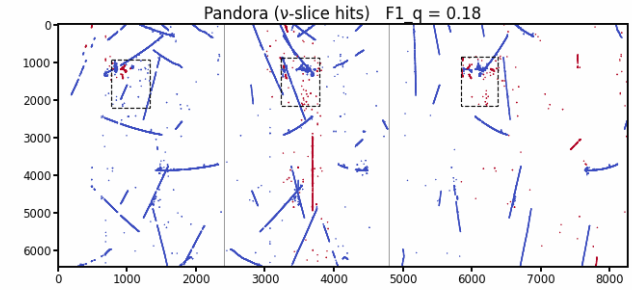
Truth: νμ CC, E = 1.50 GeV, vtx = (19, 1, 316) cm; task label: 1.50 GeV

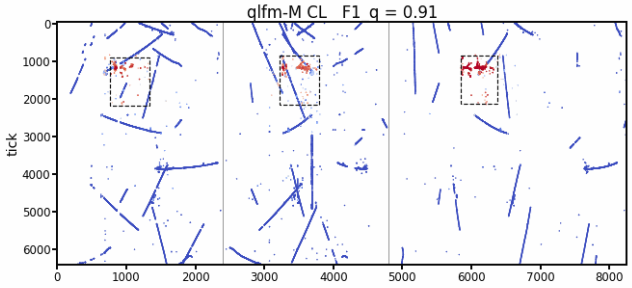
qlfm-M CL: 1.41 GeV (-6%) ✓ qlfm-S CL: 1.60 GeV (+7%) ✓ Panda: 1.27 GeV (-16%) ✓ PoLAr-MAE: 1.31 GeV (-13%) ✓
FM4NPP: 1.22 GeV (-19%) ✓ qlfm-SD CL: 1.35 GeV (-10%) ✓ Pandora: 1.29 GeV (-14%) ✓ Pandora+BDT: 1.23 GeV (-18%) ✓

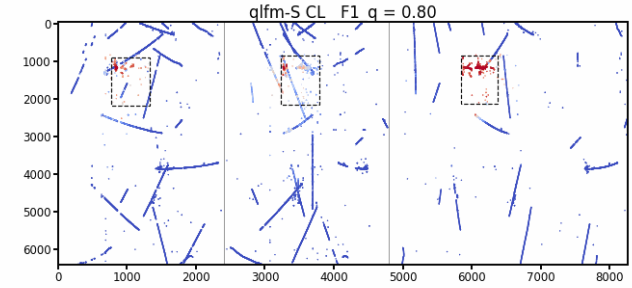


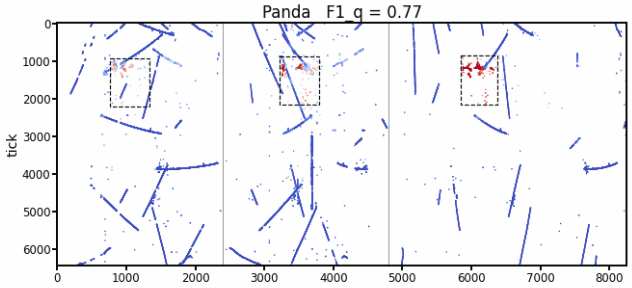


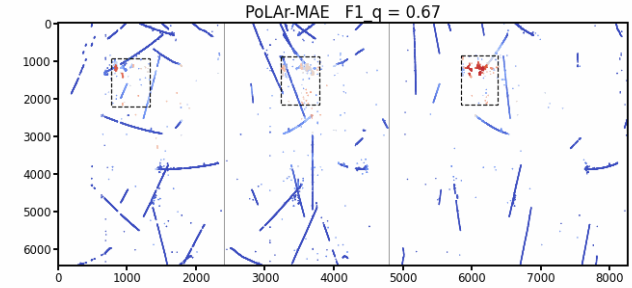


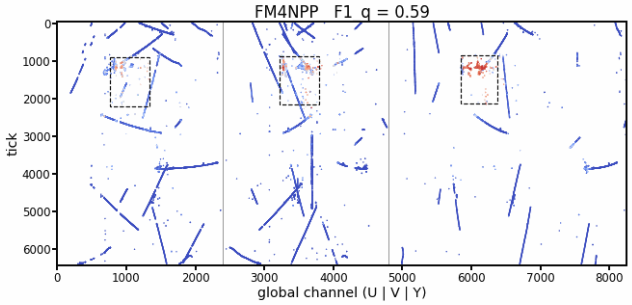


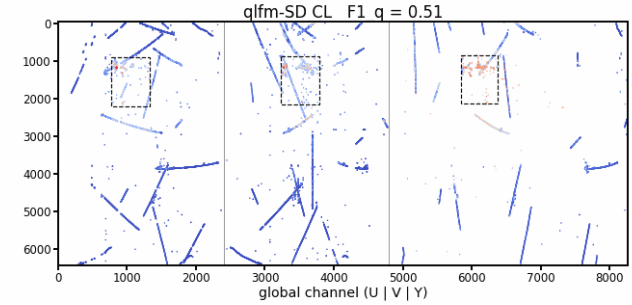










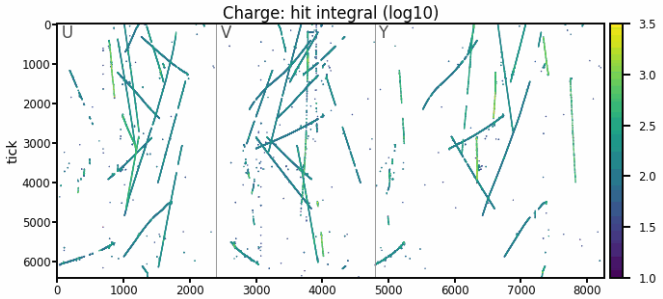


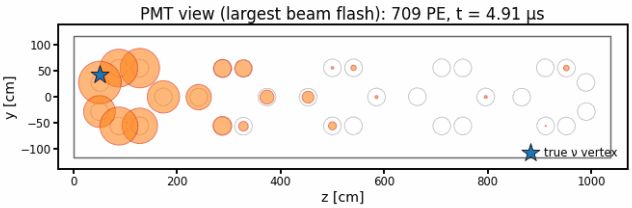
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

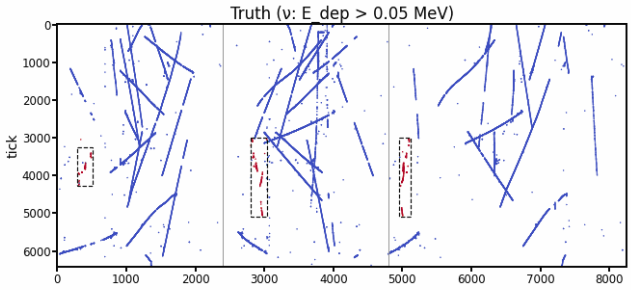
T4 energy — easy #7 (correct: 5/8) run 7020 / subrun 101 / event 5054 [gidx 517767]

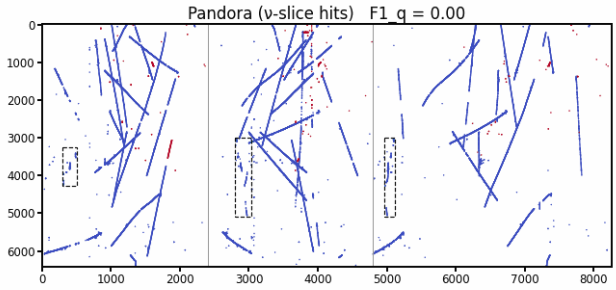
Truth: νμ CC, E = 0.62 GeV, vtx = (188, 43, 50) cm; task label: 0.62 GeV

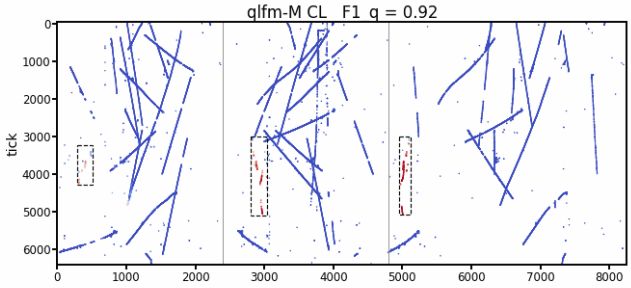
qlfm-M CL: 0.72 GeV (+17%) ✓ qlfm-S CL: 0.70 GeV (+13%) ✓ Panda: 0.72 GeV (+16%) ✓ PoLAr-MAE: 0.92 GeV (+48%) ✗
FM4NPP: 0.85 GeV (+36%) ✗ qlfm-SD CL: 0.84 GeV (+35%) ✗ Pandora: 0.69 GeV (+11%) ✓ Pandora+BDT: 0.60 GeV (-3%) ✓

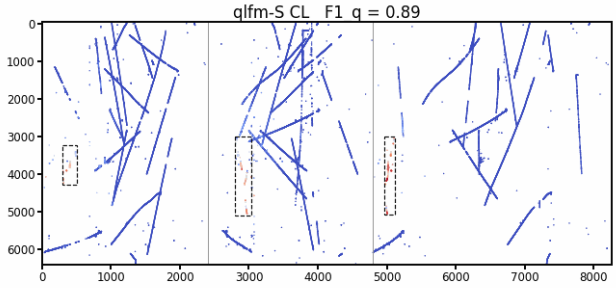


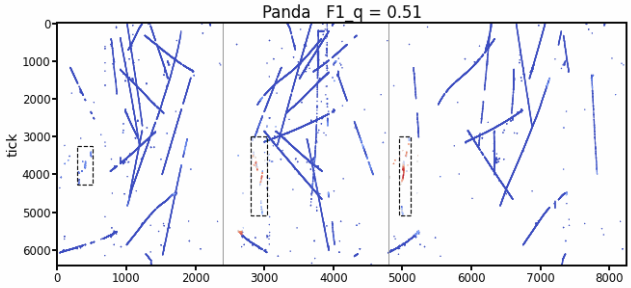


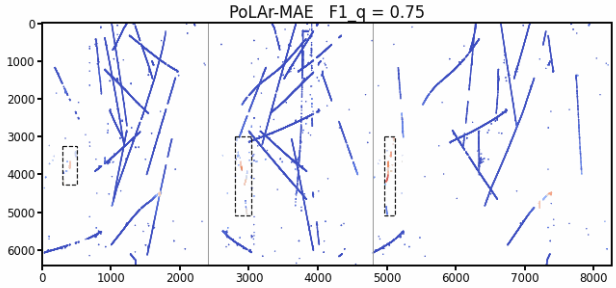


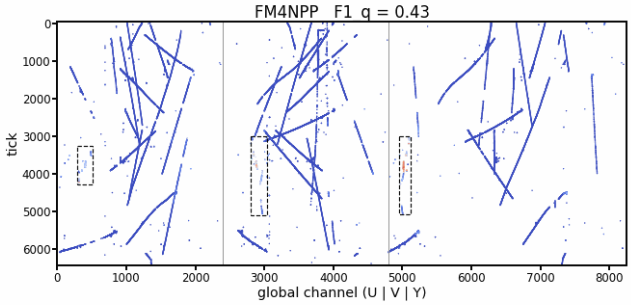


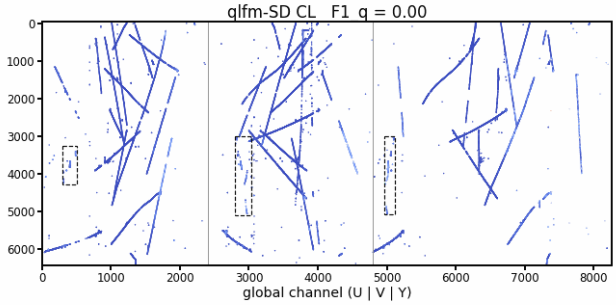










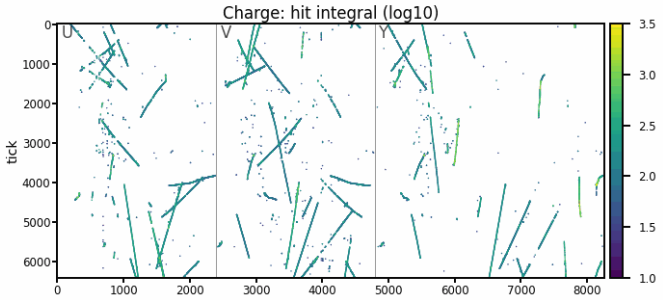


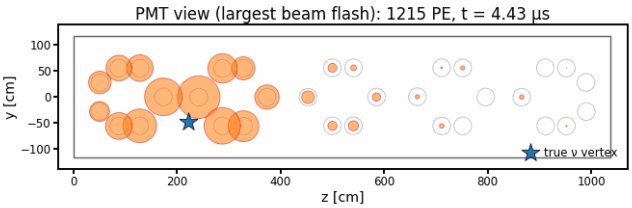
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

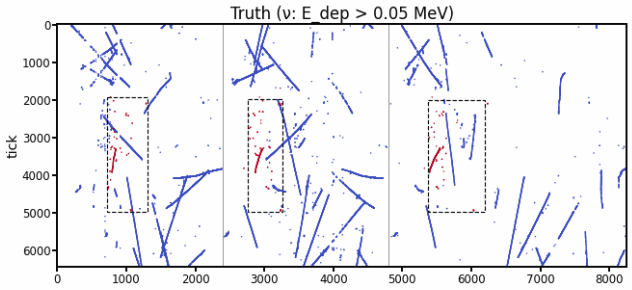
T4 energy — easy #8 (correct: 8/8) run 5384 / subrun 83 / event 4188 [gidx 422679]

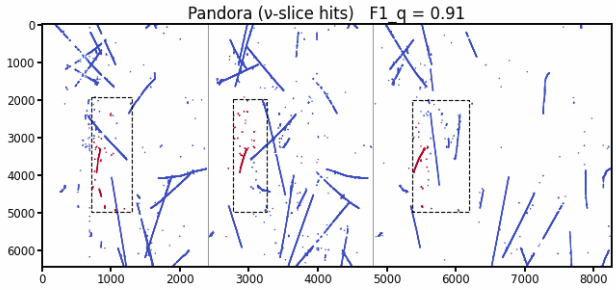
Truth: νμ CC, E = 0.83 GeV, vtx = (134, -48, 222) cm; task label: 0.83 GeV

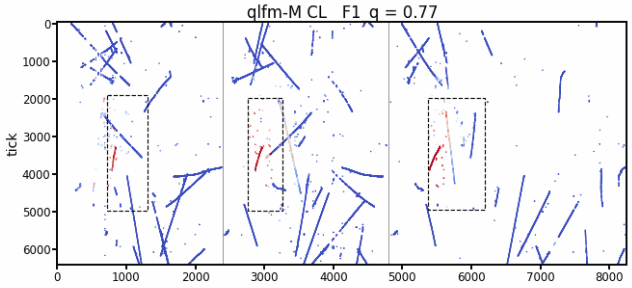
qlfm-M CL: 0.90 GeV (+8%) ✓ qlfm-S CL: 0.95 GeV (+15%) ✓ Panda: 0.75 GeV (-10%) ✓ PoLAr-MAE: 0.94 GeV (+13%) ✓
FM4NPP: 0.92 GeV (+10%) ✓ qlfm-SD CL: 0.92 GeV (+10%) ✓ Pandora: 0.98 GeV (+18%) ✓ Pandora+BDT: 0.90 GeV (+8%) ✓

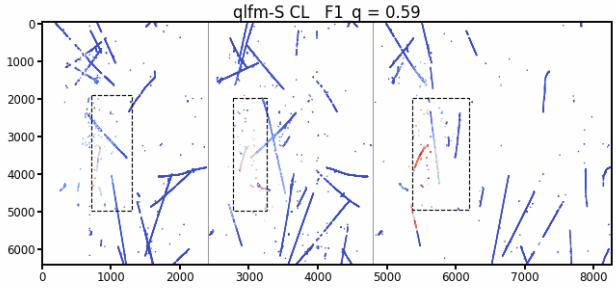


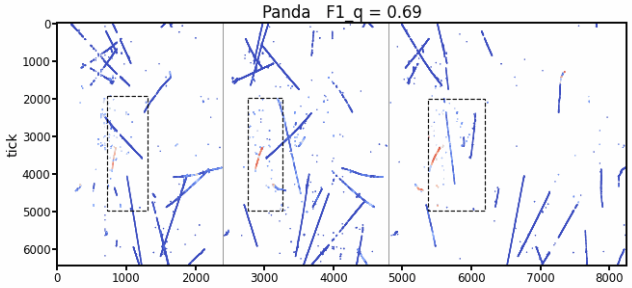


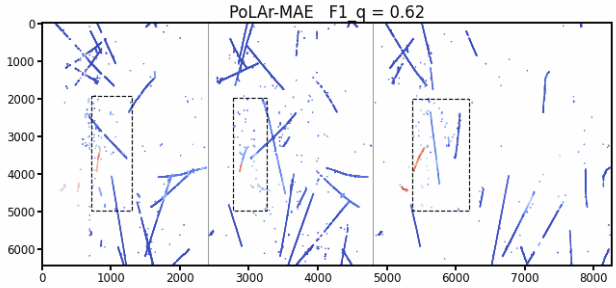


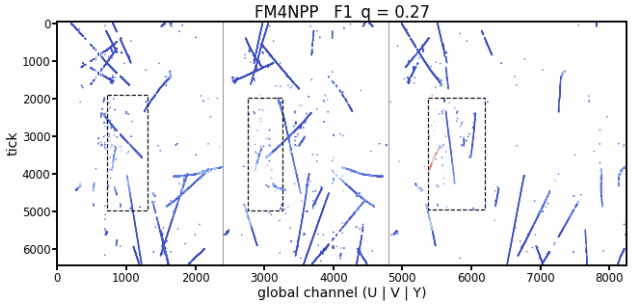


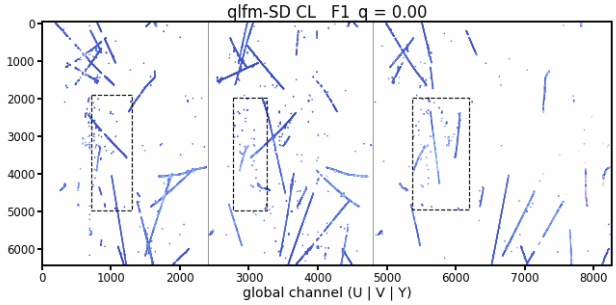










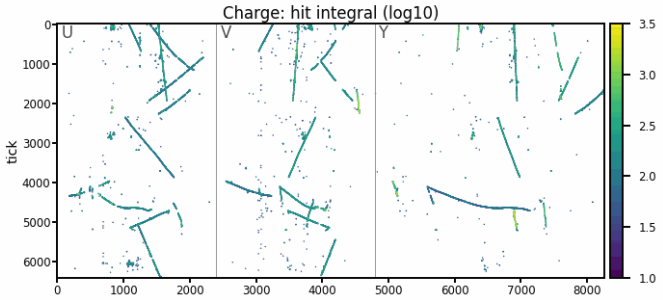


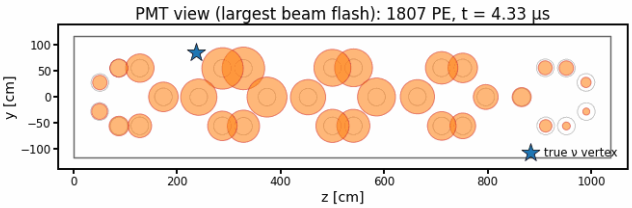
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

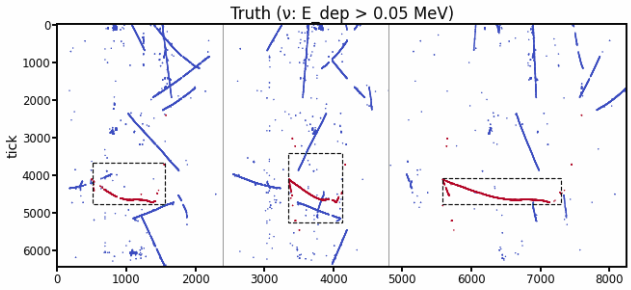
T4 energy — easy #9 (correct: 8/8) run 7004 / subrun 1715 / event 85759 [gidx 407536]

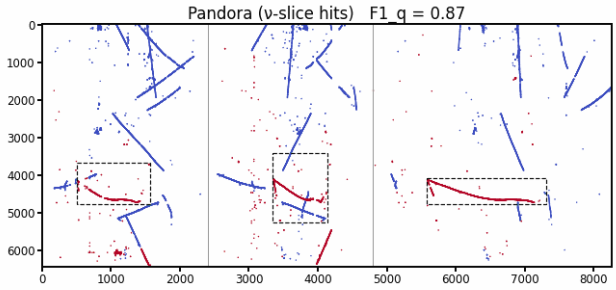
Truth: νμ CC, E = 1.63 GeV, vtx = (180, 85, 236) cm; task label: 1.63 GeV

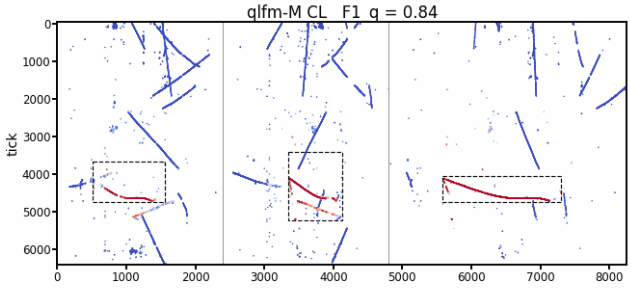
qlfm-M CL: 1.85 GeV (+13%) ✓ qlfm-S CL: 1.64 GeV (+1%) ✓ Panda: 1.55 GeV (-5%) ✓ PoLAr-MAE: 1.70 GeV (+4%) ✓
FM4NPP: 1.35 GeV (-17%) ✓ qlfm-SD CL: 1.55 GeV (-5%) ✓ Pandora: 1.43 GeV (-12%) ✓ Pandora+BDT: 1.68 GeV (+3%) ✓

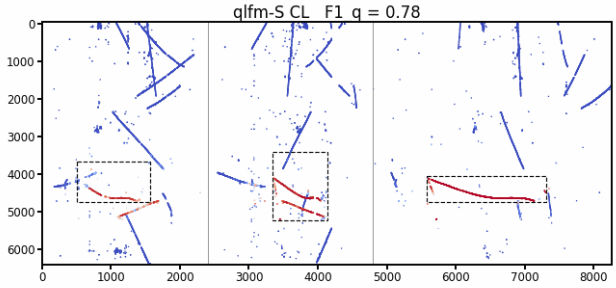


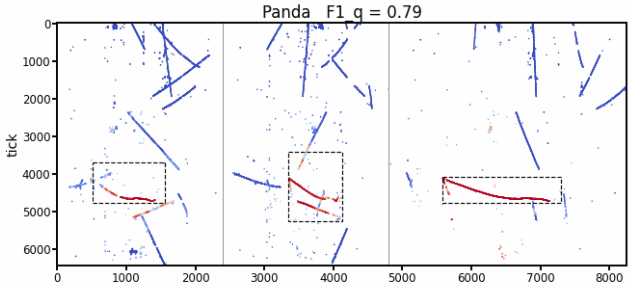


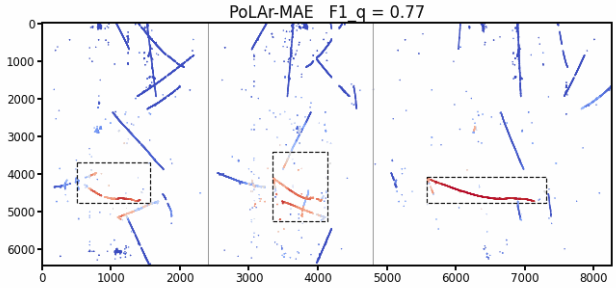


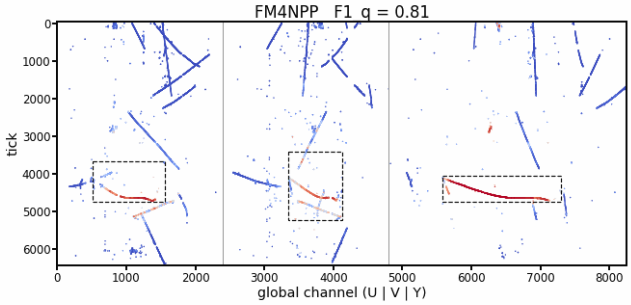


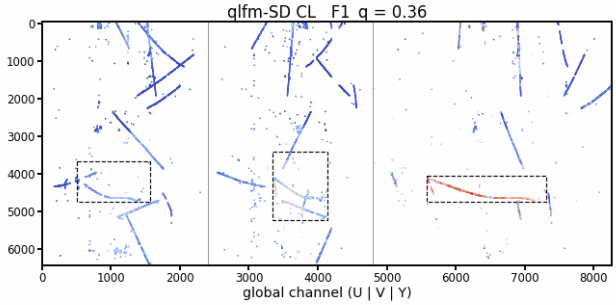










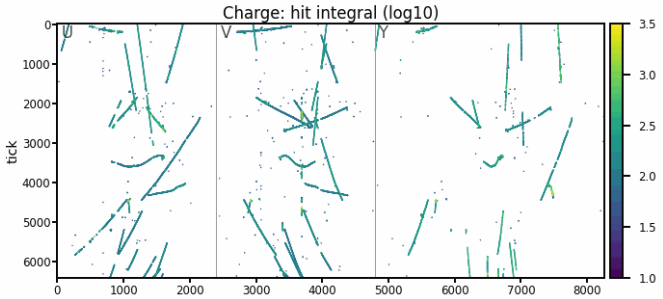


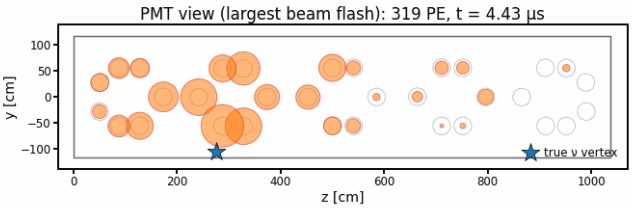
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

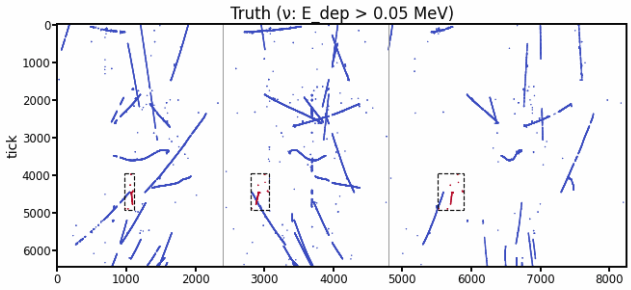
T4 energy — easy #10 (correct: 5/8) run 6151 / subrun 26 / event 1339 [gidx 133888]

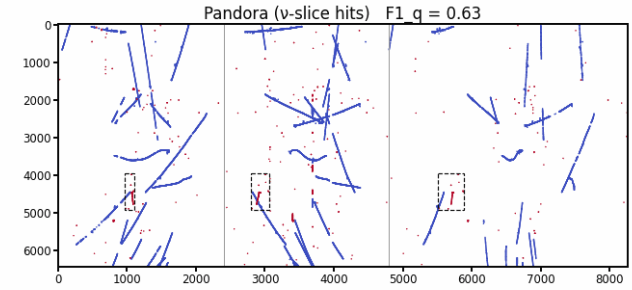
Truth: νμ CC, E = 0.69 GeV, vtx = (199, -105, 276) cm; task label: 0.69 GeV

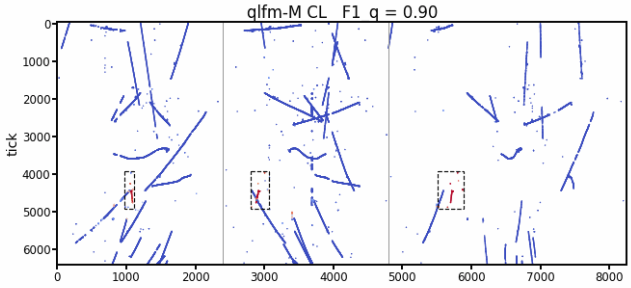
qlfm-M CL: 0.61 GeV (-12%) ✓ qlfm-S CL: 0.84 GeV (+22%) ✗ Panda: 0.63 GeV (-8%) ✓ PoLAr-MAE: 0.91 GeV (+32%) ✗
FM4NPP: 0.81 GeV (+18%) ✓ qlfm-SD CL: 0.83 GeV (+20%) ✓ Pandora: 0.94 GeV (+36%) ✗ Pandora+BDT: 0.63 GeV (-9%) ✓

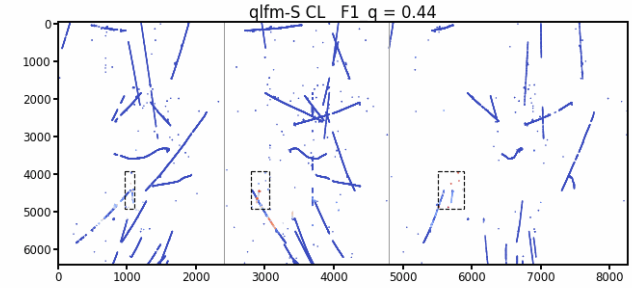


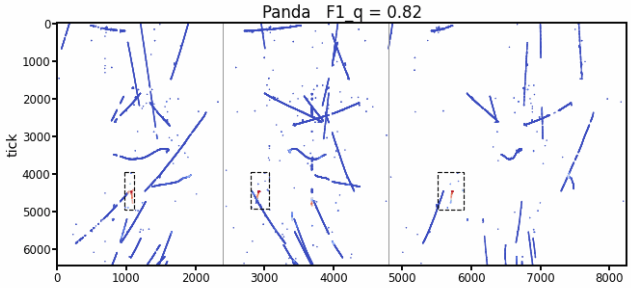


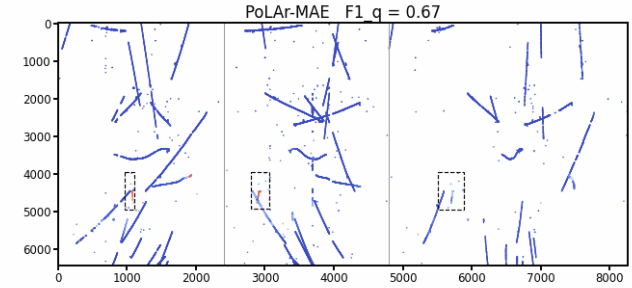


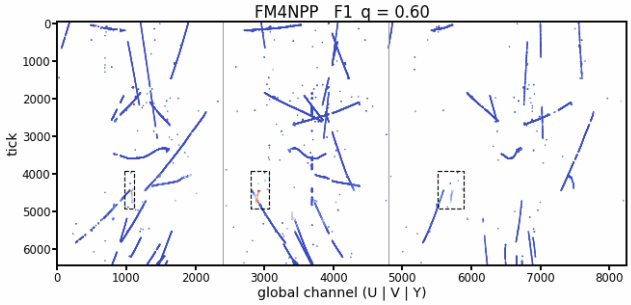


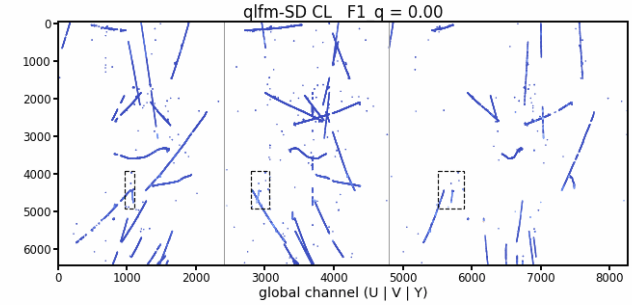










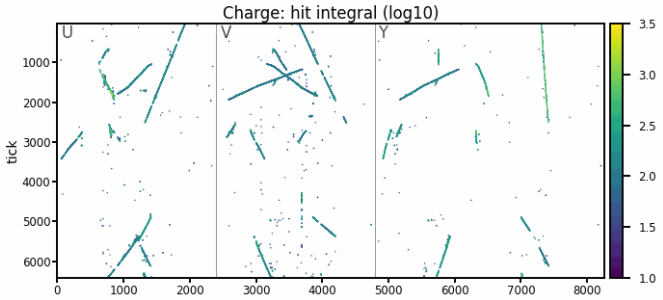


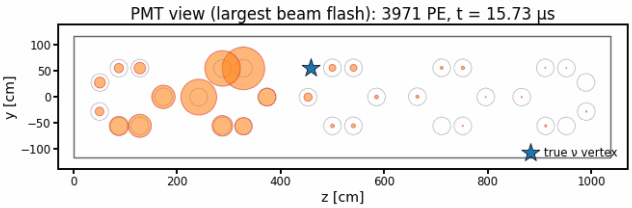
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

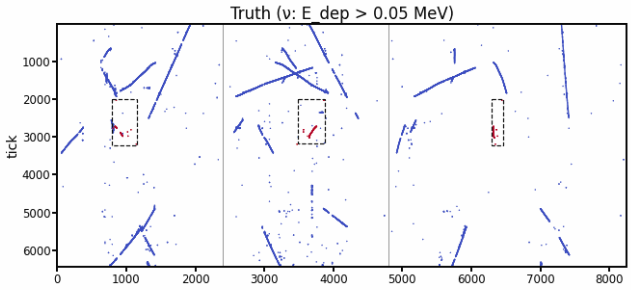
T4 energy — hard #1 (correct: 0/8) run 7014 / subrun 1758 / event 87948 [gidx 388609]

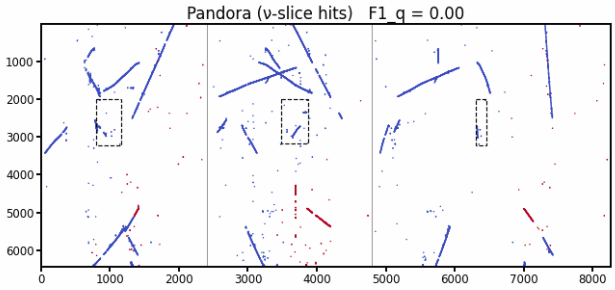
Truth: νμ CC, E = 0.34 GeV, vtx = (117, 55, 458) cm; task label: 0.34 GeV

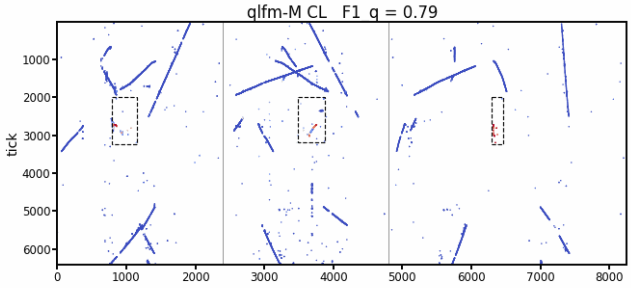
qlfm-M CL: 0.69 GeV (+103%) ✗ qlfm-S CL: 1.00 GeV (+196%) ✗ Panda: 0.49 GeV (+44%) ✗ PoLAr-MAE: 0.90 GeV (+165%) ✗
FM4NPP: 0.86 GeV (+153%) ✗ qlfm-SD CL: 0.82 GeV (+142%) ✗ Pandora: 0.92 GeV (+170%) ✗ Pandora+BDT: 0.68 GeV (+99%) ✗

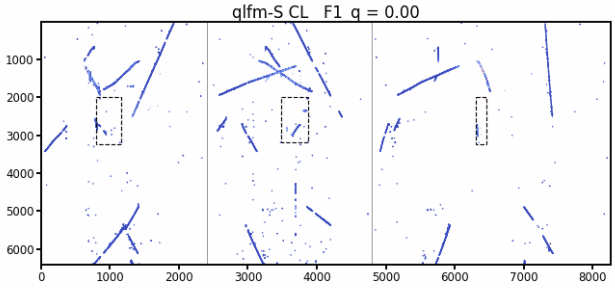


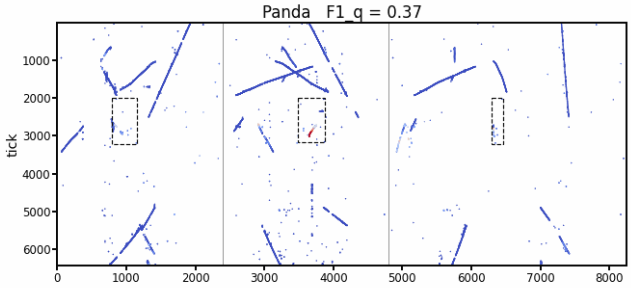


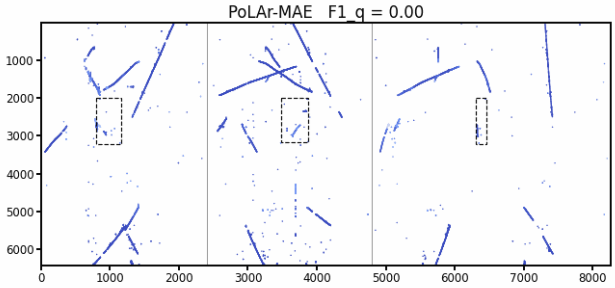


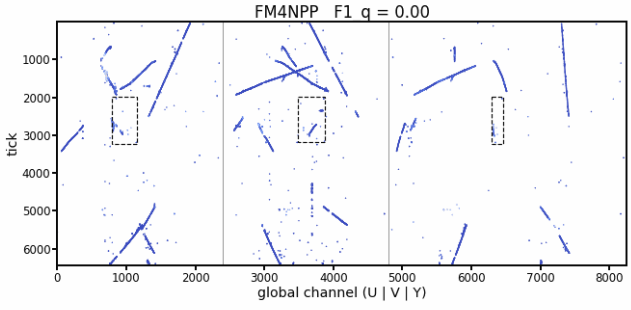


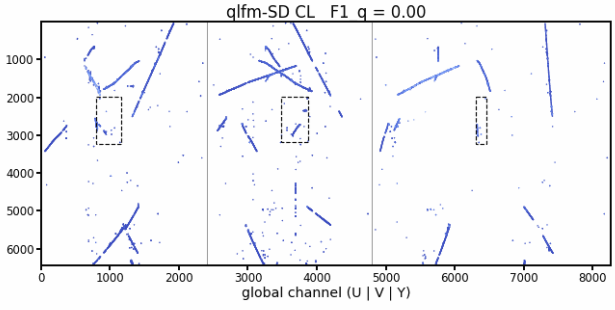










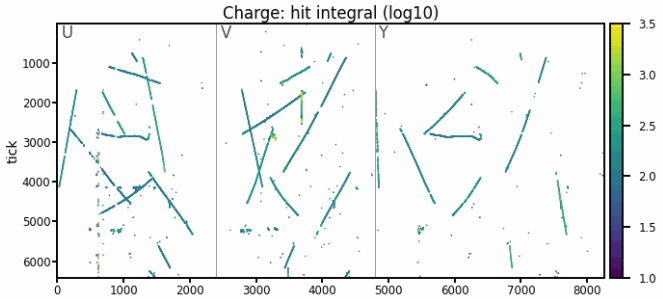


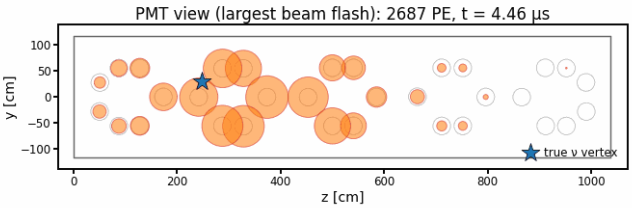
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

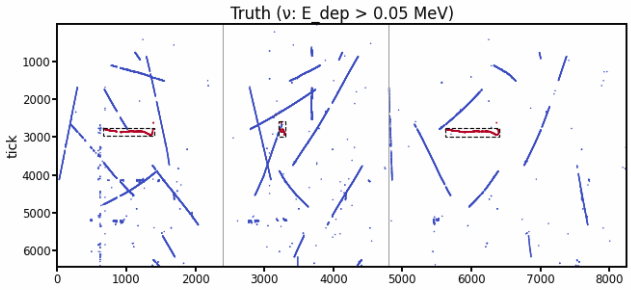
T4 energy — hard #2 (correct: 0/8) run 6830 / subrun 86 / event 4310 [gidx 684170]

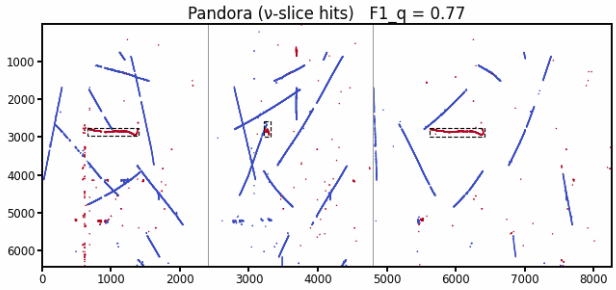
Truth: νμ CC, E = 0.80 GeV, vtx = (105, 29, 248) cm; task label: 0.80 GeV

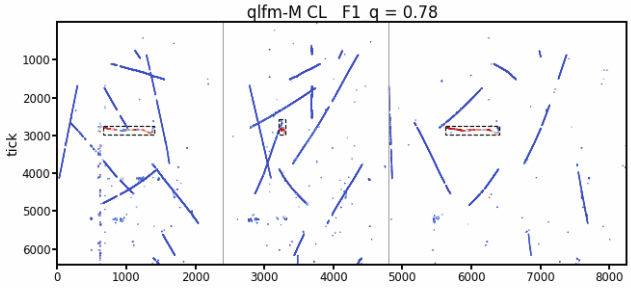
qlfm-M CL: 0.96 GeV (+21%) ✗ qlfm-S CL: 1.13 GeV (+41%) ✗ Panda: 1.14 GeV (+42%) ✗ PoLAr-MAE: 1.06 GeV (+32%) ✗
FM4NPP: 1.06 GeV (+33%) ✗ qlfm-SD CL: 1.12 GeV (+40%) ✗ Pandora: 1.29 GeV (+61%) ✗ Pandora+BDT: 1.07 GeV (+34%) ✗

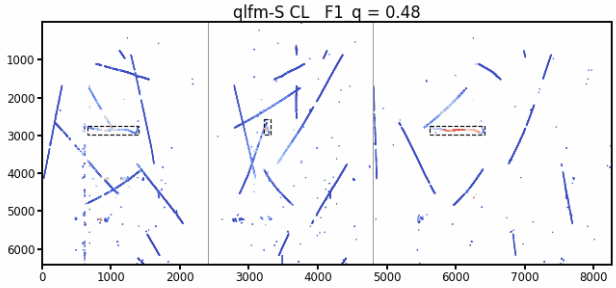


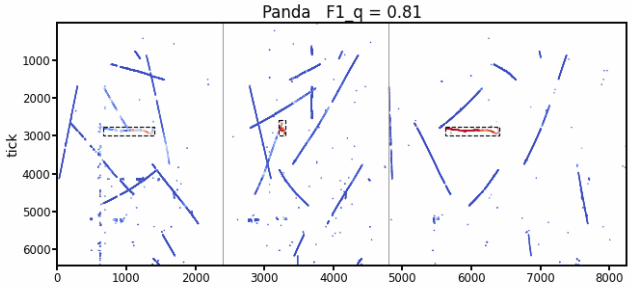


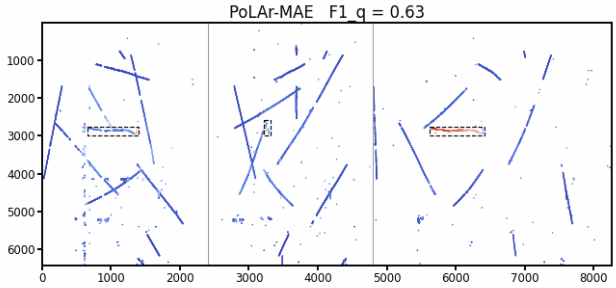


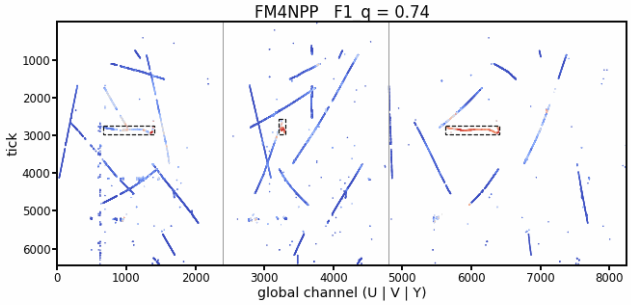


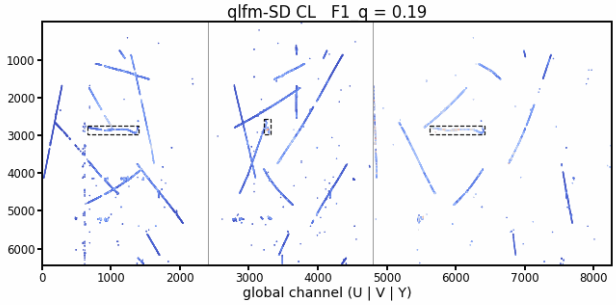










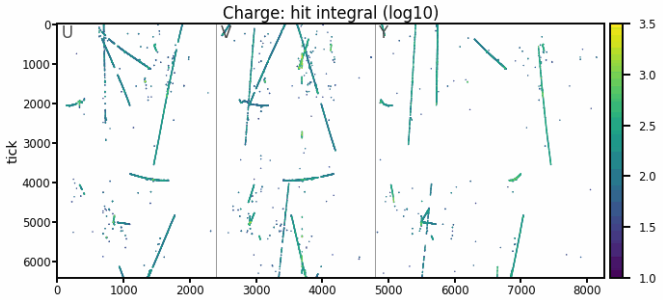


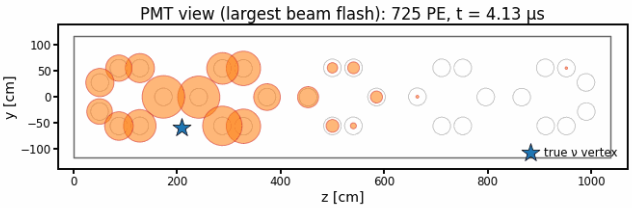
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

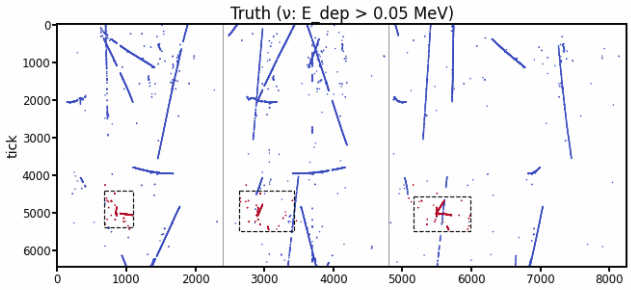
T4 energy — hard #3 (correct: 0/8) run 7023 / subrun 1211 / event 60575 [gidx 508179]

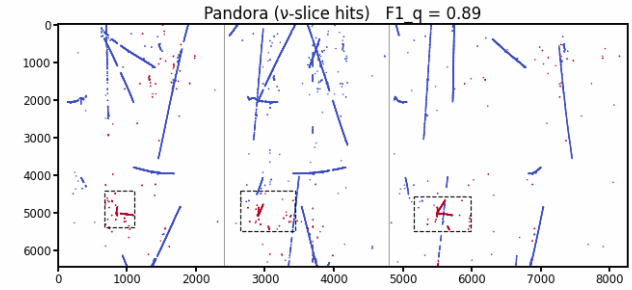
Truth: νμ CC, E = 1.89 GeV, vtx = (228, -60, 208) cm; task label: 1.89 GeV

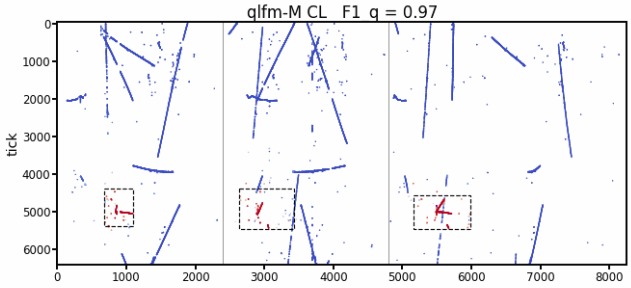
qlfm-M CL: 1.15 GeV (-39%) ✗ qlfm-S CL: 1.06 GeV (-44%) ✗ Panda: 1.37 GeV (-27%) ✗ PoLAr-MAE: 0.91 GeV (-52%) ✗
FM4NPP: 0.85 GeV (-55%) ✗ qlfm-SD CL: 0.91 GeV (-51%) ✗ Pandora: 1.24 GeV (-34%) ✗ Pandora+BDT: 1.10 GeV (-42%) ✗

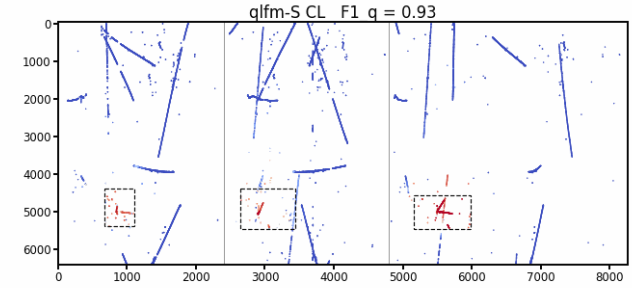


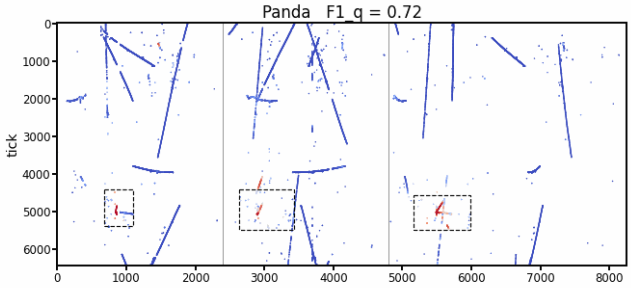


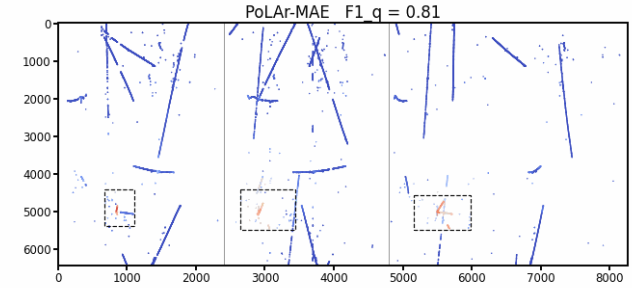


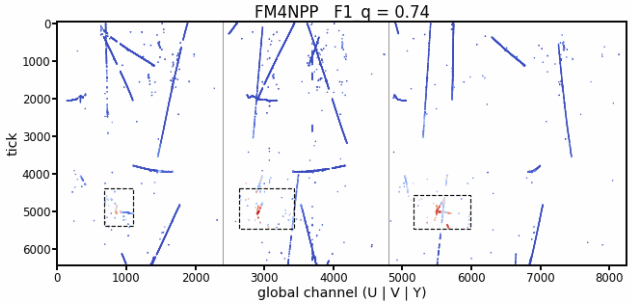


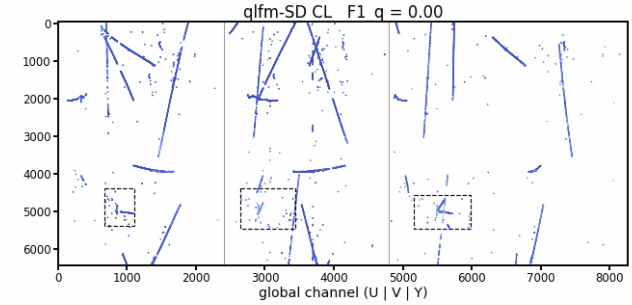










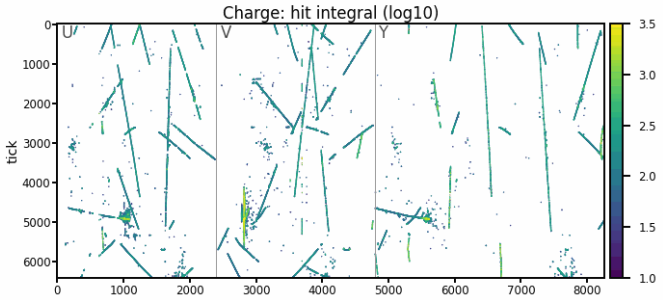


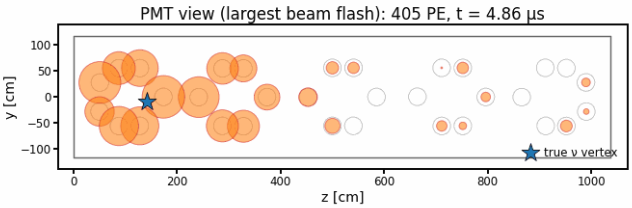
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

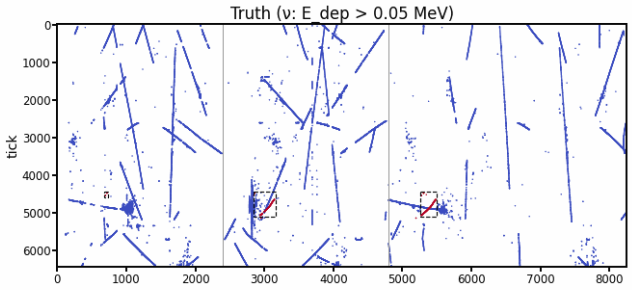
T4 energy — hard #4 (correct: 0/8) run 7023 / subrun 528 / event 26413 [gidx 734290]

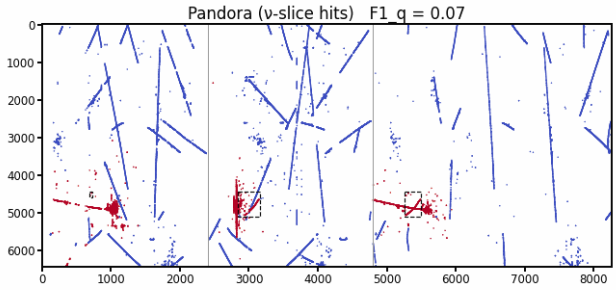
Truth: νμ CC, E = 0.35 GeV, vtx = (232, -9, 142) cm; task label: 0.35 GeV

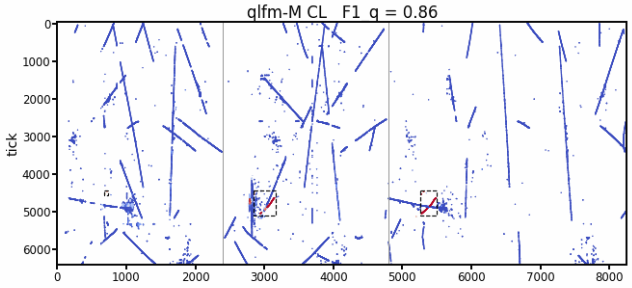
qlfm-M CL: 0.64 GeV (+81%) ✗ qlfm-S CL: 0.72 GeV (+103%) ✗ Panda: 0.90 GeV (+154%) ✗ PoLAr-MAE: 0.87 GeV (+146%) ✗
FM4NPP: 0.82 GeV (+134%) ✗ qlfm-SD CL: 0.83 GeV (+135%) ✗ Pandora: 1.87 GeV (+429%) ✗ Pandora+BDT: 1.28 GeV (+262%) ✗

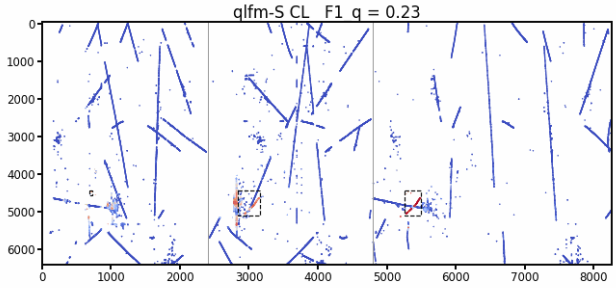


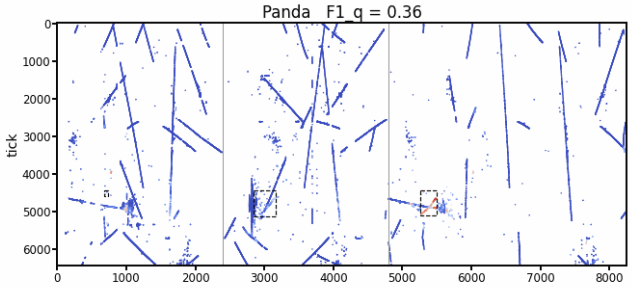


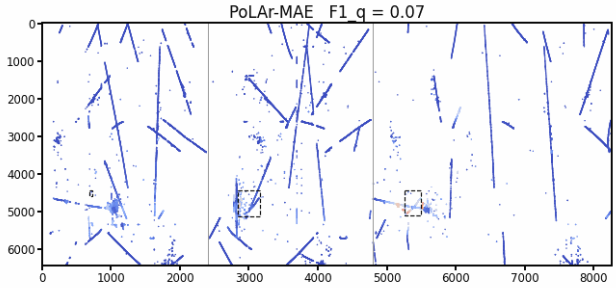


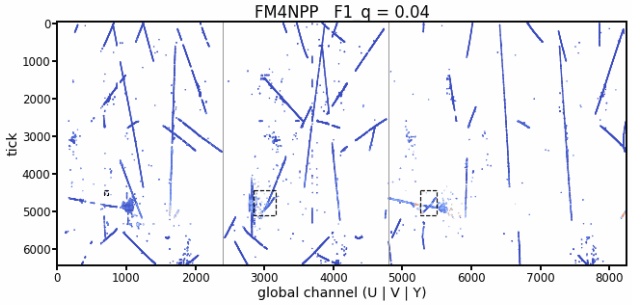


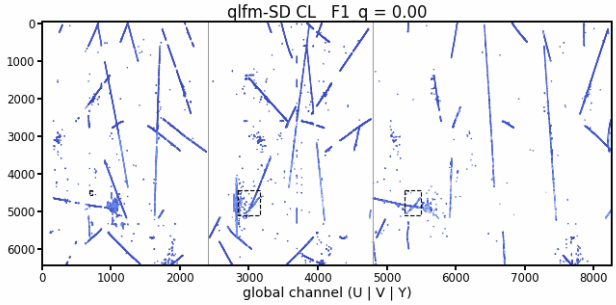










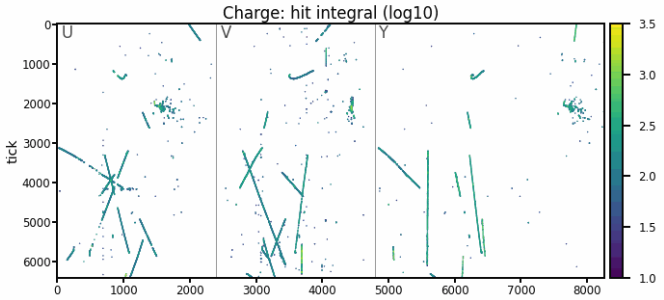


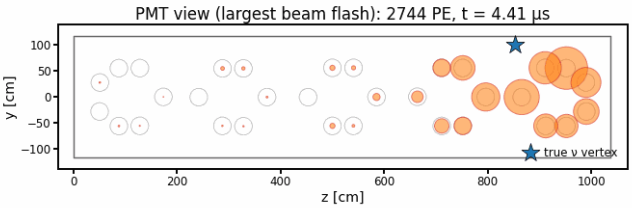
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

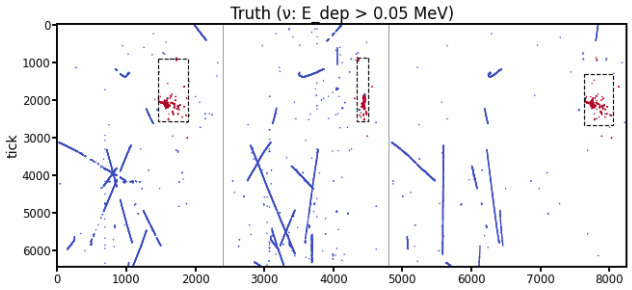
T4 energy — hard #5 (correct: 0/8) run 5005 / subrun 169 / event 8499 [gidx 790352]

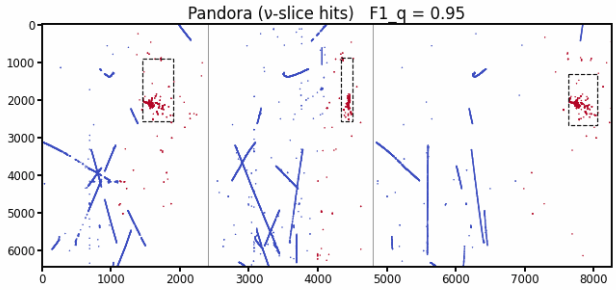
Truth: νe CC, E = 0.80 GeV, vtx = (67, 100, 853) cm; task label: 0.80 GeV

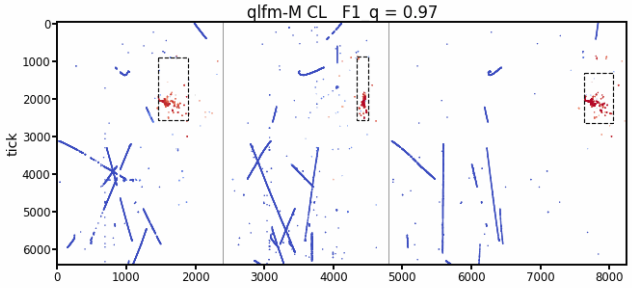
qlfm-M CL: 1.30 GeV (+62%) ✗ qlfm-S CL: 0.99 GeV (+24%) ✗ Panda: 1.01 GeV (+26%) ✗ PoLAr-MAE: 1.16 GeV (+45%) ✗
FM4NPP: 1.21 GeV (+51%) ✗ qlfm-SD CL: 1.00 GeV (+25%) ✗ Pandora: 1.27 GeV (+58%) ✗ Pandora+BDT: 1.02 GeV (+28%) ✗

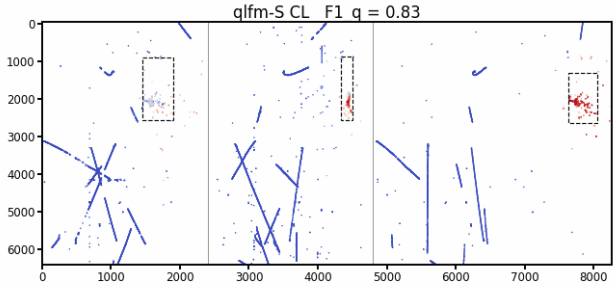


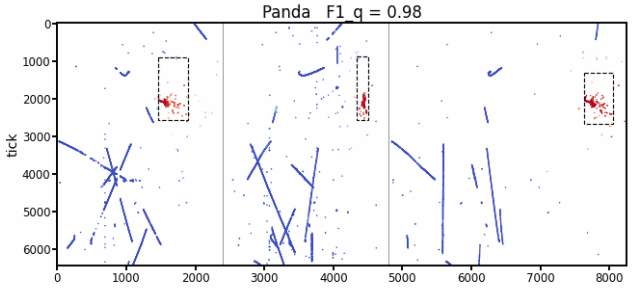


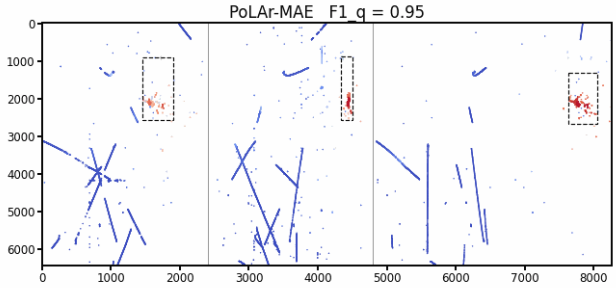


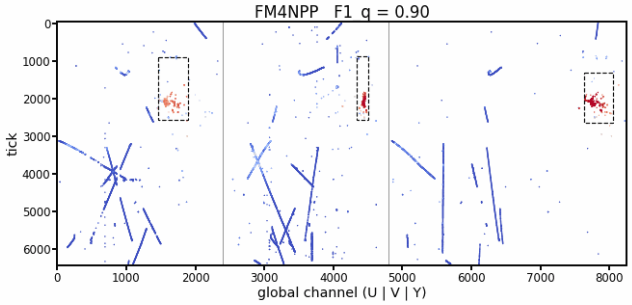


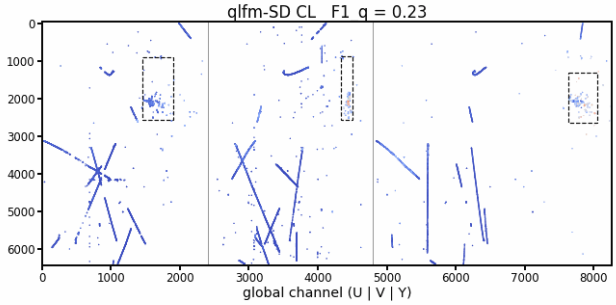










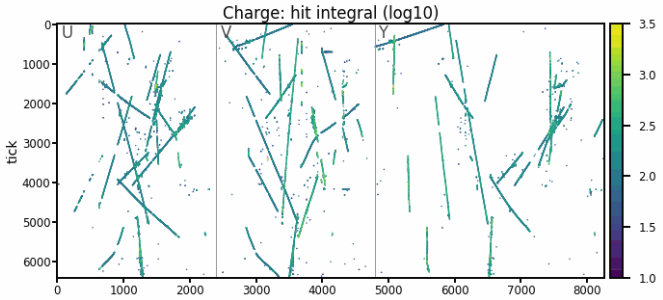


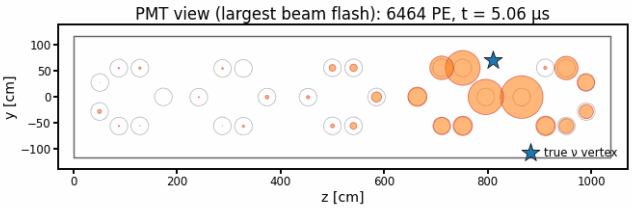
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

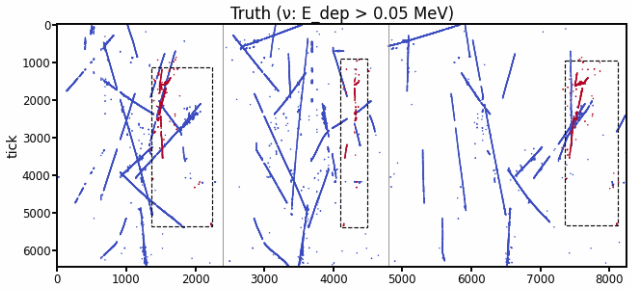
T4 energy — hard #6 (correct: 0/8) run 7003 / subrun 34 / event 1747 [gidx 8868]

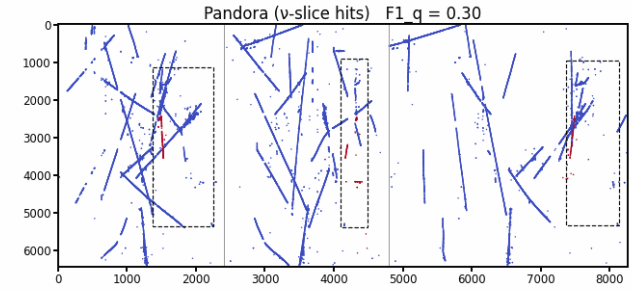
Truth: νμ CC, E = 2.09 GeV, vtx = (43, 71, 811) cm; task label: 2.09 GeV

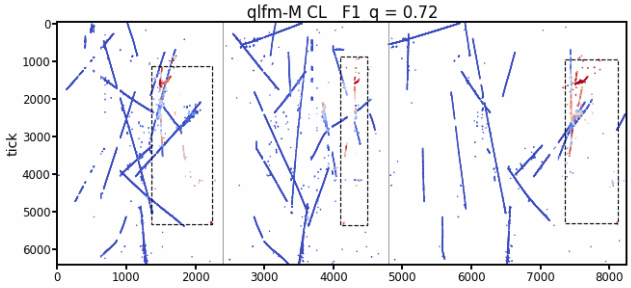
qlfm-M CL: 1.45 GeV (-31%) ✗ qlfm-S CL: 1.66 GeV (-21%) ✗ Panda: 1.17 GeV (-44%) ✗ PoLAr-MAE: 1.45 GeV (-30%) ✗
FM4NPP: 1.39 GeV (-34%) ✗ qlfm-SD CL: 1.40 GeV (-33%) ✗ Pandora: 0.88 GeV (-58%) ✗ Pandora+BDT: 1.15 GeV (-45%) ✗

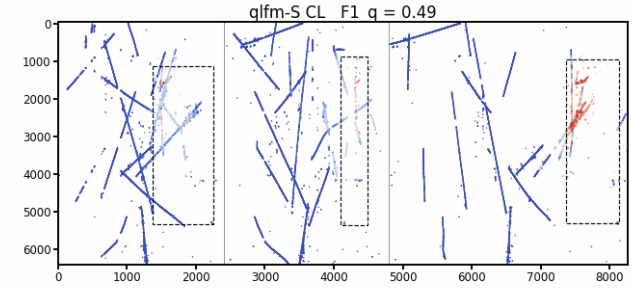


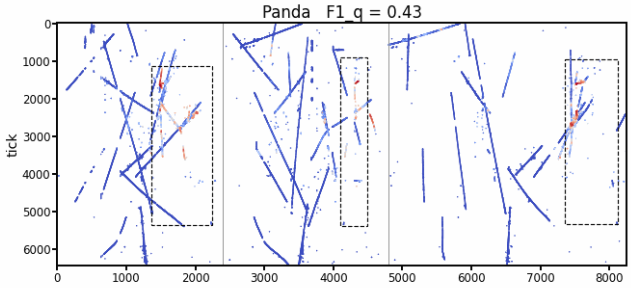


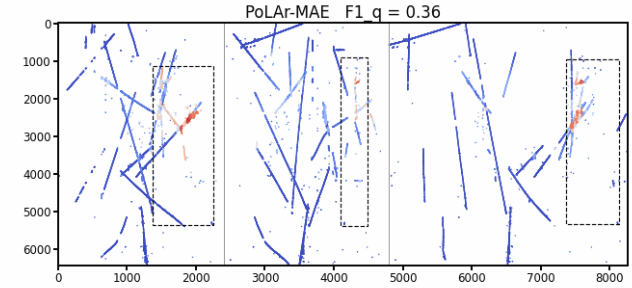


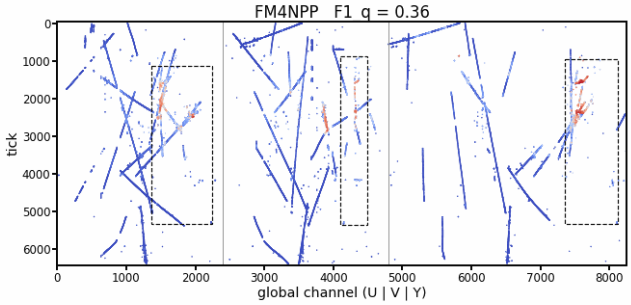


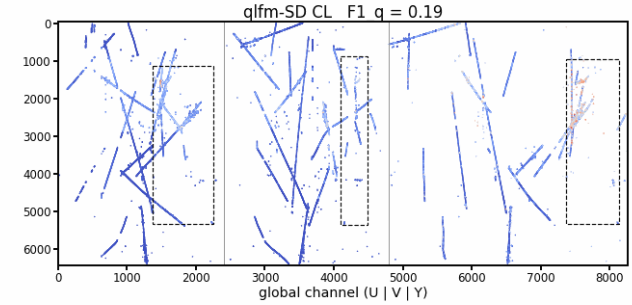










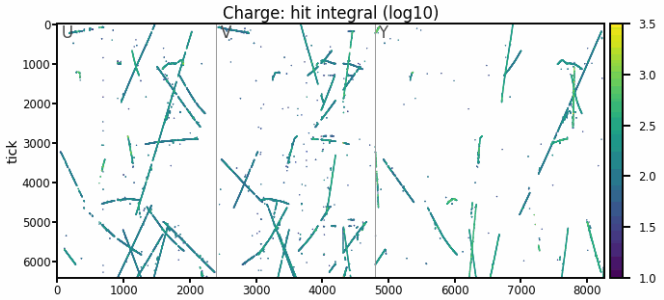


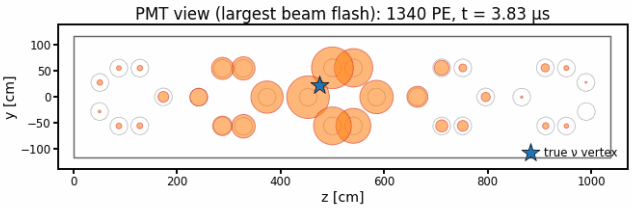
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

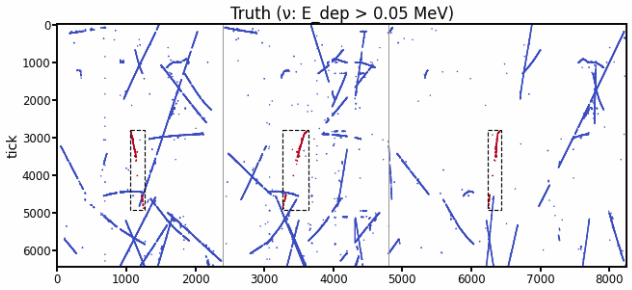
T4 energy — hard #7 (correct: 0/8) run 7048 / subrun 89 / event 4488 [gidx 763685]

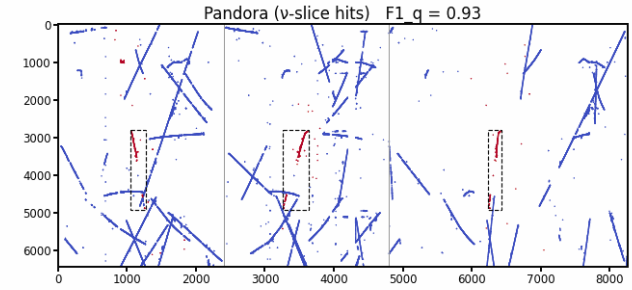
Truth: νe CC, E = 0.47 GeV, vtx = (110, 22, 476) cm; task label: 0.47 GeV

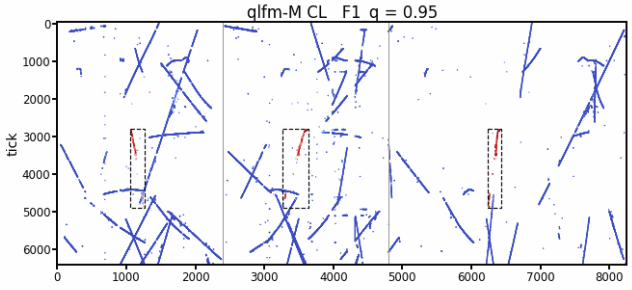
qlfm-M CL: 0.64 GeV (+37%) ✗ qlfm-S CL: 0.82 GeV (+76%) ✗ Panda: 0.60 GeV (+29%) ✗ PoLAr-MAE: 0.96 GeV (+106%) ✗
FM4NPP: 0.92 GeV (+97%) ✗ qlfm-SD CL: 0.88 GeV (+90%) ✗ Pandora: 1.08 GeV (+132%) ✗ Pandora+BDT: 0.72 GeV (+54%) ✗

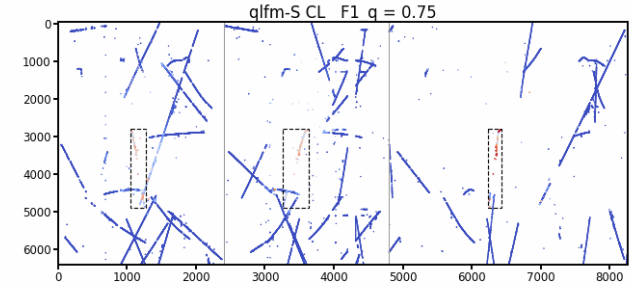


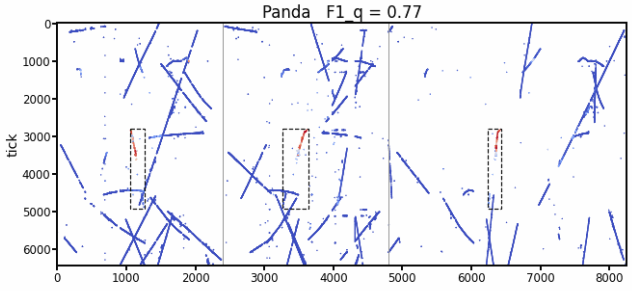


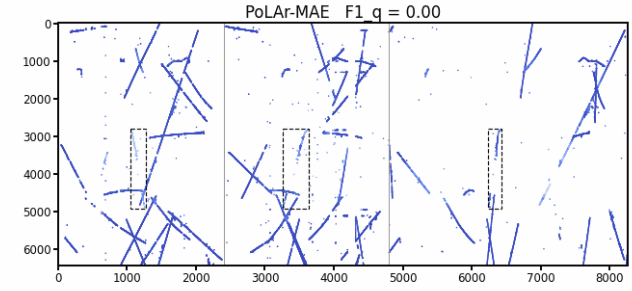


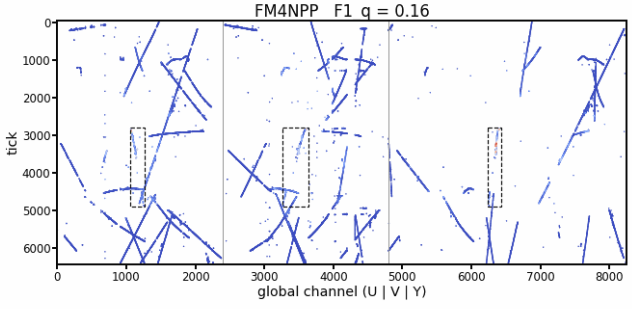


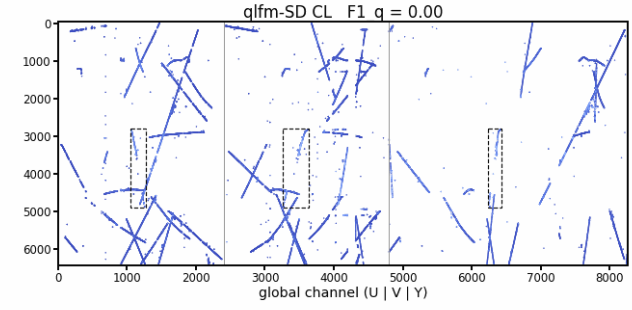










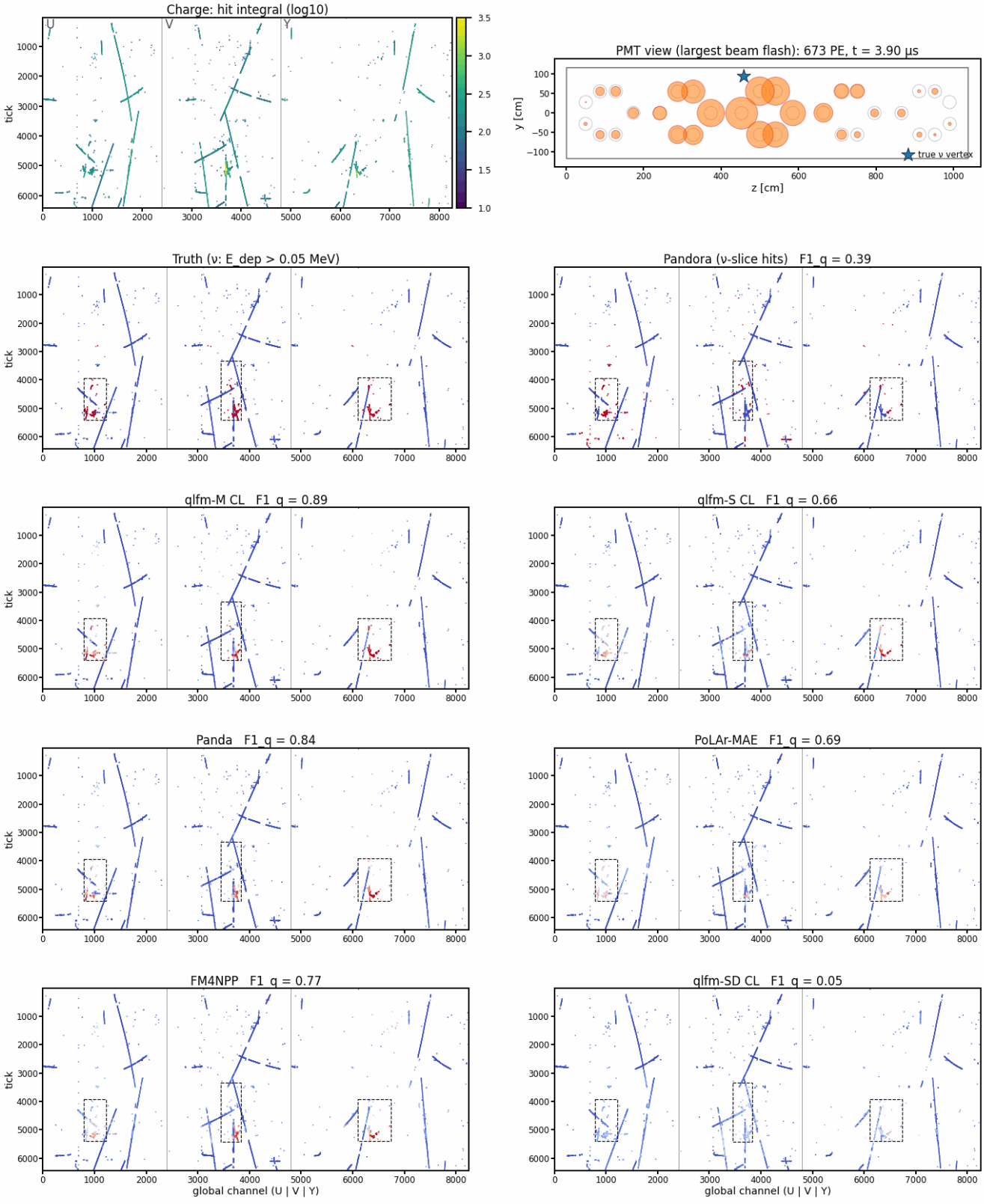


hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

T4 energy — hard #8 (correct: 0/8) run 7017 / subrun 1508 / event 75440 [gidx 481915]

Truth: νμ CC, E = 1.49 GeV, vtx = (239, 94, 459) cm; task label: 1.49 GeV

qlfm-M CL: 1.15 GeV (-23%) ✗ qlfm-S CL: 1.19 GeV (-20%) ✗ Panda: 0.99 GeV (-34%) ✗ PoLAr-MAE: 0.90 GeV (-39%) ✗
FM4NPP: 0.85 GeV (-43%) ✗ qlfm-SD CL: 0.87 GeV (-42%) ✗ Pandora: 0.85 GeV (-43%) ✗ Pandora+BDT: 1.01 GeV (-33%) ✗

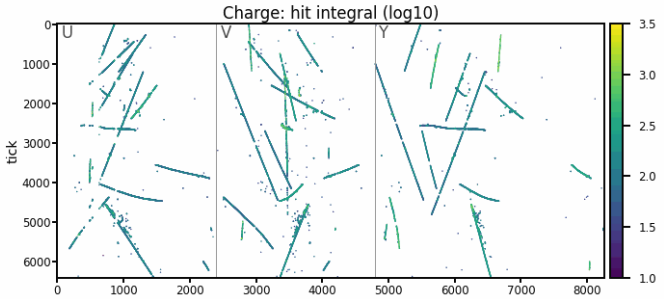


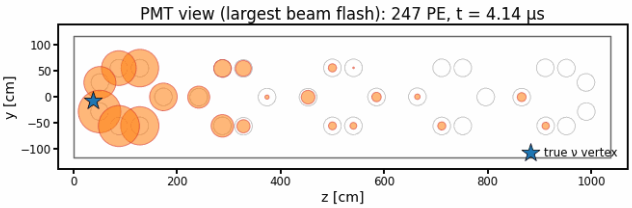
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

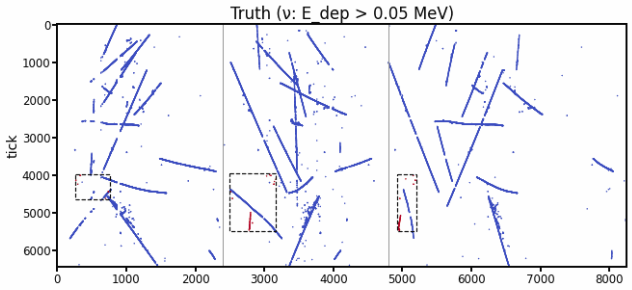
T4 energy — hard #9 (correct: 0/8) run 7018 / subrun 161 / event 8082 [gidx 710013]

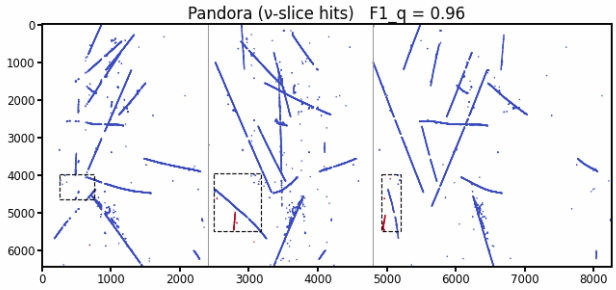
Truth: νμ CC, E = 1.52 GeV, vtx = (252, -7, 38) cm; task label: 1.52 GeV

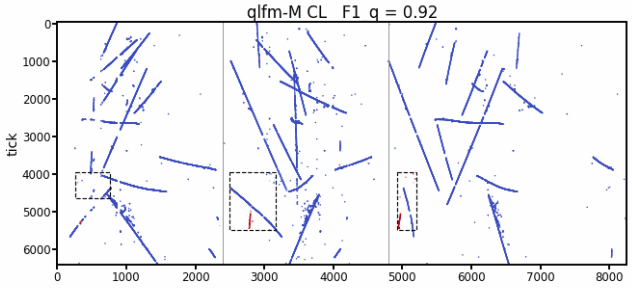
qlfm-M CL: 0.93 GeV (-39%) ✗ qlfm-S CL: 0.92 GeV (-39%) ✗ Panda: 1.18 GeV (-22%) ✗ PoLAr-MAE: 0.90 GeV (-41%) ✗
FM4NPP: 0.81 GeV (-46%) ✗ qlfm-SD CL: 0.83 GeV (-46%) ✗ Pandora: 0.79 GeV (-48%) ✗ Pandora+BDT: 0.84 GeV (-44%) ✗

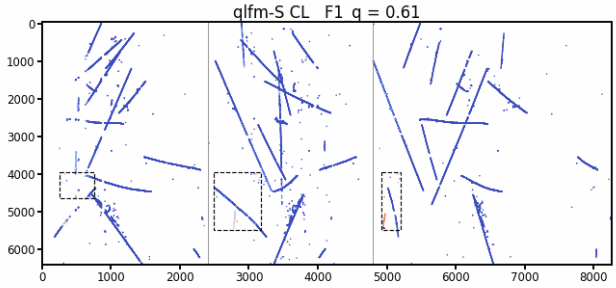


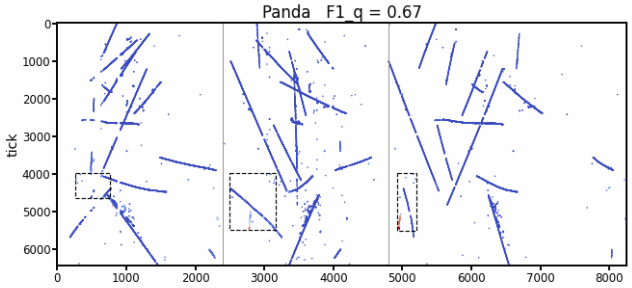


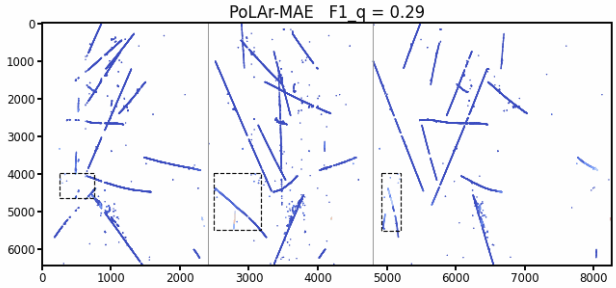


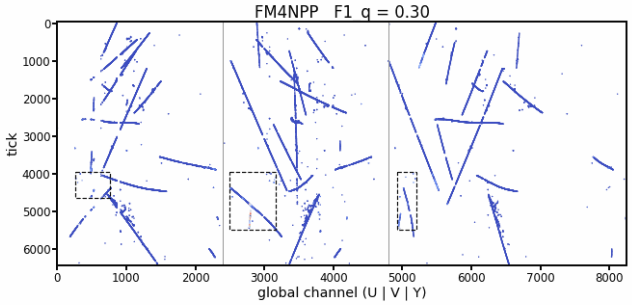


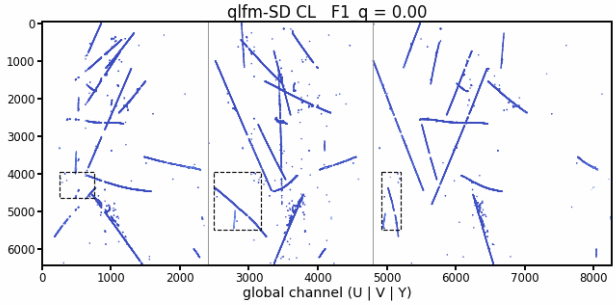










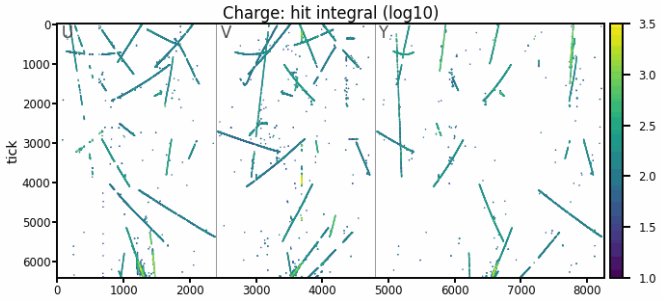


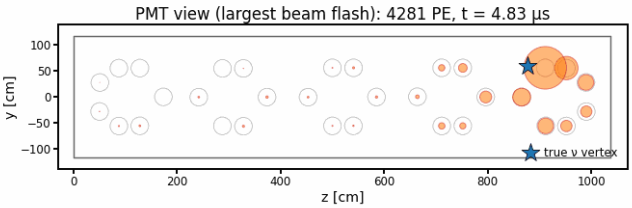
hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane

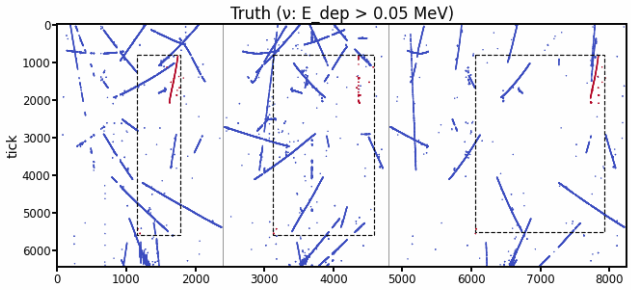
T4 energy — hard #10 (correct: 0/8) run 7055 / subrun 50 / event 2538 [gidx 16508]

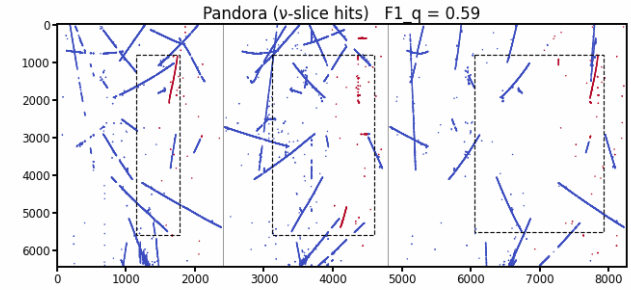
Truth: νμ CC, E = 0.61 GeV, vtx = (64, 59, 877) cm; task label: 0.61 GeV

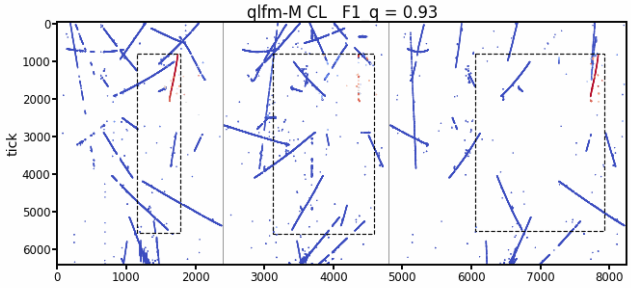
qlfm-M CL: 0.74 GeV (+22%) ✗ qlfm-S CL: 0.85 GeV (+39%) ✗ Panda: 1.03 GeV (+68%) ✗ PoLAr-MAE: 1.17 GeV (+93%) ✗
FM4NPP: 1.02 GeV (+68%) ✗ qlfm-SD CL: 1.01 GeV (+66%) ✗ Pandora: 1.00 GeV (+65%) ✗ Pandora+BDT: 0.80 GeV (+31%) ✗

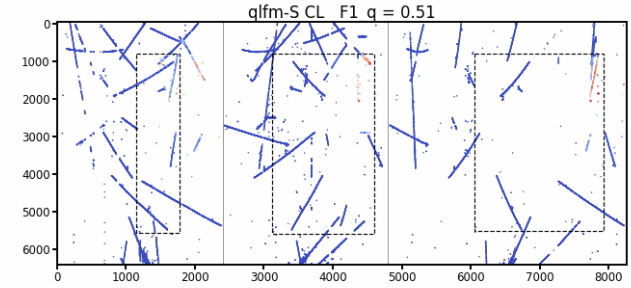


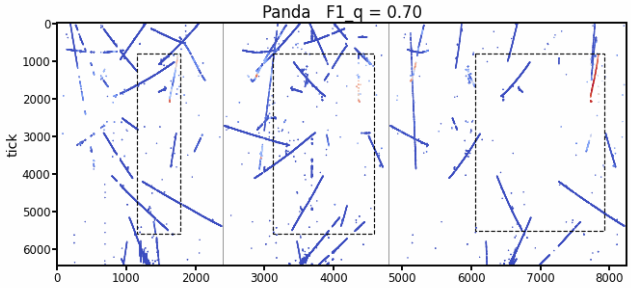


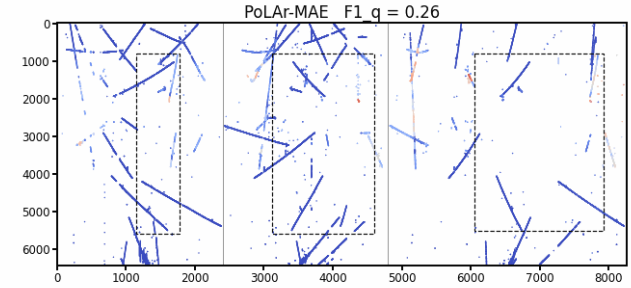


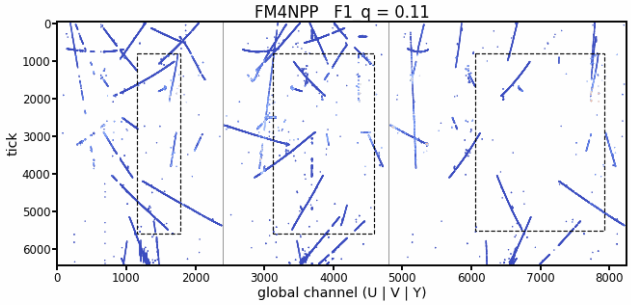


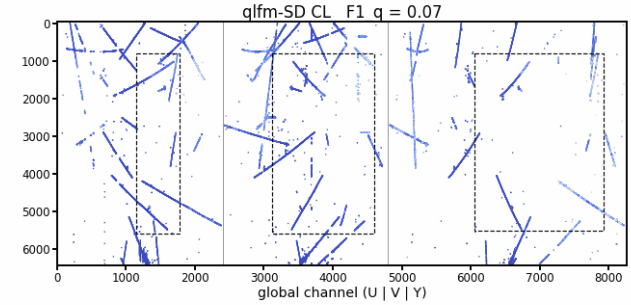












hit colour: p(ν), blue = cosmic, red = ν; dashed boxes: 1–99 % extent of true-ν hits per plane